

NDP3® Therapy manual

Nuffield Centre Dyspraxia Programme

3rd Edition 2004-2012 (NDP3)

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Please note The Nuffield Centre Dyspraxia Programme (NDP3) is intended to be used under the supervision of a qualified speech and language therapist who will apply clinical judgement in selecting appropriate activities and methods for an individual child.

Acknowledgements

Here is the third edition of the Nuffield Centre Dyspraxia Programme (1985; 1992; 2004), which we hope will build on the success of the previous two editions. In producing this edition, we have been both contributors and editors, but we would like to acknowledge the contributions made by many different people:

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Pam Williams and Hilary Stephens
Nuffield Hearing and Speech Centre
2004

Introduction

I am delighted to be finally writing this introduction to the third edition of the Nuffield Centre Dyspraxia Programme.

As far back as 1996, my former job-share partner, Juliette Corrin, and I began to think about the possibility of producing a new edition of the NDP. These preliminary ideas gradually took shape and now, at last the new product is in print.

In 1996, the NDP was already a successful product. Several thousand copies of the previous editions had been sold, both in UK and abroad, such that the NDP had become a standard resource for paediatric speech and language therapists. Adaptations had also been produced in Dutch and Swedish.

Despite this success, we knew our product could be improved and so decided to work on a new edition rather than just reprinting the previous edition. In doing so, we had a number of aims:

To improve the overall “look” of the NDP, by creating new designs and presentations and by both improving and updating the artwork.

To expand the contents of the therapy resources, by adding additional sections to create a complete programme from early sound making to connected speech activities.

To strengthen the assessment and make it more accessible to clinicians. In particular, we wanted to improve the evaluation of the data obtained from the speech and oro-motor tasks, to support treatment planning.

To extend the previous manual, to accompany the assessment and therapy materials, by setting it in a background of published literature and to create a major information resource on developmental verbal dyspraxia.

To make the NDP generally more user-friendly to the busy clinician, by providing a combination of materials to use in therapy sessions and in practice programmes; by providing cross-referencing between the manual and the therapy materials and by providing explanations and resources.

This new NDP package includes:

- 3 Ringbinders of photocopiable therapy worksheets
- A detailed manual contained in a separate ringbinder
- A separate assessment ringbinder, including photocopiable assessment forms and 6 assessment flipbooks
- A large box of therapy cards (4 of each item of single sounds, CV, CVC, CVCV etc pictures)

Undoubtedly, this will not be the end of NDP development. However, for the next few years, I hope that this new product will provide clinicians with a flexible, comprehensive assessment and therapy package, to use in the management of children with severe speech disorders.

Pam Williams
Principal Speech and Language Therapist

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- Mixed Vowel lottos
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Chapter 1: Developmental verbal dyspraxia – a review of the literature

As Shriberg, Aram and Kwiatkowski (1997) pointed out: “The validity of developmental verbal dyspraxia as a childhood speech disorder is one of the most controversial nosological (classification) issues in speech pathology” (p273). Yet, it is clear that “Dyspraxia” seems to provide clinicians with a category or label for children, whose profiles differ from those with straightforward speech delay, who fail to progress in an expected manner and are in some way a “puzzle”.

The term Apraxia or Dyspraxia was first used with reference to adults, who had acquired a problem in programming motor speech output, as a result of brain damage (Broca 1861). Typical characteristics included: observable groping for speech sounds in words, breakdown in production of multisyllabic words, inconsistent and variable production of words and frustration at incorrect speech production.

Almost 100 years later, the term was first applied to the developmental population by Muriel Morley in the 1950s and 1960s, when she identified a “dyspraxic” group of 12 children, aged 4-10 years. She identified the condition as follows: “a defect of articulation which occurs when movements of the muscles used for speech appear normal for involuntary and spontaneous movements....but are inadequate for the complex and rapid movements used for articulation and reproduction of sequences of sounds used in speech” (Morley 1965).

Since this early work, a number of **controversial issues** about dyspraxia have concerned practitioners:

- (a) Does it exist as a distinct disorder?
- (b) What should it be called?
- (c) How should it be defined?
- (d) What is the theoretical understanding of the condition?
- (e) What causes dyspraxia?
- (f) What are the symptoms/characteristics of the condition? i.e. How do we recognise the condition?

To consider each of these individually:

(a) Does it exist as a distinct disorder?

There has been conflicting evidence over the existence of a “dyspraxic subgroup” as distinct from other groups of children with speech disorders. For example the now classic study by Yoss and Darley (1974) distinguished between children with dyspraxia and those with other speech disorders on the basis of neurological ratings (e.g. difficulties with fine motor co-ordination, gait and with diadochokinetic control), volitional control of isolated oral movements and speech errors. However, an attempt to replicate this study by Williams, Ingham and Rosenthal (1981) was unsuccessful. Although the authors suggested a number of plausible explanations for the different findings, it gave support

to Guyette and Diedrich's (1981) conclusion from their literature review: "developmental apraxia of speech is a label in search of a population".

Despite this damning finding, research continued into the condition. Answering Guyette and Diedrich (1981), Milloy (1986) reported her findings from a study into children with speech disorders: "although small, there is evidence of such a population" (i.e. of those with dyspraxia).

However, whether or not a distinct dyspraxic subgroup exists remains debatable and some authors have commented on the difficulties in differentiating this group from other groups of speech disordered children e.g. Stackhouse (1993) observed that the differentiation between the subgroups of speech disorders: phonological disorder and developmental verbal dyspraxia, has proved particularly problematic.

In concluding her review of the literature, Stackhouse (1992a) observed that no dyspraxic subgroup had been empirically identified in the speech-disordered population. Ten years later that position remained unchanged (Williams 2002).

(b) What should the condition be called?

Muriel Morley first used the term developmental articulatory dyspraxia (DAD), but other terms are used to describe the same condition. These include:

- Developmental apraxia of speech (DAS)
- Childhood apraxia of Speech (CAS)
- Developmental verbal dyspraxia (DVD)

In general, DAS or CAS are the preferred terms in the American literature, whereas DVD is now the preferred term in the British and Australian literature.

McCabe, Rosenthal and McLeod (1998) suggest that the terminology used reflects the orientation of the authors. Thus, DAS reflects an oro-motor orientation, DVD a linguistic orientation and DAD, a speech motor orientation. To avoid this issue, some authors have elected to use the global term "developmental dyspraxia". However, this term can also lead to confusions as it is often used to describe the co-ordination difficulties of children with generalised or body dyspraxia, but who do not necessarily have speech problems.

The authors of the Nuffield Dyspraxia Programme used the term developmental articulatory dyspraxia in the first edition, in line with typical UK practice of the time. By the second edition, the term developmental verbal dyspraxia was used instead, reflecting the change of terminology then favoured in the UK. This continues to be our favoured term.

The components of the term "developmental verbal dyspraxia" were described by Morgan Barry (1994) as follows:

- Developmental – the condition was present before speech onset.
- Verbal – the condition has both speech and language aspects.
- Dyspraxia – a difficulty in achieving purposeful, sequential movements in the absence of muscular paresis.

(c) How should the condition be defined?

Numerous attempts have been made to define developmental dyspraxia, which indicate the author's view of the nature of the disorder. For example:

Darley, Aronson and Brown (1975) "Developmental articulatory dyspraxia is defined as a breakdown in the motor planning required to initiate articulatory movements."

Aram and Glasson (1979) "Developmental apraxia of speech is not confined to the articulatory or motor control aspects of speech – rather all levels of expressive language are affected including lexical, syntactic and phonemic aspects".

Byers-Brown and Edwards (1989) "Developmental dyspraxia is an impairment in the selection, planning and programming of linguistic and of motor schemata for production of language."

Milloy and MorganBarry (1990) "A difficulty in initiating, in directing and in controlling the speed and duration of movements of articulation, in the absence of muscular paresis".

Crary (1993) "Developmental Apraxias of Speech are a group of phonological disorders resulting from disruption of central sensorimotor processes that interfere with motor learning for speech".

Ripley, Daines and Barrett (1997) "Developmental verbal dyspraxia is a condition where the child has difficulty in making and co-ordinating the precise movements which are used in the production of spoken language, although there is no damage to muscles or nerves."

In the first two editions of the Nuffield Dyspraxia Programme, no precise definition of dyspraxia was offered. However, it was described as a "neuromotor difficulty in producing the learned patterns of movement required for speech." This would suggest that the authors considered it to be primarily a neurological deficit, affecting output speech processing. However, the change to using the term developmental verbal dyspraxia, rather than articulatory dyspraxia would suggest a shift towards recognising both motoric and linguistic deficits. Connery (1994) writing about the Nuffield Dyspraxia Programme described dyspraxia as "a disorder in neuromotor control of the speech apparatus with resulting difficulties in communication".

(d) What is the theoretical understanding of the condition?

The literature has been dominated by two theoretical approaches:

Developmental verbal dyspraxia is a disorder of motor planning/programming (see definitions above: Darley, Aronson and Brown 1975; Milloy and Morgan-Barry 1990 etc).

Developmental verbal dyspraxia is a linguistic/phonological impairment (see definition: Aram and Glasson 1979 and papers by Ekelman and Aram 1983 and Martin 1974).

In more recent years, some authors have recognised that the symptoms presenting in DVD may arise from a combination of the two approaches i.e. the child with developmental verbal dyspraxia has both a motor and linguistic deficit. (See definition by Byers-Brown and Edwards 1989 and Crary 1993). Crary conceptualised DAS as a group of disorders on a motolinguistic continuum, which could embrace both motoric and linguistic deficits. The weighting of these two components for any particular child will vary, depending on their individual profile of difficulties.

As described above, the approach taken by the authors of the first editions of the Nuffield Dyspraxia Programme would suggest that they viewed articulatory dyspraxia essentially as a motor disorder. By the second edition of the Nuffield Dyspraxia Programme, there was more recognition of the linguistic deficits as part of the condition, developmental

verbal dyspraxia. Therefore, a broader based theoretical understanding recognising both motoric and linguistic deficits was adopted and this continues to be our viewpoint.

(e) What causes the condition?

To date no definite cause has been found, but neurological and genetic factors have been implicated:

Genetic factors

Both in the literature and in clinical practice, it has been noted that there is often a strong family history, with other family members also presenting with speech, language and/or learning difficulties.

One particular family has been studied in depth – the KE family – a large 3-generation family, in which about half the members are affected by a severe orofacial and verbal dyspraxia, as well as expressive language difficulties. Recently, a linkage study was carried out, which indicated that the affected members had damage to an area of chromosome 7, and it was confirmed that the condition was inherited, through autosomal dominance. (Fisher, Vargha-Khadem, Watkins, Monaco and Pembrey 1998)

Further work is now required to replicate these findings in other families, in order to generalise the genetic origin of the condition.

Neurological factors

In the acquired condition, apraxia is associated with lesions in Broca's area and the sensorimotor cortex. In contrast, it has been difficult to find evidence of "hard" neurological aetiology in the developmental condition. Some authors have hypothesised possible brain sites which may be implicated in dyspraxia e.g. the posterior parietal cortex and the corpus callosum, (Rapin 1982).

The study of the KE family (referred to above) employed the use of MRI scans (to investigate structural brain abnormalities) and PET scans (to investigate functional brain abnormalities). The findings implicated the caudate nucleus as being the site of primary pathology, which the authors thought could plausibly account for the presenting orofacial and verbal dyspraxia. Further work is required before any generalisations can be made.

In contrast to the lack of "hard" neurological evidence, "soft" neurological signs have often been reported e.g. clumsiness. For example: In Crary's (1984) review of 25 case files: 52% had generalised motor problems (dyspraxia).

This would seem to match our clinical impression at the Nuffield Centre. Approximately half of the children have generalised motor co-ordination difficulties in addition to their speech difficulties.

(f) What are the symptoms/characteristics of the condition? i.e. How do we recognise it?

If it is accepted that Developmental Verbal Dyspraxia is a discrete clinical entity, which can occur independently of other speech disorders, there should be one or more diagnostic features or characteristics, which can be used to identify the condition.

Despite numerous attempts over 40 years (Davis et al 1998), little agreement has been reached on this subject. Possible diagnostic markers have included: performance on oro-

motor sequencing tasks (Rosenbek and Wertz 1972), performance on diadochokinetic tasks (Yoss and Darley 1974; Riley 1984), inappropriate stress (Shriberg, Aram and Kwiatkowski 1997), slow progress in therapy (Macaluso-Haynes 1978; Jaffe 1984; Pannbacker 1988) etc.

Due to the difficulties in identifying a diagnostic marker, it has been suggested that developmental verbal dyspraxia can instead be recognised through a "Symptom Cluster" consisting of a range of features and characteristics. However, the diversity of possible symptoms further highlights the heterogeneity of the "dyspraxic" population.

Hall (1992) proposed that symptomatology lists should be used to identify "dyspraxic features" in a child's speech rather than to categorise him as having, or not having, the condition "dyspraxia". Hall (1992) argued that by adopting such an approach the ongoing debate over the existence of developmental verbal dyspraxia, as a discrete disorder, then becomes irrelevant.

A number of authors have produced symptomatology lists/characteristics of the disorder, through literature reviews (Jaffe 1986; Pollock and Hall 1991; Stackhouse 1992a; Davis et al 1998; McCabe et al 1998). In addition to speech characteristics, these checklists include the complex range of language, learning and motor difficulties that can occur at different points in time.

Such an approach was adopted by the authors of the first two editions of the Nuffield Dyspraxia Programme.

In the 1992 edition, 8 core characteristics were proposed, covering speech (segmental and supra-segmental), oro-motor and language aspects.

In recent years, the Speech and Language Therapists working at the Nuffield Hearing and Speech Centre have expanded this list into sections of: Speech characteristics, Treatment characteristics and Other co-occurring characteristics:

Speech characteristics

- Limited range of consonant and vowel sounds
- Vowel distortions
- Favourite articulations
- Errors of omission and substitution
- Glottal stop insertion and substitution
- Inconsistent production
- Breakdown in sequencing at: CV, VC, CVC, CCV, CVCV, multisyllabic word levels, and/or connected speech levels
- Prosodic problems (with rate, stress, intonation, rhythm)
- Resonance difficulties (hypernasal, hyponasal, mixed nasality)
- Voice difficulties (lack of control over volume, pitch, length; atypical quality)
- Unintelligible speech

Treatment characteristics

- Slow progress in therapy
- Resistance to therapy
- Difficulties with generalisation

Other co-occurring characteristics

- Family history of speech and language difficulties
- Delayed language development
- Delayed development of early speech skills
- Feeding difficulties
- Oral dyspraxia (affecting lips, tongue, palate, larynx)
- Generalised developmental dyspraxia (co-ordination difficulties)
- Reading, Spelling, Writing difficulties

To examine these characteristics further:

Speech characteristics

The **segmental speech errors** presenting in developmental verbal dyspraxia result from difficulties in eliciting (single sounds), in sequencing (between sounds) and in connecting (sounds, syllables, words) (Morgan-Barry 1994).

Typically, the child presents with a limited phonetic inventory, with both vowels and consonants affected. Many children at four or five years may have only 5 or 6 consonants and a few vowels in their sound repertoire. In addition, they often have difficulty in imitating adult productions.

Speech system: typical errors include: vowel distortions (imprecise realization of vowels, neutralizations or actual vowel substitutions); errors of omission and substitution (place, manner and voice errors, and combinations of feature errors); glottal stop insertions and substitutions; favourite articulations (Ingram 1976) i.e. one consonant used for virtually all others e.g. /d/. It has often been noted that errors increase with word length and complexity of the production i.e. these add to the articulatory load. Inconsistency of production has also sometimes been reported: both one word pronounced differently on different occasions (e.g. bat: bap, bak, bab, baʔ etc) and one sound realised by a different substitution in different words, depending on the phonetic environment (e.g. /k/: car, tat, etc). Although inconsistency can be an encouraging sign of progress in therapy (indicating that the child's system is widening, enabling him to change error forms to correct productions (Dodd 1995), inconsistency for many children with dyspraxic difficulties is not such a positive feature.

Speech structure: there is a breakdown in control of sequencing at one or more levels of phonotactic structure: CV, VC, CVC, CCV, CVCV, CVCVC, multisyllabic word levels and/or connected speech levels. Children with dyspraxic difficulties not only have difficulties in producing individual sounds, they also experience difficulties in combining sounds together to produce words (at different levels of complexity). Typically, this results in more complex phonotactic structures being reduced to simpler structures; weak syllables being deleted or reduced and structural errors e.g. epenthesis and metathesis occurring. Whatever the exact presentation, speech structure disturbances will have a significant effect on intelligibility.

See also Appendix 1: Speech error patterns in children with DVD, for further information.

Suprasegmental or prosodic difficulties

It has been noted increasingly in the literature that many children with developmental verbal dyspraxia have difficulties with one or more aspects of prosody e.g. rate, rhythm, intonation and stress patterns. Different presentations have been reported e.g. fast or

slow rates of production; a lack of rhythmic organisation, leading to jerky or staccato production; and a lack of stress and intonation, leading to monotonous production.

Theoretically, it has been debated whether these presenting features result as a consequence of motoric difficulties (Darley et al. 1975) or are evidence of core linguistic organisational deficits (Shriberg et al 1997; Velleman and Strand 1991). Shriberg has suggested that for one subgroup of developmental verbal dyspraxia, a stress pattern known as "Excess equal stress", characterised by forceful, monostressed utterances, may be the diagnostic marker for the condition. Velleman and Strand (1991) have also highlighted the importance of intonation, proposing that the core deficit is a difficulty in utilising linguistic frames, which underlie syllabic structure.

Resonance effects can be quite a common feature in DVD, resulting from incoordination of the palatopharyngeal sphincter. For example: in Shriberg, Aram and Kwiatkowski's (1997) review, 6/14 of their participants, aged 4-14 years had resonance effects. Both hypo and hypernasality can occur. Hypernasality can be severe and give the impression of a structural abnormality e.g. cleft palate, but on examination no structural abnormalities are detected. Clinical experience suggests that management of hypernasality can be particularly difficult and may require surgical intervention.

Incoordination of the vocal tract

Since Edwards (1984) suggested that dyspraxia could affect respiratory and laryngeal control as well as oro-motor control, there has been increased recognition of laryngeal incoordination in developmental verbal dyspraxia. Clinical findings indicate that for some children there are effects on: control of breathing for speech, timing and ordering of vocal tract sequences (e.g. leading to voicing errors) and production of voice in terms of volume, length, pitch and overall quality. Typical voice characteristics include: overloud volume or overquiet volume, inability to sustain vocalisation, lack of pitch variation or limited pitch range, harsh or "gravelly" or husky voice quality.

Unintelligibility

The combination of speech system errors, speech structure errors and suprasegmental features results in a loss of clarity for the listener. The individual profile of difficulties, the severity of those difficulties as well as context will affect the degree of unintelligibility.

Speech Processing Deficits

Stackhouse (1992a) considered the surface speech difficulties, characteristic of developmental verbal dyspraxia in speech processing terms and proposed that they could be the result of a single deficit or a combination of 3 levels of breakdown:

- Phonological misrepresentations in the lexicon
- Inability to plan the speech output
- Vocal tract incoordination

This view of developmental verbal dyspraxia as a multi-deficit disorder involving programming, planning and vocal tract control was supported by Ozanne (1995).

(See Chapter 3 - Assessment for further information on speech processing.)

Treatment characteristics

Slow progress in therapy

Clinicians often report that children with developmental verbal dyspraxia progress at a slower rate than children with other speech disorders. In particular, a comparison is made with children with “pure” phonological disorders, who often make rapid progress in therapy and generalise skills learnt rapidly into connected speech. In the literature, some authors have suggested that the diagnosis of developmental verbal dyspraxia should be based not only on speech symptoms, but also on the longevity of their rehabilitation process (Court and Harris 1965; Macaluso-Haynes 1978; Pannbacker 1988).

An analysis of the difficulties facing a child with developmental verbal dyspraxia illustrates why progress is likely to be much slower: they have to “unlearn” incorrect patterns, they have to learn new motor programs, which requires endless repetition and practice, they need to learn to combine individual motor programs together, they need to apply their new motor skills to their phonological acquisition and they require much support to generalise new skills from one level to another (see Treatment planning and therapy chapters for further information).

Generalisation

Clinicians frequently report that although children make progress in “controlled” therapy sessions, they have difficulties in transferring newly learned skills to their spontaneous speech. Alternative explanations for this observation can be proposed, based on different learning theories. Based on motor learning theory this is likely to occur because the new skills are not sufficiently automatized and therefore require more practice before carryover can occur. Carefully graded therapy steps, increasing the level of complexity of task demands, can help the generalisation process.

The therapy approach offered by the Nuffield Dyspraxia Programme includes work not only at a single word level, but also at a phrase and clause level, thereby helping the child bridge therapy exercises and connected speech. Coupled with the development of automatic movement patterns through frequent repetitive practice, this graded therapy approach should enable the child to make the carry-over into connected speech (Connery 1994).

(Also see chapter 2 on Therapy approaches and chapter 4 on Treatment Planning for further discussion of some of these points).

Other co-occurring characteristics

Family history of speech and language difficulties

As noted above, a strong family history is often noted, at case history interview. In clinical practice, it is common for clinicians to treat siblings or other relatives.

At the Nuffield Hearing and Speech Centre, many families are seen with more than one family member affected. In the case of siblings, it is noteworthy that the individual diagnostic profiles of the children seen often differ considerably, which highlights the heterogeneity of this group as a clinical population.

Delayed language development

As reported earlier, it has been well documented that many children with developmental verbal dyspraxia have delayed language development. The exact presentation varies, but verbal comprehension is often less severely affected than expressive language. In some cases, language disorder rather than just delay occurs.

In the literature, two particular areas of linguistic delay/disorder have been investigated: (a) Phonology and (b) Syntax.

(a) Phonology

As Crary (1993) commented: "There is a common acknowledgement that the primary symptom of dyspraxia is a severe phonological deficit".

In his case file review of 25 subjects, he identified 25 phonological errors, which accounted for 90% of all the speech errors noted: final consonant deletion, cluster reduction, stopping, gliding, fronting, backing, vowelisation, weak syllable deletion, prevocalic voicing, initial consonant deletion, vowel neutralisation, labialisation and glottal replacement.

Both paradigmatic processes (sound system processes) and syntagmatic processes (syllabic structure processes) have been identified. There has been some debate in the literature as to which are most prevalent in the speech of children with developmental verbal dyspraxia. Crary (1984a) identified more syntagmatic errors in his subjects, whereas Parsons (1984) identified more paradigmatic errors. Stackhouse (1992a) suggested that this discrepancy may be explained by the different ages of the children in the studies (Parsons: 3.11 – 7.9 years and Crary: 3.9 – 13.11 years). Paradigmatic processes result mainly from articulatory immaturity and syntagmatic processes from higher level programming skills. The younger the child, the more likely it is for both types of processes to occur. However, as the child with developmental verbal dyspraxia matures, paradigmatic processes are likely to be overcome, leaving syntagmatic processes.

(b) Syntax

A small body of work has investigated the syntactic difficulties of children with developmental verbal dyspraxia. Ekelman and Aram (1983) collected 50 utterances from 8 children, previously diagnosed as having developmental verbal dyspraxia and in age range 4.4 years to 11.11 years. They found them to have difficulties with sentence structure, grammatical categories (e.g. main verb, personal pronouns etc) and grammatical markers (e.g. copulas, modals, auxiliaries, past participles, regular and irregular past tense, 3rd person singular, inversion of question forms etc.)

In seeking to explain the findings for grammatical marker errors, the authors suggested that they may occur because the elements are usually unstressed and are said quickly or alternatively may arise from a more basic underlying organisational deficit affecting both speech and language. However, Adams (1990) noted that children with developmental verbal dyspraxia are very variable in their syntactic abilities. This would certainly be the view from clinical practice.

Delayed development of early speech skills

The majority of children with developmental verbal dyspraxia are reported to have been slow to speak. In particular, there is often limited early vocalisation, late development of

babbling (or lack of this usual preverbal skill), and late emergence of first words. Imitation skills are often reported to be poor. Such children are often described by their parents as having been quiet, “good” babies.

Feeding difficulties

Some children with developmental verbal dyspraxia are reported to have been slow in sucking as babies and to have had difficulty moving on to solid food. They are sometimes described as slow, messy eaters and may have difficulties chewing harder foods e.g. meat.

A smaller number of children have persistent difficulties and require food to be pureed for some years.

It is debatable whether feeding difficulties (early or ongoing) are linked with oral difficulties (Ripley, Daines and Barrett 1997). There is also debate over the possible carry-over of improved control over feeding behaviours to speech performance. Although some studies have suggested that facilitating improved feeding in babies and young children may improve later speech performance (Edwards 1984), other authors have found such activities to be pointless, as motor programming for feeding does not involve the same neural circuits as for speech (Stark 1985). Marshalla (1985) advised that it was important to remember that “whether the child is speaking or swallowing the principles that govern the movement of the oral articulators are the same. In other words, the young child learns to form sounds and swallow in certain patterns because of the way the tongue is designed to move” (p317).

Oral dyspraxia

“Oral dyspraxia is present when the child is unable to copy volitional movements of oral organs i.e. tongue, lips, jaw and face on command or demonstration in the absence of attempts to articulate” (Milloy 1991).

Typically, the child has difficulties with carrying out oral movements on verbal request/or through copying – may be single movements or sequences of movements. The literature suggests sequences of movement are particularly difficult. Drooling/dribbling may be indicators of poor control of oral movements.

The key presentation of oral dyspraxia is that the observed difficulties are with volitional control. The same actions can be performed automatically e.g. although a child may be able to lick an ice cream, if it is placed close to his mouth, he is unable to protrude his tongue on request (stick out your tongue) or by copying a therapist’s demonstration.

It remains debatable within the literature whether oral dyspraxia always accompanies verbal dyspraxia. Morley’s (1965) original description was that “a concomitant oral apraxia is frequently, but not always present”. Since then, some authors have proposed that the two always co-exist (Eisenson 1972), whereas others consider that the two conditions can occur independently (Yoss and Darley 1974).

In Crary’s (1984b) case file review of 25 children, with apraxia of speech, aged 3.2 years – 4.10 years, showed 13 to have oral deficits. (No data was available for 7 of the cases.) Crary (1993) summarises the literature position as follows: “Oral apraxia is considered a dominant, although not mandatory, attribute of developmental apraxia of speech”.

Historically, the approach taken at the Nuffield Hearing and Speech Centre in the early 1980s regarded oro-motor control difficulties as being at the heart of speech motor

control difficulties. For this reason, therapy for children with developmental verbal dyspraxia, using the Nuffield Dyspraxia Programme usually started with oro-motor exercises (for lips, tongue, palate and larynx), which underpinned subsequent speech work. In the light of more recent theory and evidence, we have modified our approach and generally recommend working only on oro-motor skills underlying specific articulatory features, in order to facilitate sound production. (See chapter 5, Expanding the Phonetic Inventory.)

The assessment of oral skills is not a simple procedure (see chapter 3, Assessment: oro-motor skills and Williams and Stackhouse 2000 for further detail). Despite frequent references in the literature to children having oro-motor control difficulties, there has been surprisingly little normative information available to compare the performance of young children with speech disorders on oral tasks (See Normative information in Appendix 7).

Observation in clinical practice as well as evidence from research studies has highlighted the variability in clinical presentation – some children with typical features of verbal dyspraxia have severe oro-motor difficulties, but others do not (Evans 1994). In addition, it has been noted that there is not a clear cut correspondence between severity of oro-motor difficulty and severity of speech difficulty. Some children with very poor speech skills have surprisingly good oro-motor skills and some children with quite good speech skills have poor oro-motor skills.

Such findings have led some authors to suggest that control of oro-motor skills and speech motor skills are independent (Lancaster and Pope 1989). However, other authors e.g. Crary (1993) report that there is a close neurological relationship between non-speech oral movement and speech production.

There is also evidence that oro-motor skills respond well to training (Stackhouse 1992a), which may be relevant and also age factors may be implicated – for example, poor oro-motor skills (in the absence of muscular paresis) may be more common in younger children and improve with age.

Traditional oro-motor work concentrated on exercises for the lips, tongue and palate. In recognition that other levels of vocal tract control may be involved e.g. laryngeal control and respiratory/phonatory control (Edwards 1984) and control over facial expressions etc, exercises for the larynx and breathing for speech, as well as exercises for facial control, have been included in therapy approaches (Connery 1992; 1994; Milloy and MorganBarry 1990; Ripley, Daines and Barrett 1997).

Performance on diadochokinetic tasks (DDK)

This is a very frequently cited characteristic in the literature and, as Ozanne (1995) pointed out, has been the most commonly used criteria for subject selection in research projects. Recent work (Ozanne 1995) has questioned the role of DDK tasks in the differential diagnosis of developmental verbal dyspraxia, as many children with speech disorders have difficulties performing these tasks (i.e. not only those with developmental verbal dyspraxia). The search has been on to provide norms for comparison purposes (Fletcher 1978; Henry 1990; Williams and Stackhouse 1997; 2000).

Generalised developmental dyspraxia

As noted earlier, some children with developmental verbal dyspraxia also have generalised motor incoordination difficulties (Gubbay 1978; Kirby 1999). The exact relationship between the spoken and generalised conditions is unclear – clinical practice suggests that not all children with developmental verbal dyspraxia also have generalised difficulties. Equally, not all children with generalised incoordination difficulties have speech difficulties.

Reading, Spelling and Writing difficulties

There is increasing recognition that there is a relationship between spoken and written language disorders (Stackhouse 1992a).

The child with developmental verbal dyspraxia has a breakdown in phonological processing which leads to difficulties in acquiring phoneme-grapheme correspondence rules. In addition, the child with developmental verbal dyspraxia has inconsistent and variable speech output which hampers successful segmentation and blending skills needed in reading and spelling.

Bishop and Adams (1990) put forward the "Critical Age Hypothesis" proposing that if a child continued to demonstrate marked speech and/or language problems beyond age 5 years, he would be at risk for literacy difficulties. Bird, Bishop and Freeman (1995) augmented this proposal by reporting their findings that children who have severe expressive phonological impairments (including developmental verbal dyspraxia), at the time they start school are particularly at risk for reading and spelling problems. Subsequent work has supported this view (Stackhouse, Goulondris and Snowling 2002).

In addition to these phonological processing difficulties, handwriting and drawing skills will also be affected if the child has generalised motor dyspraxia.

Auditory processing

In concluding this list of characteristics, a comment should be made about auditory processing in children with developmental verbal dyspraxia. Historically, the view was held that the primary difficulty for children with dyspraxia was motoric and that listening and auditory discrimination skills should not be affected (Macaluso-Haynes 1978). As Eisenson (1984) explained: "Most children with oral apraxia who are also of normal intelligence are likely to do as well as their peers in tests for auditory discrimination, such as Wepman's..." (p.235).

More recently, this rather unitary view has been challenged, by studies which have demonstrated that children with dyspraxic type difficulties can have poorer auditory processing skills than age-matched peers (Bridgeman and Snowling 1988; Stackhouse and Snowling 1992), as part of their speech processing profile. Recognition of the different speech processing demands of auditory tasks has also contributed to a better understanding of auditory processing in children (Stackhouse and Wells 1997 and see Chapter 3, Assessment this volume for further discussion).

Historically, the view held by the Speech and Language Therapists at the Nuffield Hearing and Speech Centre, was the traditional one e.g. as described by Eisenson. The list of typical characteristics of developmental verbal dyspraxia concluded with the statement: "whilst usually having normal auditory discrimination" (1986). This was stated for the main purpose of indicating that a therapy approach for "phonological disorders" focusing

on listening and auditory discrimination was not addressing the primary motoric deficit for children with developmental verbal dyspraxia.

In more recent times, it has been acknowledged that children with speech disorders, including developmental verbal dyspraxia, can have different profiles of speech processing breakdown (see diagnostic issues below and Chapter 3: Assessment for further discussion of this topic). Some children with developmental verbal dyspraxia will have auditory processing difficulties and therefore a therapy programme will need to be designed to address these difficulties.

Diagnostic issues

(a) Symptomatology lists

The use of symptomatology lists in diagnosing speech disorders is one common approach used in clinical practice. The clinician attempts to identify features from the child's speech pattern and other areas of performance from a checklist of characteristics, proposed for a particular disorder e.g. Jaffe 1984; Pollock and Hall 1991; Stackhouse 1992; McCabe et al 1998 have all produced checklists for developmental verbal dyspraxia. Also see this chapter for the characteristic lists used by speech and language therapists at the Nuffield Hearing and Speech Centre.

An extension of the symptomatology approach is to compare the characteristics of two different but related speech disorders e.g. dysarthria/dyspraxia; dyspraxia/phonological impairment in an attempt to make a differential diagnosis (See Tables 1 and 2 for examples of this dichotomy approach).

Table 1: Differential diagnosis: Dyspraxia - Dysarthria

Dyspraxia: is a difficulty in initiating, in directing and in controlling the speed and duration of movements of articulation, which occurs in the absence of obvious neuromuscular abnormality (Milloy and MorganBarry 1990).

Dysarthria: is an impairment of movement and co-ordination of the muscles required for speech, due to abnormal muscle tone (Milloy and MorganBarry 1990).

Comparison between dyspraxia and dysarthria (Johns and Darley 1970)

Dyspraxia	Dysarthria
Inconsistency in articulation performance; difficulty in predicting errors	Speech errors are consistent and predictable
Most common types of articulation errors are substitutions and repetitions	Articulation errors are primarily distortions
Discrepancy between voluntary, purposeful and spontaneous reflexive performance	Accuracy of articulation does not vary with situations (spontaneous speech, reading or imitation)
Prosody - rate, rhythm, and stress of speech are adversely affected by repetitions, hesitations and groping articulation	Slow laboured speech is present with evidence of strain and tension, especially for difficult consonant clusters
Articulation is poorer as words increase in length	Articulation is poorer as words increase in length

Table 2: Differential diagnosis: Dyspraxia - Phonological Disorder

Verbal dyspraxia: is a disorder of the implementation of articulatory gestures and/or programmed sequences of articulatory movements. Thus the child is not able to develop a normal phonological system.

Phonological disorder: is a disorder of the systematisation of meaning contrasts at word and morphemic level, dependent on a consistent inventory of consonants and phonotactic structures (i.e. CV, CC, CVCV etc) and a knowledge of the restrictions of their occurrence in English.

Comparison between verbal dyspraxia and phonological disorder

Verbal dyspraxia	Phonological disorder
Unable to develop a system of speech sounds	Restricted system of speech sounds
Problems with vowels.	No vowel problems
Problem sequencing speech sounds	No problem of sequencing sounds
Do not generalise new sounds into a system	On learning a new sound, generalise it into their system
Prosodic problems common	Prosodic problems rare
Extremely slow progress in therapy and does not respond well to breaks	Progresses well with therapy, improves with maturity/consolidation
Oro-motor difficulties	No oro-motor difficulties
Word production can be worse in imitation than spontaneous production	Word production better in imitation than spontaneous production

Dodd (1995), in particular, has adopted a Subgrouping approach to the identification of speech disorders. She proposed 4 subgroups, based on detailed descriptions of speech output: delayed phonological development, consistent deviant phonological disorder, inconsistent deviant phonological disorder, and articulation disorder. An additional label of developmental verbal dyspraxia might be appropriate for children who have a multi-deficit motor speech disorder (Ozanne 1995).

In such an approach, a speech sample is analysed and evaluated and an attempt is made to find the best match for the identified speech characteristics.

However, the use of this subgrouping approach is not without pitfalls: the clinical validity of such diagnostic subgroupings (e.g. dyspraxic, phonological etc) depends on sound knowledge of the underlying theoretical distinctions, and a treatment protocol that is equally distinctive i.e. one assumes these form different treatment groups.

Although such an approach can clearly be helpful at times, it does rely on clear cut matching between an individual child's speech profile and the labels. Difficulties can occur if the child does not seem to match any of the categories exactly. This is well illustrated by symptomatology lists proposed for developmental verbal dyspraxia, which have not proved simple to use. Questions over their use include:

- How many of the characteristics are required to make a diagnosis of developmental verbal dyspraxia?
- Are some symptoms more important than others?
- Will the presentation be different at different ages?

These points have been taken up by authors such as Jaffe (1984) and Stackhouse (1992a):

1 How many symptoms are required? and which symptoms are most important?

Jaffe (1984) described the complications of this approach for developmental verbal dyspraxia: "Not all symptoms must be present; no one characteristic or symptom must be present; and the typically reported symptoms are not exclusive to developmental apraxia of speech."

Stackhouse (1992a) proposed that evidence is required from all four perspectives (clinical, cognitive, linguistic and phonetic) before a diagnosis of dyspraxia can be made. Evidence from speech characteristics alone (i.e. phonetic perspective) should not be used as a basis for diagnosis.

Shriberg et al (2003) The standard inclusionary guideline is that a child has to be attested as positive on a minimal number or percentage of items on the checklist (e.g. over 50%) to be classified as having suspected apraxia of speech.

2 Change over time?

Jaffe (1984) recognised the changing nature of the characteristics of developmental verbal dyspraxia as children mature: "Compounding the problem (with symptomatology) is the observation that children change over time."

This point was taken up by Stackhouse (1992a), who advocated the need to adopt a developmental perspective in the identification/diagnosis of developmental verbal dyspraxia. By adopting such an approach, key features of the disorder will vary with the developmental stage of the child and the range of problems experienced will unfold as the child progresses and more demands are placed upon him.

The following example illustrates this approach: For a young pre-school child, there may be virtually no expressive speech to use as a basis for analysis, with the suspected diagnosis resting upon case history factors and the obvious difficulty making both speech and other oral movements. For the older pre-school child who has acquired a good body of expressive speech, it may be the pattern of speech error itself that is distinctive of dyspraxia. For the primary school child, speech may be largely clear, but the subtle problems with multisyllabic words and learning to read and spell are the telltale signs of a dyspraxic history (Williams and Corrin 1998).

Shriberg et al (2003) advise caution in the use of checklists. In their view, long lists of clinical features "cast too wide a net" and limit specificity. They suggest that as a deficit in praxis is, by definition, central to diagnosis of dyspraxia, 8 core defining speech features (5 segmental and 3 suprasegmental) are sufficient. These are:

Segmental

- Articulatory struggle (groping) particularly on word onsets
- Transpositional (metathetic) substitution errors, reflecting sequencing constraints on adjacent sounds
- Marked inconsistencies on repeated tokens of same word type
- Proportionally increased sound and syllable deletions relative to overall severity of involvement
- Proportionally increased vowel/diphthong errors relative to overall severity of involvement

Suprasegmental

- Inconsistent realization of stress (i.e. prominence on syllables or words)
- Inconsistent realization of temporal constraints on both speech and pause events
- Inconsistent oral-nasal gestures underlying the percept of nasopharyngeal resonance

(b) Speech Processing models

An alternative approach to using symptomatology lists in making diagnoses of speech disorders is to employ the use of speech processing models (Waterson 1987; Hewlett 1990; Stackhouse and Wells 1997). Such Psycholinguistic models account for speech difficulties as resulting from breakdown at one or more levels of speech processing: within the input system, within the output system and/or within the stored representational systems.

Through the use of psycholinguistic models it is possible to distinguish between speech disorders in broad terms by speech processing breakdown e.g. dysarthria as affecting the execution level of speech output; dyspraxia as affecting the motor programming and planning levels of speech output and phonological impairments as affecting the representational level of speech processing.

However, in addition, and perhaps more usefully, adopting a psycholinguistic approach allows a clinician to profile the difficulties demonstrated by an individual child. Systematic assessment enables a detailed profile of the child's relative speech processing strengths and weaknesses to be drawn up, which then help to provide therapy goals. Adopting such an approach can also enable the speech and language therapist to "transcend labels" e.g. dyspraxia, phonological disorder etc. Instead an umbrella term "Speech disorder" can be used, provided it is supported by a detailed profile of the child's speech processing performance.

(See Chapter 3, Assessment for a more detailed discussion of a psycholinguistic approach to the assessment of speech disorders)

At the Nuffield Hearing and Speech Centre, we have been utilizing this approach for some years – in particular, adopting Stackhouse and Wells Psycholinguistic Approach (1993; 1997). This involves administering and evaluating a detailed speech processing assessment and then drawing up a child's individual speech processing profile. Although in some cases a straightforward diagnosis and label of developmental verbal dyspraxia may still be given, more commonly a description is given e.g. "Jo has a (mild, moderate, marked, severe, very severe etc) speech disorder, characterised by dyspraxic type features, such as: a limited speech sound inventory, vowel distortions, frequent errors of omission and substitution, a breakdown in sequencing beyond CV level, prosodic difficulties and marked unintelligibility."

In conclusion

Identification and diagnosis of developmental verbal dyspraxia continues to be debated in the literature but as Shriberg, Aram and Kwiatkowski (1997) observed, no consensus has been reached – "Currently there is no descriptive, diagnostic or theoretical consensus on the nature and origin of developmental apraxia of speech". To date, this position remains unchanged.

However, despite all the controversy, it appears that for many clinicians this diagnostic category or descriptive label (dyspraxic) continues to be useful. The results of clinicians surveys (Williams, Packman, Ingham and Rosenthal 1980; Murdoch et al 1984) suggested that clinicians have definite views concerning features of developmental verbal dyspraxia but they do not always mirror those reported in the literature (McCabe et al 1998). Observation of clinicians' practice led Shriberg et al (1997) to comment that the label of "suspected DAS" may provide clinicians with a functional solution to children whose profile differs from those with straightforward speech delay, who fail to progress in an expected manner and who are in some way a "puzzle". However, Davis et al (1998) warned that clinicians may in fact apply this term rather too freely to any child who presents with multiple articulation errors and a very slow rate of progress.

At the Nuffield Hearing and Speech Centre, the speech and language therapists continue to find "developmental verbal dyspraxia" a highly valuable clinical description which focuses service delivery in terms of assessment, treatment and educational provision to the considerable benefit of the children and families concerned.

For more recent reviews of the literature, please see:

1. ASHA (2007) Position Statement and Technical Report on Childhood Apraxia of Speech.
2. RCSLT (2011) Policy Statement on Developmental Verbal Dyspraxia.

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Addendum

Subsequent to the writing of this chapter, ASHA (American-Speech-Language and Hearing Association) Ad Hoc Committee on Childhood Apraxia of Speech, published a draft Position Statement and Technical Report in February 2006. The committee was set up in 2002 and undertook narrative reviews of peer-reviewed literature published in the past 10 years with additional sources consulted as needed for coverage of certain topics.

Following extensive peer-review by researchers and clinicians, the committee published a second draft of the Position statement and Technical report in November 2006. The finalised versions are still awaited at the time of writing (March 2007). Readers are directed to the ASHA website for further information. A summary of the key findings from the 2nd draft of the Position Statement is included after the Appendix for our readers' information.

March 2007

Chapter 2: Management of developmental verbal dyspraxia - a review of the literature

Therapy approaches

Just as there have been numerous papers written on the theoretical aspects of developmental verbal dyspraxia, there have also been many attempts to propose therapy approaches and strategies for managing the condition. The confusions over the existence of the disorder and its exact presentation have both helped and hindered management. What distinguishes one therapy approach from another is the author's theoretical understanding of the condition: as a motor programming/planning disorder or as a phonological/linguistic disorder or as a combined motor/linguistic disorder (see theoretical review). Principles from a number of sources have been included e.g. management of adults with acquired apraxia of speech, management of other motor speech disorders e.g. dysarthria, management of functional articulation disorders and management of phonological disorders.

The Nuffield Dyspraxia Programme (1985; 1992) was produced as a clinical tool to address the speech difficulties of those children attending the Nuffield Hearing and Speech Centre, thought to have dyspraxic difficulties. Although no specific theoretical framework was adopted, the authors were clearly influenced to some extent by the accepted theoretical views and management advice of the time (late 1970s and early 1980s). In this review of the literature on management approaches, reference will be made to the Nuffield Dyspraxia Programme and its design, where appropriate, as well as to the clinical work at the Nuffield Hearing and Speech Centre and at the Nuffield Speech and Language Unit.

A focus on speech production

Macaluso-Haynes (1978) reviewed therapy approaches from the available literature of the time and summarised them into the following headings:

Concentrated drill on performance, both in imitation and on command, of tongue and lip movements.

Imitation of sustained vowels and consonants followed by production of simple syllable shapes (e.g. CV, VC, CVC).

Movement patterns and sequences of sounds – focus should not only be on the production of sounds in isolation; the clinician should attempt a rapid transition from individual vowels and consonants to the production of syllables and words.

Avoidance of auditory discrimination drills – these are not generally necessary as most children with apraxia do not have specific auditory discrimination difficulties.

Slow rate and self-monitoring – heightening of self-monitoring skills and a slower rate (often utilizing an even stress pattern) help to increase intelligibility.

Core vocabulary – the introduction of a core set of simple words that the child can articulate accurately (or nearly accurately) can help develop intelligible verbal communication.

Carrier phrases – the introduction of stereotyped “carrier phrases” e.g. “I want a ...”; “I see the ...” can help to extend sequencing efforts.

Rhythm, intonation and stress paired with motor involvement (e.g. foot/finger tapping) – singing, incorporating rhythm and stress, has been found helpful in facilitating motor sequencing.

Intensive, frequent and systematic drill – ideally, daily drill sessions are advised to develop habituation of accurate motor planning for speech.

Orosensory perceptual awareness – multisensory stimuli should be used to increase awareness of the orosensory mechanism. Icing, brushing, rubbing, touching and deep pressure and resistance techniques should be employed. Principles from Proprioceptive neuromuscular facilitation (PNF) can be useful to improve sensori-motor patterning and feedback.

Physical therapy – for children who also have generalised motor difficulties, collaborative working with physiotherapists and occupational therapists is advised to work on general motor planning and co-ordination.

Macaluso-Haynes’ (1978) review illustrates that the prevailing view of the time was that developmental verbal dyspraxia was a motor programming or planning disorder. The therapy approaches advocated were therefore essentially “bottom-up” in focus i.e. aimed at improving the execution of speech. As Crary (1993) p200 explains: “Initial emphasis is placed on non-speech oral movements and/or single consonant or vowel production and the clinician and child attempt to build a complex and functional speech production system from that therapeutic groundwork.”

The Nuffield Dyspraxia Programme (1985; 1992) shares many of these bottom-up features e.g. work on lip and tongue movements and orosensory awareness, single sound production, transition from single sounds to words of gradually increasing complexity, drilling techniques, control of prosodic features, development of a core vocabulary etc.

Ten years later **Pannbacker (1988)** carried out an updated review of the literature on treatment approaches and intervention strategies advised for developmental verbal dyspraxia. Although many of the approaches continued to reflect a motor programming/bottom-up approach, there was an increasing focus on a manual component being included in treatment regimes. There was also recognition by a few authors of the linguistic difficulties shown by children with developmental verbal dyspraxia and the need to include syntactic work in therapy regimes (Aram and Nation 1982; Ekelman and Aram 1984), and of phonological approaches (Crary 1984).

Another therapy approach given attention in the literature involves using augmentative communication:

A focus on manual communication

Different approaches have advised: (a) the use of signing systems and/or gestures (Jaffe 1984; Harlan 1984; Helfrich-Miller 1984) and (b) the use of tactile and gestural cues (Chumpelik 1984; Bashir et al 1984; Klick 1985; Shelton and Garves 1985).

Brief reviews of some of these approaches follow:

Jaffe (1984) Total Communication

"Total Communication, speech combined with all modes of communication, but primarily manual communication, may be a viable treatment approach with some apraxic children." Jaffe proposes extensive use of manual sign language and the manual alphabet treatment programmes for children with moderate – severe developmental verbal dyspraxia. In the early stages, sign language can be the major communication system, and in the later stages, signs and finger spelling can augment verbal communication.

Harlan (1984) A dual approach to therapy

A single case study is presented of a young child, Mark aged 2.7 years, diagnosed as having developmental verbal dyspraxia. A dual therapy approach was shown to be effective, combining the introduction of manual signs (American Sign Language) with an oral, traditional therapy approach.

This paper also illustrates the importance of early intervention and the involvement of parents in therapy (See Service delivery issues this chapter).

Strand and Skinder (1999) recommend augmentative communication for children with DAS. They advise that signing is appropriate providing there is a suitable signing environment. Otherwise natural gesture, picture exchange systems and communication books may be more appropriate.

The Nuffield Dyspraxia Programme does not specifically involve the use of signing as in the approaches advocated above. However, clinical experience from work at the Nuffield Hearing and Speech Centre and particularly at The Nuffield Speech and Language Unit supports the helpful role that signing can play in a remediation programme for a child with developmental verbal dyspraxia. At the Unit, Paget Gorman Signing is introduced to all the children and is used on an everyday basis by the staff. In line with findings in the literature (Moore 1980; Harlan 1984) signing has been shown to increase communication, improve verbal output and to be dropped spontaneously when speech becomes functional for the child (Harris and Vanderheiden 1980).

Chumpelik (1984); Square-Storer and Hayden (1989) PROMPT

"Prompts for Restructuring Oral Muscular Phonetic Targets" is a therapy approach involving the use of tactile cues to help the child reshape individual and connected articulatory phonemes. Each of the English phonemes (consonants and vowels) has its own individual prompt, given by the clinician touching specific places on the child's face and neck. The aim of the prompts is to provide locations/end-points to be reached by the articulators in producing a sound or a sequence of sounds. One of the important aspects is that prompts are dynamic rather than static, providing the child with information on both the timing and manner of articulation. For example, the prompt for /k/ involves the clinician applying pressure over time to help the child elevate the back of the tongue.

This therapy approach requires speech and language therapists to receive training before attempting to use it and relies on good knowledge of anatomical structures and of physiological phonetics.

Bashir, Grahamjones and Bostwick (1984) A touch-cue method

This is another tactile based cueing system, used to help children produce and use phonemes correctly. This system is not as complex as PROMPT as topographic indicators designated by touching a particular area on the lower face and neck have been devised for only 8 consonants (/b, d, g, s, sh, f, n, l/) + an additional cue for voicing. Tactile cues can be used simultaneously with auditory and visual cues and the approach can be combined with a language programme.

Shelton and Garves (1985) Signed Target Phoneme Therapy (STP)

These authors used hand shapes from the American Manual Alphabet as cues to support work on individual phonemes.

Klick (1985) Adapted Cueing Technique (ACT)

Cueing techniques were adapted from the manual alphabet for the deaf to enhance oral stimuli and increase the frequency of correct articulatory responses.

Passy (1985; 1990) Cued Articulation

A method of cueing articulation by using simple hand signs, designed "to help those who find it difficult to remember how to pronounce and/or sequence the sounds of spoken English". This gestural cueing approach is used widely by speech and language therapists in the UK, in the treatment of children with speech disorders, including developmental verbal dyspraxia.

At the Nuffield Hearing and Speech Centre, some therapists have found it successful to combine this approach with the use of pictorial symbols from the Nuffield Dyspraxia Programme, to maximize the cues when working on speech sounds (Blakeley 1983).

Some authors have favoured linguistic approaches to developmental verbal dyspraxia:

A focus on language (syntax)

Ekelman and Aram (1983; 1984) Incorporating syntactic work

Aram proposed that children with developmental verbal dyspraxia did not only have motor speech difficulties, but also had language difficulties. Her guidelines for intervention (Aram and Nation 1982), advised that therapy goals should incorporate: vocabulary development, lexical retrieval and syntactic development.

Syntactic areas requiring attention included: copulas and auxiliaries, verb tenses, regular third person singular, pronouns, past participle (-ed), passive, yes/no and wh- questions and complex sentence constructions.

Helfrich-Miller (1984; 1994) Melodic intonation therapy

This is another approach which incorporates language goals. It has been adapted from a therapy programme designed for treating adults with acquired apraxia. It aims to help a child develop language structures of increasing grammatical complexity in stages, through the use of intoned sequences.

It is recommended for children of about 7-8 years who have good attention skills and a moderate to marked apraxia. Signed English, is advised to help the child remember the sequences and to support language development, rather than the accompanying hand tapping used in the adult approach.

A focus on language (phonology)

Crary (1984) Dyspraxia as a phonological disorder

Crary suggested a challenge to the view of apraxia as a motor speech disorder and proposed that children with developmental verbal dyspraxia should instead be classified as having phonological disorders. He considered that such children demonstrated disordered organisation of their output speech processing systems.

Therefore, Crary proposed that therapy approaches designed for working on children's phonology were appropriate to use with children demonstrating developmental verbal dyspraxia. In particular, he incorporated the use of phoneme contrast therapy (Fokes 1982) in two studies (Crary and Hunt 1982; Weckler and Crary 1982).

See further review of Crary in next section (Crary 1993).

Lancaster and Pope (1989) Working with Children's Phonology

"The broad definition of phonological disorder as a speech disorder which results in reduced ability to signal contrasts must include the disorder referred to as developmental dyspraxia" (p10). 3 different approaches have been used to address phonological disorders: (a) Meaningful Minimal Contrast Therapy, (b) The Auditory Input Approach and (c) Articulation Training. Clinicians are advised to adopt an eclectic approach to therapy: "all three may be used in conjunction, either concurrently or at different stages of treatment" (p31).

Lancaster and Pope (1989) describe the Nuffield Dyspraxia Programme (1985) as a "highly controlled method of articulation therapy" (p113). In their review of the materials, they report: "Many of the worksheets, such as those illustrating minimal pair words and the simple paper and pen games, are very useful. The programme materials are therefore a useful resource where articulation training is included in a treatment programme which recognises, in addition, the importance of phonological factors" (p115).

An eclectic approach

In the 1990s, papers/books on the subject of developmental verbal dyspraxia have also included therapy approaches. Although some of these continue to favour a particular focus (Hall 1993/1995), the overriding view is that a therapy package for an individual child should incorporate strategies from different approaches, to address his/her particular profile of difficulties. An individual treatment package drawn from a variety of sources fits well with the Psycholinguistic Framework for the Assessment and Management of Speech and Literacy Disorders (Stackhouse and Wells 2001). As Rees (2001) explains "There are no specifically designed psycholinguistic materials for intervention. Almost all teaching and therapy material can be used in a psycholinguistic way" (p66).

Brief reviews of some of the therapy approaches reported in the literature during the 1990s follow:

Milloy and MorganBarry (1990) Therapy Guidelines

General therapeutic guidelines for working on developmental dysarthria and developmental verbal dyspraxia are given in the appendix to this chapter p129-30.

Marquardt and Sussman (1991) An eclectic approach

A single case intervention study is reported of a child N, diagnosed with developmental verbal dyspraxia. The intervention over a three year period involves a number of different therapy approaches: Sensory-Motor Approach (McDonald 1964), Traditional Articulation therapy (Van Riper 1978), Multiple Phoneme Approach (McCabe and Bradley 1975), Speech Pattern Remediation (Hodson 1982), Contrast Therapy (Costello and Onstine 1976), and Melodic Intonation (Helfrich-Miller 1984).

The reason given for such an approach is that the diversity of symptoms in developmental verbal dyspraxia "demands that treatment be eclectically developed" (p358).

Hall (1993; 1995) A motor programming approach

Hall continues to favour a motor programming approach designed to facilitate the development of the memory of speech movement patterns, allowing them to be accurately performed during future speech attempts. Therapy features should therefore include:

- many repetitions (drill orientated)
- fine-tuned hierarchy of task difficulties
- the use of multi-sensory modalities
- manipulation of prosodic features
- intensive therapy programmes

Dodd (1995) Appendix VIII Management strategies for developmental verbal dyspraxia p276-281 in "Differential diagnosis and treatment of children with speech disorders" provides a helpful review of therapy approaches to date and summarises the recommendations under 3 main headings:

General management issues

- a) Spend part of the therapy session working on language skills; not just speech intelligibility.
- b) Consider the use of augmentative and alternative communication (AAC) systems.
- c) Remember other associated deficits e.g. generalised motor dyspraxia and ensure they are included in treatment approaches.

Specific techniques that have been used: work on non-verbal oral movements; work on phonetic placement; the use of cueing techniques (graphic, pictorial, gestural, visual, tactile, kinaesthetic); the use of core vocabulary/functional communication techniques; prosody-based therapies.

Issues to consider in therapy: choosing production targets; the need to grade complexity of tasks; the need for repetition drill; the use of gross motor movements to support speech work.

Dodd (1995) refers to the Nuffield Dyspraxia Programme in her review under two headings: (a) Cueing – "Extensive graphic –visual cues have been developed for the Nuffield Dyspraxia Programme" and (b) The need to grade complexity of tasks - ... "it is important to use small graded steps in a therapy programme.... Programmes, such as the Nuffield Dyspraxia Programme.... provide therapy guidelines."

MorganBarry (1995) The use of instrumentation

A study is reported which employs Electropalatography (EPG) successfully in the remediation of children with developmental dysarthria and developmental verbal dyspraxia.

EPG is a computerised visual feed-back device that records real-time, tongue-palate contact via an artificial plate worn in the mouth. It has been used successfully to support work on phonetic placement with children with structural, neurological and persistent functional difficulties (Hardcastle and Gibbon (1997).

Electropalatography has been used successfully at Nuffield Hearing and Speech Centre to treat children with persistent speech difficulties, involving inaccurate phonetic placement (Friel 1998). In some cases, an approach combining activities from the Nuffield Dyspraxia Programme and EPG has been adopted (Williams et al 1997).

Crary (1993) A Multi-focal approach

Crary proposed a motolinguistic model to provide a theoretical understanding of motor speech disorders. A continuum of clinical impairment was proposed ranging from those dominated by executive or motor deficits to those that are more planning or language based. He suggested that dyspraxia might be either executive or planning depending on the individual profile of a particular child.

Crary rejected bottom-up approaches as being the only way to treat children presenting with developmental verbal dyspraxia and advocated that top-down planning strategies should also be incorporated into therapy programmes.

He proposed a "multi-focal program" (Weckler and Crary 1982; Crary and Hunt 1982; Crary 1993) for the treatment of developmental verbal dyspraxia, to address the 3 key aspects of difficulties involved in motor speech disorders: (a) oral motor deficits and motor deficits, (b) disorganization in the phonological system (c) disorganization in the expressive language system (syntactic system).

Velleman (1994) Including work on prosody

Velleman suggested that the difficulties that children with developmental verbal dyspraxia demonstrate in combining smaller units into larger wholes (sounds into syllables, words into sentences etc) can be seen as deficits in the generation and use of linguistic hierarchies i.e. as a problem of "bridging among elements" (p69). A therapy approach focusing on the following elements was recommended:

- movement patterns and building larger elements, with the basic unit being the syllable
- use augmentative/alternative communication, to reduce frustration and "communication pressure"
- teach grammatical word endings, if required
- address prosody very directly, beginning at a young age: include work on rhythm, stress, intonation, loudness and pitch
- include movement activities (e.g. to music)

Ripley, Daines and Barrett (1997) A combined approach

In the chapter on Speech and Language, these authors review the following approaches used to facilitate the development of a sound system:

- The Nuffield Dyspraxia Programme (Connery et al 1985; 1992)
- Metaphon (Howell and Dean 1994)
- Letterland (Wenden and Carlisle 1985)
- Cuing (use of manual gestures)
- Signing (e.g. Makaton; Signed English)
- Using reading to reinforce therapy

The authors suggest that speech and language therapists would be using all or any combination of these programmes/approaches to help children develop their speech sound systems. In addition, depending on the child's individual profile of difficulties, work on feeding, oro-motor skills, prosody, language and social skills would be included.

Ripley, Daines and Barrett (p50) describe the Nuffield Dyspraxia Programme's use in improving the "motor programming" aspects of developmental verbal dyspraxia: "All the sounds used in the programme have visual symbols, and the system is comprehensive enough for parts of it to be used to suit a child's individual needs. It combines oral skills work with sound production and sequencing. This enables the therapist to extend the child's skills at several levels simultaneously. Repetition is used to help achieve and establish success, and frequent exercise to build up patterns. It also enables the child to use these skills eventually in spontaneous speech."

In conclusion

Therapy approaches proposed for developmental verbal dyspraxia have been influenced by the differing theoretical views of the condition and have been drawn from a number of sources. The overriding view from recent work in the literature appears to be that an eclectic approach to treatment is advised involving articulatory, phonological, semantic, syntactic and manual elements. The exact weighting of these components in an individual therapy programme will be influenced by the particular child's speech and language profile and by the clinicians preferred treatment options.

Although there is increasing recognition of the need to demonstrate the effectiveness of treatment approaches, there have been few, efficacy studies published involving interventions for children presenting with developmental verbal dyspraxia. A number of single case studies have been published demonstrating progress following a particular treatment regime e.g. Harlan (1984), Chumpelik (1984), Shelton and Garves (1985), Helfrich Miller (1984) etc, but no large scale group studies. In part, this may be attributed to complications in carrying out controlled efficacy studies with a group who cannot be easily identified and where eclectic therapy approaches are used (Marquardt and Sussman 1991).

The Nuffield Dyspraxia Programme

The Nuffield Dyspraxia Programme (1985; 1992) is one of the few published intervention packages for managing speech disorders in the UK. Its contribution can be considered in two main ways:

As a resource

It offers over 500 black and white line drawings of pictures, representing sounds, words, phrases and sentences which can be photocopied, coloured, and/or cut out to be used in games and activities designed by speech and language therapists. In this sense it provides a valuable and flexible clinical tool. In addition to the original production tasks, tasks can be designed for a range of other applications e.g. auditory discrimination, segmentation and blending, rhyme and literacy work.

As offering an intervention approach

As can be seen from some of the reviews above, the Nuffield Dyspraxia Programme (1985; 1992) was designed to address the speech difficulties demonstrated by children with developmental verbal dyspraxia and recommends a systematic approach to therapy. There is clearly a “bottom-up” focus, particularly in the early part of the programme, involving work on speech motor skills, single sound production and transition to simple syllable structures. Principles from motor learning theories are involved e.g. frequent repetition (drills), the need for practice to enable movement patterns to become automatic, the use of feedback etc. However, the phonological and syntactic (“top-down”) aspects involved in developmental verbal dyspraxia are also recognised and addressed, particularly in the later stages of the programme e.g. through the use of minimal pair worksheets and the word combinations sections, involving phrase and clause level syntactic structures. The therapy approach described is recommended to be both multi-level (working on several “levels” at the same time though with varying degrees of difficulty) and multi-target (within each level a range of sound contrasts are taught at the same time).

Appropriacy with other client groups

As described, the Nuffield Dyspraxia Programme was designed to provide an intervention approach for children with developmental verbal dyspraxia. However, it has become clear that many children with other speech disorders e.g. developmental dysarthria, phonological impairments, cleft palate, can benefit from therapy activities drawn from the Nuffield Dyspraxia Programme.

A Programme?

Although it has been called a “Programme”, this is something of a misnomer. Strict use of this term implies a hierarchical organisation of activities, with criteria set to enable progression from one stage to another. The Nuffield Dyspraxia Programme does not advise such an approach – rather clinicians are expected to draw on their clinical skills and judgements to design tailor-made treatment plans for individual children using the available resources. Principles are advised to help with treatment planning (see Top Tips sections in therapy chapter), but are not prescriptive.

Experience from meeting numerous clinicians on our training courses, suggests that therapists do not want to be given “Cook-book” therapy approaches, but they do want therapy advice and to be provided with comprehensive materials to be used in therapy. This is what we are hoping to provide through this new edition of the Nuffield Dyspraxia Programme.

An eclectic approach

As has already been described, the recent literature recommends an eclectic approach to therapy and experience from meeting many speech and language therapists on our training courses suggest that this is the way many therapists choose to work. Activities from the Nuffield Dyspraxia Programme are successfully combined with a variety of other approaches to provide a multi-stranded approach. These include:

- Signing (e.g. Makaton, Signed English, Paget-Gorman etc)
- Cueing (e.g. Cued Articulation, Use of letters, Use of articulograms etc)
- Phonological therapy approaches (e.g. Metaphon, Hodson and Paden etc)
- Electropalatography

Service delivery issues

Historically, therapy was delivered to most children with speech difficulties (including those with developmental verbal dyspraxia) as a one to one, face to face contact with a speech and language therapist, on a once weekly basis. An open-ended policy for the termination of treatment, depending on the child’s speech becoming intelligible was often adopted.

In more recent times, the range of options for managing children with speech and language disorders in general has widened to include: individual vs. group treatment; direct delivery vs. indirect delivery; involvement of parents as facilitators; involvement of teachers, classroom assistants etc.

In addition, waiting list pressures and limited staff resources have demanded more stringent controls on therapy time e.g. offering blocks of care rather than open-ended treatment; the careful measurement of progress in therapy and the employment of criteria for terminating treatment, before speech is 100% intelligible.

In the literature on developmental verbal dyspraxia, some comments have been made on these service delivery issues. These will be presented along with reports of clinical practice from the Nuffield Hearing and Speech Centre:

Type of intervention

Strand and Skinder (1999) advise that there is no experimental evidence to say individual treatment is better than group treatment. However, for approaches requiring frequent practice, logic indicates that individual treatment will provide more opportunities for practice than group treatment and therefore it is to be preferred. Hall (1995), Laxerson and Velleman support this view. Group treatment is only acceptable if the other members of the group have the same diagnosis and a similar level of speech difficulty (Velleman).

Despite this advice, clinicians often favour at least some group therapy, especially for young children and/or for those who are very aware of their difficulties. An example of a published case study reporting a successful intervention programme involving both individual and group treatment can be found in Watson and Leahy (1995).

At the Nuffield Hearing and Speech Centre, we have provided both individual and group therapy for many years, with some children receiving only one intervention type and others receiving combinations of treatment types. Single case studies have demonstrated progress as a result of both individual and group intervention. The selection of intervention type for a particular child at a particular time is likely to be influenced by a range of factors e.g. the child's age, the child's speech profile, distance from the Centre, the child's awareness of his difficulties/personality, family issues etc.

Frequency of Intervention

In the literature, frequent and intensive speech and language therapy is recommended for children with developmental verbal dyspraxia in order to address their core motor planning/programming difficulties. For example, Strand and Skinder (1999) advise that as it is widely accepted that motor skill is only acquired through practice, frequent speech and language therapy sessions are necessary to allow motor learning to take place. For speech learning specifically, mass practice (involving fewer but longer sessions) is not as valuable as distributed practice (same amount of total practice, but more sessions).

Other authors who have commented on frequency of therapy include:

Velleman who advises that therapy should be as intensive and as often as possible e.g. 5 short sessions of 30 minutes duration a week, rather than two 90 minute sessions.

In addition, she comments that long breaks in therapy (e.g. over the Summer holidays) are not advised and may lead to regression.

Hall, Jordan and Robin (1993) who advise that intensive services are needed for the child with developmental verbal dyspraxia, involving a great deal of professional input.

Although such advice can be found in the literature (mainly from U.S.A.), observation of practice in the UK, indicates that there is considerable variation in the input children with dyspraxia receive. Although small numbers do receive intensive (daily or twice daily) input in Language units, the majority of children receive far less therapy. As for other children with speech difficulties, the most common NHS therapy regime is provided in blocks of treatment (lasting 5-8 weeks on a once weekly basis) with breaks (lasting 6-12 weeks +). Some children receive no more than a few sessions per year delivered directly by a speech and language therapist. Privately funded therapy continues to be provided on a more traditional model of once to twice weekly sessions over long periods of time.

Although it is frequently claimed that provision of therapy in blocks/episodes of care with breaks is an appropriate model of intervention for children with speech disorders, it would appear that limited resources are often key in such decisions. There is a need to spread limited resources as widely as possible and there appears little room for prioritising the child with developmental verbal dyspraxia. In the UK, one of the only ways to secure an enhanced level of support for an individual child has been through the educational statementing process, but even this is currently under change.

At the Nuffield Hearing and Speech Centre, we are also subject to resourcing constraints and are no longer in a position to offer ongoing treatment to individual children. Our therapy services are complementary and supplementary to local provision. Children may be offered a group intensive therapy course, an individual intensive therapy course or a block of individual therapy over 10 weeks. At the Nuffield Speech and Language Unit, intensive help (2-3 sessions daily) is offered to children with severe difficulties.

Practice

In motor learning theory, repetition and practice is essential for motor skills to be established (Ruscello 1993). Therefore daily practice of speech targets is usually recommended (Connery 1994). If a child is receiving daily speech and language therapy, practice will be incorporated into his programme. If he receives therapy on a less frequent basis, practice needs to be provided by parents, teachers, classroom assistants etc.

Clinical experience at Nuffield Hearing and Speech Centre supports the literature (Morley and Fox 1969; Harlan 1984; Velleman 1994) that involving parents in therapy regimes is essential to ensure maximum success. It is advised that parents both observe and participate in therapy sessions and that they are provided with "homework" to consolidate activities introduced in therapy sessions. Our experience is similar to Harlan's (1984) that many parents are highly creative and ingenious in finding novel ways to make speech work fun for their children e.g. a parent originally created the "stories" included in this edition at CVCV level in order to help her child work on words at this level of complexity.

Direct/Indirect intervention

Law, Harris, Boyle and Harkness (1998) report in their systematic review of the literature on speech and language delay, that direct, clinician-administered treatment was found to be more effective for articulation/phonology difficulties than indirect, parent-administered treatment. However, only 7 studies were included and non-standard treatments were involved.

Other authors have supported the recommendation that children with developmental verbal dyspraxia require therapy from a speech and language therapist e.g. Velleman suggests that at least some of the therapy needs to be delivered by a qualified Speech and Language Therapist e.g. once a week. Other professionals need to be supervised and should meet up with the Speech and Language Therapist regularly, i.e. once a week.

Hall et al (1993) believe that it is essential for a high level of therapy to be given by a trained speech and language therapist, supplemented by support from parents/other professionals.

The above advice appears to be at odds with the current trend within the UK for speech and language therapy services to become more consultative, with therapists providing programmes and training to other professionals and parents, who are then responsible for delivering the programme.

Parents and other professionals clearly have an important role to play in a child's intervention, especially with regard to practice and consolidation of skills learned in the therapy sessions, but they should not be expected to take the therapist's place.

Clinical experience at Nuffield Hearing and Speech Centre from children who attend parental based therapies to enhance communication e.g. Hanen; Parent-Child Interaction, is that these approaches are helpful in a general way, but do not usually impact specifically on speech intelligibility. Our experience is generally that children require direct work on their speech to make significant progress.

Early identification and intervention

It is generally recognised in the literature that the earlier a child's difficulties are recognised and the earlier treatment is started the better the prognosis (Ruscello 1993, Eisenson 1972; 1984, Watson and Leahy 1995, Harlan 1984).

At the Nuffield Hearing and Speech Centre, this view is supported. The Nuffield Dyspraxia Programme provides materials suitable for young children from about the age of 3 years. In the manual, suggestions are made for activities to adapt the approach to be suitable for younger children (see Appendix 5).

Prognosis

It is generally recognised that children with developmental verbal dyspraxia do not overcome their difficulties without help and do not grow out of them. They require professional support to help them progress, but most do become intelligible and communicate effectively over time. The interaction of a number of factors will determine an individual child's rate of progress – these include child specific factors e.g. severity of difficulties, co-morbidity of other conditions, general health, cognitive level, personality, motivation etc and intervention factors e.g. age at which intervention starts, appropriateness of therapy, frequency of therapy, parental involvement and support, and other professional support (Hickman).

Outcome measurement

It is commonly acknowledged that it is important to measure the outcome of any treatment given. Different tools have been employed to provide such measurements – see Enderby and John (1997). Clinicians may choose to measure a change in the impairment e.g. increase in % of consonants correct on a test; increase on subtests of a tool such as the Nuffield Dyspraxia Assessment, but may also choose a functional outcome measure. In the case of children with speech disorders, a suitable measurement might be a change in intelligibility.

Campbell (1999) used parental ratings of their children's speech intelligibility to unfamiliar listeners, pre and post treatment. Participants in the study included:

49 (mean age 50 months) children with moderate-severe phonological disorders and 11 children (mean age 48 months) with developmental apraxia of speech.

At the start of treatment, 84% parents of children with phonological disorders, estimated that unfamiliar listeners understood their children less than 50% of the time and the other 16% estimated that they were understood about half the time. By the end of treatment, 94% estimated their children could be understood three quarters of the time by unfamiliar listeners.

For the children with DAS, the picture was also positive but not so significant: At the start of treatment, all the parents estimated their children could be understood less than half the time by unfamiliar listeners. By the end of treatment, 8 of the 11 estimated that their children could be understood three quarters of the time and 3 about half the time. However, these children had received an average of 151 individual, 45 minute sessions, 3 times a week over a 360-420 day period in comparison to the group of children with phonological disorders, who had received an average of 29 individual, 45 minute sessions, twice weekly over a 90-120 day period. Unfortunately, little information was provided on the nature of the treatment given.

In conclusion

Advice offered in the literature suggests that children with developmental verbal dyspraxia require frequent individual therapy sessions delivered directly by a speech and language therapist and supported by regular practice sessions to help them overcome their difficulties. Unfortunately, current resourcing and practice in the UK makes such a regime difficult to provide under the National Health Service. More research is urgently needed to prove treatment efficacy and service delivery effectiveness.

For an update on current intervention approaches for DVD, please see:

1. ASHA (2007) Position Statement and Technical Report on Childhood Apraxia of Speech.
2. RCSLT (2011) Policy Statement on Developmental Verbal Dyspraxia.

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Chapter 3: Assessment

Assessment of a child, presenting with a speech disorder, begins with a detailed case history, and consideration of his overall communication skills and development. Areas investigated may include: play, social skills, communication skills, verbal comprehension, expressive language, oral examination, phonology, articulation, voice, prosody, auditory skills, gross and fine motor skills, visual/tactile skills, non-verbal skills and emotional well-being (Clinical Guidelines by Consensus for Speech and Language Therapists, 1998). In many cases, other professionals, such as paediatricians, psychologists, occupational therapists, audiologists, will contribute to the assessment.

Initial assessments will suggest the nature and cause of the child's difficulties, and may indicate the presence of developmental verbal dyspraxia (See chapter 1 for discussion of characteristics of developmental verbal dyspraxia).

However, detailed investigation of the child's speech is necessary in order to plan intervention. Therapists typically employ a range of formal and informal assessment tools, to elicit single words, connected speech and a phonetic inventory, depending on the level of the child's speech and expressive language. Phonological analysis of this data reveals his sound system, at various positions in syllable structure, and relationships between the adult target and the child's production.

The Nuffield Dyspraxia Programme Assessment takes this process a stage further, recognising the dyspraxic child's increasing difficulty with longer words and phrases, by sampling and analysing separately words and sentences of increasing phonotactic complexity. Vowels are sampled systematically, as well as consonants. Evaluation of oral motor skills, diadochokinetic skills (DDK), and connected speech are also included. Closely related to the treatment programme, the Nuffield Dyspraxia Programme Assessment also provides a sensitive measure of progress.

Psycholinguistic approach

Since the Programme was last published in 1992, understanding of complex speech disorders has benefited from the development of psycholinguistic models of speech processing. Psycholinguistic models attempt to explain the process of speech, and therefore speech difficulties, as opposed to linguistic analysis, which describes the end result. Therapists at the Nuffield Centre have particularly been working with the model developed by Stackhouse and Wells (1997), and the discussion and diagrams below are based on this.

The basic tenet of a psycholinguistic approach is that speech processing involves the routing of information between ear, brain and mouth.

The first distinction to be made is whether information is being routed from the ear up to the brain (input processing), or from the brain down to the mouth (output processing). These can be thought of as two information channels: on the input side, speech information is received and decoded, and on the output side, speech information is encoded and transmitted.

In addition to the two information channels, the psycholinguistic approach assumes that there must be a store of information (a representation), which serves as a basis for recognising speech as well as generating speech output. The representation would include details of a word's meaning (semantic representation), its sound structure

(phonological representation), and instructions for articulation (motor programme). An important clinical distinction to make, is whether it is the stored information (lexical representation) which is faulty, or on-line processes, such as discrimination and motor programming. Assessment tasks must therefore be analysed to ascertain whether lexical representations must be accessed, may be accessed, or cannot be used, to be performed successfully.

It should also be noted that information may be routed in a top-down direction (utilising stored information) and/or a bottom-up direction (utilising peripheral sensory input) in order to perform a task. This applies to either side of the model. For example, as incoming speech is being processed (bottom-up), the brain may predict a part of it (e.g. the final syllable of "caterpi..." caterpillar). Equally the basically top-down processing of the output side of the model may be influenced by kinaesthetic feedback from the articulators. This is an important consideration in the design of therapy programmes, where it may be possible to utilise stronger processes to support weaker ones, and it may be necessary to experiment with various processes and processing routes in order to elicit a target.

When applied to clinical assessment, it is clear that children with speech disorders may have difficulty at one or more points along this processing route. Assessment tasks can be selected to check specific processes. In this way a profile of the child's processing strengths and weaknesses can be created, avoiding the need for diagnostic categorisation, where this is unhelpful, and facilitating well-targeted treatment.

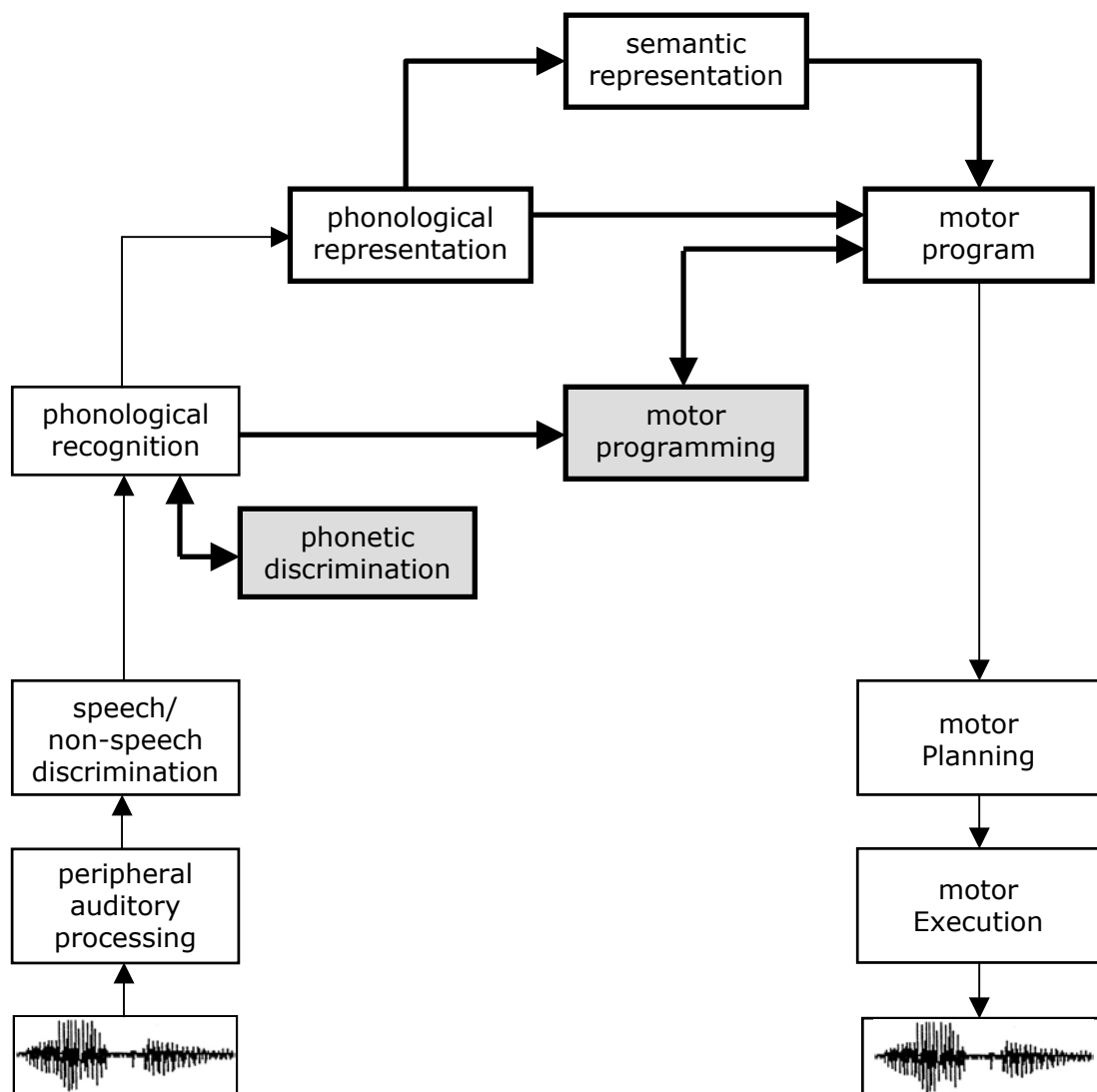
Psycholinguistic Model

The Stackhouse and Wells speech processing model is reproduced below, with permission. In the notes on each level of processing, which follow, we have combined the speech processing model with the questions from Stackhouse and Wells' speech processing profile. However, it should be noted that the questions do not strictly map onto particular processing levels, but relate to processing routes, (including or bypassing particular levels) required by types of assessment task. Two further questions (X, Y) have been added on the output side of the model, to clarify issues particularly relevant to children with features of developmental verbal dyspraxia.

Levels of processing are represented by boxes, and possible processing routes by arrows. Although not strictly hierarchical, there is a sense of processing order. For instance, the child must have discriminated the phonetic characteristics of a word before recognising it as a stored phonological representation. The levels do not differ in terms of absolute ease or difficulty, but in terms of relative cognitive sophistication. Off-line processing units appear as shaded boxes, indicating that processing at this level does not occur during the real time performance of the task i.e. on-line, but at some point afterwards i.e. off-line. Broad arrows indicate that information flows between boxes as part of a learning process, rather than in the on-line processing of familiar input.

For a detailed, and very readable, discussion of the speech processing model, its relation to normal speech acquisition and speech disorders, and analysis of assessment procedures, readers are recommended to refer to the original text (Stackhouse and Wells, 1997).

Speech Processing Model



© Whurr Publishers Ltd. From Stackhouse and Wells (1997): Children's Speech and Literacy Difficulties

Input processing

Peripheral auditory processing

The child must first hear the incoming speech signal, before any further decoding can take place. Hearing should be assessed as a prerequisite to any assessment of speech or language.

(A: Does the child have adequate auditory perception?)

Speech/nonspeech discrimination

Having perceived the signal, he must then recognise it as speech.

Phonological recognition

The phonological recognition level includes the recognition of the speech signal as a familiar language, and its segmentation first into words, and then into phonological units. Discrimination is thought to involve articulatory rehearsal. This may account for discrimination difficulties in children with difficulties on the output side of the model.

Phonological recognition, without reference to the child's stored representations, is tested by same/different judgements on nonwords (e.g. foss/fozz). More complex words, including multisyllabics and clusters should be used when testing older children.

(B: Can the child discriminate speech sounds without reference to lexical representations?)

Stackhouse and Wells included in their assessment framework a question specifically about the child's ability to discriminate legal/illegal non-words in a particular language. They suggest that this may be important in assessing bilingual children, and those with severe and persisting speech processing difficulties.

(C: Does the child have language specific representations of word structures?)

Phonological recognition may also be assessed by real word discrimination tasks, involving same/different judgements. However, these may be achieved by accessing stored phonological representations (see section under Lexical representations), avoiding the need to segment the signal. Again the child's age should be taken into account when selecting appropriate materials. Auditory memory loading can be high for these tasks (e.g. selecting odd one out from 3-4 spoken words) and this should also be considered when selecting tests.

(D: Can the child discriminate between real words?)

Phonetic discrimination

When the input contains novel phonetic material e.g. from an unfamiliar accent or language, this level of processing is required. Phonetic discrimination is also required for routine therapy tasks e.g. asking the child to discriminate at a phonetic level between sounds he substitutes e.g. [s] vs. [ʃ] – "did you hear a hissing sound or a slushy sound?"

Lexical representations (stored information)

Stored lexical representations for words include information on semantics, grammar and eventually the written form of the word, as well as the phonological representation and motor programme.

Phonological representation

The phonological representation includes the information necessary to identify the word, as produced by a range of speakers. Information is probably stored hierarchically, in terms of syllable, onset + rime, nucleus + coda, with phonemes specified by distinctive features.

In early speech development, representations would be specified only in sketchy terms (e.g. the /t/ in "tea" might be specified only as {+lingual, +plosive}), until the addition of "key" to the child's receptive vocabulary required that the term {+ alveolar} was added. Phonological representations must therefore be continually up-dated as the child extends his speech and language skills.

The accuracy of phonological representations can be checked, using real word discrimination tasks involving picture stimuli (i.e., asking the child to point to the picture named: e.g. pear/bear), or by right/wrong judgement tasks (e.g. "Is this a nake?", when looking at a picture of a snake). Of course, difficulties with peripheral auditory processing or phonological recognition may also impair performance on these tasks.

(E: Are the child's phonological representations accurate?)

The accuracy of phonological representations can also be checked without the spoken presentation of the item. In this case, the child is presented with picture items and asked to carry out a sorting task silently e.g. by onset, by coda etc.

(F: Is the child aware of the internal structure of phonological representations?)

Motor program

The motor program contains a series of gestural targets for the articulators, designed to achieve an acceptable pronunciation of the word. For example, accurate production of the word "tea" would require the following instructions: initiate airstream, close velopharyngeal sphincter, make closure with margins of tongue/alveolar ridge, release closure, assume raised front tongue body posture, spread lips and initiate voice.

Spontaneous speech and picture naming are used to assess accuracy of motor programs, but it should be remembered that these activities are also dependent on motor planning and execution.

(G: Can the child access accurate motor programs?)

Output processing

Motor programming

The motor programming level is thought to include a store of phonological units, probably onsets and rimes rather than individual phonemes, which can be combined to create new motor programs for unfamiliar words.

Motor programming is involved in early sound production and babble, and is crucial in early speech development, in creating and revising motor programs for a rapidly increasing vocabulary. Motor programming continues to be important, after the basics of speech are established, in dealing with new words, (e.g. for curriculum areas and hobbies, and the names of people and places). The manipulation of sounds is also involved in the development of alphabetic reading and spelling skills.

Motor programming can be tested by nonword repetition (although phonological recognition is also involved in these tasks), by rhyme production and in older children, by spoonerisms.

(X: Can the child create new motor programs?)

(H: Can the child manipulate phonological units?)

Motor planning

Once a motor pattern for a word is retrieved, or a new one created, a final plan for the utterance is formed, assembling gestural targets in the correct sequence, and taking account of the intended grammatical structure, phonetic environment and prosodic features. This is the level at which “slips of the tongue” occur in normal speakers. In children with developmental verbal dyspraxia, inconsistent realisation of targets occurs at this level, and there may be marked breakdown of articulation/phonology in connected speech.

Motor planning is assessed by comparing production of single words with production in sentences and/or connected speech, and by observation of inconsistent realisation of established motor programs. Motor planning is also assessed by DDK tasks, where the child is asked to repeat a target item on a number of occasions.

(Y: Can the child assemble a motor plan for an utterance effectively?)

Motor execution

Motor execution difficulties relate to problems with the structure or innervation of the articulators due, for example, to dysarthria or velopharyngeal insufficiency.

This level is assessed by means of an oral examination, observation of feeding skills, and DDK tasks. Consistent errors are observed in speech, which are often paralleled by difficulties with nonspeech functions.

(K: Does the child have adequate sound production skills?)

Self-monitoring

To complete the process, the child should be able to monitor his speech, identifying errors by comparison with stored phonological representations, and making an attempt to correct them.

(L: Does the child reject his own erroneous forms?)

Phonological awareness

Awareness of the phonological structure of words, and the ability to manipulate this at a conscious level, develops as the child matures. 3-4 year olds are able to demonstrate awareness of words and syllables, 4-5 year olds show awareness of onset and rime. Awareness of individual phonemes develops with exposure to letter sounds, at school (Goswami and Bryant 1990).

The child's developing awareness of the phonological structure of words is an important tool in therapy, enabling him to understand instructions such as "put a /t/ on the end".

Phonological awareness is assessed, as appropriate to the child's age, using both input and output tasks. Readers are recommended to refer to Stackhouse and Wells (1997), Chapter 3, for a detailed discussion of these issues.

Onset or rhyme matching, or "odd one out" discrimination tasks can be used to assess the child's awareness of the phonological structure of words. Tasks involving pictorial rather than spoken stimuli and responses ensure that phonological representations are used.

(F. Is the child aware of the internal structure of phonological representations?)

The ability to manipulate phonological representations can be tested by tasks such as production of rhyming words, segmentation, blending and deletion/addition tasks, (and in older children, spoonerisms). The use of nonword stimuli ensures that manipulation skills are tapped, and avoids the possibility that responses have been learned.

(H. Can the child manipulate phonological units?)

The Nuffield assessment from a psycholinguistic perspective

Stackhouse and Wells state that psycholinguistic assessment does not require new tests, but rather provides a framework for making decisions about what areas to assess and what materials/methods should be used.

The Nuffield Dyspraxia Programme Assessment is a test of output processing, providing information about the levels of motor programs, motor planning and motor execution. Levels targeted are shown below:

Pre-speech motor skills	Motor execution Motor programs (non-speech level)
DDK	Motor programming Motor planning Motor execution
Single sound imitation	Motor programs Motor programming Motor execution
Single word naming	Motor programs
Phrases and sentences	Motor planning
Connected speech	Motor planning with linguistic formulation

Skills are sampled and analysed separately at different levels of phonotactic complexity:

- single sounds
- CV and VC words
- CVCV words
- CVC words
- multisyllabic words
- consonant cluster words
- phrases and sentences

The assessment allows detailed analysis of what articulatory skills the child has, and what he is not yet able to do, and what phonological contrasts are maintained, at relevant levels of syllable structure.

Vowels are sampled systematically, as well as consonants. Evaluation of oral motor skills, DDK and connected speech are also included. Closely related to the treatment programme, the NDP Assessment also provides a sensitive measure of progress.

Most of the items are designed to be achieved by a 4 year old, but materials are also relevant to older children.

Psycholinguistic Profiling

It is recognised that while children with a possible diagnosis of developmental verbal dyspraxia are likely to have significant difficulties with one or more of the output components on the speech processing model, many also have inaccurate phonological representations, some discrimination difficulties, and poorly developed metaphonological skills.

In order to arrive at an overall profile of the child's speech processing, the Nuffield Dyspraxia Programme Assessment must therefore be supplemented by measures which assess other areas of the psycholinguistic model. A variety of test materials is available, including: The Auditory Discrimination and Attention test (MorganBarry 1989); Wepman's Auditory Discrimination Test (Wepman and Reynolds 1987); Aston Index (Newton and Thomson 1982); the Phonological Awareness Battery (Frederikson (ed) 1995); the Phonological Awareness test (Muter, Hulme and Snowling 1997); The Children's test of Nonword Repetition (Gathercole and Baddeley 1994). The selection of tests will depend on the child's age, and on the breadth and depth of assessment required.

Creating a microprofile

Although an individual child will have general patterns of strength and weakness in processing (e.g. phonological representations may be better than motor programs), difficulties in processing may vary between sound groups and levels of complexity (Corrin 2001). For example, a child who is unable to imitate either /ʃ/ or /k/, may have difficulty discriminating /s-ʃ/ but not /t-k/, if these are his typical substitutions. Similarly, he may have accurate phonological representations for shorter, but not for longer words, or motor execution difficulties affecting only tongue tip articulation.

Analysis of assessment data permits likely target areas for treatment to be identified, (for example: eliciting velar plosives, or establishing /p t k/ in CV structures). See Chapter 4: Treatment Planning for discussion. Target areas can then be examined in terms of the speech processing model, to identify strengths and weaknesses. These will determine the design of therapy tasks and the type of facilitatory techniques employed (see case studies).

Formal or informal probes may be carried out to tap different aspects of processing:

Auditory discrimination	Phonetic discrimination Phonological recognition Phonological representations
Imitation of real words to compare with naming	Motor programming, (probably without accessing phonological representations)
Non-word imitation	Motor programming
Wider vocabulary	Phonological representations Motor programs
Stimulability of missing items	Various routes: top-down/bottom up
Self monitoring	Feedback loop Phonological representations
Segmentation, matching	Awareness of the internal structure of phonological representations

NDP Assessment: subtests and speech probes

A description of each subtest, and the processing routes involved, is given below, followed by possible probes.

Oral motor assessment

This subtest assesses the motor execution level. Pure motor skills are elicited via non-verbal routes, at an automatic level, or by imitation. Additional information on motor skills in speech sound production is obtained from the single sound imitation section, which should be completed prior to administering the oro-motor assessment.

The oro-motor checklist is designed as a simple assessment for observing the child's functional skills, relating pre-speech oral skills to articulation of single sounds. The assessment is divided into 5 sections:

- Lips and jaw
- Tongue
- Airstream
- Voice
- Palate

It is not necessary to assess each of these areas in detail. Questions are posed at the beginning of each section to help advise whether assessment of a given area is advised.

An individual child's performance on oro-motor skills can be compared with normative information available (see Appendix 7 on normative issues: oral motor skills).

The assessment identifies motor skills underlying specific articulatory features, which may need to be developed in order to elicit specific sounds, and skills which the child already has, which might be used as a basis for extending his sound system. (See Appendix 4 for suggestions on eliciting specific sounds). For example:

- d) The child substitutes alveolars for labial sounds:
Does he achieve lip closure at an automatic level (feeding/controlling saliva)?
What is muscle tone like?
Can lip closure be achieved by imitation, instruction or with a physical prompt?
- e) Articulation of tongue tip sounds is imprecise:
What range of tongue tip movements are achieved at an automatic level (feeding, licking)?
What can be achieved with imitation, mirror, prompts?
What is muscle tone/appearance of the tongue like?
- f) Hypernasality/nasal emission is heard on single sounds:
Does the child achieve an oral airstream for blowing tasks?
Any abnormalities of muscle tone or structure?
Has the child been referred for palatal investigation?
- g) The child has a very restricted range of consonant and vowel sounds:
Investigate feeding skills and spontaneous sound-making.
What key movements/postures of the lips and tongue can be elicited by imitation or with prompts?
Observe structure of lips/tongue/palate, and muscle tone.
Observe wider motor skills.
- h) Voice quality and intonation are poor:
Is vocal tone better at an automatic level?
Can better voice be elicited with prompts and models?
Is there any sound or symbolic noise produced at a spontaneous level, which has better voice?
Is the child able to use basic pitch, volume and length contrasts?

The comments box should be used to note observations on oral structure, muscle tone, symmetry, involuntary movements, feeding, drooling, vocalisation, general motor skills.

Children with significant oro-motor difficulties may benefit from more detailed assessment of their feeding skills, and advice from an occupational therapist or physiotherapist on seating and techniques to normalise tone.

DDK

Syllable and word sequencing skills are assessed by imitation of a model once and then subsequently on five repetitions. For syllable sequences, the processing route is: auditory perception, phonological recognition, motor programming, motor planning and motor execution. For word sequences, following phonological recognition, the route is: motor program, motor planning and motor execution.

Six sequences are assessed involving two different types of stimuli: syllable sequences and real words (which can be supported with picture cues). Sequences involving two and three syllables are included. Two syllable sequences include two examples of sequences involving the same consonant in each syllable and two examples involving a change of consonant. Three syllable sequences involve a change of consonant in each of the three syllables. A practice example of each type of sequence is included to introduce the task. Each target is modelled once and the child is asked to imitate the model. He is then asked to repeat it 5 more times.

Scoring is based on accuracy, consistency and rate of production (see scoring section). Comparisons can be made between normal and atypical performance on such DDK tasks (see Appendix 7 on normative issues: DDK section-Williams and Stackhouse 2000).

Single sound inventory

The child's phonetic inventory is assessed by imitation. This requires the following processing route: auditory perception, phonetic discrimination, motor programming (or motor program, if the child has developed one during therapy!), motor planning and motor execution.

Consonants are divided on the assessment form into a basic range /p b m t d n k g f s h w j/ and harder sounds. It may be appropriate only to attempt the basic range with young children or those with very limited skills.

An individual child's performance on single sounds can be compared with normative information available (see Appendix 7: normative issues: speech sound emergence).

Vowels considered in terms of type (long/short/diphthong), and basic (i.e. those involving more extreme lip/tongue/jaw postures) as opposed to those requiring half-open jaw postures. It is recommended that all vowels are checked, as it is difficult to predict which will be easier for an individual child. Details of postures required for a standard set of English vowels are given in Appendix 3: Articulograms. Regional accents will require some modifications.

Probing speech processing

- Check consistency (motor planning), by attempting repetitions of a sound e.g. repeat "b" five times. Consonant sequences can be assessed through diadochokinetic tasks (see above). Vowel sequences can be assessed as appropriate e.g. can you say /i - u/, /a - i/ etc five times after me.
- Check stimulability of "missing" sounds, i.e. ways in which the motor program may be elicited. This involves experimenting with various processing routes: auditory models, visual cues (such as articulograms, lip reading and mirror), physical prompts (e.g. supported lip closure to elicit bilabials), semantic cues (e.g. food picture to help elicit /m/), and verbal instructions (e.g. "put your tongue on your teeth"). See also Appendix 4 for methods of eliciting sounds.
- Investigate underlying motor skills (motor programs and motor programming). Can the component movements/features of the target sound be elicited at a voluntary level? Do they occur in other sounds in the child's repertoire? Do they occur during a non-speech activity (feeding, crying, babble, animal noises, play, tooth-brushing)? If not, can more mature/wider range of movement patterns be facilitated? Liaise with Occupational therapist.
- Investigate discrimination of target from substituted sound(s).

Word Lists

Single words are elicited by picture naming, using the following processing route: semantic representation, motor program, motor planning, motor execution. Where the child is unable to name a picture, the word may be elicited by imitation. However, this should be noted, as the processing route now involves auditory discrimination and motor programming, rather than a stored motor program. For some children it may be very helpful to compare performance on naming tasks with that on repetition tasks and in such cases, the word lists can be presented separately for naming and repetition on different occasions.

20 words are elicited, through picture naming, at each of the following levels of syllable structure:

- CV and VC
- CVC
- CVCV (and CVCVC)
- Multisyllabics.

(Words of 3-4 syllables, with a range of syllable structures and stress patterns. Compound words and consonant clusters are also included.).

- Consonant clusters.

(Initial, medial and final clusters of various types are sampled.).

The pictures are organised into separate assessment flip-books by syllable structure.

With only 20 items per syllable structure, it is not possible to sample all sounds in all positions. The word lists have therefore been constructed to include as far as possible:

- *Vocabulary familiar to a 4 year old.*
- *A wide range of consonants and vowels in each list.*
- *Greater emphasis on the basic range of consonants at lower levels.*
- *Sequences of sounds which frequently cause difficulties for children with DVD.*

Children with developmental verbal dyspraxia have difficulty co-ordinating sequences of movement. This can result, at a phonological level, in syntagmatic processes, such as vowel or consonant assimilation. It may also result in more subtle articulatory/phonetic errors, where particular sounds/groups of sounds occur adjacent to one another. For example, when diphthongs are followed by nasals, the second element of the diphthong is often omitted. Sequences which may need specific teaching are listed on the assessment forms.

Data is analysed in terms of articulatory and phonological skills. Questions are included on the assessment forms as a guide. In depth assessment of speech processing, in relation to particular areas to be targeted, should be used as necessary to clarify the nature of the child's difficulties.

Probing speech processing

1 Check stimulability of target (i.e. ways of facilitating motor programming).

imitation of auditory model (e.g. boat)	Motor programming, attempting to by-pass phonological representations
imitation of split model (e.g. bow – t)	
imitation of non-word model (e.g. boit)	
Draw attention to target sound sequence (e.g. "It's got a /t/ at the end!")	Phonological awareness
Provide articulatory cues (verbal/visual) (e.g. "Put your tongue on your teeth"; use of mirror; articulogram)	Articulatory awareness and manipulation Motor execution

Check discrimination of substituted sounds

Minimal pairs: child selects picture to auditory model	Phonological recognition, routed via phonological representations
Same/different judgements (real words)	Ditto, but memory loading is higher
Same/different judgements (non words)	Phonological recognition, by-passing phonological representations

Check phonological representations

Right/wrong judgements of target word and errors, produced by therapist	Phonological representations
---	------------------------------

Check motor programs for a wider group of words with target structure/sounds (e.g. wider range of CVC words with velar and alveolar plosives). Motor programming difficulties often mean that motor programs are not updated spontaneously.

Naming pictures	Motor programs
-----------------	----------------

Check motor programming

Imitating non-words	Motor programming
---------------------	-------------------

Check motor planning (consistency of target)

Produce repetitions or sequences including target	Motor planning
---	----------------

Check motor execution

Look for effects of limited range, speed, accuracy and strength of movement of lips, jaw, tongue, palate, respiration and larynx.	Motor execution: neuro-muscular control
Look for effects of structural problems (e.g. velo-pharyngeal insufficiency; dental anomalies; severe tongue tie)	Motor execution: structure

Phrases and sentences

Phrases and sentences are elicited by imitation, supported by pictures. The processing route includes input processing (auditory perception, phonological recognition) accessing lexical representations and motor programs for familiar words, followed by motor planning and motor execution. However, grammatical morphemes may be processed via a non-lexical route (discrimination and motor programming).

20 sentences are modelled, 4 - 12 syllables in length, and incorporating a range of sounds and syllable structures. This level of the assessment provides the opportunity to observe how the child deals with word joining, prosodic features and grammatical morphemes.

Organising utterances of increasing length represents a higher level of complexity for children with DVD. Breakdown is frequently seen at this level, despite adequate production of individual words. This may be due to difficulties at the motor planning stage of processing. Sentences include some words from the single word lists, in order to facilitate comparison.

Difficulties can also be seen at this level in children with greater impairment of motor execution, who have difficulty sustaining accurate articulation in longer utterances.

Because of the complexity of the task, however, difficulties may also result from problems with input processing, auditory memory and expressive language.

Noting strengths, weaknesses and strategies

The following features are noted specifically at phrase/sentence level:

- 1 Assimilation of consonants or vowels, simplifying the motor plan.

/ə taɪdə i? ɡɪɡɪn .../ - A tiger is digging...

Final consonant deletion, which may be the result of difficulty with word joining.

/ə kʌ? ə ti/ - A cup of tea.

Cluster reduction

/kætəpɪləz teɪndz ɪntu bʌtʌfaɪz/ - Caterpillars change into butterflies.

Imprecise vowels, probably due to motor execution difficulties.

/ni: maɪ? ə na:ʔha:θ/ - Lee made a lighthouse.

Imprecise consonants

/xɛt̪ɪ pɑɪ/ - Katy's party.

Syllable deletion

/daɪsə zɪjʌm/ - Dinosaur museum.

Junction. Various types of normal junction processes are modelled. A brief key is given here, but see Chapter 5: Connected speech, for more detailed information. (# is used to indicate a word boundary, and ' to mark syllables, where necessary).

- /j/ - A /j/ is inserted between 2 vowels, if the first ends in a high front/spread /i/ posture. For example, "Daddy is talking" is produced as /dædɪjɪz'tɔːkɪŋ/.

- C#V – Where a final consonant is followed by a vowel, the syllable structure of the sequence is reorganised so that these 2 sounds become a CV syllable. Thus “A cup of tea” is produced as /ə'kʌ'pʌv'ti/.
- C#C – Where a final plosive is followed by an initial plosive, the first is not released, and may be assimilated to the second. For example, “Two sad cats” may be produced as /tusæd'kæts/ or /tu sæg'kæts/. However, the second C#C sentence involves a fricative followed by a plosive, and both consonants should be heard: “I saw five camels...” /aɪsəfaɪvkæm'lz.../.
- /n/ - “A” becomes “an” before a vowel, and syllable structure is reorganised. For example, “Get an apple” is produced as /'gɛ'tə'næ'pl/.
- /w/- A /w/ is inserted between 2 vowels if the first ends in a high back/rounded /u/ posture. For example, “Go in the house” is produced as /gəʊwɪnðəhaus/.
- /r/ - A /r/ is inserted between 2 vowels when the first vowel sound could be spelt with an “r” digraph (ar, er/ir/ur, or (aw). For example “Peter is eating his dinner” is produced as /'pi'tə'rɪ'zi'tɪŋ'hɪz'dɪ'nə/.

Rate of speech/stress patterns. Slow, syllabic speech, rapid, dysfluent or variable pace and rhythm should be noted. Although sometimes said to be an unfortunate consequence of therapy approaches such as the NDP, a slow regular pace can be a very useful strategy for some children to achieve intelligibility, despite very severe speech difficulties. Some discover this for themselves, so this possibility should be considered when a child presents with slow/syllabic speech.

Voice quality, pitch and volume problems should be noted, as at lower levels.

Nasality. Hyponasality and hypernasality/nasal emission should be noted.

Intonation. Note anything that sounds “odd”: monotone/very narrow pitch range, sing-song quality, only one intonation contour/nuclear tone, inappropriate intonation pattern (sounds like a question/statement/command!), wrong focus (stress on the wrong word). Does the child imitate your intonation readily? See also speech probes, below.

Compensatory strategies. Is the child doing anything that helps his intelligibility? For example, pacing his speech, producing one word at a time. Is he aware of doing it? It may still be that he should be helped to use a more effective strategy, or that he is doing something that doesn't help at all, such as exaggerating intonation.

Probing speech processing

Further assessment of intonation can be carried out by asking the child to imitate a sentence several times with different intonation (nuclear tones/focus). More information on junction and intonation can be found in Chapter 5: From sentence level to connected speech section.

Connected speech checklist

The connected speech checklist provides a framework for looking at the child's connected speech, monitoring progress, and determining what aspects of connected speech may be impairing intelligibility. The scoring system also produces a severity rating, which we feel is appropriate for children of 4-5 years upwards.

In processing terms, spontaneous connected speech involves linguistic formulation, accessing lexical representations and motor programs, motor planning of complex grammatical/articulatory/prosodic sequences.

The checklist is based on a sample of the child's typical (spontaneous) connected speech. It may be elicited by means of 1:1 conversation/play, or by a story retell task. The topic can be manipulated to ensure that age-appropriate challenges are included (e.g. complex vocabulary from a school project or hobby, explanation of an activity or concept, increasing linguistic loading).

The sample should be tape recorded, and analysed in the following ways:

Listen first at a broad level, noting

- Pace, rhythm, voice, resonance (see Phrase/Sentence level).
- Consistency.

There are often slower, better articulated sections and faster, less clear parts, as the child learns to pace and monitor his speech effectively. Alternatively variation may be associated with different linguistic demands.

- Any strategies the child appears to be using to improve intelligibility.

Then listen more specifically for features of phonology and articulation

- Range of syllable structures, and what happens to final/medial consonants, clusters and weak syllables.
- Range of consonants and vowels used.
- Maintenance of basic place, voice and manner contrasts.
- Precision of articulation.
- Production of grammatical morphemes. These are difficult because they usually have to be unstressed, but children are often unsure of the phonological structure they are trying to reduce.
- Junction: how the child deals with joining sounds at word boundaries (e.g. omit final consonants, articulate one word at a time, or are they developing appropriate junction processes? (see under Phrases and Sentences this chapter).

The 80% use criterion should be estimated rather than calculated formally, and reflects features which are generally present in the child's speech.

By totalling the score on the connected speech checklist, an approximate severity rating, for children of 5 years upwards, can be given:

0 – 7	severe
8 – 15	moderate
16 – 20	mild

By deducting the one item on the checklist which would not be expected of a 4 year old (full range of fricatives and affricates, apart from /θ ð/), the rating could also be applied to 4 year olds:

0 – 6	severe
7 – 15	moderate
16 – 19	mild

Intelligibility Rating

Having considered various methods of rating intelligibility, we have found the disability scale from the phonological disorders section of Therapy Outcome Measures (Enderby and John, 1997) to be the simplest and most relevant. This is reproduced below, with permission.

Identify descriptor that is “**best fit**”: The patient does not have to have each feature mentioned. Use 0.5 to indicate if patient is slightly better or worse than a descriptor.

- 0 Completely unintelligible to familiar and nonfamiliar listeners.
- 1 Partly intelligible to familiar listeners in known context, communication partner bears the burden of responsibility.
- 2 Intelligible to familiar listeners in context, partly intelligible in context with nonfamiliar listeners, single words clear connected speech poor.
- 3 Usually intelligible to familiar listeners in and out of context, variable intelligibility in context with nonfamiliar listeners /free spontaneous speech often unintelligible.
- 4 Minor problems but intelligible to everyone, occasionally loses intelligibility at times e.g. when excited or speaking against noise etc.
- 5 Intelligible at age-appropriate level to familiar listeners and nonfamiliar listeners.

Administration, analysis, scoring and interpretation

Top Tips

- Investigate articulatory skills and phonological system at each level of complexity
- Note problems with voice and prosody
- Identify speech skills to be targeted at particular levels in therapy
- Investigate underlying motor skills for possible causes of sounds not present
- Investigate the speech processing system as it affects skills to be targeted
- Investigate the stimulability of possible targets
- Investigate the consistency of skills already evident
- Assess phonological awareness, as appropriate to the child's age

Administration

The NDP assessment is intended to be used flexibly. There is no need to assess higher levels if the child is clearly going to be working at single sound and CV levels. However, it may be useful to try just a couple of items from harder sections, to enable comparison with subsequent assessments. The assessment may be carried out over several sessions or completed in a single session.

Instructions for administration and scoring are included on the assessment forms.

Oral motor assessment

This section should be administered after the speech assessment, as it utilises information from the single sound section. It is not necessary to complete all sections- questions are posed at the beginning of each section (lips and jaw, tongue, airstream, voice, palate) to help the clinician decide whether a particular area should be assessed. Instructions for tasks are given verbally and demonstrated. Comments should be recorded regarding both structure and function. This section is not scored as a whole.

Single sounds

Single sounds are elicited by imitation. It is recommended that articulatory features (e.g. dental placement of alveolars, hypernasality, weak closure, breathy voice) are noted, as these are often important factors in identifying the nature of the child's problems and need to be incorporated in treatment planning.

Diadochokinetic skills

The target is modelled once and the child is asked to imitate the sequence. He is then asked to repeat it five times. Picture cues may be used for the real word items (from the CVCV section of the Programme).

It is recommended that each production is transcribed phonetically on the scoring sheet. It may be necessary to tape record this section to ensure an accurate measure is made. Note inaccuracies, inconsistency and unusually fast/slow rates of production. Young children may need help to understand when they have produced 5 productions e.g. using a tick box chart or by the therapist holding up a finger each time a production is said. The time taken to produce 5 repetitions should be recorded using a stop watch.

Word lists

These are elicited by picture naming. Words which the child cannot name should be imitated, and noted accordingly (see under speech processing above).

It is recommended that responses are transcribed, and articulatory patterns noted, including suprasegmental features: vocal tone, pitch, volume, resonance.

Phrases and sentences

The phrases and sentences are elicited by imitation, while the child is looking at the appropriate picture. They should be modelled at a slow, normal pace (approximately 3-4 syllables per second), with normal intonation and junction between words (i.e. not extra clearly, with pauses/glottal stops between every word).

Specific junction strategies are marked in some phrases, and care should be taken to include these in the model, if they are to be assessed in the child's production.

It is recommended that the sentences are tape recorded for later transcription (see notes this chapter for features to observe).

Connected speech

A taped sample of the child's typical (spontaneous) connected speech is elicited, by conversation or a story retell task. The sample is analysed, (see description, this chapter), and the checklist completed. The score can be related to a severity rating scale (see this chapter) for children of 4-5 years upwards.

Analysis of data

Data are analysed as appropriate, at each level, in terms of:

- Sounds, or sequences of sounds, which are correct
- Sounds, or sequences of sounds, which are incorrect
- Patterns in incorrect sounds/sequences
- Problems with syllable structure and junction between sounds
- Voice and prosodic features

Questions are included on the assessment forms as a guide to analysis. Problem areas, which are identified as next treatment targets, can then be probed for processing strengths and weaknesses (see above). The results of these probes will guide the choice of therapy approach(es) for each target. See assessment/treatment planning exercise below.

Scoring

A simple scoring system is included, in order that data can be compared and presented in numerical form, if desired, or in the form of a graph:

Single sounds	1	=	local adult model
	0	=	any significant error
DDK	1	=	can repeat sequence accurately once
	1	=	can repeat sequence accurately five times
	1	=	can produce 5 identical repetitions, even if inaccurate
	1	=	can repeat 5 repetitions in 3 seconds (2 syllable items) or in 4 seconds (3 syllable items)

NB Only syllable initial consonant accuracy is scored e.g. in “patacake”, only accuracy of p-t-k is scored.

Single words:	1	=	local adult model
	0	=	any significant error
Phrases/sentences	2	=	local adult model
	1	=	less than specified number of errors (see assessment form for details)
	0	=	more than specified number of errors

A significant error would normally be one which potentially compromised intelligibility. This could either be one which neutralised a phonemic contrast (e.g. /s/ → /ʃ/), or one which reduced clarity of articulation to a significant extent (e.g. /s/ → /t/). Another way of looking at this is to decide whether the child’s production falls outside the adult phoneme boundary, and is unacceptable production. For example, a dental [ʃ] falls within the phoneme boundary, and would be scored as correct, whereas [t ʃ θ] would all be incorrect. Where several attempts are made at one item, the best attempt should be scored. Further examples of scoring can be found in the assessment exercise, at the end of Chapter 4: Treatment Planning.

Scoring systems never deal adequately with all situations. It may be appropriate to modify this system for particular purposes. However, where such modifications are made, this should be noted clearly.

Percentage scores can be calculated for each section of the assessment. Care should be taken in interpreting scores when all items in a section have not been attempted.

Scoring instructions for each section are included on the assessment form.

Interpretation

The NDP assessment is not a standardised tool. However, data collected from typically-developing 3 and 4 year old children (Thompson 2009; Walrawens 2010) can be used to interpret the performance of children with features of developmental verbal dyspraxia:

- 3 & 4 year olds scored 85% or above on: single sounds, CV/VC words, CVCV words; CVC words and cluster words.
- 4 year olds scored 80% and above on multisyllabic words, phrases and sentences.
- 3 year olds scored 70% and above on multisyllabic words, phrases and sentences.
- As might be expected, there was more individual variation amongst 3 year olds.

Notes on assessment of younger children - further suggestions:

(i.e.: children with very limited language, vocabulary, attention, cooperation on formal testing)

- transcribe words child uses
- get parent to list child's words and approximate articulation
- name familiar objects/toys, including a variety of syllable structures and sounds
- imitation of a basic range of consonants /p b m t d n k g f s h w j (ʃ) (l)/
- imitation of as many vowels as possible, but start with basics /i a u ai iə au/
- symbolic noises
- imitation of CV syllables in play
- attempt 2-3 words from CVCV, CVC, multisyllabic and cluster lists, if possible, for comparison with subsequent assessments
- observe oral skills and feeding, and attempt imitation of oral movement in play
- note use of voice
- bear in mind child's age/developmental level, when interpreting results

Notes on assessment of older children - further suggestions:

(i.e. children who cope well with most items on the word lists, but still have problems with intelligibility)

- Start by listening to connected speech, and assessing intelligibility
- Work backwards to discover reasons for breakdown in intelligibility
- Check wider range of words at each level
- Check harder consonants /ʃ tʃ dʒ v z ʒ θ ð r/
- Check imitation of complex unfamiliar words and nonwords
- Check vowels, especially half-open postures
- Check problematic sound sequences, shown on treatment records
- Check oro-motor and sequencing skills
- Check voice and airstream
- Check prosodics: intonation, stress, rate of speech, fluency
- Check syllable/word joining strategies
- Note compensatory strategies
- Bear in mind the influence of language difficulties
- Assess phonological awareness and manipulation

Assessment and treatment planning exercise

Background information

The data below comes from a 6; 9 year old girl (E). Initial assessment shows that she has wider language and communication difficulties, but age appropriate literacy skills (including reading/spelling of non-words). She has a history of severe oro-motor dyspraxia. Her hearing is normal.

Analyse the assessment data, using the questions on the assessment forms as a guide. After reading the Treatment Planning section below, identify next treatment targets for E, and note any probes you would wish to make. A solution is given at the end of Chapter 4: Treatment Planning.

Diacritics used in the transcription are as follows:

- Ç - dental articulation
- Ç ʋ - more open articulation
- Ç ʋ̥ - devoiced

Name E

Date _____

Age 6;9 years

Single Consonants & Vowels

Elicited by picture naming.

Scoring: 1 = correct (local adult model)
 0 = incorrect/unable to attempt
 NT = not tested

Target	Response	
Easier consonants	p	<i>p</i> 1
	b	<i>b</i> 1
	m	<i>m</i> 1
	t	<i>d</i> 0
	d	<i>d</i> 1
	n	<i>n</i> 1
	k	<i>t</i> 0
	g	<i>d</i> 0
	h	<i>h</i> 1
	w	<i>w</i> 1
	j	<i>j</i> 1
	f	<i>f</i> 1
	s	<i>θ</i> 0
Harder consonants	ʃ	Silent posture 0
	l	<i>ðə</i> 0
	ŋ	NT
	v	NT
	z	NT
	tʃ	<i>t</i> 0
	dʒ	<i>d</i> 0
	θ	NT
	ð	NT
	r	<i>ð</i> 0
	ʒ	NT

i	<i>i</i>	1
a	<i>a</i>	1
u	<i>u</i>	1
ɔ	<i>ʊ</i>	0
ɜ	<i>ɜ</i>	1
εə	<i>æ:</i>	0
aɪ	<i>aɪ</i>	1
ɑʊ	<i>ɑʊ</i>	1
ɪə	<i>ɪə</i>	1
əʊ	<i>əʊ</i>	1
ɔɪ	<i>ɔwɪ</i>	0
eɪ	<i>aɪ</i>	0
ɪ	<i>ɪ</i>	1
æ	<i>æ</i>	1
ʊ	<i>ʊ</i>	1
ɒ	<i>ʊ</i>	0
ʌ	<i>ʌ</i>	1
ε	<i>ʌ</i>	0

TOTAL consonants /24	(9)
TOTAL vowels /18	12
TOTAL /42	
%	

What consonants are imitated correctly?

What consonants are incorrect? Any patterns?

Can any be elicited in other ways/ occur spontaneously?

What vowels are imitated correctly?

What vowels are incorrect? Any patterns?

Can any be elicited in other ways or occur spontaneously?

Any voice features to comment on?
Slightly hoarse/breathy + low, static pitch

Name EDate _____ Age 6;9 years

CV & VC Words

Target	Response	
moo	<i>mu</i>	1
door	<i>tu</i>	0
bee	<i>bi</i>	1
go	<i>tau</i>	0
knee	<i>ni</i>	1
deer	<i>dɪə</i>	1
boy	<i>bɔɪ</i>	1
car	<i>ta:</i>	0
cow	<i>dau</i>	0
two	<i>du</i>	0
pay	<i>bəɪ</i>	0
four	<i>fʊ</i>	0
sea	<i>θi ti θi</i>	0
shoe	<i>fu</i>	0
chair	<i>θɛ teə</i>	0
lie	<i>ðai</i>	0
out	<i>au?t</i>	1
up	<i>ʌ? ʌ?bʌ ʌp ʌp</i>	1
arm	<i>a:m</i>	1
ice	<i>aɪθ</i>	0
	TOTAL /20	8
	%	40

Elicited by picture naming.

Scoring: 1 = correct (local adult model),
 0 = incorrect/unable to attempt
 NT = not tested

What consonants are used correctly?

What consonants are incorrect? Any patterns?

What vowels are used correctly?

What vowels are incorrect? Any patterns?

Are both CV and VC syllable structures used?

Are sounds joined smoothly?

Any voice/prosodic features to comment on?

Level pitch, sing-song quality

Name E

Date _____

Age 6;9 years

CVC Words		
Target	Response	
cake	<i>d̥aɪ?tθ</i>	0
boat	<i>bəʊ?tθ</i>	0
bird	<i>bɜ:d̥</i>	1
down	<i>d̥aʊn</i>	1
moon	<i>mu:n</i>	1
dog	<i>d̥ɒ?t</i>	0
girl	<i>d̥ɜ:n</i>	0
tap	<i>t̥æ?pʰ</i>	0
pig	<i>b̥ɪ?d̥</i>	0
cat	<i>d̥æ?t̥</i>	0
duck	<i>d̥ʌ?t̥</i>	0
farm	<i>fɑ:m̥</i>	0
horse	<i>hɔ?θ</i>	0
leaf	<i>ð̥i?f</i>	0
sun	<i>fʌn</i>	0
fish	<i>fɪ?θ</i>	0
light	<i>ð̥aɪt̥</i>	0
watch	<i>wɒ?θ</i>	0
jam	<i>d̥æm</i>	0
web	NT	
	TOTAL /20	3
	%	

Elicited by picture naming.

Scoring: 1 = correct (local adult model),
 0 = incorrect/unable to attempt
 NT = not tested

What consonants does the child use in initial position?

What consonants does the child use in final position?

What processes are seen?

Stopping

Prevocalic voicing

Fronting

Gliding

Backing

Final consonant deletion

Assimilation

Metathesis

Glottal substitution

Favourite articulation

Inconsistency

Imprecise placement

Weak articulation

Inadequate airflow

What vowels are used?

long vowels

diphthongs

short vowels

Any processes affecting vowels?

Are final consonants joined smoothly?

Lengthening

Lack of half-open vowels

Shortening

Neutralisation

Reduction of diphthongs

Nasalisation

Any voice/prosodic features to comment on?

Slow, deliberate articulation

Name E

Date _____

Age 6;9 years

CVCV Words		
Target	Response	
mummy	<i>mʌmɪ</i>	1
baby	<i>bɛɪbɪ</i>	1
teddy	<i>dædɪ</i>	0
cooker	<i>tʊ?tʌ</i>	0
dirty	<i>dʒ?dɪ</i>	0
dinner	<i>dɪnə</i>	1
party	<i>bɑɪ</i>	0
table	<i>tɛɪ?bʊ</i>	0
picking	<i>bɪ?tɪn</i>	0
cowboy	<i>taʊbɔɪ</i>	0
kettle	<i>dætu</i>	0
garden	<i>dɑ:dʌn</i>	0
tiger	<i>taɪdə</i>	0
water	<i>wʌ?dʌ</i>	0
fire	<i>faiɹʌ</i>	0
jelly	<i>dædʒɪ</i>	0
kitchen	<i>dɪ?dʌn</i>	0
coffee	<i>tʊ?fɪ</i>	0
tissue	<i>dɪ?dʒu</i>	0
sitting	<i>fɪ?fɪm</i>	0
	TOTAL /20	3
	%	15

Elicited by picture naming.

Scoring: 1 = correct (local adult model),
 0 = incorrect/unable to attempt
 NT = not tested

Are both syllables marked?

What consonants does the child use in initial position?

What consonants does the child use in medial position?

What processes are seen?

Stopping

Prevocalic voicing

Fronting

Gliding

Backing

Imprecise placement

Weak articulation

Inadequate airflow

Omission

Assimilation

Metathesis

Glottal substitution

Favourite articulation

Inconsistency

What vowels are used?

long vowels

diphthongs

short vowels

Any processes affecting vowels?

Are syllables joined smoothly?

Lengthening

Lack of half-open vowels

Shortening

Neutralisation

Reduction of diphthongs

Nasalisation

Any voice/prosodic features to comment on?

Glottal insertion, substitution

Omission of sound, or part of

Deletion of weak syllable

Syllables often produced with equal stress

Name EDate _____ Age 6;9 years

Complex & Multisyllabic Words

Target	Response	
pocket	<i>bʊ?dæ̃t</i>	0
ticket	<i>d̃ɪ?dæ̃t</i>	0
chocolate	<i>d̃ɒ?dæ̃t</i>	0
banana	<i>bæ̃næ̃næ̃</i>	0
birthday cake	<i>bɜːd̃eɪ d̃eɪ?t</i>	0
dinosaur	<i>d̃aɪnəʊθə̃</i>	0
caterpillar	<i>t̃æt̃æpɪj̃l</i>	0
helicopter	<i>hæ̃jɪtə?t̃l</i>	0
aeroplane	<i>hau?peɪ̃n</i>	0
crocodile	<i>t̃u t̃u?daɪ̃v</i>	0
television	<i>θ̃eɪwə̃tɪt̃un</i>	0
ambulance	<i>(refused)</i>	0
fire engine	<i>væ̃væ̃ væ̃vɪ̃n</i>	0
washing machine	<i>wə̃θɪ̃n mæ̃θɪ̃n</i>	0
computer	<i>t̃əʊp̃uɪ̃t̃l</i>	0
butterfly	<i>b̃l̃t̃lθ̃aɪ̃</i>	0
animals	<i>?æ̃nɪ̃..?æ̃nɪ̃..?æ̃nɪ̃</i>	0
umbrella	<i>ʌ̃m̃d̃ɛ̃ñl</i>	0
hospital	<i>p̃f̃ɒ?tɪ?t̃l</i>	0
sausages	<i>f̃ɒ?f̃ɒ?hɪ?</i>	0
	TOTAL /20	0
	%	0

Elicited by picture naming.

Scoring: 1 = correct (local adult model)
 0 = incorrect/unable to attempt
 NT = not tested

How many syllables can the child signal correctly?

Are plosives & nasals included, with place, voice and manner contrasts maintained?

Are fricatives and affricates included appropriately?

Are clusters included, initially, medially & finally?

Any vowel errors/distortions?

Are syllables joined smoothly?

Are stress patterns appropriate?

Tends to produce one syllable at a time, with equal stress

Any voice/prosodic features to comment on?

Name EDate _____ Age 6;9 years

Consonant Cluster Words		
Target	Response	
star	<i>tʒ</i>	0
sky	<i>taɪ</i>	0
snake	<i>deɪ?t</i>	0
swing	<i>wɪn</i>	0
blue	<i>bu:</i>	0
flower	<i>fjaʊwə</i>	0
clock	<i>dɒt</i>	0
tree	<i>tri:</i>	0
pram	<i>tæm</i>	0
frog	<i>dɒt</i>	0
fast	<i>fa?θ</i>	0
boats	<i>bəʊt</i>	0
hand	<i>hæm</i>	0
crisps	<i>dɪ?p</i>	0
biscuit	<i>bɪ?dɪt</i>	0
lift	<i>vɪ?θ</i>	0
jump	<i>dʌmp</i>	0
monkey	<i>mʌn?dɪ</i>	0
ice-cream	<i>aɪ?ɪm</i>	0
Christmas	<i>(refused)</i>	0
	TOTAL	0
	%	0

Elicited by picture naming.

Scoring: 1 = correct (local adult model)
 0 = incorrect/unable to attempt
 NT = not tested

How does the child deal with initial clusters?

/s/ +C	C+ /l/	C+ /r/

How does he deal with medial clusters?

Nasal + C	/s/ + C	C + /t.d/

How does he deal with final clusters?

Nasal + C	/l/ + C	/s/ + C
C + /s.z/	C + /t.d/	

Are clusters articulated smoothly? Any intrusive [ʔ]
 or [ə]?

Any voice/prosodic features to comment on?

Name EDate _____ Age 6;9 years

Phrases & Sentences (1)

Elicited by imitation, supported by pictures. (Model at a slow normal rate, in normal conversational speech. Take care to model the junctions marked.)

Target/Response		
My baby.	<i>maɪ beɪbɪ</i>	2
Katy's party.	<i>dɛɪtɪθ baɪ..baʔdɪ</i>	0
Tommy cutting paper.	<i>tʊmɪ tʌtɪnd dʌ peɪpə</i>	0
Farmer sitting.	<i>fɑmɪ sɪʔsθɪn</i>	0
Daddy is talking. (/j/)	<i>dædɪ ʔɪç tʊʔtɪn</i>	0
A cup of tea. (C#V)	<i>ə tʌt pɒð ti</i>	0
Two sad cats. (C#C)	<i>tu sæd tæt</i>	1
Get an apple. (n)	<i>dɛt æd æpu</i>	0
Go in the house. (/w/)	<i>dəʊ ʔɪn də haʊθ</i>	0
Dinosaur museum.	(refused)	0
Do you like jelly? (intonation)	(refused)	0

Scoring: 2 = correct (local adult model)
1 = only 1 error

0 = more than 1 error/unable to attempt
NT= not tested

<i>Assimilation</i>	<i>Junction</i>
<i>Final consonant deletion</i>	<i>Rate of speech/stress patterns slow very careful production, equal stress</i>
<i>Cluster reduction</i>	<i>Pitch/voice quality</i>
<i>Imprecise vowels</i>	<i>Resonance</i>
<i>Imprecise consonants</i>	<i>Intonation</i>
<i>Syllable deletion</i>	<i>Compensatory strategies</i>

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Chapter 4: Introduction to the Nuffield therapy approach and treatment planning

The Nuffield Dyspraxia Programme has been developed to meet the needs of children with severe speech disorders, and specifically those with significant difficulty with motor programs/programming and motor planning stages of the speech processing model.

The Programme focuses on building up articulatory skills, in small graded steps, through frequent, systematic practice. Phonological contrasts are also taught or reinforced. The materials can also be used to support other aspects of speech processing.

A comprehensive range of materials is included in the programme, to enable speech and language therapists to design appropriate intervention for individual children. The programme should be used flexibly, and supplemented, as necessary, with other resources and approaches.

The materials are most suitable for children between 3 – 7 years, but can be adapted for younger and older children (See Appendix 5). Activities drawn from the NDP have also been used with children with a wider range of speech difficulties (phonological disorder, dysarthria, cleft palate) and with adults and children with learning difficulties.

Aims of therapy

- 1 To build motor programs for speech in small graded steps, working from single sounds to complex words.
- 2 To facilitate speech work by working on auditory discrimination, as necessary.
- 3 To practise skills at each level until production is automatic.
- 4 To establish and/or reinforce phonological contrasts.
- 5 To work towards establishing a full range of psycholinguistic processing skills at each level.
- 6 To extend skills to sentence level and connected speech.

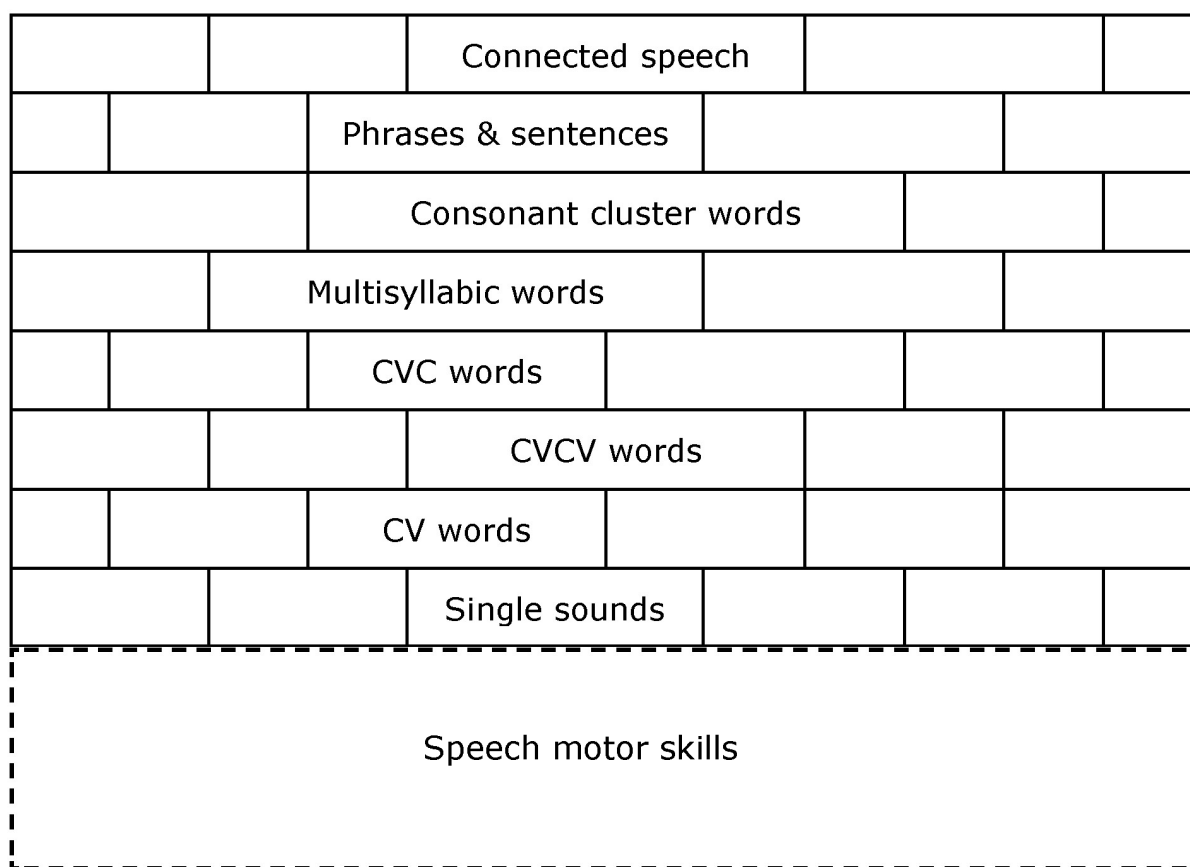
Nuffield therapy approach

Clinical experience at the Nuffield Centre of working with children with developmental verbal dyspraxia, and in keeping with views from the literature (see chapter 2: Therapy approaches) has demonstrated the importance of a motor skills learning approach in therapy.

Articulation is seen as a complex, hierarchical motor skill: the development of skills at higher levels being dependent on the mastery of lower level components. Skills are established by means of frequent repetition, elicited by cues and reinforced or modified with the support of specific feedback. Success is dependent on a thorough and cumulative approach, and frequent, consistent practice.

Motor speech skills can be conceptualised as a wall (fig 1), with single sounds at the bottom and connected speech at the top.

The aim of treatment is to “build the wall”, basically bottom up.



The focus is on establishing as full as possible a set of **motor programs** at each level of the wall, but also on supporting the development of a full range of **psycholinguistic processing** skills. Work on **eliciting specific sounds** or on **auditory discrimination** may be required in order to facilitate sounds which cannot be imitated. Where there is a dysarthric element, **motor execution** may be improved by oro-motor exercises. **Motor planning** is practised by means of repetition and sequencing tasks, and ultimately by sentence level work.

As speech is not only a complex motor skill, but also a linguistic medium, motor programs at each level must be contrastive. The sounds, syllables or words at each level are practised in contrast with each other, in order to establish a **contrastive system**. Sounds/words that the child can already produce are therefore included in activities, and form the starting point for building a contrastive system at each level.

While working on motor programs, **phonological representations** are also clarified, by making the structure of words explicit, through cueing with symbol pictures, and developing phonological awareness. Work on input (**phonological discrimination**) is included, as necessary.

Motor programming skills are practised while establishing programs for a wide range of vocabulary at each level, and in practising sequencing tasks.

Although the emphasis is generally on output practice, frequent modelling of targets, and the development the child’s awareness of sounds and articulatory features, also play an important part in treatment.

Top Tips

- Work on several targets at the same time
- Start from the sounds/words that the child is already able to produce
- Incorporate frequent repetitive practice, in order to establish motor programs
- Start from single sounds and CV syllables
- Work on eliciting sounds or discrimination, as necessary
- Use sequencing practice to consolidate motor programs and develop motor planning skills
- Create a contrastive system at each level of complexity
- Aim for consistent and accurate production of a wide range of sounds/words at each level
- Clarify phonological representations, and develop awareness of phonological structure
- Establish at sentence level before expecting generalisation to connected speech
- Incorporate work on voice/resonance from the earliest stages

Treatment planning

First treatment targets will involve “filling gaps” at single sound level i.e. extending the child’s speech sound range (consonants and vowels), and using the sounds that the child can produce to extend the range of CV syllables. It should be noted that a limited range of consonants /p b m t d n k g h f s w j (ʃ) (l)/ is targeted initially, with harder consonants added later. Long vowels are usually targeted first, then diphthongs and short vowels.

Single sounds and CV syllables are the basic building blocks. Consistent and accurate production is established by means of repetition, contrasting groups of sounds/words and, eventually, sequencing tasks. A wide range of CV syllables should be practised, including real words and nonsense syllable “babble”.

The single sound and CV building blocks are later used to build motor programs at more complex levels. Therapy activities are designed for each target, according to the processing strengths and weaknesses shown, (see assessment probes).

A full range of psycholinguistic processes should be facilitated at each level. At each level of the speech wall, therefore, the child should:

- have a sound system, initially including all vowels and a basic range of consonants /p b m t d n k g h f s w j (ʃ) (l)/, and eventually a full range of sounds
- discriminate between sounds
- have accurate phonological representations for a wide range of sounds, syllables and words
- have accurate motor programs for these
- execute motor programs as accurately as possible
- access required motor programs readily, and incorporate in sequences/phrases
- imitate unfamiliar real words and non-words
- follow instructions to modify production
- sort by onset and rime, or be working towards this
- segment and blend sounds, or be working towards this

The treatment programme will generally involve working concurrently on several targets at a particular level, and also targets at different levels of the motor speech wall. For example, velars and fricatives at single sound level, voicing contrasts at CV level, and CVCV words with easier sounds, could be worked on concurrently (See typical treatment plans below).

Developing specific psycholinguistic processing skills, using the NDP

Motor execution

Motor execution refers to peripheral speech output processing. Factors affecting motor execution include structural problems (e.g. velo-pharyngeal insufficiency), and difficulties with the range, strength and coordination of oral movement (dysarthria).

Where there is a dysarthric element to the child's speech difficulties, it may be appropriate to target range and strength of movement through non-speech oromotor activities. For children with a diagnosis of DVD practising non-speech activities is unlikely to result in improved speech production: intervention should be focused on sound production. See chapter 5 for further discussion.

Motor programs

The child with DVD has difficulty learning new motor programs. Motor programs must therefore be established in small stages, combining individual elements to create more complex programs. This involves learning programs for individual sounds, and sometimes articulatory features, and eventually combining these to establish programs for syllables and words of increasing complexity. It is likely to be necessary to teach programs for a wide range of sounds/words/syllables at each level, because of the child's difficulty in creating motor programs spontaneously.

The acquisition of motor programs is supported by the use of specific instructions and cues, (e.g. "put your tongue on your teeth") relating to movements which the child is already able to produce, and specific feedback on his production (e.g. "Careful! Your tongue didn't touch your teeth that time"). Cues and feedback may also be visual (e.g. a visual model and mirror for feedback; symbol pictures; articulograms-see Appendix 3). Selection of cues/feedback is based on the needs and preferred learning style of the individual child.

Suggestions for establishing motor programs at specific levels are included in Chapter 5: Therapy.

Storage and retrieval of motor programs is supported by the use of pictures and, if the child's speech and language skills allow, by using semantic and syntactic links (i.e. talking about what the word means; using it in a sentence).

In establishing motor programs, frequent repetition is vital. The NDP provides resources which can be used to play a range of simple games, (see Appendix 6 for games and activities), sustaining motivation for practice. Repetition can easily be built into these games, by saying each word 3 or 5 times, instead of once.

Motor programs which the child is working on should be practised daily, until production is consistent (say 95%) and as accurate as possible. Once this criterion has been reached, frequency of practice can be reduced. Newly acquired motor programs can then be incorporated into a set of items at that level, which the child is able to produce. These can then be used for contrasting and sequencing activities.

Phonological representations

Children with DVD frequently have inaccurate phonological representations for more complex, and less familiar, words. Some children show marked difficulties with this level of processing (see Chapter 7: case study Paul).

Phonological representations are clarified, while working on motor programs, by:

- Making the phonological structure of words explicit, through the use of sound symbols (symbol pictures/Cued Articulation (Passy 1990) to cue and check production
- Providing frequent auditory models, using slow, clear speech
- Practising discrimination of the child's errors, either using minimal pairs, or right/wrong lexical judgements
- Using picture cues, discussing word meanings and putting words in simple phrases/sentences, in order to consolidate the whole lexical representation, and aid retrieval
- Building up and practising a set of words at each level, and including a wider range of words, if the child is struggling to generalise his skills
- Developing phonological awareness (see Chapter 6: Developing literacy skills)

Auditory discrimination

Many children with DVD have specific, rather than pervasive, difficulties with auditory processing. It is therefore appropriate to work on discrimination skills relating to specific sounds/contrasts, (at the relevant levels), which are being targeted for treatment.

Auditory discrimination and motor programs are worked on "back to back", incorporating both input and output targets in practice. For example, a child who substitutes /s/ for /ʃ/, and is beginning to work on /ʃ/, will practise discriminating between /s/ and /ʃ/, at single sound level, while learning the correct tongue posture and placement, attempting /ʃ/ production, and ultimately /s - ʃ/ contrasts.

Care should be taken, when selecting auditory discrimination tasks, that the demands on memory and processing are appropriate to the child's age and ability. Examples of useful auditory discrimination tasks are included in Appendix 6: tasks, games and activities. These include: minimal pair judgement tasks, same-different judgement tasks, lexical judgement tasks and segmentation tasks.

In more general terms, auditory processing can be supported by providing frequent auditory models. For children experiencing difficulties in this area, models should be produced quite slowly, with target sounds emphasised, as necessary. Visual cues should

be used to support auditory input (e.g. symbol pictures, articulograms, Cued Articulation (Passy 1990), lip reading etc).

Motor programming

Motor programming is the “on-line” process of creating new motor programs, and is a core area of difficulty for children with DVD. These children’s motor programming skills must be supported in order for them to establish motor programs for words, (see Motor Programs, above).

Motor programming skills are developed/maximised, through the process of learning motor programs, by:

- establishing the basic building blocks of single sounds and CV syllables
- practising building more complex programs from these basic units
- developing the segmentation and blending skills to tackle unfamiliar words independently
- practising alternate sequencing tasks, where the child needs to create a motor program for the sequence

Further opportunities for practising motor programming skills can be created by:

- including a wider range of vocabulary at each level
- practising imitation of nonwords at each level
- production of rhyming sets of words
- other tasks requiring manipulation of words (e.g. phoneme or syllable deletion)

Where children have major difficulties with motor programming, they will need to learn strategies for learning new words. The development of segmentation skills enables the child to break words into syllables and phonemes, with a greater or lesser degree of support. Children who have developed basic phonic reading/spelling skills can then use a written “phonic” form to help them to practise saying the word (e.g. “e – leck – tri – si – ty”). Children without this level of literacy will need auditory models, broken into syllables, to repeat until the program is established.

Motor planning

Motor planning involves the assembly of the actual neurological instructions for an utterance, based on the motor programs for individual words (retrieved from the lexicon), in conjunction with the required grammatical structure. Assembly at the motor planning stage requires the modification of motor programs to incorporate the prosodic features of the utterance, and to facilitate word joining. An increased processing load is also created by the demands of linguistic formulation, and length of utterance.

This can also be a core area of difficulty for children with DVD, leading to inconsistent production of the same word, and a disproportionate breakdown of articulation/phonology at sentence or connected speech level.

Motor planning skills are practised from the earliest stages of the Nuffield Dyspraxia Programme, through sequencing activities. At each level, consistent repetition of targets is established first, followed by alternating sequences of 2, and later 3 items. Finally,

random sequences are practised. The speed of sequencing is increased, in order to strengthen motor planning skills.

Once a range of CV words is established, motor planning is also practised in short (2 word) phrases and sentences. Planning may need to be supported, initially, by picture cues for each word, e.g. Bee go (see Word combinations worksheets). Alternatively, the child may be required to access motor programs independently, in response to a composite picture, e.g. Four bees (see Word combinations worksheets). Once more complex structures have been established (e.g. CVCV and multisyllabic words) motor planning can be practised using the later worksheets in the Word combinations section e.g. SV, VO, SVO sentences, Everyday phrases and Complex sentences.

Prosodic features can also be practised through sequencing and phrase/sentence level activities (see Chapter 5: Therapy-Oro-motor/voice section, and also Connected Speech/prosodics).

Phonological awareness

Many children with developmental verbal dyspraxia have poorly developed phonological awareness (see Chapter 6: Developing literacy skills for further discussion).

The child's awareness of the phonological structure of words needs to be built up, alongside the development of motor programs, by:

- the use of sound symbols to cue production
- simple blending and segmentation tasks
C + V = CV
CV + CV = CVCV
- sorting to onset/rime, using:
CV lotto pictures
CVC rhyming sets
- syllable segmentation (clapping/jumping/counting syllables)

A complete programme to build phonological awareness skills, based on the NDP symbol pictures is described in Chapter 6: Developing Literacy Skills.

Self-monitoring

The ability to self-monitor and modify production is part of the process of developing consistent articulation, at each level of the motor speech wall.

Monitoring of auditory feedback takes place "after the event", and self-correction in this case is obvious. Self-correction is a positive step at each level, and should be commented on. This is likely to form an important part of the child's eventual strategies to maximise intelligibility.

However, the development of on-line monitoring of kinaesthetic/tactile feedback is necessary to allow fluent and accurate speech. From our experience with children with DVD, pacing of speech is crucial in maximising their planning and monitoring skills. Children with DVD have less efficient speech processing systems, and need to give themselves more time to produce accurate speech.

These skills are developed by aiming for fluent, consistent and accurate production, at each level, and extended by work on sequencing and sentence level, and, ultimately, by developing explicit strategies, if necessary.

Motivation

Establishing and sustaining motivation are important considerations when working with children with DVD. Treatment is likely to continue over a period of 2 years or longer, and frequent practice is essential. We generally recommend at least 15-20 minutes daily practice, where possible.

Success is probably the most important factor in motivating older children, and the adults working with them. Children tend not to find the repetitive practice boring, but rather find satisfaction and confidence in something they know they can do. There should be, at least, steady, perceptible progress in the skills the child is working on. If this is not achieved, the following issues should be considered:

- **Is the child working at too high a level?** Has he really consolidated lower level skills, so that they are achieved consistently and automatically (i.e. without effort)? If the child is struggling at sentence level, consider going back to word level (practise sequencing, and learn motor programs for a wider range of words). If he is struggling with CVC, CVCV or more complex words, go back and check single sounds and CV syllables, and practise sequences. Build up slowly to the next level (See Chapter 5 for specific treatment strategies).
- **Are skills being practised frequently enough?** It could be worth practising twice a day, for a period, and monitoring progress. Or consider the suggestions regarding motivation below.
- **Is the child receiving sufficient therapy input?** Although much straightforward practice can be carried out by an appropriately trained parent or support worker, children who are having difficulty moving on to the next stage of treatment may need more regular, or intensive, therapy sessions. Other children may benefit from short periods of intensive therapy from time to time, in order to master specific skills.

The following suggestions may also be useful in sustaining motivation and establishing effective practice:

- a) Establish a routine for practice. We all do certain repetitive things, such as cleaning our teeth, on a regular basis, without being “bored”, because we are so used to it that we don’t worry about it!
- b) Ensure the parent/support worker understands the reasons for repetitive practice.
- c) Ensure the parent/support worker knows what to expect from the child and is giving helpful feedback and plenty of praise.
- d) Ensure that the child achieves plenty of success, by including words and sounds that he knows he can do, as well as his “new” or “difficult” ones.
- e) If unfamiliar or difficult work is given, ensure that the parent knows what to do if it goes wrong, and has permission to “give up”. Explain to the child that “this is an experiment” or that “it might work some days, but not others”.

- f) Acknowledge the child's difficulty. Explain the reason, in appropriate terms, and how he will overcome it. Explain that it will take a long time and a lot of practice, but it will get easier (e.g. "Tongues take a long time to learn!"). Children can feel that they are responsible for the problem, and at the same time be unable to express their anxiety. We often explain to the children at the Unit that "when you were a baby, you wanted to learn to talk, but that bit of your brain didn't work properly and you couldn't do it. So now you are a bit older, we/you can teach your tongue and your lips what to do. But they do need a lot of practice, don't they?"
- g) Include functional vocabulary, adapting articulatory targets, if necessary, to enable the child to attempt words which he finds meaningful. For example, a child beginning to attempt CVCV words could practise his brother Robert's name, as /wɒbə/. For more information on adapted targets: see Appendix 5 Ideas for working with younger children.
- h) Work on a few harder, "key" words. One child, struggling with severe speech difficulties, and still working on single sounds and CV words, learned to say "banana", by practising everyday at snack time. This gave him the confidence that he would eventually manage longer words, and an understanding of how he would use his "little words" to make longer ones.
- i) Exploit the child's interests. If he's keen on football, devise a football-related presentation for practice etc. We have included football and train worksheet formats in this edition, and blanks are also available!
- j) Use a variety of presentations and activities. See Appendix 6 for additional games.
- k) Work from the child's strengths. Start a session or an activity with a skill that he is confident with (e.g. a motor task or an input activity).
- l) Encourage parents to focus on enjoying the activity with the child, rather than purely on progress, (which can often seem to be a case of two steps forward and one step back!).

Treatment plans

The following treatment plans show a typical progression of therapy for a child with developmental verbal dyspraxia:

Stage 1

- Establish a full range of vowels, at single sound level
- Establish a basic range of consonants /p b m t d n k g f s h w j/, at single sound level
- Monitor voice/resonance
- Work on basic pitch, volume and length control
- Establish a range of CV words using the single consonants and vowels the child can produce
- As new sounds are learned, incorporate them into CV words
- Begin sequencing single sounds and CV words

Stage 2

- Consolidate vowels, basic consonants and CV syllables/words, by sequencing
- Elicit next 2/3 consonants (probably /l ʃ ŋ/)
- Establish CVCV words with full range of vowels and easier consonants
- Establish CVC words with full range of vowels and easier consonants
- Incorporate new sounds, once established at single sound level
- Consolidate basic pitch, volume and length control
- Monitor vocal tone, pitch, resonance at CV, CVCV and CVC levels
- Monitor lip, jaw, and tongue postures, and airstream at CV, CVCV and CVC levels

Stage 3

- Establish later sounds (probably /tʃ dʒ v z/)
- Increase speed of sequencing and retrieval of CV, CVCV and CVC words
- Consolidate CVCV and CVC words with current range of sounds, and incorporate new sounds and wider vocabulary
- Practise 2/3 word phrases, using CV, CVCV and CVC words only
- Introduce first multisyllabics
- Introduce first clusters
- Extend prosodics to practise simple intonation patterns at phrase level
- Monitor vocal tone/resonance and articulation, as above

Stage 4

- Establish acceptable articulation of last consonants /r θ ð/, and phonological contrasts at single sound and word level
- Establish a wide range of multisyllabics and words with consonant clusters
- Establish consistent and accurate production for a range of phrase and sentence level tasks, including articulation of grammatical words and word joining strategies
- Extend control of voice and intonation, to include changing focus and shouting
- Establish strategies for learning new words
- Establish strategies for maximising intelligibility: pacing, monitoring, communication skills

Treatment records

Treatment record forms are included, to facilitate planning of treatment and recording of progress. The treatment records highlight key skills, and common problems, at each level of syllable structure. For treatment strategies, see Chapter 5: Therapy. Boxes can be shaded as targets are achieved.

Oro-motor skills

The oro-motor record charts the development of non-speech and speech skills for each structure (lips/jaw, tongue, palate, respiration/voice), including the acquisition of basic articulatory features.

Single sounds

Consonants and vowels are listed, as well as key contrasts, which need to be established in order to develop the phonological system. These can also be practised, later, as sequences, in order to develop motor planning skills. Sounds are grouped according to articulatory features.

CV syllables/words

Consonants and vowels are listed, and can be checked off as they are incorporated into CV syllables. Key contrasts are also listed.

CVCV words

The most common difficulty at this level involves sequences of consonants with different place, manner or voice, within words. Consonant sequences are listed, according to articulatory features (e.g. labial-alveolar, plosive/nasal). Fricatives, liquids/glides and affricates are listed separately, and can be checked off once they are included at this level. Vowels are grouped as long, short or diphthong. Present tense verbs with -ing are also recorded at this level. Acquisition of a wider vocabulary of CVCV words can be recorded, along with the child's ability to imitate unfamiliar structures (motor programming).

CVC words

The same consonant sequences are listed at this level, in addition to vowel groups, wider vocabulary, and unfamiliar structures. Final consonant groups are also listed separately at this level, along with two items which tend to cause problems at this level: diphthongs followed by a nasal consonant, and final /l/.

Multisyllabic words

The number of syllables is listed, in addition to the range of consonants, accuracy of vowels, and inclusion of consonant clusters. Stress patterns and junction between syllables is noted, plus the child's ability to deal with wider vocabulary and unfamiliar words.

Consonant clusters

Basic categories of initial, medial and final clusters are listed. The child's ability to deal with wider vocabulary and unfamiliar words is also noted.

Word combinations

Early word combinations are listed by syllable structure/range of consonants, plus some simple grammatical elements and junction strategies.

Sentence level

A range of phonological, articulatory and prosodic features are highlighted, to facilitate planning of work at this level. The use of compensatory strategies and the child's ability to combine linguistic formulation with motor planning, are also included.

Connected speech

The same list is reproduced at connected speech level, for children who are working at monitoring these features.

Name _____

D.o.b. _____

Oromotor Skills Record***Shade sections when achieved*****Lips/jaw**

Automatic: feeding, open/closed, smile, blow, kiss	
Imitation: smile, blow, open, closed	
Movement sequences	
Bilabials /p b m /	
Labio-dentals /f v /	
Basic vowels (smile/open/round) / i ʌ u ɪ æ ʊ /	
Diphthongs & glides / aɪ ɔʊ ɪə w j /	
Half-open vowels & diphthongs / ɜ ɔ ɛə ʌ ɔ ɛ əʊ eɪ ɔɪ /	

Airstream

Basic blow	
Controlled blow	
Plosives /p b t d k g /	
Fricatives /f s ʃ /	

Voice

Vocalisation/laugh/cry/babble	
Imitates sounds/words	
Intonation in babble/jargon	
Imitates rise/fall tones	
Imitates loud/quiet voice	
Imitates long/short voice	

Tongue

Automatic: chewing, swallowing, licking	
Imitation: protrude, retract, lateral, elevation, depression	
Sequences: in/out, up/down, left/right	
Front/alveolar /t d n /	
Mid/palatal /j /	
Back/velar /k g /	
Grooved/alveolar /s z /	
Slit /pal-alv / ʃ tʃ dʒ /	

Front/high /i ɪ /		Back/high /u ʊ /
Front/half-open / ɛə ɛ /	Mid/ half-open / ɜ ʌ /	Back/half-open / ɔ /
Front/low /æ /		Back/low /ɑ /
Diphthongs / aɪ ɔʊ ɪə əʊ eɪ ɔɪ /		

Palate

Basic blow	
Harder/longer blow	
Nasal consonants /m n /	
Open vowels /æ ɑ /	
Close vowels /i u /	
Plosives /p b t d k g /	
Fricatives /f s /	

Comments

Name _____

D.o.b. _____

Treatment Record (1)***Shade sections when achieved*****Single Sounds**

p	b	m	f	v			w
t	d	n	s	z	θ	ð	l
tʃ	dʒ		ʃ	ʒ			r
k	g		h				j

i	a	u	ɜ	εə	ɔ
ɪə	aɪ	ɑʊ	əʊ	eɪ	ɔɪ
ɪ	æ	ʊ	ʌ	ε	ɒ

Single Sound Contrasts/Sequences

Max contr's	h/b...
Plos—nas	b/m, d/n
Lab—alv	p/t, b/d
Lab—vel	p/k, b/g
Alv—vel	t/k, d/g
+vcd/-vcd	p/b, t/d, k/g
Fric—plos	f/p, s/t
Nas—liqu	n / l
Grooved/slit	s / ʃ
Fric—affric	ʃ/tʃ, ʒ/dʒ

Basic LV's	i/a/u
Basic Sh V's	ɪ/æ/ʊ
Basic Diph's	aɪ/ɑʊ/ɪə
Long/short	i/ɪ, u/ʊ, a/ɒ, ɜ/ʌ, εə/ε
Long/diph	i/ɪə, a/aɪ, u/ɑʊ εə/eɪ, ɔ/ɔɪ, ɜ/əʊ
Close/ half-open	i/εə, ɪ/ε, u/ɔ
Open/ half-open	æ/ε, a/ɔ, a/ɜ, əʊ/ɑʊ, aɪ/eɪ

CV Syllables/Words

p	b	m	f	v	w
t	d	n	s	z	l
tʃ	dʒ		ʃ		r
k	g		h		j

i	a	u	ɜ	εə	ɔ
ɪə	aɪ	ɑʊ	əʊ	eɪ	ɔɪ
ɪ	æ	ʊ	ʌ	ε	ɒ

CV+CV Contrasts/Sequences

Max contr's	h/b...
Plos—nas	b/m, d/n
Lab—alv	p/t, b/d
Lab—vel	p/k, b/g
Alv—vel	t/k, d/g
+vcd/-vcd	p/b, t/d, k/g
Fric—plos	f/p, s/t
Nas—liqu	n / l
Grooved/slit	s / ʃ
Fric—affric	ʃ/tʃ, ʒ/dʒ

Basic LV's	i/a/u
Basic Sh V's	ɪ/æ/ʊ
Basic Diph's	aɪ/ɑʊ/ɪə
Long/short	i/ɪ, u/ʊ, a/ɒ, ɜ/ʌ, εə/ε
Long/diph	i/ɪə, a/aɪ, u/ɑʊ εə/eɪ, ɔ/ɔɪ, ɜ/əʊ
Close/ half-open	i/εə, ɪ/ε, u/ɔ
Open/ half-open	æ/ε, a/ɔ, a/ɜ, əʊ/ɑʊ, aɪ/eɪ

Name _____

D.o.b. _____

Treatment Record (2)***Shade sections when achieved*****CVCV Words**

Consonant Sequences

same place	vel—lab	+vcd/-vcd
lab—alv	alv—vel	/f s ʃ/ +
alv—lab	vel—alv	/l w j/ +
lab—vel	plos/nas	/v z tʃ dʒ/ +

[]ing verbs

Wider range CVCV

Imitates unfamiliar words/ nonwords

Vowels

long vowels

diphthongs

short vowels

CVC Words

Consonant Sequences

same place	vel—lab	+vcd/ -vcd
lab—alv	alv—vel	/f s ʃ/ +
alv—lab	vel—alv	/l w j/ +
lab—vel	plos/nas	/v z tʃ dʒ/ +

Final Consonants

- vcd plos	fricatives	Diph + nasal
+ vcd plos	nasals	//

Vowels

long vowels	diphthongs	short vowels
-------------	------------	--------------

Wider range CVC

Imitates unfamiliar words/ nonwords

Consonant Clusters

Initial clusters

/s/ +C	C+ //	C+ /r/
C+/j.w/	/s/ + CC	

Medial clusters

Nasal + C	// + C	/s/ + C
C + /s.z/	C + /t.d/	C + //

Final clusters

Nasal + C	// + C	/s/ + C
C + /s.z/	C + /t.d/	C + //

Wider range cluster words

Imitates unfamiliar words/ nonwords

Multisyllabics

1 2 3 4 syllables	+ clusters
plosives + nasals	vowels correct
+ easy frics	syllable junction
+ harder frics	stress patterns
wider vocab	unfamiliar words

Wider range multisyllabic words

Imitates unfamiliar words/ nonwords

Name _____

D.o.b. _____

Treatment Record (3)***Shade sections when achieved*****Word Combinations**

CV + CV (plos, nas, glides)	CV + CV (+ frics/affrics)	3 element sentences
CVCV + CV (plos, nas, glides)	CVCV + CV (+ frics/affrics)	Gramm. is, the
CVCV + CVCV (plos, nas, glides)	CVCV + CVCV (+ frics/affrics)	Junc: /j/, /w/, C#V
CVCV + CVCing (plos, nas, glides)	CVCV + CVCing (+ frics/affrics)	

Sentence Level

Maintains place, manner voice contrasts appropriately	<u>Appropriate junction</u> : V#V- /w/, /j/, an, /r/, C#V, C#C
Includes final consonants	<u>Appropriate intonation</u> : nuclear tones, use of focus, basic intonation pattern
Includes clusters	Acceptable voice quality
Includes weak syllables, and multisyllabics	Appropriate rate of speech
Appropriate lexical stress patterns	Acceptable resonance
Accurate articulation of consonants and vowels	Appropriate volume control
Able to formulate own sentence while sustaining articulation	
Appropriate articulation of grammatical morphemes	Compensatory strategies

Connected Speech

Maintains place, manner voice contrasts appropriately	<u>Appropriate junction</u> : V#V- /w/, /j/, an, /r/, C#V, C#C
Includes final consonants	<u>Appropriate intonation</u> : nuclear tones, use of focus, basic intonation pattern
Includes clusters	Acceptable voice quality
Includes weak syllables, and multisyllabics	Appropriate rate of speech
Appropriate lexical stress patterns	Acceptable resonance
Accurate articulation of consonants and vowels	Appropriate volume control
Utterances of 20+ words, while sustaining articulation	
Appropriate articulation of grammatical morphemes	Compensatory strategies

Solution to assessment and treatment planning exercise

Analysis of data and notes on scoring

You will have noted from the background information that E's speech difficulties appear to be on the output side of the speech processing model (history of oro-motor dyspraxia and age appropriate reading/spelling of non-words).

Single sounds

- She has a limited range of consonants /p b m d n w j h f/
- She has no velars
- Alveolars are produced with interdental placement
- /t/ is realised as /d/, although there does not appear to be a general pattern of voicing at this level
- Harder tongue tip sounds /s ʃ l/ are not achieved
- She manages a basic range of vowels, but few requiring a half-open jaw posture
- She shows problems with voice and prosody

Scoring

- 1 Dental placement [d̪] for /d/ is scored as correct, as it falls within the phoneme boundary.
- 2 Vowels transcribed as the target phoneme, but noted to be too open /ɜ ʌ/, were scored as correct. The transcription reflected the fact that E's production, although inaccurate, probably fell within the phoneme boundary. Where a different vowel phoneme was heard, this was scored as incorrect.
- 3 A number of harder consonants were not tested. This makes it difficult to calculate a percentage score. Her production might be better summarised by saying that she was able to imitate 9/13 of the easier consonants, and 12/18 vowels.

CV and VC words

- Both CV and VC structures are attempted
- Voicing contrasts /p – b/ and /t – d/ are inconsistent
- Otherwise articulation reflects single sound production
- Her attempts to improve her articulation of "up" and "sea" show active self-monitoring

Scoring:

- 1 The devoiced [b̥] in "pay" would be scored as an error, because it neutralises the voicing contrast p/b. However, E scores 0 for this item, in any case, because of the vowel error.
- 2 E is not penalised for her intrusive glottal stop in "out" [aʊʔt], as this falls within the phoneme boundary. However, it remains a consideration in therapy, if it is a consistent feature of her articulation, as she needs to be able to join sounds smoothly, in order to produce fluent and acceptable speech.
- 3 She made several attempts at "up", and her best attempt is therefore scored.

CVCV words

- Both syllables are marked
- The range of consonants and vowels reflects that used at CV level
- Place contrasts (alveolar/labial), and manner contrasts (plosive/nasal), are achieved, in line with her single sound inventory
- Insertion of glottal stops between syllables, equalised stress, and some assimilation of consonants, suggests that she is having greater difficulty with sequencing at this level

Scoring:

Vocalisation of syllabic /l/, e.g. table [tɛɪbʊ] is accepted as correct, as this is essentially a CVCV list. Fronting of "ing" endings to [ɪn] is also accepted, as this is such a common variant. See transcription list for further acceptable variants. Remember that you may add your own, according to local accents.

CVC words

- The range of consonants and vowels reflects use at CV level
- Place and manner sequences are achieved, in as far as she is able to articulate the sounds concerned
- Final voiceless plosives, fricatives and nasals are used
- Substitutions are systematic, apart from some confusion over fricatives and occasional assimilation
- Articulation is effortful

Scoring:

The scoring of articulation errors such as "farm" [fɑm] and "boat" [bəʊtθ] is difficult. In the end the decision has to be a subjective one, based on how acceptable the child's production is judged to be.

Complex and multisyllabic words

- E signals up to 4 syllables, fairly reliably
- Sequencing breaks down at this level, with frequent assimilation and metathesis (e.g. /θɛwɪtɪtʊn/ - television, /ʊæʊæʊæʊɪn/ - fire engine)
- Stress is frequently equalised

Consonant clusters

- Final and medial nasal clusters are beginning to be signalled
- Otherwise, clusters are reduced to a single element

Phrases and sentences

- Her articulation is much as at word level
- Production is slow and careful, and may not therefore reflect E's connected speech, or may be a compensatory strategy
- You will have noted her reluctance to complete the task, and surmise that she is struggling!

Treatment plan

Work simultaneously on:

1 Consolidating and extending single sound system

Probes:

- You will want to investigate her oro-motor skills, and attempt to elicit /t k g s l/ and the missing vowels
- You may wish to check the consistency of the sounds that she did manage to imitate, and motor planning for sequences
- You will want to check her auditory discrimination of substituted sounds /t – k/ /eɪ - aɪ/ etc, although this is probably not the problem, given her strengths in phonological awareness
- You will want to investigate vocal control

Therapy targets:

- Specific oro-motor skills to develop tongue tip control to elicit /t s l/
- Specific oro-motor skills to develop tongue back control to elicit /k g/
- Work on half-open lip/jaw postures
- Discrimination, as necessary, to support work on the above sounds
- Sequencing of current range of sounds, in order to consolidate motor programs, motor planning, and phonological system

2 Consolidating and extending CV words/syllables

Probes:

- You will want to investigate imitation of voiced/voiceless minimal pairs, response to verbal instructions and discrimination between minimal pairs

Therapy targets:

- /p t/+V = CV transitions, and repetition of /p t/ CV's
- Voiced/voiceless CV discrimination
- Attempt voiced/voiceless output contrasts
- Sequencing of current range of CV words, in order to consolidate motor programs, motor planning, and phonological system

3 Developing control of voice and prosodics

Probes:

- You may wish to investigate her use of pitch, volume and length control in spontaneous vocalisation.

Therapy targets:

- Practise basic pitch, length and volume contrasts on easier vowels and CV's (*See Chapter 5: Therapy*)

Discussion

E still needs to develop basic skills at Stage 1 of the treatment plan.

She has an unusually pure motor speech disorder. Many children with DVD need to work at developing accuracy and awareness of phonological representations, alongside output work (see above).

E shows good use of the sounds and CV syllables available to her in attempting CVCV and CVC words. Other children require more support with sequencing in order to establish motor programs at higher levels (e.g. toe + bee = Toby).

References

Passy, J. (1990) Cued Articulation. STASS Publications.

Chapter 5: The therapy approach

Expanding the phonetic inventory

Contents

Early sound making activities (new)

Lip shapes and articulograms

Voice worksheets

Eliciting sounds (Appendix 4)

Aims

- 1 To establish motor programs for a range of consonants and vowels
And, as appropriate,
- 2 To develop motor programs underlying specific features of speech sounds
- 3 To develop awareness and control of lips, tongue, airstream and voice for speech
- 4 To increase the accuracy of articulation

Eliciting sounds

The first challenge when working with children with DVD is to elicit a range of sounds and syllables. At the beginning of therapy, children are frequently unable to imitate several of the earlier developing consonants and vowels.

Various strategies can be used to attempt to elicit sounds. Bernthal, Bankson and Flipsen (2009, p 287) identify four approaches:

- imitation, from an auditory/visual model
- phonetic placement, where the SLT instructs the child where to place the articulators to produce a specific sound
- successive approximation, where a new sound is shaped from one already in the child's repertoire
- contextual utilisation, where a sound is isolated from a particular phonetic context in which the child happens to produce it correctly.

Imitation is the simplest and therefore preferred method. Frequent clear models should always be given when working on sound production.

Examples of each of the other approaches can be found in Appendix 4 – Eliciting Sounds, where strategies for eliciting specific sounds are described.

Young children with DVD are often not only unable to imitate sounds but also have difficulty organising oral movements to follow instructions regarding phonetic placement. They may have a very limited phonetic repertoire, and find it difficult to modify existing motor programs. Hence, both successive approximation and contextual utilization may be of limited use.

Further strategies used to elicit sounds in children with DVD include assisted movement, which ranges from helping the child to achieve lip and jaw positions (examples can be found in Appendix 4 and also in the Early Sound Making worksheets in Therapy Resources p6), to the complex system of tactile cues used in PROMPT (Hayden, Eigen, Walker and

Olsen, 2010). For older children with persisting articulation difficulties with lingual consonants, use of an electropalatograph (EPG) may be considered.

The role of oral motor exercises

The NDP has traditionally used nonspeech oral motor exercises (NSOMES), i.e. in the main, licking and blowing activities, to develop awareness and control of oral movements, as a precursor to phonetic placement techniques. Indeed oral motor skills have been viewed as the foundation for speech motor control, and therefore a necessary component of intervention for at least some children with DVD (Williams and Stephens 2004).

NSOMES are defined as any non-speech intervention that aims to “influence tongue, lip and jaw resting postures, increase strength, improve muscle tone, facilitate range of motion and develop muscle control” (Ruscello 2008). Clark (2010) summarises as follows:

“...the underlying assumption (of NSOMES) is that facilitating sensorimotor function of the speech musculature during nonspeech activities will enhance sensorimotor control for speech sound production”.

However, recent reviews of the theory and evidence base for NSOMES (Lof 2003; Lass and Pannbacker 2008; McCauley, Strand, Lof and Schooling 2009) have challenged both the logic and the effectiveness of targeting oro-motor skills with a view to improving speech. A position paper on childhood apraxia of speech produced by the American Speech-Language Hearing Association (2007) concludes that the consensus opinion among researchers is that nonspeech oro-motor therapy is not necessary or sufficient for improved speech production. Two recent surveys of the literature conclude that there is currently insufficient evidence to support or refute the use of NSOMES to produce effects on speech (McCauley, Strand, Lof and Schooling 2009; Ruscello, 2010).

In the light of these conclusions, the recent RCSLT UK Policy Statement on Developmental Verbal Dyspraxia (2011) advised caution in the use of such approaches.

Having examined the arguments and the evidence afresh (see below), we have modified our recommended approach.

Theoretical issues

Ruscello (2008) points out that “when a practitioner employs a specific treatment, there is an implicit acceptance of the theory underlying the treatment”. The rationale for the use of NSOMES in speech sound disorders is based on the following assumptions:

- developmental speech disorders are caused by neuromuscular impairment
- neuromuscular control for speech and nonspeech actions is the same
- learning can be facilitated by working on components of a skill separately.

These assumptions are disputed in the light of recent research and current theoretical understanding.

The nature of DVD

While children with a diagnosis of DVD may have difficulty with the control of oral movements, muscle weakness and disorders of muscle tone are not consistent with this diagnosis. Children with DVD should not therefore need exercises to strengthen muscles or normalize muscle tone. Where children have a mixed dysrathric/dyspraxic presentation, a case might be made for targeting these areas.

It should also be noted that it has been shown that relatively little force is required for speech. Clark (2010) indicates that forces produced during normal speech are around 20% of the maximum force that can be produced by the articulators in nonspeech movements.

Specificity of neuromotor control

If the control of articulatory structures is the same for speech and non-speech activities, it is logical to assume that skills/control will carry over from one to the other. Unfortunately, the evidence does not support this view. Speech has been found to require finer levels of coordination and lower levels of strength in comparison with nonspeech movements (ASHA 2007). Lof (in Bowen 2009) cites studies of various articulatory structures including lips, jaw, tongue and palate, which demonstrate that organisation of speech and nonspeech oral movement within the nervous system is different. For example, a Study by Bonilha, Moser, Rorden et al (2006) demonstrated by means of functional magnetic resonance imaging (fMRI) that different parts of the brain are activated during speech and nonspeech movements.

This makes it unlikely that practising nonspeech movements, such as licking and blowing will result in improved speech. Further, it challenges the usefulness of practising skills such as lateral tongue movements which are not components of speech.

Motor learning theory

Research into the learning of motor skills has suggested that complex and highly integrated skills are more efficiently learned as whole units, rather than broken down into component skills which are practised separately.

This has been used to argue that it may be counterproductive in speech interventions to break skills into smaller components, and specifically to target oro-motor skills (Forrest 2002). However, it should be remembered that the research cited did not investigate speech skills or speech interventions. Moreover, the participants being trained presumably had intact neuromotor control. Whereas normal speech development does involve learning whole words, the child with DVD may be unable to learn in this way. Indeed it is suggested that a core component of DVD is a difficulty in learning speech motor skills.

Effectiveness of NSOMs as intervention for speech disorders

Two systematic reviews of the evidence for the effectiveness of NSOMs have been carried out recently by McCauley, Strand, Lof and Schooling (2009) for ASHA, and Ruscello (2010). Both conclude that there is currently insufficient evidence to support or refute the use of NSOMs to produce effects on speech. It should be noted that mixed interventions, where NSOMs were combined with speech interventions, were not investigated in these reviews.

Conclusions on the use of NSOMs

- Practising non-speech activities such as licking and blowing, in isolation, is unlikely to change speech.
- Targeting strength or muscle tone in children with a diagnosis of DVD is not necessary.
- Any non-speech movements targeted should be closely linked to the target sound, i.e. a component movement, such as lip closure, should be followed by production of / m / where possible. (See Early Sound Making sheets, Therapy Resources 1, p6)

Thomas Campbell (2012), writing on the Apraxia Kids website provides a sensible summary of current recommended practice in relation to the use of NSOMs for this population:

"If the treatment objective is to improve speech-sound production, then speech tasks should be used whenever possible. For young children with severe speech motor deficits and limited speech-sound inventories, you must start somewhere. For these children, imitation of non-speech movements may be all they can achieve initially and could serve as a foundation for future sound production. It makes conceptual sense to choose non-speech tasks that are components of speech gestures and directly linked in time and space to a speech-sound production. These non-speech tasks should be faded or eliminated as soon as the child is able to produce the speech sound of interest. In addition, some focus on non-speech oral movements at the start of a therapy session (2-3 minutes) may serve to increase awareness of the speech system and establish a starting point or anchor for a group of related speech sounds."

Early sound making sheets

The early sound making sheets (Therapy Resources 1, p6) provide instructions on developing consonants and vowels which occur early in typical development, by eliciting articulatory postures and movements associated with specific features, and combining these to create speech sounds.

The first two worksheets focus on encouraging sound making and establishing voluntary control of vocalization. The next three sheets combine vocalization and a voiceless airstream (blow) with basic lip shapes and tongue movements to produce / m b i a u / and / h p t k /. An activity is provided to practice placement for a basic range of plosives and nasals / p b m t d n (k) (g) /. A further sheet focuses on developing placement for the velars / k g /. Two worksheets develop a basic set of long and short vowels and diphthongs, and the glides / w y / using three basic lip shapes. Instructions for developing the vowels requiring intermediate jaw positions are provided on a separate sheet. Early CV syllables are also practised.

Encouraging sound making	page 6
Vocalisation	page 8
Exploring sound production: / m b ee ah oo /	page 9
Exploring sound production: / d n /	page 10
Exploring sound production: / h s p t /	page 11
Basic lip shapes: / ee ah oo m / and short / i a oo /	page 12
Lip shape sequences: / eye ow ear / and / w y /	page 13
Placement for sound production: / p b m t d n (k) (g) /	page 14
CV babble	page 16
Half-open lip shapes: / air or er ay oy oh / and short / e u /	page 17
Velars: / k g /	page 19

Sheets include suitability criteria, and notes for therapists, and the option of customizing instructions for a particular child. These sheets replace the Advice Sheets for Parents which focused on NSOMs.

Children who have little or no speech

Children who have not yet begun to develop speech, (and who are older than the age where speech typically emerges), present particular challenges. The diagnosis is often unclear, and even when there is evidence of oro-motor difficulties or a family history of DVD, the child's lack of speech may be due to wider motor, cognitive or language delays. These children may go on to develop speech relatively easily, but with greater difficulties

in other areas. Alternatively, they may indeed turn out to have DVD. Even with intervention, a small number (usually with additional difficulties) will not develop functional speech.

Such children will need SLT intervention including communication and language development as well as speech. Children with age-appropriate cognitive skills or verbal comprehension, from around the age of 2½ - 3 years, should try the early sound making sheets, and then progress through the levels of the programme. Some of the suggestions on the sheets may be appropriate for younger children. The key is to try, but not to persist for months if unsuccessful. It is important to ensure that young children with severe speech disorders have support for communication and language development. This may take the form of signing, a communication book, home-school diary or voice output communication aid.

For children with additional difficulties, we would suggest that some sound making or babble play is included in their therapy program along with work on other skills.

Children with severe oro-motor difficulties may require assessment and intervention for feeding skills. It may be appropriate to try some simple blowing activities, voice production and lip shapes (see the Early Sound Making sheets, and therapy suggestions which follow here). However, we would not recommend that these are continued for more than 6 weeks unless there is definite progress towards speech sounds.

Therapy ideas

Therapy ideas for developing particular aspects of motor control for speech (airstream, voice, lips, tongue and palate) are included below. These are designed for children who have difficulty with these aspects of speech production, and may be helpful in expanding the child's phonetic repertoire or increasing accuracy of production. Suggestions start with non-speech activities to elicit the movement required, followed by activities designed to incorporate this into speech production. Verbal cues such as "lips together" and/or visual cues such as the articulograms (Therapy Resources 1, p21) should be linked to articulatory features to facilitate access to motor programs.

Bearing in mind the comments above, therapists should select the appropriate activities for a particular child. It is not intended that these are used with children who do not have difficulties in these areas. Where a child is not progressing, there is no point continuing with these activities.

Further suggestions for eliciting specific sounds can be found in Appendix 4. See also Early Sound Making sheets (Therapy Resources 1, p6).

Top Tips for expanding the phonetic inventory

- Establish production of a range of early consonants and vowels
- Practise skills in the context of speech sounds wherever possible
- Explore a range of elicitation strategies
- Include frequent clear models
- Elicit the movements or postures which produce particular articulatory features
- Link with verbal and/or visual cues
- Combine movements and postures to form speech sounds
- Practise new sounds frequently
- Link sounds to picture cues
- Practise simple syllables in babble play
- Ensure maximal oral control through appropriate seating and posture
- Be aware of fatigue and hypersensitivity

NB For simplicity, the child is referred to as "he" in all the therapy exercises, but they are of course applicable to both boys and girls.

Therapy

Respiration/Larynx

Clinical work at the Nuffield Hearing and Speech Centre suggests that many children with dyspraxic type difficulties in their speech, present with atypical voices: - husky, gravelly, creaky and/or breathy type presentations can occur. On ENT examination, there is rarely any pathological finding, suggesting that the difficulty lies in the controlled timing of the vocal folds and the expiratory airstream, leading to phonatory and vocal difficulties. In addition, it is clear that some children have particular difficulties controlling their respiration for speech. The following activities have been found useful to improve control over respiration, phonation and aspects of voice production.

Aims

- 1 To develop adequate breath support for speech
- 2 To develop control of initiation of voice
- 3 To extend vocal control: length, volume, pitch
- 4 To improve voice quality

Equipment

Pipe and ball, candles, bubbles, whistles, pipes, recorders, straws, plastic tubing, toy aeroplane/helicopter, toy slide, tightrope, cotton reels, roads/paths.

Early Sound Making sheets (Therapy Resources 1, p6).

Voice worksheets (Therapy Resources 1, p24).

Computerised equipment such as the laryngograph is also useful for many aspects of voice work.

1 Developing adequate breath support

The expiratory phase of the normal breathing pattern can sometimes be shortened, or poorly co-ordinated in children with dyspraxic difficulties, resulting in only one or only a few syllables per breath. As a consequence, their speech can sound jerky and stilted, and intonation patterns cannot be used to help convey meaning.

Begin by trying to extend the expiratory phase during a non-speech activity e.g. using blowing activities: aim for long gently controlled blows and for an increasing number of short blows on one breath. For long controlled blows, try:

- Pipe and ball. Aim to keep the ball spinning a couple of inches above the cradle, for as long as possible.
- Blowing through a straw into a bowl of water with some washing-up liquid (very rewarding, but beware if child does not have blow/suck contrast fully established).
- Blowing big bubbles with a bubble pipe.
- Any blowing toy which requires a strong and sustained blow helps to strengthen respiratory muscles.

For younger children, and those with very severe difficulties, it is often worth starting the exercises with plastic tubing rather than a straw, as the tubing is less easily crushed, thus success is more likely. Wine-making tubing is a good size; different sizes of tubing can be found in DIY and hardware stores, for plumbing of washing machines etc.

For several blows on one breath:

- Blow out candles on a board (one blow per candle). Gradually increase the number of candles to be blown out from two to five. Larger candles are much harder to blow out, so to succeed as soon as possible, use birthday cake candles.
- Use a whistle or recorder. Give two blasts on one breath, and increase the number of blasts.

2 Control of Initiation of Voice

- Once the child can initiate voice, help him to be aware of this. Show pleasure and give praise for any sound he makes.
- Make him aware of the sound that he has made, and encourage the use of any different sounds he is able to make.
- The aim at this stage of voice initiation, is to encourage the child to produce voice on request, and in a sequence i.e. not random voice production, which is often what the child is doing. He needs to learn to vocalize in response to a cue e.g. "switch your voice on".
- Use voice worksheets with vowel symbols or simple games, using coloured dots to represent vocalisation e.g. model for the child a repeated vocalisation of a long vowel ah /a,a,a,a/ and see if he can copy that model. Encourage gentle initiation, good vocal tone and resonance, together with an even rhythm. Start slowly and increase speed only when the child is able to sustain vocal tone, rhythm etc.
- There are a range of toys available which respond to noises or talk back -these can be very stimulating for some children and are worth trying to motivate voice production.

3 Improving voice quality

a) Principles:

- Make sure work is concrete and visual to aid the child's comprehension of the rather abstract concepts involved.
 - Always provide a vocal model when working on voice e.g. demonstrate high-low loud-quiet sounds, etc. Much of the work is aimed at eliciting an accurate imitation of the therapist's model.
 - Different textures may help to explain e.g. rough-sandpaper, smooth-stone, etc.
 - Fluid, easy movements can help to represent smooth quality.
 - "Good/Bad" should not be used to describe voice. Try more accurate descriptors e.g. too loud/too quiet; too high/too low; too long/too short etc.
 - Use long vowels initially for voice work before attempting any work at word level.
 - Incorporate work on voice into speech work from the beginning and continue monitoring voice as therapy progresses.
 - A set of symbols to represent different voice parameters may be helpful with some children, e.g.:
- | | | | | | |
|---------------|---|-------------------|--------------|---|------------------|
| ➤ <u>loud</u> | - | gorilla | <u>quiet</u> | - | ladybird |
| ➤ <u>long</u> | - | caterpillar | <u>short</u> | - | frog |
| ➤ <u>high</u> | - | mosquito (in air) | <u>low</u> | - | cow (lying down) |

But be careful to avoid confusions, if you intend to use "concepts" to describe speech sounds e.g. long/short sounds, loud/quiet sounds (Howell and Dean 1994).

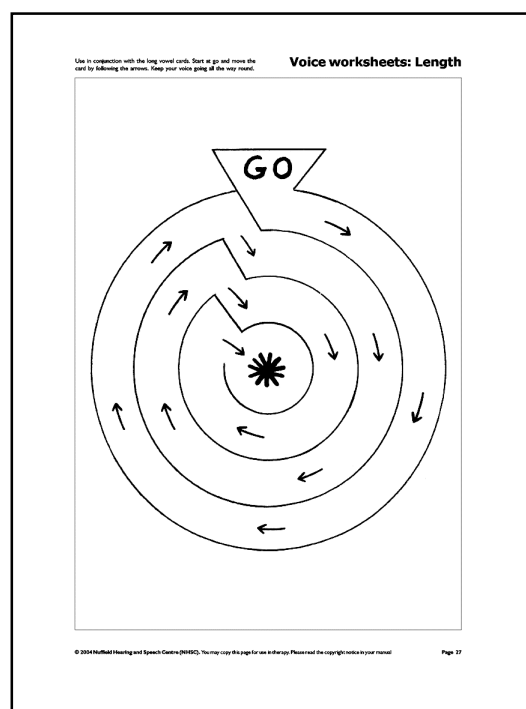
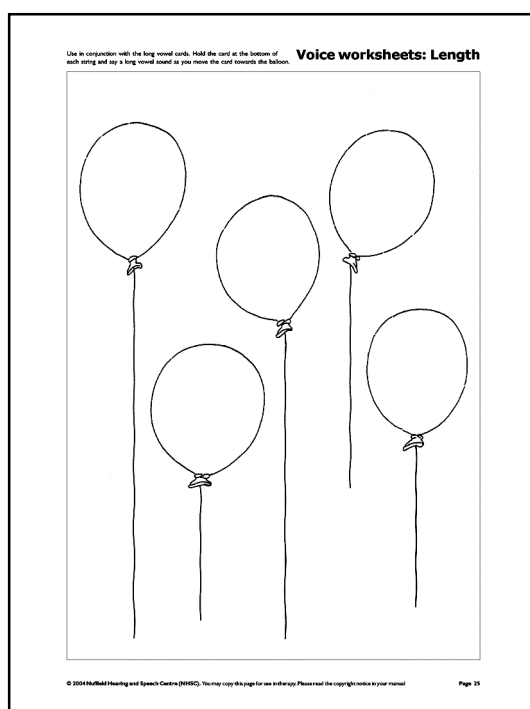
- Begin by introducing a simple contrast e.g. producing a high and low sound, playing with a helicopter/aeroplane etc. then, as the child develops more control, aim for graded productions e.g. very low - low - slightly higher - high etc. Finally, try to elicit a smooth continuum from one pitch to the next. Speechviewer III can be very useful to support such activities.
- Be aware of the other dimensions of voice quality when working on one particular aspect e.g. make sure when trying to increase length of production that pitch and volume are constant.
- Encourage humming, singing or rhythmic chanting of songs and rhymes. It does not matter whether he manages to get the words or the tune, as long as he begins to use his voice for the length of vocalization demanded by the song, and with some rhythmic pattern.
- These activities should be done for “fun” and in a relaxed way, so that both child and adult can enjoy them. It can be hard for therapists and parents to persevere with this type of work, when the child’s response is almost non-existent, and parents may need support and encouragement.

(b) Activities

As a general principle work on length is advised first, then work on volume and finally work on pitch.

Length of Production

- Tightrope. Tie a length of string around two blocks. Use a small teddy or doll to move across the tightrope - the doll only moves when the child uses his voice. Aim to get the doll across in one breath, while producing a long vowel.
- Cotton Reels. Use a reel of cotton or string with a bead at one end. The child pulls the bead, making a vowel sound. See how far the child can move away from the stationary reel.

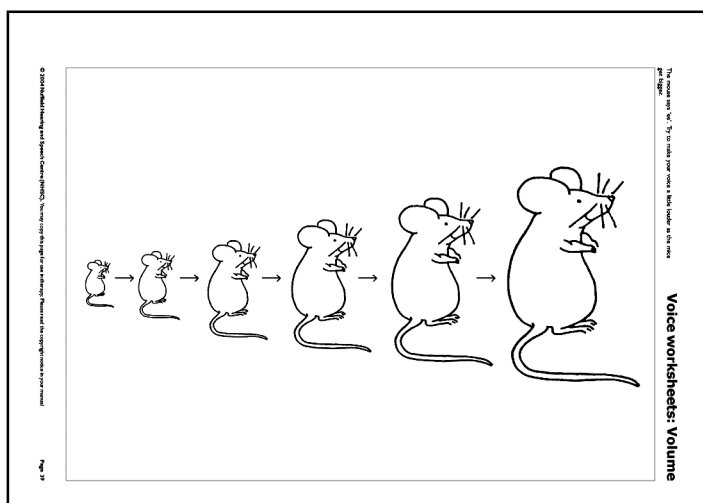


- Draw a long path. The child is to take an animal/doll home to tea. The child can only move along the path when using his voice. See how many breaths it takes to reach home.

- Worksheets –use ladders, balloon strings, Go wheel, snail spiral, roads (see first Ringbinder) and long vowels to encourage long vocalizations etc.
- Make the child aware of long vs. short vocalisations. The voice worksheet (see first Ringbinder) with different length roads can be useful.

Volume control

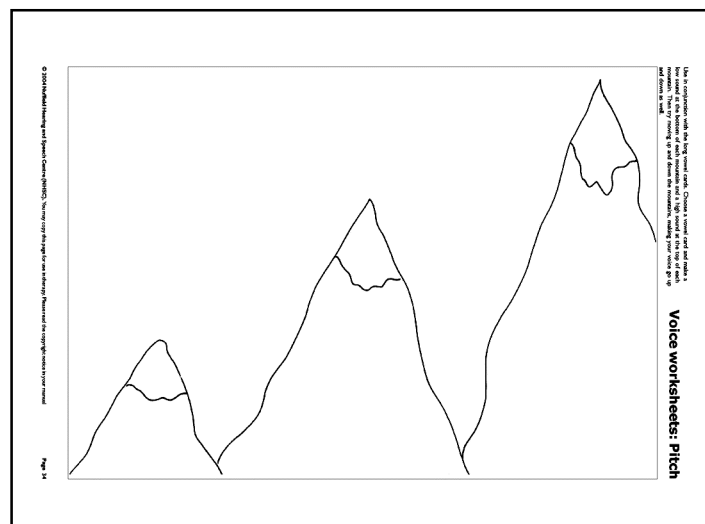
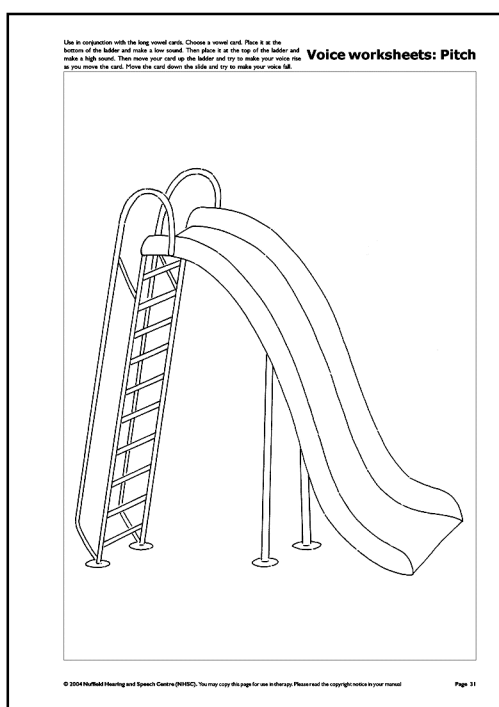
- Introduce the concept of loud vs. quiet sounds.
- Use different sized balls of play dough to encourage the child to produce loud and quiet sounds. The child bangs the large ball and produces a loud sound, and then a small ball is patted for a quiet sound, etc.
- Imitate the sounds of pairs of animals, e.g. big/little pigs, big/little lions, etc.
- Use the voice worksheets which show a gradual increase in size to increase the volume, e.g. for /i/ the pictures increase from a little mouse to bigger ones (see first Ringbinder).
- Find animals or environmental sounds that differ in volume and try to imitate them e.g. mouse-lion, dog barking-cat purring, drill-clock ticking etc.



Pitch control and intonation

- Introduce the concept of high vs. low, e.g. produce a high pitch whilst standing on a chair, a low pitch while standing on the floor, etc.
- Slide. Start teddy at the top of the slide (a model or a picture) and elicit a high pitched sound. When the teddy reaches the bottom, produce a low pitched sound. When enough control is achieved, aim for a continuum of high to low or low to high.
- Worksheets of slides, mountains, diving boards, ladders, helter-skelters (see first Ringbinder) can be used with a doll or teddy, or small pictures of the long vowels, to elicit rising or falling intonation patterns.
- Snakes and ladders worksheet (in first Ringbinder): produce variations of pitch as the child has to move up the ladder or down the snake.
- Try walking downstairs or upstairs producing graded changes of pitch.
- Play games using different voices - a baby, a giant, a man, a little girl/boy, a mouse, a bull etc. This encourages awareness as well as control.
- Use stories and books to model different voices and patterns. Tones such as happy, sad, angry, surprised, scared etc can all be imitated.

- For further ideas on intonation at a higher level, see from Sentence level to
- Connected speech section at the end of this chapter.



Lips

Aims

- 1 To achieve accurate lip shape postures at a non- speech level in isolation and in sequence
- 2 To achieve correct lip shape for individual speech sounds, both in isolation and in sequence

Equipment

Mirror, tubing and straws, blowing toys, lipstick, slide, puppets, lip shape pictures (Therapy Resources 1, p23). Also see Early Sound Making sheets (Therapy Resources 1, p6).

1 Non-speech level

- Introduce the four basic lip postures. Start with open and closed, followed by spread and round postures. As control improves the quality of all 4 basic postures should be monitored for accuracy, strength and speed/ease of elicitation, and these aspects worked on as necessary.

a) Mouth open

The mouth should be opened fully but not beyond the point where subluxation of the temporo-mandibular joint occurs. A rounded rather than spread shape can be encouraged, to facilitate the production of ah /a/ for accents where this is a rounded vowel. This can be supported by holding the corners of the child's mouth in with finger

and thumb as the mouth is opened. However, if this proves very difficult accept a more spread shape in the early stages.

b) Lips together

Both lips should be visible (not sucked in, or held between the teeth). Closure should be firm and symmetrical with equal pressure from top and bottom lip, and must be bilabial rather than labio-dental. Support if necessary by pushing upwards with one finger under the lower lip. Strengthen and encourage even pressure by asking the child to hold a piece of card between his lips, keeping it horizontal. Try to avoid excessive tension.

c) Smile

The posture should be symmetrical, and the lips should be parted to show his teeth. Support by pushing the corners of the child's mouth outwards. Ideally, the upper and lower teeth should be slightly apart. Excessive tension should be discouraged. This posture may be elicited by getting the child to hold a spatula crosswise between his teeth.

d) Round mouth

Lips should be rounded and slightly protruded. The hole should be approximately the diameter of a drinking straw. Reduce the size of the opening by letting the child close his lips round a straw, then slowly removing the straw while he watches in a mirror and tries to retain the posture. A child who is unable to imitate lip rounding may be able to achieve closure round a 1/2" - 1/4" diameter plastic tube or a blowing toy. The tube is withdrawn gradually as he learns to sustain or anticipate the posture.

- Encourage the child to sustain postures for up to 4 seconds.
- Once several of the basic lip shapes are achieved fairly readily, you can begin to attempt sequences of two. Practise these slowly at first. Continue to monitor the quality of the postures and encourage the child to monitor them himself, using the mirror. Use the four lip shapes to make simple games, books or worksheets and see Appendix 6 for other ideas e.g. fishing, pairs etc.
- Other activities which encourage a range of different lip movements include: licking and chewing with a variety of food tastes and textures, toothbrush play and blowing toys.

2 Linking lip shapes to speech sounds

a) Basic shapes and sounds

Once the four basic lip shapes are achieved easily in response to a verbal or visual cue, try combining them with vocalisation to produce the sounds /i a u m/. Play games to reinforce the association between individual sounds and lip shapes e.g. matching lip shapes to sound pictures etc. (See Appendix 6: games).

b) Bilabial sounds /m/, /p/, /b/

Once a good "lips together" posture is achieved, bilabial sounds e.g. /m/ /p/ and /b/ can be attempted (See Appendix 4 for elicitation ideas).

c) Basic diphthongs

Assuming that tongue postures for /i u/ are reasonable, the basic diphthongs /aɪ əʊ ɪə/ can be taught as a sequence of two basic lip shapes, plus vocalisation. These will need to be taught in an exaggerated fashion e.g. /a/ → /i/ blended together slowly to make "eye" /aɪ/. Symbol pictures on sticks/straws, which can move and slides can be useful for these vowel blending activities.

d) Harder lip shapes /ɔ/, /ɜ/

For these vowels, establish an intermediate degree of jaw opening (try between 1/2" - 3/4" between upper and lower teeth), and combine this with spread and rounded lip postures. If jaw control is difficult, get the child to practise holding the handle of a small toothbrush edgewise between his teeth.

e) Short vowels

Lip/jaw postures for short vowels can be shaped from their corresponding long vowels: e.g. Shorten /i/ to produce /ɪ/. Also see sound elicitation ideas (Appendix 4).

Repetitions of each sound should be practised, and then sequences of two sounds, monitoring lip shape and any other features as necessary. Use vowel lottos and worksheets as appropriate.

Tongue

Aims

- 1 To develop tongue protrusion
- 2 To develop awareness and control of tongue movements
- 3 To develop correct placement for alveolar sounds
- 4 To develop placement for velar sounds

Equipment

Mirror, torch, toothbrush, spatulas, small lolly sticks, plates, egg cups, spoons, sugary/sugar-free foodstuffs*, gummed envelopes/stars/stickers.

Foodstuffs*

Sugary foods: 100s and 1000s, chocolate flakes, sherbet, icing sugar, chocolate spread, honey, jam, treacle, ice cream, ice lollies, mints, small sweets.

Sugar-free foods: marmite, ground almonds, grated apple, grated carrot, mashed banana, yoghurt, sugar-free sandwich paste, sugar-free lollies (e.g. Chuppa Chups).

Oral Health Note:

It is recommended that sugary foods and drinks are used only at mealtimes. At other times, sugar-free foods should be used to avoid the risk of tooth decay. It is not recommended that teeth are cleaned immediately after eating sugary foods. Children's teeth should be brushed routinely for 2 minutes, morning and night, using a toothpaste containing no less than 1000ppm of fluoride. Children under 7 years should be supervised. It is important to arrange regular check ups with a dentist who will then be able to offer individual oral health advice. (Our thanks to: Mrs K Kerr, Senior Oral Health Promoter, Bucks Priority Dental Service.)

In the exercises below, reference is made to some sugary foods. Similar sugar-free alternatives may be used.

1 Encourage tongue protrusion

- This can be elicited by placing a spatula vertically in front of the child's mouth.
- Encourage him to touch the spatula with the tongue.
- Initially any protruding movements should be rewarded by moving the spatula to reach the tongue.
- Gradually increase the distance between the mouth and stick to encourage further protrusion.
- Licking sugar strands off the plate whilst it remains on the table can also be useful to elicit tongue protrusion. Be careful to ensure there is no lip involvement to facilitate picking up the sweets

2 Develop awareness and control of tongue movements

Note that these movements are not necessary for speech. If the child is able to attempt them, they can be an enjoyable way of developing awareness and control of tongue movements. However, some children cannot achieve this range of movement. This activity should not be continued if the child is unable to achieve any tongue movement outside his mouth.

A licking activity can be used to develop awareness and control of the front of the tongue. Place the spatula at various positions around the mouth:

- To develop lateral movement the spatula is placed at the corners of the mouth.
- Depression of the tongue outside the mouth is encouraged by placing the spatula at right angles to the chin with sugar strands facing upwards.
- To develop elevatory movement begin by asking the child to open his mouth and protrude his tongue. Place the spatula vertically in front of the mouth and encourage the child to lick up the stick. Further elevatory movements can be encouraged with the spatula at right angles on the upper lip with sugar strands on the underside. The spatula should always be placed just outside the child's range of movement. When the maximum movement has been achieved the spatula should be moved to meet the tongue to provide reinforcement and feedback.
- Compensatory movements of the head and jaw should be discouraged, and the task modified so that it can be achieved without them. It may be necessary to hold the child's head in place initially, or place a finger firmly on the chin to discourage any concomitant jaw movement.
- The above tasks can be varied by placing the sugar strands directly on, or around, the lip area.

3 Encouraging tongue tip development and linking it with correct placement for alveolar sounds

- To encourage accurate tongue placement for alveolar sounds tell the child to open his mouth, and either hold a spatula (sweet side down) under the upper teeth, or place some sweets behind his upper teeth, on the alveolar ridge. Place a finger firmly on the chin to resist the jaw closing and ask him to lick the sweets.

- Once this movement can be achieved, alveolar sounds e.g. /t/, /d/, /n/, /l/ can be introduced. Begin by placing the sweets on the alveolar ridge and encouraging the child to "find" the sweets by saying any one of the alveolar sounds. Once the correct pattern of movement has been established try producing the sounds with no sweets. If it is difficult to move on to this level an intermediate level may be introduced. This involves the child touching his own alveolar ridge with a finger to increase sensory feedback and then producing the sound.
- The majority of tongue tip sounds can be produced acceptably with a limited range of movement, although they are likely to be dental in placement.

4 Encouraging movement of the tongue body for velar consonants /k/, /g/ and /ng/

- Velar consonants may be elicited by holding the tongue tip down with a spatula or finger. Then ask the child to produce /k/, /g/ and /ng/.
- Alternatively, velar consonants may be elicited from babbling, whilst chewing on a toothbrush. As the front of the tongue is held down by the toothbrush, velar sounds are often produced.
- For further suggestions see Early sound Making sheets and Appendix 4.

Palate

Aims

- 1 To encourage development of an oral airstream
- 2 To develop a sustained oral airstream
- 3 To develop an awareness of oral/nasal airstreams
- 4 To produce oral/nasal speech contrasts, beginning at a single sound level and working up to control in connected speech

Equipment

Candles, bubbles, paper fish, cotton wool, pipes, whistles, tubes, straws, pipe and ball, diving boards etc.

The role of blowing exercises

Children with severe speech difficulties, including those with dyspraxic type difficulties may fail to produce palatal closure because they have not established control over velopharyngeal musculature or because of a structural deficiency or a combination of both. If a structural deficiency is suspected, it is essential that a referral is made for a palatal investigation. Such a referral is also advised if the child has persisting difficulties in establishing an oral airstream.

There is no evidence currently that improving velopharyngeal control for blowing or other non-speech activities, carries over to speech. However, it is acknowledged (Shprintzen 1995) that blowing exercises may be useful to introduce the concept of an oral vs. nasal airstream to a child.

Blowing exercises are introduced independent of speech sound production. However, once some basic movement patterns are established, speech sounds should be associated with these patterns, as soon as possible. Thus voiceless plosives and fricatives are attempted by combining a "blow" airstream with other articulatory movements. For example /p/ can be produced by combining a blow airstream with lip closure.

1 To encourage development of an oral airstream/awareness of oral/nasal airstreams

- From the beginning, blowing is linked with a palatal cue, such as "make the air come through your mouth, not your nose". Older children may benefit from a more detailed explanation of how they can prevent the air coming down their nose.
- If the child is unable to achieve palatal closure assist him by blocking the nose with your first finger and thumb beneath his nostrils - there is no need to pinch the nose. This will increase awareness of nasal airflow, and may trigger velopharyngeal closure.
- Once good oral blow has been achieved with the child's nose blocked, (indicating that the palate is raised, though not necessarily with complete closure), this support can be gradually withdrawn. Initially, choose blowing activities which require a relatively weak airflow e.g. blowing candles. As the child's palatal control improves, activities requiring a stronger blow can be used e.g. whistles, pipes, blowing bubbles through a wand and through a straw in water.
- To achieve sustained palatal closure for non-speech activities, gradually increase the length of blow required e.g. blowing paper fish into a pond using a straw. Using a pipe and ball: a short blow will make the ball jump out of the cradle, but a long blow will make it spin above the cradle or arrange candles some distance apart on a horizontal line so that a long blow is necessary to blow the candles out. As control increases, contrast with short blows.

2 Incorporating speech sounds

- Once the child can achieve an oral airstream on a blowing task (with feathers, candles, bubbles, paper etc), a speech sound can be used instead of a blow e.g. /h/, /p/, /f/ to cause movement.
- Fricatives are particularly useful for working on sustained length (e.g. say /h/ slowly and make the cotton-wool move along the table) and aspirated plosives for quick, explosive actions (e.g. try to make each of the candles go out as you say /p/).
- Alternating an oral blow with a /p/ sound (e.g. make the paper move along the table as you say /p/ and then blow) can help to increase the level of control required.

3 Developing oral-nasal contrasts (speech sound level)

- Try introducing oral and nasal speech sound contrasts by using the sounds /m/ and /b/. (If the plosives are difficult, ask the child to say /m/ while you block his nostrils, thus producing a weak /b/. If the child has achieved palatal closure for blowing, use whatever cues you have established to help raise the palate. Gradually decrease this support until the child is able to produce /b/ on his own). Once the child can readily produce strings of /b b b b b/, introduce the contrast /m-b-m-b/. You can also try to contrast /n/ and /d/, working through the same steps to contrasting /d-n-d-n/.

- Once accuracy is achieved at a single sound level, contrasts can be introduced at a CV level e.g. /bee-me/, /dough-no/. If this stage is difficult go back to a C + V level ensuring correct resonance on the consonant and following vowel whilst aiming towards blending the two sounds. Once contrast is established at a CV level, introduce CVCV words. Initially, introduce words which involve no manner changes e.g. "money", and then work up to an oral-nasal manner change within a word e.g. "bunny".
- Gradually aim to establish the contrast in longer words, short phrases and finally connected speech.

Single sounds

Contents

Single sound reference grids (**new**)
Single sound lottos (2 sounds per page – **new**)
Mixed consonant lottos (some additional **new** symbols)
Mixed vowel lottos (some changes to vowel symbols - **new**)
Sequencing worksheets (**new designs**)
Articulograms (Appendix 3 – **new**)
Eliciting single sounds (Appendix 4 – **new**)
More games and activity ideas (Appendix 6 – **new**)

Aims

- 1 To help the child learn the picture symbols
- 2 To help the child learn how to produce new sounds in isolation, thereby extending his speech sound repertoire
- 3 To improve discrimination skills, as necessary
- 4 To consolidate motor programs and to practise retrieval
- 5 To reinforce contrastive nature of sound system
- 6 To develop sequencing and motor planning skills

Children with speech difficulties often have difficulties in producing speech sounds, even in isolation. In the case of children with developmental verbal dyspraxia, one of the identifying features of the condition is a limited range of consonant and vowel sounds (see Chapter 1: Literature Review). The explanation for this characteristic is phonetic rather than phonological i.e. it is not simply the case that the child uses a small repertoire of consonants and vowels within their contrastive speech sound system, but that they are unable to produce more than a handful of sounds, even by imitation of an adult model.

Children who present with single sound production difficulties require help to increase their speech sound inventory. For children with dyspraxic type difficulties, this is likely to involve work on both consonant and vowel sounds.

As already described in Chapter 4: Treatment Planning, single sounds are seen as the core building blocks or foundations from which speech is constructed and therefore work at this level holds the key to later success. Work on auditory discrimination or on the motor skills underlying specific articulatory features may be required to facilitate production of a wider range of sounds.

Use of symbol pictures

Speech and language therapists have traditionally used a variety of visual and manual cues to help children learn speech sounds. For example, picture symbols which represent speech sounds, are used commonly, especially with young children. The use of such symbols helps the child to create a representation for that sound and thereby aids retrieval. In teaching the child symbols, a shared system between the child and the therapist/parent is created. In the case of the Nuffield Dyspraxia Programme, this also allows ease of access to work at later stages of the Programme.

The Nuffield Centre Dyspraxia Programme was one of the first schemes to publish their symbols in 1985, and remains one of the few to offer both consonant and vowel symbols.

Consonant symbols

Following the revision of the Nuffield Centre Dyspraxia Programme in 1992, some modifications were made to the consonant symbols to ensure the onset of the picture symbol name matched the sound it represented e.g. /m/ for motorbike; /f/ for firework; /k/ for camera etc.

In this third edition, the consonant symbols have remained unchanged, although a small number of additional consonants have been included for completeness: /ŋ/, /ʒ/, /θ/, /ð/. In the first two cases, these sounds do not occur in initial position in English words and therefore the onset sound of the symbol – sound match cannot apply.

Reference grid		Consonant symbol pictures	
	p		ng
	b		w
	t		l
	d		r
	k/c		y
	g		h
	m		f
	n		v
			th
			th
			s
			z
			sh
			as in measure
			ch
			j

The pictorial symbols can be combined with other cueing methods e.g. manual cues such as Cued Articulation (Passy 1990); symbol cues such as Articulograms (Stephens and Elton 1986) a system of diagrammatic symbols used to help the child recognize and utilize the key features of a sound-(see Appendix 3); visual cues such as colour coding systems and/or orthographic cues such as letters to support phoneme-grapheme links.

Vowel symbols

In recognition of the importance of sound – letter links in early literacy acquisition, careful consideration has been given to changing some of the vowel symbols to provide a match between the onset sound of the picture symbol name and the vowel sound represented, in this third edition of the Nuffield Centre Dyspraxia Programme. Unfortunately, vocabulary constraints have limited this possibility across the vowel system. However, we have decided to make the following changes:

Short Vowels

Short vowels are particularly important in early literacy work and therefore it has been decided to change 4 of these symbols to enhance phoneme-grapheme links:

/ɪ/ represented by Incy (spider)

/ɛ/ represented by elephant

/ɒ/ represented by octopus

/ʌ/ represented by umbrella

Of the other 2 short vowels: /æ/ is already represented by ant; /ʊ/ continues to be represented by a monkey, due to vocabulary constraints, but it is suggested that the monkey is given a name "Oofy"/"Oogy", to help make a link.

Reference grid		Vowel symbol pictures	
	ee		ay
	ah		oy
	oo		ow
	oar		oh
	er		air
	eye		ear
			oo (as in 'look')
			o
			e
			u

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Diphthongs

The symbols for /eɪ/ and /ɛə/ are being changed as there are suitable alternatives which are more child-friendly, and easier to remember:

/eɪ/ represented by angel

/ɛə/ represented by aeroplane

The remainder of the symbols are unchanged, but it is recommended that some sound – symbol matches can be enhanced through the use of character names e.g.

Arthur crocodile says /ɑ/

Eddie mouse says /i/

Ernie bear says /ɜ/

Obie fish says /əʊ/

or through carrier phrases e.g.

The wind blows and says /u/

Row the boat with an /ɔ/

I can see with my /aɪ/

I can hear with my /iə/

The boy bumps his head and says /aʊ/

Noisy parrot shouts /ɔɪ/

Top Tips: Single sounds in isolation

- Introduce and work with several sounds at once
- Introduce consonants and vowels at the same time (use picture symbols as cues)
- Start with sounds the child can make already (i.e. from his strength)
- Gradually introduce new sounds
- Accept “approximations” of new sounds and refine them gradually
- Combine production and discrimination “back to back” to maximize progress
- ####Develop motor skills underlying specific articulatory features
- Utilize early sound making sheets (therapy resources 1) and elicitation ideas (see Appendix 4) to teach new sounds
- Provide opportunities for frequent repetitive practice in order to establish motor programs
- Make activities fun through the use of reinforcing games (see Appendix 6)

Therapy

1 Single sound production

- Start by introducing picture symbols for about 4-6 consonant and vowel sounds that the child can produce/imitate. Ideally, these should involve sounds from different phoneme classes and/or involve strong contrasts (Dodd 1995) - e.g. an open long vowel, a close long vowel, a voiceless fricative, a nasal, a voiced plosive etc.
- Include vowels from the beginning – remember that they carry a heavy information load (Dodd 1995) and that the distinction between working on consonants and vowels as distinct “categories” is more likely to be an issue for the therapist than for the child. Clinical experience at the Nuffield Centre suggests that children have no difficulties in working on consonants and vowels simultaneously, using picture symbols.
- Devise tasks to improve discrimination (input work, with the child as listener) and reinforce production (output work, with the child as speaker) thereby establishing and consolidating the picture symbol/sound links (see Appendix 6 for ideas).
- Once some sounds have been learned (so that the child can both discriminate and produce the sound consistently in response to the symbol) more can be introduced. In choosing the next sounds to introduce, consider: (a) stimulability – which may need to override all other considerations (Crary 1984; Dodd 1995) and (b) typical developmental acquisition order: plosives, nasals, long and short vowels, approximants (except /r/) and diphthongs, fricatives and affricates.
- Each child will, of course, learn at his own pace, but aim to teach the child 10-15 symbols and sounds in the first few weeks of therapy. The aim in introducing many sounds (rather than focusing on one sound at a time as in traditional articulation therapy), is to help the child develop a speech system of contrastive parts. This is consistent with normal speech development: children do not learn one sound to perfection before taking on another, rather they acquire a number of sounds simultaneously and gradually develop a system.

2 Eliciting speech sounds

Many children with speech difficulties can imitate a single sound in isolation, when given a model. However, some children including those with dyspraxic type difficulties may need to be taught how to produce individual sounds. Techniques for teaching sounds involve a variety of cues and support e.g. visual, verbal, tactile and kinaesthetic, oro-motor, other speech sounds etc, used either singly or in combination.

In this third edition of the Nuffield Centre Dyspraxia Programme, we have included detailed ideas for eliciting speech sounds (both consonant and vowel sounds) in Appendix 4, in response to requests from many speech and language therapists.

- Elicit the movements and postures of lips, jaw, tongue palate and voice underlying specific articulatory features. Aim to increase control and awareness of movement and structures. Link movements/features to verbal cues, such as “lips together” or “voice on”, and use these to facilitate production of speech sounds.

- Voiceless plosives /p/, /t/, /k/ should be taught without an accompanying schwa sound.
- Check that appropriate tone, pitch, volume and length are incorporated even at this level. Deliberately prolonging long vowels can be practised to increase control of prosodic features. Poor voice quality may indicate the child needs specific work on vocal control (See "Improving Voice Quality" in oro-motor section).

3 Single sound sequencing

Aims

- 1 To consolidate motor programs for single sounds
- 2 To reinforce phonological contrasts
- 3 To increase the speed and accuracy with which programs can be accessed
- 4 To develop motor programming and planning skills
- 5 To maintain sound/symbol links, so that these can be used to support work at higher levels

Top Tips for single sound sequencing

- Introduce sound sequencing work as early as possible
- Continue work on sequencing skills throughout therapy programme
- Start with repetitive sequencing of the same sound
- Start slowly, but increase speed when the child is ready
- Move from repetitive sequences to contrasting (alternating and random) sequences
- Start with contrasts that are very distinct and easy for the child to achieve
- Incorporate pitch work into single vowel sequences
- Incorporate rhythmic (e.g. p-p /p-p/ etc) and stress patterning (e.g. b-m /b-m/ etc) into the exercises gradually and support this with manual activity e.g. finger tapping
- Make sequencing activities fun by using games (see Appendix 6)

Children with speech difficulties are often reported to have difficulties in sequencing sounds together. In particular, difficulty with sound sequencing has been identified as a key feature of developmental verbal dyspraxia. Such difficulties may arise from unstable motor programs or motor planning difficulties, affecting the control of the timing and co-ordination of speech. Sound sequencing skills are thought to underlie the production of consistent and accurate sequences of sounds in connected speech, and practise of these skills is therefore continued throughout the course of treatment.

The Nuffield Centre Dyspraxia Programme offers many sequencing activities, starting at the single sound level. In motor terms, sequencing of single sounds can be seen as the next step up from controlled production of sounds in isolation. The ability to produce a sound sequence requires the child to retrieve the correct motor program(s) and to utilize motor planning skills to maintain accurate production over the sequence.

The Nuffield Centre Dyspraxia Programme offers a fine-tuned graded approach to ensure the child achieves success and is provided with support as the tasks become increasingly challenging.

a) Repetitive sound sequencing

- While continuing to work on single sound production, introduce some sequencing by building up repetition of the same sound e.g. /p-p-p-p/; /i-i-i-i/. This will need to be done slowly at first with a gradual increase in speed as the child gains facility. Such tasks require the child to maintain consistent production of a sound on a number of occasions.
- Lotto bases with small cards are very useful for repetitive sound sequencing. The aim is for the child to carry out a manual activity (covering each picture on the lotto base one by one with the cards) while producing a controlled sequence of sounds (producing a sound each time in response to the picture on the lotto base and then to the picture on the card).
- For some children (especially young children) it may be helpful to introduce the task silently first (to familiarize him with the procedure) and then to progress through graded supported steps e.g.
 - 1 therapist completes a 4 picture lotto in silence
 - 2 child completes a 4 picture lotto in silence
 - 3 therapist completes a 4 picture lotto with speech sound added
 - 4 therapist hands child a card one at a time – child says sound and places card
 - 5 child places his own cards one at a time and says the sounds

Some children may find it very difficult to manipulate the cards or become over absorbed in the manual activity – in which case, the therapist should continue to support the activity (as described above).

- To maximize sequencing opportunities, this activity can be extended from a 4 sound sequence to an 8 sound sequence, by also taking off the cards one by one and saying the sounds.
- Other sequencing activities from Appendix 6 can also be used for this work e.g. collect-a-sets; roads; paths etc.

b) Sequencing contrasting sounds

1) Simple contrasts

As soon as a number of sounds have been established and the child can manage sequences of the same sound, contrasts can be introduced. Work at this level challenges the child to retrieve two different motor programs and to utilize motor planning skills to maintain accurate production of the individual sounds over the sequence.

Some children find retrieval of one motor program as opposed to another very similar program very difficult. For this reason, the starting point for work at this level should involve very distinct contrasts i.e. involving at least 2 or more feature changes e.g. “m-s”; “b-k” etc and a slow pace should be expected. Gradually, more contrasts can be introduced, providing increasingly challenging contrasts for a particular child. Speed can be increased gradually.

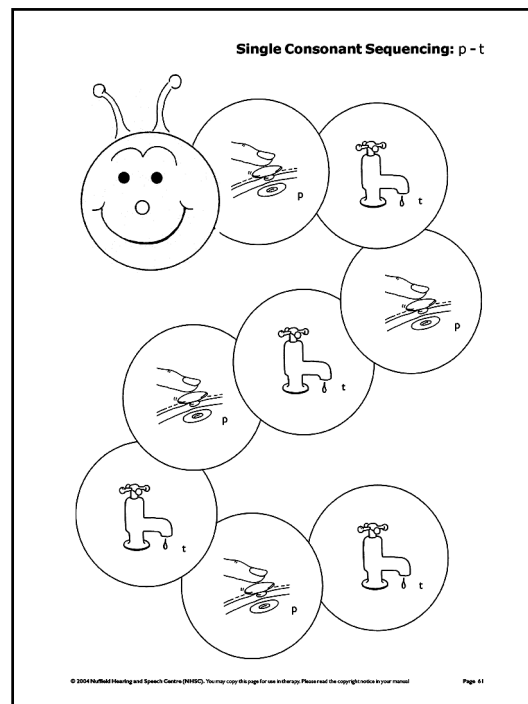
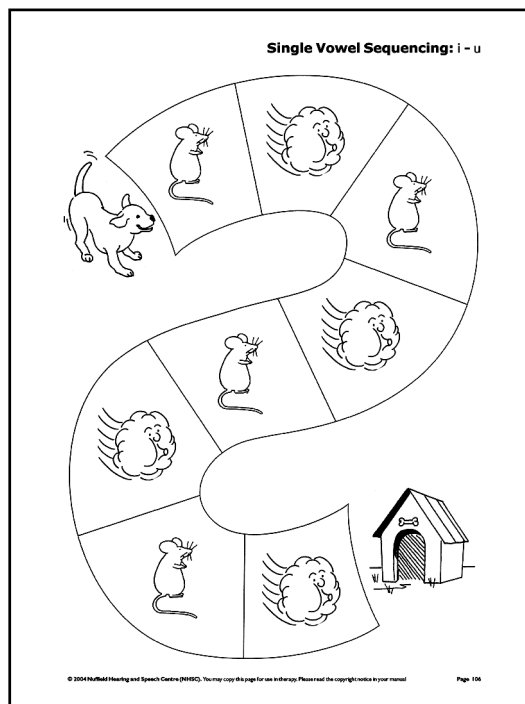
- a change in place (e.g. /p-t/ or /t-k/)
- a change in manner (e.g. /b-m/ or /s-t/)
- a change in voice (e.g. /k-g/ or /s-z/)

- a change in degree of (jaw) openness/tongue height (e.g. /a - i/)
- a change in place/part of tongue raised (e.g. /i - u/)
- a change in lip shape (e.g. /u - ɔ/)
- a change in vowel length (e.g. /i - i:/)
- a change from a static to a moving vowel (e.g. /i - iə/)

2) More complex sequences

Alternating changes of two sounds e.g. /p-t-p-t-p-t/
Random changes of two sounds: e.g. /i-u-u-i-u-i-i/
Alternating changes of three sounds: e.g. /p-t-k-p-t-k/
Random changes of three sounds: e.g. /b-d-b-g-g-d-b/

- Activities for sequencing include: Lotto bases with a mixed pack of small cards on top; Sequencing worksheets; and games (See Appendix 6 e.g. collect a sets, dominoes, bricks, stepping stones, television, pairs etc).



- Aim for the child to practise the sound sequence shown on a worksheet/lotto with cueing and/or modelling by therapist or parent as appropriate. As the child gains accuracy and ease at producing sequences, encourage an increase in speed in preparation for the next stage of combining sounds into words.
- Remember, it can be helpful to continue working on oro-motor skills in conjunction with all sound work.
- Avoid patterns of movement becoming so automatic that the child is unable to modify them and they become an isolated skill; practise sequencing each sound with a wide range of others; practise the inverse sequence of the worksheets provided; use a variety of games and materials; vary prosodic features such as speed, volume and intonation patterns.

3) Consonant and Vowel sequences

In addition to sequences involving combinations of consonant or vowel sounds, it is also possible to practise sequencing a combination of consonant and vowel sounds. The aim of sequencing initially is to introduce the child to word-type patterns, through artificial syllable sequences. This is later extended by attaching linguistic meaning to these patterns (see Transition from single sounds to CV and VC words section).

“Artificial” sequencing (i.e. without any meaning attached) can be practised using similar procedures described above e.g. lotto bases and cards, worksheets, collect a set games, etc.

Transition from single sounds to CV and VC words

Contents

C + V = CV build up worksheets (**new** designs)

C + V = CV hiding worksheets

C + V = CV onset +rime worksheets (**new**)

V + C = VC build up worksheets (**new**)

V + C = VC hiding worksheets (**new**)

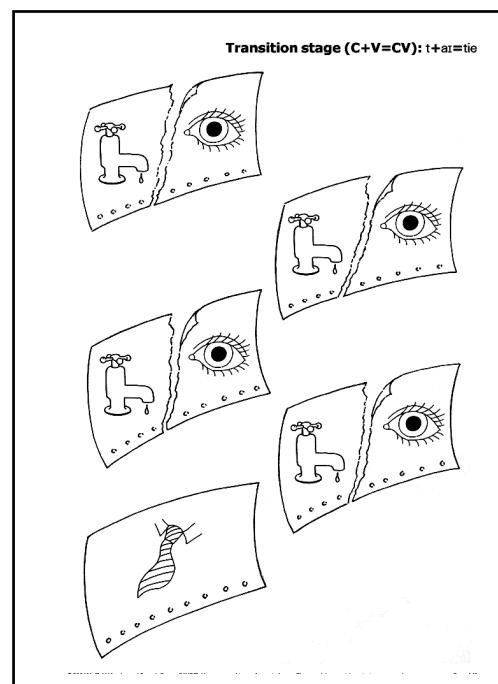
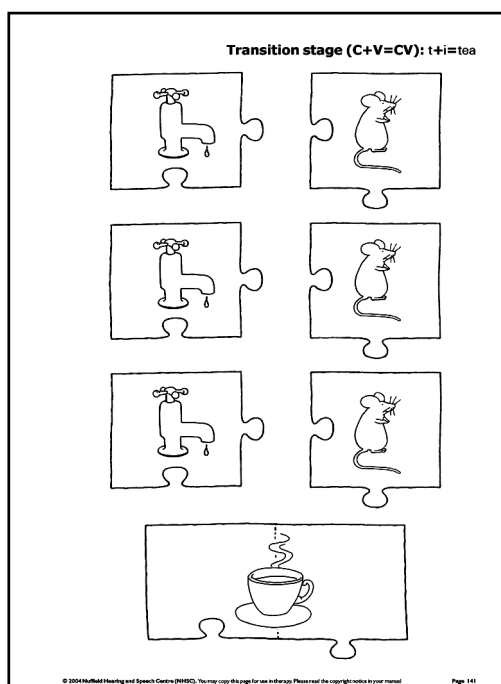
At this stage, motor programs for consonants and vowels, established at single sound level and practised in C + V and V+ C sequences, are combined to form CV or VC syllables and given linguistic meaning as CV or VC words.

Making the transition from single sounds to CV syllables (C + V = CV) can be a very difficult process for children with developmental verbal dyspraxia as it involves the creation of new motor programs (i.e. the process of motor programming) – a core difficulty in dyspraxia.

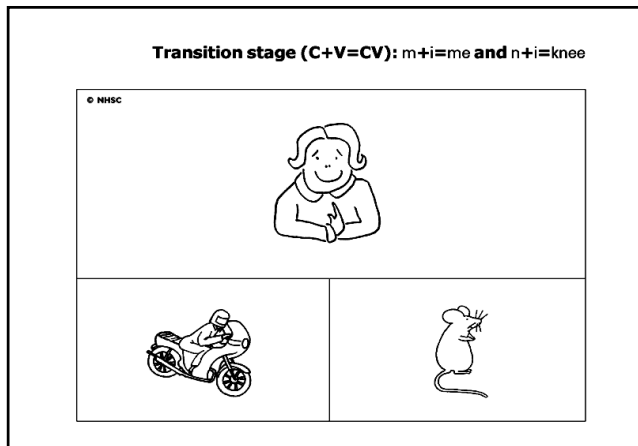
Combining motor programs, without the “gap” left earlier in sequencing tasks, involves modifying the two existing programs so that they join smoothly. Children may have difficulty combining sounds at all to begin with, or may need help just with particular sounds or groups of sounds.

Therapy

- Once the child can produce “artificial” C + V sequences, C+V=CV worksheets can be attempted. These should be modelled first by the therapist with demonstration/explanation that the two sounds are to be joined to make the word. Utilise the joining presentations e.g. two jigsaw pieces to explain this to the child. The child can then attempt this himself. If his production of the CV word reverts to his own motor program, nonsense syllables should be practised as an intermediate stage (Use consonant and vowel lottos and cards).



- The hiding picture worksheets also provide motor programming practice. In this case, the therapist hides the CV picture; the child is then asked to say the consonant and vowel sounds he can see and blend them together to produce a CV word. Once he has said it, the cover is removed and the child and therapist can check that his attempt matches the CV picture.



Strategies for dealing with common problems at this level:

1 The child cannot produce any CV syllables.

Start by asking the child to say /m/ and to keep his voice going as you help him to open his mouth. Once he can manage /ma/ on his own, try to extend it to other vowels. Use the appropriate symbol pictures to support and prompts/articulograms to help him achieve the lip shape for the vowel as necessary.

Move on to attempt /w, j, n/ + vowels. These voiced continuants are easiest as there is no change of phonation or airflow between consonant and vowel.

The fricatives /f, s, h, ʃ(sh)/ can also be attempted. These are also continuants but care should be taken to combine the sounds slowly and to say vowels gently at first to avoid intrusive plosives or glottals.

2 The child can produce /m/ in CV words but not /b/

Ask the child to say a CV word starting with /m/ e.g. "me". Tell him you will change it by magic into "bee" by gently blocking his nose. If this works, try the same to obtain /d/ from CV words starting with /n/.

3 The child has voiced plosives in CV words but cannot achieve the voiceless counterparts

If you have practised /p/, /t/, and /k/ almost as affricates at a single sound level, rather than with a whispered schwa, this will pay off now. Tell the child that it is the little puff of air from the plosive that must "fill the gap" between the sounds. Say the sounds slowly, prolonging the friction/aspiration segment.

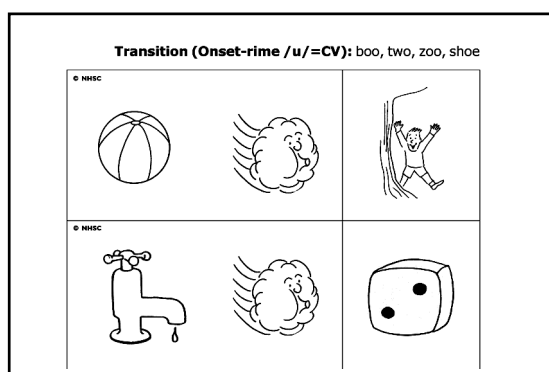
4 The child has voiceless plosives in CV words but cannot achieve the voiced counterparts

Voicing can often be achieved by either babbling a CV syllable involving /p, t or k/ e.g. /papapa/ or "shouting" a CV syllable.

5 The child needs help to understand about blending

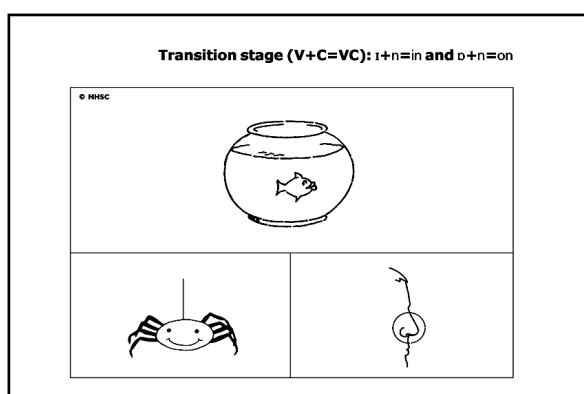
It may help the child to listen to the therapist blending e.g. using the hiding pictures: the CV picture is covered; the child listens to the therapist blending the C + V slowly and tries to find the appropriate CV card from a selection on the table.

The hiding picture worksheets and onset-rime worksheets can also be used to develop the child's awareness of sounds in words and allow for work on segmentation at a CV level, appropriate to the child's age. In a segmentation task, the consonant and vowel symbol pictures would be covered and the child asked to find the appropriate symbol pictures, from a choice, which make up the word (See Appendix 6: games and also Chapter 6: Developing literacy skills, for further activities). NB: It may be necessary to clarify the phonological representations for some CV words by means of minimal pair discrimination or lexical judgement tasks.



VC syllables

Although CV syllables are regarded as the core syllabic units in a building approach, motor programming skills can also be practised by creating VC syllables from V+C components. Vocabulary constraints limit the number of items, but in this edition examples of both sequencing and hiding presentations have been included. For some children V+C=VC may well be easier than C+V=CV and it may be particularly useful as a lead into CVC level work. Similar principles apply as described for C+V=CV level.



CV and VC Words

Contents

CV lottos (2 per page)

VC lottos (2 per page)

CV sequencing worksheets (**new** designs)

Mixed CV lottos, by onset (**new**)

CV and VC cards (**new**)

Aims

- 1 To establish a range of functional CV and VC words
- 2 To establish motor programs for as wide a range as possible of CV and VC syllables (linked to picture cues) to act as building blocks for “higher levels” of the Programme
- 3 To fine-tune discrimination skills
- 4 To clarify phonological representations
- 5 To develop the ability to contrast and sequence CV and VC words in minimal pair sequences (involving changes of place, manner, voice and vowels)

Children with speech disorders often present with restricted phonotactic structures, in addition to speech system limitations. Structural difficulties may affect monosyllabic production (e.g. CV, VC, CVC, CCV levels) and/or bisyllabic (e.g. CVCV, CVCVC levels) and multisyllabic levels.

Children with developmental verbal dyspraxia usually have both sound system and structural difficulties (see Chapter 1: Literature review) and need help to address both aspects. As already described, motoric therapy approaches advocate that phonotactic structures can be “built” by combining previously established “building blocks” or units e.g. single consonants and vowels can be combined to build CV or VC structures (see transition from single sounds into CV and VC words section above).

Dodd (1995) citing Eisenson (1984) suggests that the syllable should be the basis for therapy for children with developmental verbal dyspraxia, as it is often the sequencing of the articulators for phoneme combinations that is the major difficulty. CV and or VC structures are the simplest syllable structures and therefore the best starting point.

It is very important for children with speech difficulties to achieve success at a functional level, as early as possible. Therefore, in addition to early work on oro-motor skills, single sounds and single sound sequencing, it is important to establish a number of accurate (or nearly accurate) CV or VC words. However limited their speech output, the majority of children will have a few CV or VC words already in their vocabulary e.g. bye, no, more, boo, go, up, out etc.

Top Tips

- Aim to establish a meaningful set of CV words as early as possible
- Start from CV words the child can already make
- When teaching new CVs, use Cs and Vs already established at single sound level
- Accept “approximations” and refine them gradually
- Remember the general guidelines: *plosives, nasals and approximants first, combined with long vowels and diphthongs; later fricatives and affricates*
- VC words might be easier for some children – experiment. Principles are similar
- Utilize input approaches, to support output work
- Use cues e.g. single sound picture symbols or verbal cues for support, if required
- #####Keep working at single sound level
- If success is limited, return to earlier single sound and sequencing work
- Make activities fun through the use of games (see Appendix 6)

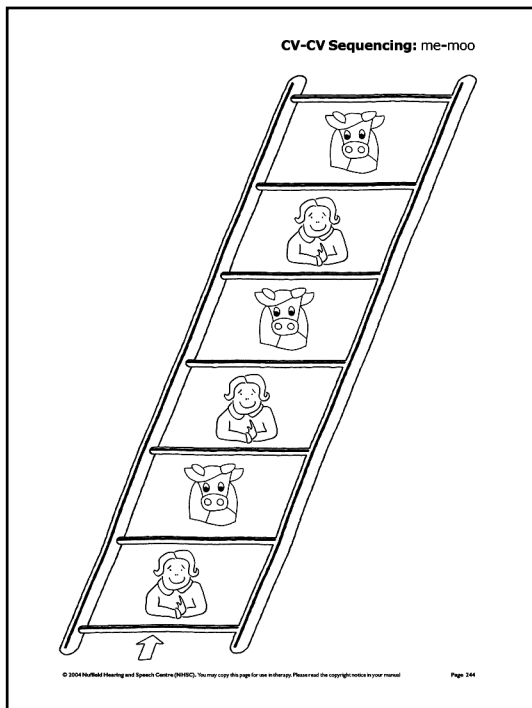
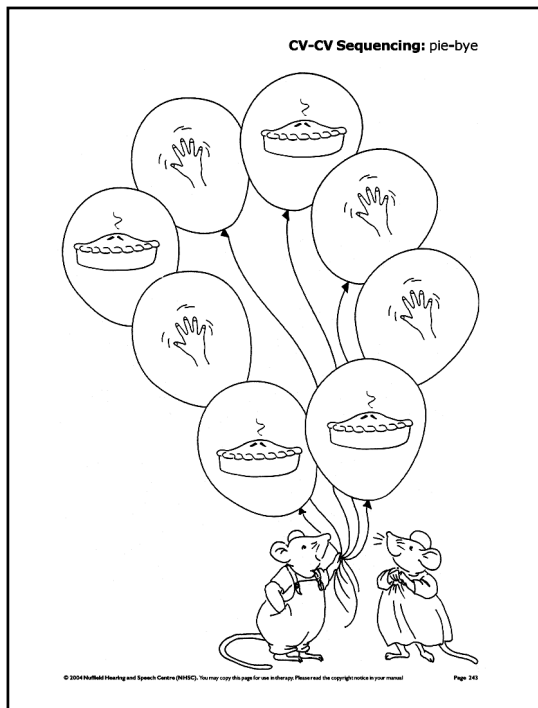
Therapy

- Introduce CV pictures to match any CV words the child can produce accurately or nearly accurately (e.g. at this stage voicing of voiceless onsets might be accepted). Use both input and output tasks: to reinforce learning of the picture cues, to check auditory discrimination and representations and to practise controlled production.
- Tasks/activities for work at this level might include: posting, snap, collect-a-set, fishing etc and discrimination activities (see Appendix 6).
- Gradually expand the child’s CV vocabulary by combining established consonants and vowels together to build new CVs (see C + V = CV Transition section). Once these become established, incorporate them into tasks with other CV words.
- In addition, as the child’s single sound repertoire expands and sounds become established, try eliciting new CVs through imitation techniques.
- The Nuffield Centre Dyspraxia Programme uses pictures to support production work – this helps to forge links in the lexicon between different representations i.e. semantic, phonological, motor program etc. Older children may also be helped by making an orthographic link e.g. by looking at written word/writing the word under the picture etc.
- Although the main aim of therapy at this level is production of CV structures, it is important to incorporate input work into the therapy programme, to ensure that the child recognizes the pictures being used to represent words and that connections are being made between new motor programs and semantic and phonological information. Minimal pairs are particularly useful for Input work at this level. However, it is not advised to delay introducing production work in favour of listening and discrimination activities. Remember, output work can actually be easier for some children, who find input work too abstract and confusing.

- Some children will find it easier to produce VC words than CV words. Unfortunately, vocabulary restrictions limit the use of picture presentations – however a number have been included in this edition of the Nuffield Centre Dyspraxia Programme, in recognition of their usefulness. The principles for working on such words are very similar to those described for CV words.
- Simple CV and VC words can form the basis of a key-word vocabulary (Chappell 1973) for children with severe speech disorders. The aim is for the child to build up a number of words, which he is able to produce accurately (or nearly accurately), within their articulatory ability, and which they can use functionally to communicate. Such words will vary from child to child, but might include: boo, bye, two, go, more, me, no, hi, up, out, eat etc. Children with severe speech difficulties can be encouraged to combine a signing system or gestures, vocalizations and “key words” to maximize their communication skills.

Top Tips for CV and VC sequencing

- Start with repetitions of one CV or VC e.g. bee, bee, bee; arm, arm, arm
- Use different rhythmic patterns to facilitate sequences of the same CV or VC e.g. bee- bee; bee-bee-bee etc. Incorporating CV words into nursery rhymes may be helpful e.g. Old MacDonald had a farm...and on that farm he had a bee: with a bee, bee here and a bee, bee there etc
- Remember the principle for sequenced picture presentations: one picture – one spoken utterance
- Aim for accuracy, at a fairly slow pace; increase speed gradually
- Monitor suprasegmental features and encourage good patterns of vocal tone, pitch, volume and resonance
- Introduce changes of sequence gradually. Start by contrasting whatever CV/VC words the child can make
- Extend the contrasts gradually. Follow the general order, but be flexible according to the child’s needs:
 - Vowel contrast e.g. pie-pea, toe-two etc
 - Placement contrast e.g. pea-tea, sea-she etc
 - Voicing contrast e.g. pea-bee, tea-dee etc
 - Manner contrast e.g. paw-four, Sue-chew etc
- Move from alternating sequences to random sequences
- Increase the length and complexity of sequences as the child progresses
- Utilize single sound cues for support, as required
- Support with Input work as necessary
- Continue to work on earlier levels
- Utilize lotto bases, sequence worksheets, path games etc (see Appendix 6) for this work



Therapy

Sequencing activities help the child to practise using newly developed motor programs in a controlled connected utterance. These activities also help to develop motor planning skills, as they involve assembly of the individual parts into a connected sequence. Tasks with a clear starting and finishing point e.g. worksheets are useful for this work.

Remember the general principle for all sequencing work – controlled production: one picture = one spoken utterance. For many children, the sequence will be more fluent and controlled if the therapist points (or moves the car, animal, character along the path) to the pictures and the child says the words. If the child is desperate to do the moving/pointing, take turns and let him point while you say etc.

Contrasts can be controlled to ensure that the child has a great deal of success, but is gradually and increasingly challenged by the phonetic demands of the individual words in the sequence. For example, staged sequencing work for a child who previously produced tar/car as tar/tar, might include the following:

- | | |
|-----------------------------------|-----------------------------------|
| o Car, car, car contrast | Repetition |
| o car, core, car, core contrast | Vowel change |
| o car, baa, car, baa contrast | Placement change (easy) |
| o tar, car, tar, car contrast etc | Placement change (more difficult) |

In all word level sequencing work, remember the suprasegmental features and encourage: maintenance of an even and appropriate rate, maintenance of appropriate rhythm, intonation, pitch, volume etc.

CVCV words

Contents

CV + CV = CVCV transition sheets (**new** designs)
CVCV noun pictures
CVCV nouns cards (**new**)
CVCV verb pictures
CVCV verbs cards (**new**)
Symbol cues (**new**)
Ending pictures (**new**)
Stories (reintroduced from 1985 edition)

Aims

- 1 To establish accurate motor programs for CVCV words and CVCing verbs
- 2 To develop concept of syllableness
- 3 To clarify phonological representations for CVCV words
- 4 To sustain articulation for sequences and short phrases/sentences

Accurate production of disyllabic words presents difficulties for many children with dyspraxic difficulties. Typically, a CVCV word is reduced to a monosyllable, by omitting the unstressed (weak) syllable e.g. /bei/ for baby or the correct number of syllables is maintained, but the onset sound of the second syllable is omitted or substituted by another phoneme or a glottal stop e.g. /beiʔɪ/.

Clinical practice at the Nuffield Centre suggests that many children can learn to “build” a CVCV structure accurately by combining two CV structures, more easily than a CVC structure. Therefore, it is recommended that, as a general principle, therapy progresses from a CV level to a CVCV level, before attempting a CVC level. However, it is important to recognize individual variation and to be flexible accordingly.

Development of accurate CVCV words involves motor programming skills to develop new motor programs and to update inaccurate old programs. Motor planning skills are involved when sequences of words, phrases and sentences are produced.

In addition to these output skills, some children will require input work to develop awareness of the number of syllables in a given word. Developmental phonology research (Stoel-Gammon and Dunn 1985) indicates that “syllableness” emerges early in a normally-developing child’s speech acquisition, typically at approximately 18-24 months. Hodson and Paden (1991) recommend that syllableness is introduced early in a therapy programme, through the use of compound words with two (and later three) syllables e.g. cowboy; cowboy hat etc. The aim is for the child to become aware of the concept of syllable number (i.e. one vs. two, using clapping and beating techniques). For further information on this, see Chapter 6: Developing Literacy Skills.

Once a number of accurate CVCV words have been established, the child’s phonological representations for these words will need to be checked and clarified.

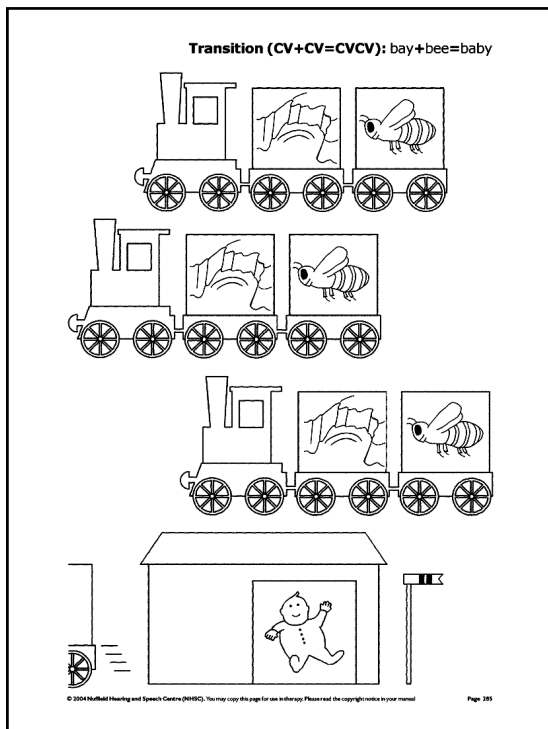
Top Tips

- Introduce CVCV before CVC
- Introduce a few CVCV words (e.g. mummy, daddy, baby, nanny) as soon as the child has established CV syllables involving /b, d, m, n/
- Start with “artificial” blending of CV + CV, before making linguistic link
- Aim for smooth fluent transitions between syllables
- Work on words with the same consonant in both syllables before words with a change of consonant
- Voiced (rather than voiceless) plosives and nasals are usually easier, as the child just needs to maintain voice throughout
- Work on verbs as well as nouns
- Incorporate work on stress patterns. Move gradually from a pattern of equal stress on both syllables to the appropriate stress pattern
- Utilize cues to support production e.g. picture symbols, CVs, manual cues etc
- Include work on concept of syllableness – one syllable vs. two. Practise “how many beats can you hear?”
- Keep working at earlier levels to support fluent, automatic production, eg: single sounds, CV levels
- Introduce sequencing work when the child has achieved some success with CVCV production
- Make activities fun through the use of games

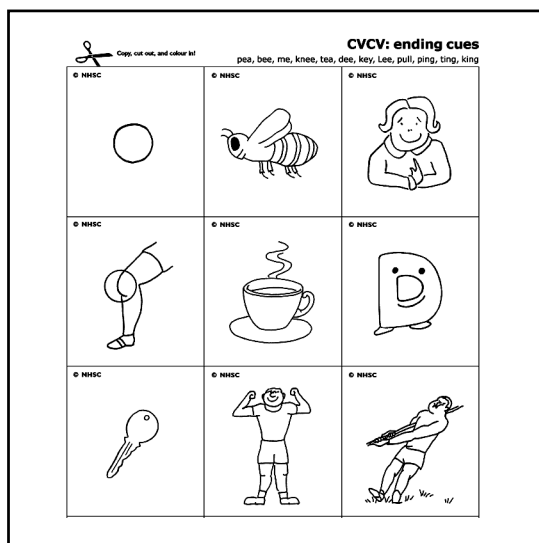
Therapy

1 Developing motor programs for CVCV words

- Introduce a few very simple CVCV words as soon as the child has established syllables with /b, d, m, n/ e.g. mummy, daddy, baby, nanny.
- Once he can produce a range of CV+CV sequences accurately, at speed e.g. bee, boo, bee, boo, bee, boo, he has the basic articulatory skills to form CVCV words
- Practise joining syllables at a non-linguistic level first, using CV+CV sequencing sheets e.g. tea, two, tea, two etc and using CV lotto bases and cards.
- Use CV+CV =CVCV transition worksheets to build CVCV words e.g. bay+bee = baby (see over).
- Start with: CVCV words where the consonant is the same in both syllables and/or CVCV words where all consonants are voiced.
- Use cues to support production e.g. consonant symbol strips and/or Cued Articulation (Passy 1990) to focus on the articulation of a particular sound –as appropriate. NB In this edition, the syllable onset symbols have not been drawn under each CVCV picture, but pages of consonant strips have been provided for therapists to cut up and use as appropriate (see second Ringbinder).

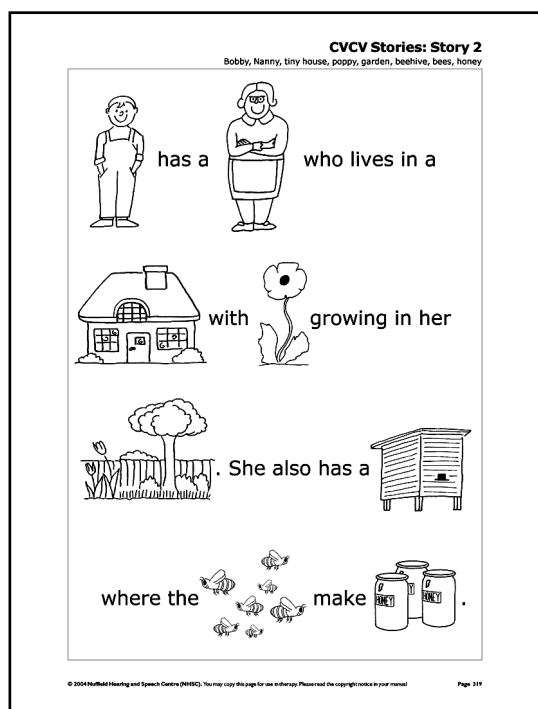


- Use the Ending cards to support production –as appropriate. Many children find production of the second syllable particularly difficult and may be helped by a CV cue card.



- This is similar to the “rebus approach” proposed by Young (1987) to help children overcome weak syllable reduction. CVCV structures are worked on through “backward chaining” i.e. the second syllable is elicited first through the use of the cue and then incorporated into the whole word e.g. “knee”, then “tiny”; “me” then “mummy”. It can be a very helpful approach in establishing CVCV word production. However, it does tend to encourage equal stress patterns and therefore it is necessary to incorporate a staged approach to ensure appropriate stress patterns are also developed (Young 1987) – see below.
- Monitor accuracy of consonants and vowels and smooth joining between sounds.
- Facilitate smooth transition between syllables with voiceless consonants by encouraging gentle articulation and plenty of aspiration/friction.

- Work on CVCing verbs in the same way as nouns. Practise /ŋ / if this is an appropriate target for the child, then ping, ting, king syllables. Use the ending cues 'ping', 'ting', 'king' to support production of CVCV verbs.
- Keep working at lower levels e.g. oro-motor, single sounds and CV sequencing in order to support fluent, automatic production.
- Extend to include a wider range of CVCV words if the child is not generalising his skills spontaneously to other words.
- Build up and practise a set of CVCV words which the child can say in order to consolidate motor programs and ensure contrasts are maintained.
- Utilise the story presentations to make practice of CVCV words fun.



- Incorporate work on stress patterns, move gradually from a pattern of equal stress on both syllables to the appropriate stress pattern. Most English CVCV words have primary stress on the first syllable (e.g. party, baby, kettle). However, a few have primary stress on the second syllable (e.g. today, tattoo, guitar).

2 Developing the concept of syllableness

- Use CV + CV=CVCV activities to introduce the child to the difference between one and two syllables.
- Use compound nouns e.g. car key, bowtie to help the child understand the difference between one and two syllables.
- Practise syllable marking activities e.g. clapping, jumping to words of one and two syllables. Do this alongside output work (clap as child says word) or to an auditory model.
- Also see Chapter 6: Developing literacy skills.

3 Clarifying phonological representations

- Develop awareness of the target phonological structure during output work through the use of sound and syllable cues.
- Use lexical judgement tasks i.e. ask the child to listen and judge whether you are saying a particular word correctly. Use the CVCV pictures to ensure that he is linking the phonological representation to the appropriate semantic representation.
- Talk about word meanings and include words in simple phrases in order to strengthen lexical representations and improve retrieval.
- Ensure that motor programs and discrimination skills are secure at single sound and CV levels.
- Work on segmentation and blending (see Chapter 6: Developing Literacy skills).

4 Developing sequencing skills

- Sequencing of CVCV words helps to practise motor programming and motor planning skills. Start with repetition of one word, then alternate sequences of two words (use CVCV noun and verb cards).
- Use different games/activities to practise alternating and random sequencing (see Appendix 6 for ideas).
- Extend with phrase and sentence level activities (see Ringbinder: Word Combinations section).
- Remember the principles for all sequencing work: controlled production (one picture = one spoken utterance). Start from achievable targets and gradually increase the challenge by increasing the phonetic demands. Encourage maintenance of controlled suprasegmental features.

CVC words

Contents

CV + C = CVC build up worksheets

CV/CVC sequencing worksheets

CVC/CVC sequencing worksheets

CVC rhyming pictures (**new**)

CVC pictures (**new**)

CVC cards (**new**)

Aims

- 1 To establish accurate motor programs for a wide range of CVC syllables and words
- 2 To ensure smooth transitions from CVs to final consonants
- 3 To clarify phonological representations for CVC words in the child's vocabulary
- 4 To extend the child's phonological system to enable a full range of contrasts at CVC level

Many children with speech disorders have difficulties with final sounds in monosyllabic structures. In the case of children with dyspraxic-type difficulties, there are probably two main reasons for this: (a) motoric difficulties with the increased articulatory load of a C+V+C structure, and unfamiliar syllable shape, and (b) phonological difficulties in suppressing the simplification process of final consonant deletion, which occurs in normal development (usually ironed out around 2 years of age). These two factors inevitably interact and lead to CVC level being particularly difficult for many children.

Work at this level can be approached in a number of different ways. The traditional bottom-up approach method recommends that CVC structures can be built by using previously established units of CV words and single sound consonants e.g. boo + t = boot; four + k = fork etc. CV structures are considered the basic syllable unit as they occur early in normal development in babble sequences and in addition, the usual simplification process is to omit a final sound from a CVC structure leaving a CV structure.

However, it is also possible to use different building blocks e.g. C+VC =CVC e.g. k + up = cup etc. Such an approach is based on units of onset and rime and may therefore be more compatible with literacy work for school aged children. As described earlier (see CV/VC section), there are some restrictions on a pictorial approach in using VC words as suitable vocabulary is limited.

Other approaches have also been advised in the literature e.g. backward sequencing of phonemes (Chappell 1973): first produce the final consonant, then add the vowel and, finally, the initial consonant e.g. t - eet – feet (Dodd 1995).

It is also important in CVC work to consider functional aspects. For this reason, it may be acceptable at the early stages of CVC work to teach a developmental error, rather than

aiming for completely correct production e.g. /tʌp/ is more functionally useful than /hʌp/ for the target "cup", if the child cannot produce velar plosives (Dodd 1995). This can later be shaped to the correct production as the child's articulatory skills increase.

Once motor programs for CVC words have been built, minimal pairs sequencing worksheets provide opportunities for establishing and consolidating newly learned words and clarifying phonological representations. The new CVC cards with short vowels can also be used for this purpose.

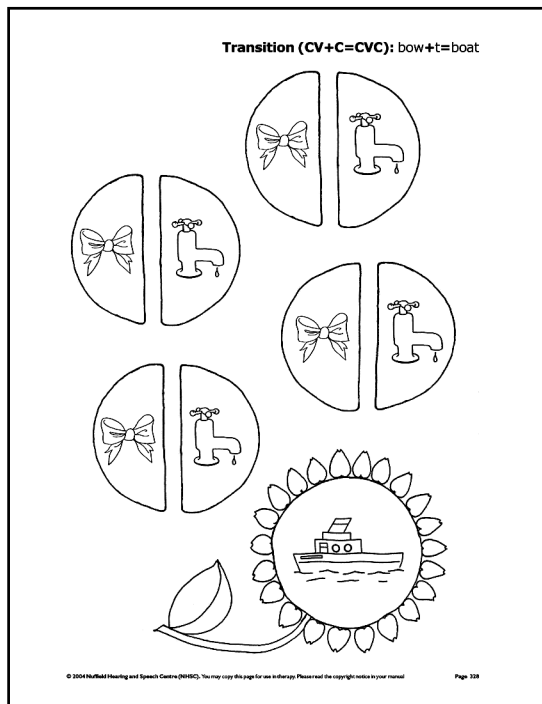
Top Tips

- Introduce work at this level after CVCV
- Build CVC structures from CV+C (older children who have a concept of onset+rime may cope with C+VC)
- Start artificially (CV + C) and then make the linguistic link (CV+C=CVC)
- Start by adding final voiceless plosives and fricatives /p, t, k, f, s/
- Ensure smooth transitions to final consonants (beware of glottals and ejectives)
- Monitor vowels carefully
- Take care in establishing velar/alveolar and voiced/voiceless contrasts within words
- Use minimal pairs to help the child clarify phonological representations
- Remember to support output work with input work
- Keep working at earlier levels e.g. single sounds, CV and VC, CVCV
- Introduce sequencing once a few CVC words are established
- Make activities fun using games etc

Therapy

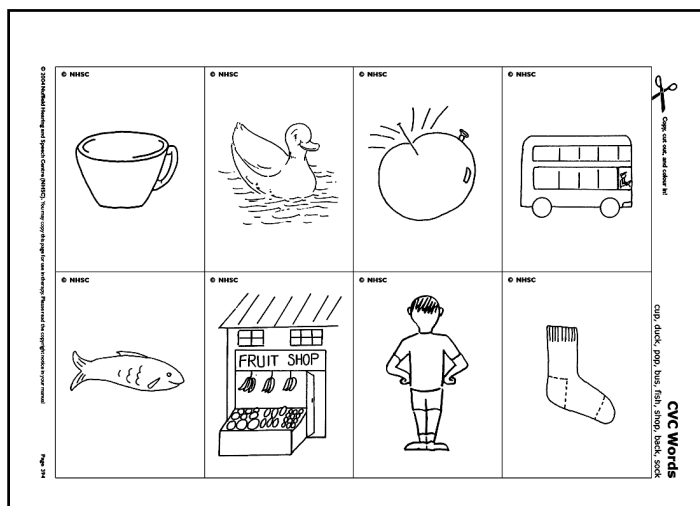
1 Developing accurate motor programs for CVC words

- Practise joining CV+C at a non-linguistic level first, using CV Lotto bases and add single sound cards on top e.g. Lotto base "bee" and single sound /p/ on top. Such activities give the child confidence in the new task by using familiar materials with which he has previously succeeded.
- Use CV+C transition worksheets to build CVC words e.g. pie + p = pipe. Choose voiceless final plosives, nasals and voiceless fricatives first /p, t, k, m, n, f, s/. Check accuracy of vowels is maintained and ensure there is smooth transition to the final consonant.
- Build up and practise a set of CVC words which the child can say in order to consolidate motor programs. Use consonant picture symbols to cue the final sound as required (see over).
- Use the CVC cards to include practise with long, diphthong and short vowels.
- Keep working at lower levels e.g. oro-motor, single sounds, CV and CVCV sequencing in order to support fluent, automatic production.



2 Clarifying phonological representations at a CVC level

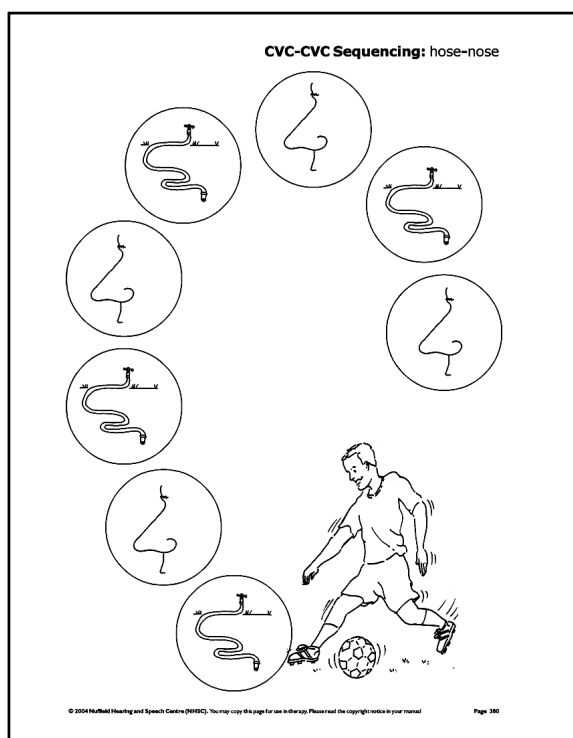
- Develop awareness of the target phonological structure during output work through the use of sound and CV cues.
- Use lexical judgement tasks i.e. ask the child to listen and judge whether you are saying a word correctly. Use the CVC picture cards to ensure that he is linking the phonological representation to the appropriate semantic representation.



- Check word meanings and use CV/CVC worksheets to clarify phonological representations.

3 Developing contrasts at a CVC level

- Sequencing CVC words helps to practise motor programming and planning skills.
- Use the minimal pair worksheets, which contrast CV and CVC words e.g. bow/boat.
- Gradually work towards sequencing CVC/CVC words, with changing consonant or vowel e.g. bat/back, hose/nose.



Strategies to deal with special problems with CVC production

1 Final voiced plosives produced with added schwa [ə] (e.g. “pig” → [pɪgə])

Final voiced plosives (and fricatives) are not actually voiced, but marked by lengthening the preceding vowel, followed by **weak voiceless** plosion. Therefore model carefully and make sure no-one models [pɪgə]. If the child does start to add a schwa, explain “make the /ɪ/ longer, then say just a little /g/ - more like a /k/ really”.

2 Diphthongs followed by nasals /m, n/

The second element of the diphthong tends to be lost e.g. “time” [taim] -> [tæm].

If this occurs, use the vowel picture symbol or manual cue to remind the child what the vowel is supposed to be. Practise CV + C, giving plenty of time to produce and hear the diphthong before adding the nasal. Input work can also aid self monitoring.

3 Final /ɪ/ (Dark or velarised ‘ɪ’)

Many children need help in the later stages of therapy to modify an immature /ʊ/ articulation to a mature /ɪ/ production e.g. tell – produced as [tɛʊ] instead of [tɛɪ]. You can help the child to reduce lip rounding by starting with /ɪt/ and /it/.

Also try explaining that the tongue tip goes up behind the teeth as for initial /ɪ/ but you need to switch voice off before allowing the tongue down.

CVCVC and Multisyllabic words

Contents

CVCVC pictures

Multisyllabic pictures

Symbol cues (**new**)

Multisyllabic cards (**new**)

Aims

- 1 To establish accurate motor programs for a wide range of multisyllabic words
- 2 To ensure smooth transitions between syllables
- 3 To clarify phonological representations for multisyllabic words within the child's vocabulary

Most children with dyspraxic type difficulties will have difficulty producing multisyllabic words. This is well recognized in most symptomatology lists of developmental verbal dyspraxia (see Chapter 1: Literature Review). Typical difficulties include both reduced syllabic structure as well as multiple sound system errors, and syntagmatic processes, such as assimilation and metathesis.

As with CVCV structures, work on "syllableness" as an input activity can be a useful introduction to production work at this level – how many beats/claps can you hear in this word?

As far as production is concerned, multisyllabic words involving 3 or 4 syllables would not generally be introduced until all simple monosyllabic (CV, VC, CVC) and disyllabic (CVCV, CVCVC) structures are achieved easily with a good range of sounds, stable contrasts and smooth joining of sounds and syllables. Work on multisyllabics uses these simple structures as building blocks.

Top Tips

- Introduce multisyllabic words only after simple syllables have been mastered
- Start with words involving plosives, nasals and approximants before those involving fricatives and approximants
- Divide words into syllables or shorter words at first, then reassemble
- Use a lot of modelling
- Use sound symbol or syllable cues to support production initially
- Monitor the accuracy of consonants and vowels
- Support the development of motor programs for longer words by continuing to practise sequencing of CV, CVCV, CVC words
- Accept inaccurate stress patterns at first; shape gradually to more acceptable pattern
- Include work on syllableness – how many beats can you hear?
- Gradually extend individual word production to sequences of words

Therapy

- Begin with plosive, nasal and approximant combinations then introduce words containing fricatives and affricates as appropriate. Most children require some cue to support their production of longer words, in the initial stages. For example, it can be helpful to place sound symbol pictures for the onset sounds of each of the syllables underneath the picture e.g. for “buttercup” the symbol pictures for /b/, /t/ and /k/ would be placed underneath to remind the child of the onset targets. In this edition of the Nuffield Centre Dyspraxia Programme, pages of multisyllabic words are included, graded in complexity. Sound symbols have not been drawn under each picture, as in previous editions, but the same support can be provided using the individual sound symbol cards or photocopyable symbol cues.
- Similarly, it may be helpful in some cases to use a syllable cue (also see CVCV section) to remind the child of one particular component e.g. for “patacake” – “cake” might be used to emphasise the final syllable. As explained in the earlier section, such strategies are very useful for supporting production in the early stages. However, they do tend to distort normal stress patterns, and therefore as the cues are gradually removed, therapy needs to extend to developing more appropriate stress patterns.
- Children who are unable to imitate multisyllabic words can also be helped by having words modelled in two parts: the first two syllables (e.g. cater-) followed by the rest of the word (pillar). In effect they then develop two motor programs, which are linked to make the longer word.
- Once the child has mastered a number of individual multisyllabic words, sequences can be gradually introduced, helping the child develop further control (see Appendix 6 for suitable sequencing activities e.g. how many? Treasure etc).

Consonant Clusters

Contents

/s/ Clusters

Transition Worksheets: /s/ + CV(C) = CCV(C) e.g. /s/+tar=star

Sequencing Worksheets: CV(C) and CCV(C) e.g. top/stop

/l/ Clusters **(Reintroduced from 1985 Edition)**

Transition Worksheets: C+CV(C)= CCV(C) e.g. b+low=blow

Sequencing Worksheets: CV(C) and CCV(C) e.g. lie/fly

/r/ Clusters **(Reintroduced from 1985 Edition)**

Transition Worksheets: C + CV(C) = CCV(C) e.g. b+red=breed

Sequencing Worksheets: CV(C) and CCV(C) e.g. red/bread

/s/, /l/ and /r/ Cluster Pictures **(new)**

/s/, /l/ and /r/ Cluster Cards **(new)**

Aims

- 1 To establish motor programs for a wide range of words involving /s/, /l/ and /r/ clusters
- 2 To ensure smooth transitions between adjacent consonants
- 3 To clarify phonological representations for cluster words in the child's vocabulary
- 4 To extend the child's phonological system to enable a range of contrasts at cluster level

Many children with speech difficulties including those with developmental verbal dyspraxia have difficulties producing consonant clusters. Cluster reduction is one of the most common and longest lasting phonological processes (Ingram 1976). The typical resulting pattern is cluster simplification in which one or more members of the cluster is deleted (Ingram 1976). For children with dyspraxic difficulties the explanation for this is likely to be due to the complex articulatory demands involved in producing a cluster.

It is generally recognized that acquisition of clusters in normal development is gradual and often occurs in stages– initially the child may delete the entire cluster, then the cluster may be realized as a single phoneme realization, then a cluster may be produced, but involving a different phoneme to the adult realization and finally the accurate cluster is produced (Greenlee 1974).

Hodson and Paden (1991) recognized that cluster reduction has a marked effect on speech intelligibility. However, if the child is deleting /s/, /s/ clusters are very easy to "build" (C+CV(C)). For this reason, Hodson and Paden advise therapy for /s/ clusters before working on /s/ as a singleton phoneme.

In recognition of the difficulties that clusters present for many children with developmental verbal dyspraxia, exercises to develop /l/ and /r/ clusters as well as /s/ clusters are included in this edition of the Nuffield Centre Dyspraxia Programme. Choosing which clusters to work on first will depend on the child's individual profile of difficulty (i.e. ability to produce /s/, /l/ and /r/ in isolation and the acceptability or unacceptability of the child's error form). As a general rule, initial /s/ + C, C +/l/ and final nasal + C clusters are good starting points. Other clusters can be gradually introduced thereafter.

As at earlier stages of the programme, a "building block principle" is applied i.e. an extra sound is added to an established syllabic structure: e.g. /s/ + tar, Ann + /d/ etc.

Top Tips

- As a general rule, initial /s/ + C and C +/l/ are good starting points
- Try /s/ clusters before working on /s/ as a singleton phoneme
- As at earlier stages of the Programme, a "building block principle" is applied i.e. an extra sound is added to an established syllabic structure: e.g. /s/ + tar, /f/ +root etc
- Once some clusters are established, use sequencing sheets to practise CV(C)/CCV(C) contrasts e.g. no/snow; rake/brake etc
- Use phonology resources for a wider range of cluster pictures

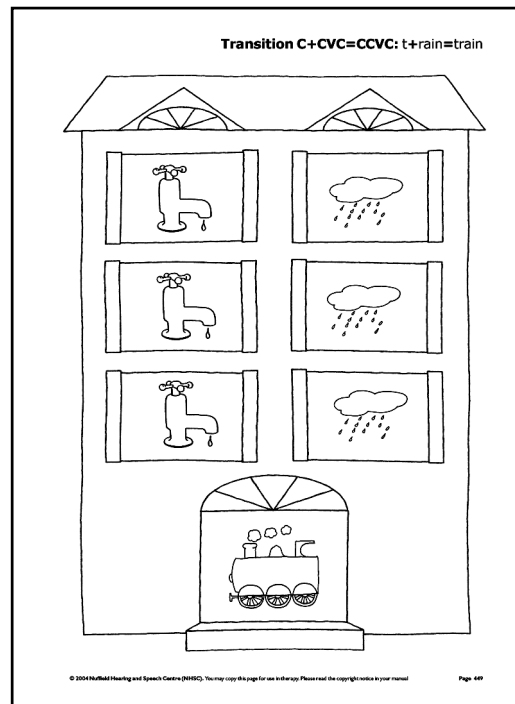
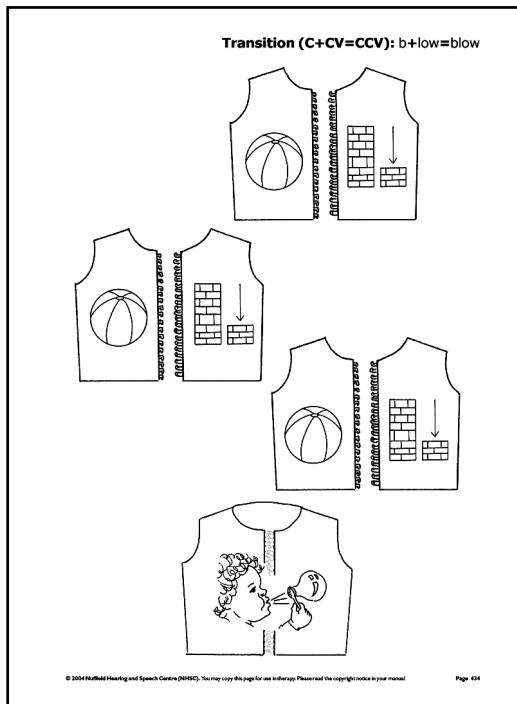
Therapy

1 Developing /s/ clusters

- Start by practising the /s/ + CV transition sheets or use /s/ lotto base and appropriate CV words.
- Gradually encourage the child to increase the speed of production whilst maintaining accuracy.
- Once this is achieved, carefully combine the two parts into a complete word. Help the child link the motor program he has produced with the word it makes, by playing matching games etc.
- Note that /p, t, k/ should not be aspirated after /s/. This adjustment will probably be made spontaneously but if necessary remind the child that only a "little /p/" is required.
- When the child is able to produce some clusters, minimal pair contrasts can be attempted using the sequencing worksheets, for example stop/top, to help the child establish clusters phonologically. Support production of clusters with symbol cues as necessary e.g. place a small /s/ symbol next to each of the "stop" pictures in a "top/stop" sequence. Gradually remove cues.
- Also, contrastive /s/ cluster sequences e.g. star/scar can be created using the /s/ cluster cards, again supporting /s/ production with cues as necessary.
- The cards can also be used for a variety of games to reinforce clusters (see Appendix 6 for ideas such as: collect- a- set, snap, pairs etc).

2 Developing /l/ and /r/ clusters

Word initial /s/ clusters, can be formed by adding an initial sound to an established syllabic structure. However, for /l/ and /r/ clusters it is not such a simple procedure, since the liquid sound occurs in the second, rather than first slot of the cluster. In motoric terms, the easiest way to help the child build /l/ and /r/ clusters, is to develop them from CV and CVC syllables, where the initial sound is realized by /l/ or /r/ e.g. log, lock, lie, low; red, root, ram etc. The process then becomes the same as for /s/ clusters in that an initial sound e.g. /k, p, b, g, f, etc/ is simply added in front of the CV(C) structure. The same graded steps can be followed as described under /s/ clusters. Use the transition worksheets, sequencing worksheets and cards accordingly.



NB it might be helpful to use a mirror to monitor tongue tip position and timing for /l/. The tongue tip must form its contact with the alveolar ridge quite quickly after the release of the plosive/fricative structure in order to avoid an intrusive schwa.

3 Developing final clusters (Not included in this edition of the Programme):

Final nasal clusters are generally the easiest of the final clusters, and the child may have begun to use some spontaneously. Make sure that placement for alveolars and velars is consistent. More practice will be needed by those children who have particular difficulty with palatal control. If practice is required worksheets can be made up for appropriate words, e.g. lamb + /p/ (lamp), pine + /t/ (pint), etc. CVC/CVCC materials can also be made e.g. pine/pint, Ben/bend, and pup/pump, bed/bend where the nasal may be omitted.

Other final clusters can be practised by using CVC cards and adding /s/ or /z/ e.g. tap+/s/=taps; light+/s/=lights; dog+/z/=dogs; bed+ /z/=beds etc. See the Connected Speech section below for further examples of final clusters, associated with grammatical suffixes.

Word Combinations

Contents

Supported (individual picture) presentations

CV – CV phrases:	Adjective noun
	Negatives (new presentation)
	Possessives
CV – CV clauses:	SV, VO
CVCV – CVCV phrases:	Possessives
CVCV – CVCV clauses:	SV, VO, SVO

Composite picture presentations

CVCV – CV phrases:	Adjective noun
CVCV – CV clauses:	VO, SV
Everyday phrases	
Complex sentences (new)	

Aims

- 1 To develop motor planning skills
- 2 To sustain accurate articulation and respiro-laryngeal control
- 3 To develop first word-joining strategies
- 4 To develop basic intonation

It is not only the complexity of sound combinations within a word that cause problems for children with dyspraxic difficulties. Words which were managed correctly in isolation may breakdown when combined to form phrases and sentences. It is important, therefore, to give the child a great deal of practice in using these longer combinations. This can be achieved using words already presented and learnt at the single word level. Words should be chosen which include sounds already established in the child's system, or sounds which are becoming stable. The language structures used need to be appropriate to the child's level of language. This is particularly important for children with language difficulties as part of their profile.

Worksheets are included at this level, where the individual elements of the phrase or clause are illustrated, as well as composite picture presentations. The former should be easier for the child as they support recall of individual motor programs. Initially the child may require a modelled response for support when attempting the composite picture worksheets.

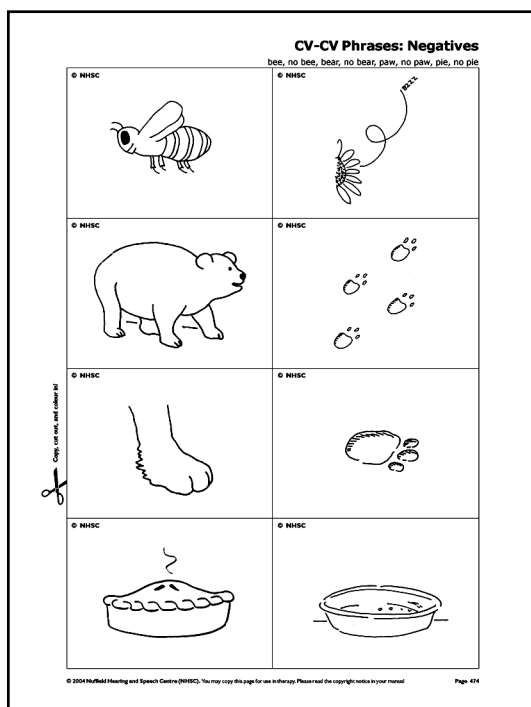
Top Tips

- Start work at this level by using simple CV /CV phrases and clauses
- Work at this level can be attempted once the child has established a good range of CV words accurately
- Use sounds in the child's system – probably plosives, nasals and approximants first before fricatives and affricates
- Encourage production of a phrase or clause on one breath – maintaining respiratory control
- Gradually move on to CV/CVCV and CVCV/CVCV phrases and clauses
- Include work on stress and intonation, as well as rhythm and rate work
- Keep working at earlier levels of the Programme to help sustain accurate articulation

Supported individual picture presentations

1. CV - CV Phrases and Clauses

Worksheets involving a number of very simple adjective -noun, negative X and noun-noun phrases (possessives) and SV and VO clauses have been created using a basic range of CV words e.g. pie - more pie(s); bear - no bear: paw - two paw(s); boy('s) tie; bee go; show saw etc.



These are particularly useful with children who have very limited language skills as they permit meaningful practice and encourage use of speech skills at a connected speech level. The worksheets or simple play situations can be used according to the child's level of understanding.

NB at this early level of practice, the child is not expected to mark the possessive "s" in the noun-noun phrases or the final "s" in the adjective-noun phrases.

An intermediary stage from CV–CV phrases and before CVCV–CVCV, can be created by using a combination of the CV and CVCV possessives worksheets to produce CV-CVCV or CVCV-CV phrases.

2. CVCV - CVCV Phrases and Clauses

This includes CVCV-CVCV noun-noun possessive phrases e.g. Popeye('s) tattoo and SV, VO and SVO phrases e.g. SV: baby peeping, Popeye peeping, teddy peeping, puppy peeping; VO: patting kitten, patting teddy, patting puppy, patting baby; SVO: farmer chasing Lassie, farmer fighting sailor, farmer giving coffee, farmer washing sofa etc.

a) Noun Noun (NN) - Possessives

First practise random combinations of two nouns so that the child is practising only length of utterance, using individual CVCV noun cards, e.g. piper-cowboy, teddy-poppy. These can be made more meaningful: e.g. Mummy('s) teapot, Bobby('s) button. One noun may be kept constant and the other altered (e.g. baby bottle, baby potty), or both nouns can be changed. If the child is able to use it, the "s" possessive ending would be added e.g. baby's bottle etc. Then try the worksheet presentations – cut up the small pictures and choose four to place on the blank boxes. Point to the character and then their possession e.g. Popeye('s) puppy; Mummy('s) poppy etc. Take them off and have another go. Select appropriate targets according to the child's speech system.

b) Subject Verb (SV), Verb Object (VO), Subject Verb Object (SVO)

Pictures are provided for words containing different sound combinations so that the word combinations can be graded to suit the child's level of sequencing skill. e.g. "Peter painting" involves only one change (place) whereas "Sophie washing" involves changes in place, manner and voicing.

Grammatical morphemes can be included if appropriate for the child's linguistic and articulatory skill e.g. auxiliary "is" and determiners e.g. "a" and /or "the". Consider whether it is appropriate to introduce a simple junction strategy "baby **y**is peeping" (see Connected Speech section below).

Further practice at this level can be given by using individual cards of nouns and verbs to produce sequences, silly sentences etc.

Although stress and rhythm work should have been incorporated at previous levels wherever possible, it is particularly important as the child moves into connected and meaningful speech. Once the child has achieved accurate sound production and is able to produce phrases fluently, prosodic features such as stress, rhythm and intonation may well be appropriate. If not achieved it may be necessary to work specifically on stress patterns and intonation through discrimination exercises and modelling /imitation exercises using visual and auditory feedback to assist the child.

Composite picture presentations

1. CVCV – CV phrases and clauses

These include adjective noun phrases e.g. dirty car, dirty toe, dirty boy, dirty door; and SV and VO clauses e.g. SV: cow pushing, boy pushing, Sue pushing, bear pushing; VO: cutting pie, cutting hair, cutting tie, cutting key etc. The complexity of these phrases and clauses is quite simple, only CV – CVCV combinations, however the presentation places more demand on the child. The individual components are no longer illustrated, rather a

composite picture incorporates both elements. Fricatives and affricates are included as well as plosives, nasals and approximants. Ensure that the child is able to produce sounds accurately at appropriate speeds before moving on to the next level.

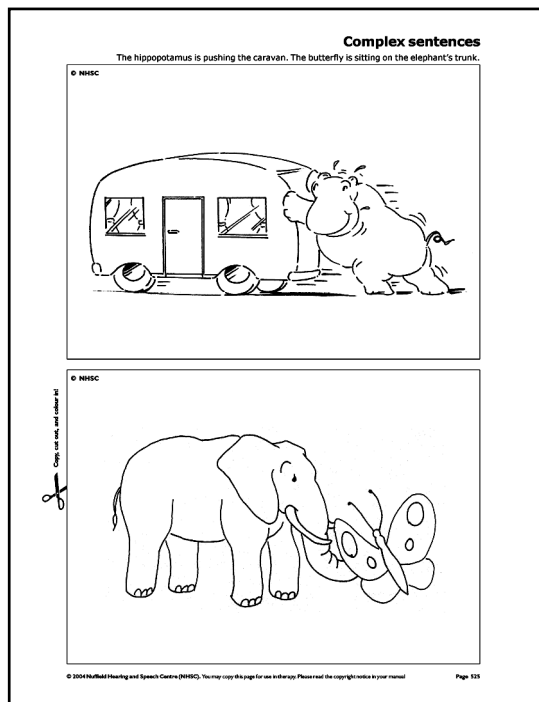
2. Everyday Phrases

These aim to give practice for both sounds and length of utterance in familiar phrases. Some contain fricatives and affricates in complex sequences. e.g. shoes and socks, jelly and custard. They are useful for work on rhythm in speech. Tapping out the rhythm or using a xylophone may help the child achieve this.

3. Complex sentences involving multisyllabic words

E.g. The elephant is pushing the piano; The hippopotamus is playing the tambourine etc.

These aim to give practice in using multisyllabic words in sentences in response to picture cues. The work is high-level, but can be very useful with children with residual type difficulties. They can also be used for sequencing work. Consider introducing junction strategies (see Connected Speech section, below).



From sentence level to connected speech

Contents

Junction phrases (**new**) – in ring binder

Therapy ideas (**new**) – see below

Aims

- 1 To extend motor planning skills to cope with increased linguistic load
- 2 To sustain articulatory control and phonological contrasts
- 3 To develop monitoring and pacing skills
- 4 To practise articulation of grammatical words
- 5 To develop word joining strategies
- 6 To develop prosodic skills
- 7 To establish acceptable articulation of remaining consonants
- 8 To develop strategies for learning new words
- 9 To maximise intelligibility

To achieve intelligible and acceptable connected speech, the child must be able to access accurate motor programs for a wide range of words, in conjunction with the linguistic processing demands of formulating a message. A motor plan for the utterance must be assembled, which takes into account prosodic features and modifications needed to facilitate a smooth transition between words. Stackhouse and Wells (1997) suggest that the refining of these skills characterises the speech development of normal 3-4 year olds, which they have called the Assembly Phase.

Many children with developmental verbal dyspraxia need to practise these skills at sentence level, before they are able to generalise to connected speech. They also need to develop strategies of pacing and monitoring their own speech, in order to maximise articulatory control, and intelligibility.

At this stage of therapy, children also tend to be working on the last remaining consonants, usually / θ ð r /, complex clusters, articulation of grammatical morphemes, and establishing strategies for learning new words.

Top Tips

- Start work at this level, once the child produces structured phrases (see Word Combinations) easily
- ##### Work on articulation and sequencing, as appropriate
- Listen carefully to the child's connected speech (see Assessment chapter), to ascertain what his problems are at this level
- Incorporate work on concepts and input, as necessary
- Work on motor programming skills may still be needed to develop motor programs for a wider vocabulary, and to establish strategies for learning new words
- Use cues established at earlier levels to help the child maintain articulation

1 Extend motor planning skills

Linguistic formulation in the early word combinations worksheets was minimised by representing each word with a separate picture, and omitting grammatical morphemes. Many children reach a stage where they can produce well-practised/supported phrases and sentences, such as these, accurately, but still have difficulty with the "on-line" motor planning required for spontaneous connected speech.

The aim at this stage is to sustain syllable structure, phonological contrasts and articulatory control, while increasing processing demands, in terms of linguistic formulation and length of utterance. Grammatical morphemes are now introduced, and an optimal rate of speech is established, to facilitate planning and monitoring (see below).

- Ensure that motor programs are secure at word level. Check that the child produces phrases, supported by picture cues for each word, consistently and accurately.
- Start by practising these, including "is" and "the".
- Produce words in a carrier phrase, such as "*I saw a _____*", or "*Can you eat a _____?*"
- Use word lists, selected on the basis of syllable structure, initial sound or semantic field/topic area, as a basis for phrase/sentence level practice.
- Use standard syntactic structure, but vary the vocabulary. For example:
- SVO The boy is climbing the ladder
The lion is eating the meat
- SVA Daddy is sitting in the garden
The spider is crawling on the table
- Ask the child to formulate his own short phrase/sentence, based on a picture. Encourage variety by using the adult's turn to model different sentence structures.
- Extend by saying 2 different things about a picture, e.g.
I like chocolate cake. You can buy chocolate in a shop.

- Formulate a complex sentence in response to a picture, e.g. *The boy fell down, because he wasn't looking.*
Ben went to the shop, because he wanted to buy some stickers.
- Use the composite pictures to practise sentences containing multisyllabic words.
- Repeat a sentence and add a word or phrase, e.g.
 - o My blue car
 - o My big blue car
 - o My big blue car is driving
 - o My big blue car is driving down the road etc
- Repeat a modelled sentence, or the child's own sentence accurately.
- Plan and produce a 2/3 sentence narrative, description or explanation.

2 Practise articulation of grammatical words

Words such as "washed", "this" and "didn't" are not included earlier in the programme, and should be checked at this level, by listening to the child's connected speech errors.

Frequent problems involve articulation of later developing consonants /θ ð r/, and the complex/less common clusters produced by the addition of grammatical suffixes (e.g. waved /vd/; asked /skt/; thinks /ŋks/).

As many grammatical morphemes tend to be heard in unstressed forms, children with DVD have often failed to work out the phonological representation or relate it securely to its stressed form, and function. These aspects may need to be taught, as well as the motor programs for both the stressed form, at word level, and the unstressed form at phrase level. It is helpful to relate these to the written form, if the child is at an appropriate level.

- Check how the child copes with the complex clusters produced by grammatical suffixes, and practise if necessary:
 - Plural "s" – cups, crisps, socks, hands, shelves
 - Possessive "s" – Alex's, Mog's, cat's
 - Third person "s" – walks, drinks, eats, helps, acts
 - Past tense "ed" – washed, jumped, winked, collapsed
- Teach phonological representations and motor programs for morphemes, such as "have", "has", "will", "of", "for" which tend to be unstressed. Listen to the child's connected speech to identify which ones are causing problems. Teach the stressed form, clarifying the meaning/use and relating to the written word if the child is at an appropriate level. Then teach the unstressed form, and practise in phrases/sentences.
- Check and teach contracted forms, such as:
 - can't, don't, doesn't, won't, wouldn't, mustn't
 - I'll, you'll, he'll, she'll, they'll, we'll
 - I'm, I've, you're, you've, he's, she's, they're, we're
 - would've, could've

3 Developing self-monitoring and pacing skills

Clinical experience suggests that self-monitoring involves not only processing of auditory feedback and self-correction of errors after the event, but also establishing a habitual rate of speech, which allows the child to use tactile/kinaesthetic feedback to adjust production “on-line”.

a) Help the child to monitor his articulation/phonology by:

- Using cues established at earlier levels (e.g. Cued Articulation).
- Challenging him to keep 10 counters, which will be taken prisoner, if the adult hears an (uncorrected) error.
- Reverse roles and see if he can spot the adult’s errors.

b) Develop optimal pacing

- Aim for a slow, normal pace (around 2-4 syllables per second), with a normal rhythm.
- Model a slower pace if the child is struggling with a faster pace, but retain rhythm and intonation if possible.
- Say the sentence with the child, to support rhythm and pace.
- For some children, a reduced rate of speech is an appropriate on-going strategy.

c) Monitor articulatory control

Clarity of articulation may break down at this level, in children with an ongoing motor component to their speech difficulties. Strategies to improve articulation at sentence/connected speech level include:

- Use of cues established at earlier levels to focus on particular aspects of production (e.g. “make sure it’s a blowy /t/”).
- Work on breath control, and finding the child’s optimum phrase length. Some children need to organise their speech in short phrases.
- Ensure adequate airflow/pressure for plosives and fricatives. Aim for slightly exaggerated articulation of consonants – but ensure that the child has adequate air for the vowels, and doesn’t reduce their length.
- Ensure articulation at word level is, consistently, as clear as possible.
- Work on sequencing at single sound and word levels, gradually increasing speed.
- Help the child to monitor specific aspects of articulation (see above).
- Establish optimal rate of speech (see above).
- Work on organising message into short phrases.

4 Completion of sound system

Children working predominantly at sentence level often also require work on the later developing sounds and sequences:

a) Complex and less common clusters

As noted above, a wider range of word final clusters are produced by the addition of grammatical suffixes. Three element initial clusters (e.g. squirrel, splash, strong, scream) should also be practised, and clusters occurring medially and in multisyllabic words (e.g. biscuit, hospital, electricity).

A range of words with initial consonant clusters, and multisyllabic words are targeted in the Programme. These can be supplemented, as necessary with other materials. Various phonological resources cover a comprehensive range of clusters (see Superduper Publications, for example). A wider range of multisyllabic words can be explored by working on key words from a current school topic, or hobby, or by using a children's picture dictionary.

Symbol pictures and/or the written word should be used to support the learning of phonological representations and motor programs (see Chapter 7: Case Studies/Paul). Build up a set of words that the child can produce, then incorporate these in phrases and sentences (see Chapter 4: Treatment planning).

b) /r θ ð/ at phonological and articulatory levels

Although there are issues about the appropriateness of correcting /θ ð r/ errors, relating to the local adult model, and age of acquisition, the following points should be noted:

- Errors other than a simple immaturity (/θ ð/ → /f v/; or /r/ → /w/), or acceptable dialectal variant, require monitoring/treatment. Some children become completely confused over these phonological contrasts, and substitute in both directions.
- Children with DVD tend not to sort out these final sounds without help.
- Where a child is unable to discriminate these contrasts, or unable to articulate the sounds distinctively, at word level, this adds to his difficulties with literacy skills.

It is therefore suggested that these sounds are established at word level, with compensatory articulation of /r/, if appropriate. See Appendix for methods of eliciting sounds. Practice can be extended to sentence level, if appropriate.

Symbol pictures and graphemes should be used to support the learning of phonological representations and motor programs. Teaching of /ð/ should be linked to grammatical morphemes (see above). The NDP now includes sound pictures for /r θ ð/, but there is little other material in the program to support work on these sounds. Other resources should therefore be used.

A basic phonological/articulatory method, adapted to the child's difficulties, is used:

- Elicit sound and link to symbol picture/grapheme.
- Check and practise discrimination of substituted sound (also with symbol picture and grapheme).
- Sequence in combination with various consonants and vowels, and especially the substituted sound.
- Practise in CV nonsense syllables and any meaningful real CV words.
- Practise the target sound in initial, final and medial position in words.

- Build up a set of words with the target sound. For some children it is also necessary to build up a set of words with the substituted sound, in order to clarify phonological representations.
- Practise discrimination and production of minimal pairs of words (e.g. write/white, thought/fought).
- Incorporate words in phrases and sentences.

5 Developing strategies for learning new words

Even when the child is able to produce intelligible connected speech, he may still have difficulties in developing motor programs for unfamiliar, and particularly complex, words (see Chapter 7: Case Studies/Paul). Children need to assimilate large numbers of new words during their education. An appropriate strategy and the necessary support need to be put in place when regular therapy is withdrawn.

The development of segmentation skills, (see Chapter 6: Developing literacy skills), enables the child to break words into syllables and phonemes, with a greater or lesser degree of support. Children who have developed basic phonic reading/spelling skills can then use a written “phonic” form to help them to practise saying the word (e.g. “e – leck – tri – si – ty”). Children without this level of literacy will need auditory models, broken into syllables, to repeat until the program is established.

Learning will take time: Paul probably learned 6 new words per week, in addition to some revision of those from previous weeks. It was therefore important to select key words either from his topic work, specific interests, or names that he wanted to be able to say accurately.

6 Developing word joining strategies

In order to produce fluent connected speech, children must learn to modify their motor programs for single words, in such a way as to facilitate smooth transition between words, without compromising too much on intelligibility. There are standard rules or strategies, which operate in particular phonetic contexts. The extent to which they are used is determined by style and rate of speech. Greater modification is made in rapid and informal speech, and less in emphatic or slower speech. See Stackhouse and Wells, (1997, Chapter 8) for a discussion of these processes in normal and disordered speech development.

Children with DVD may not readily deduce these rules from the speech that they hear, and/or may have difficulty modifying their motor programs to construct motor plans. In therapy tasks they may attempt to include all sounds at word boundaries (resulting in slow, stilted speech), but generally revert to a pattern of omitting final consonants and weak syllables, and inserting glottal stops, in spontaneous connected speech. Some children’s connected speech shows a combination of these two styles, with slow, carefully articulated sections, and rapid, unintelligible parts.

Like other aspects of speech, children with DVD are likely to need the rules (listed below) made explicit, and careful practice of the motor skills entailed.

(# is used to indicate a word boundary, and ' to mark syllables, where necessary. Examples are taken from the Nuffield Dyspraxia Programme assessment).

- /j/ - A /j/ is inserted between 2 vowels, if the first ends in a high front/spread /i/ posture: /i ɪ eɪ aɪ ɔɪ /. For example, “Daddy is talking” is produced as /dædɪjɪzˈtɒkɪŋ/.

- C#V) – Where a final consonant is followed by a vowel, the syllable structure of the sequence is reorganised so that these 2 sounds become a CV syllable. Thus “A cup of tea” is produced as /ə'kʌ'pʌv'ti/.
- C#C – Where a final plosive is followed by an initial plosive, the first is not released, and may be assimilated to the second. For example, “Two sad cats” may be produced as /tusæd'kæts/ or /tu sæg'kæts/. However, where a fricative followed by a plosive, both consonants should be heard: “I saw five camels...” /aɪsəfaɪv'kæmʌlz.../.
- /n/ - “A” becomes “an” before a vowel, and syllable structure is reorganised. For example, “Get an apple” is produced as /'gɛ'tə'næ'pl/.
- /w/- A /w/ is inserted between 2 vowels if the first ends in a high back/rounded /u/ posture: /u ʊ əv əv/. For example, “Go in the house” is produced as /gəʊwɪnðəhaus/.
- /r/ - A /r/ is inserted between 2 vowels when the first vowel sound could be spelt with an “r” digraph (ar, er, ir, ur, or (aw). For example “Peter is eating his dinner” is produced as /'pi'tə'rɪ'zi'tɪŋ'hɪz'dɪ'nə/.

These strategies are practised in short phrases, with inserted sounds written in or represented with symbol pictures. Clinical experience suggests that it is sufficient to establish use of normal junction processes within short phrases, as they basically operate within phrases. It should be noted that the same processes operate within words, at syllable boundaries, e.g. /dɒk'tə/ - doctor, /faɪə/ - fire. Some of these have been included in the phrase lists, to be tackled at the same time.

7 Prosody

The term prosody refers to suprasegmental aspects of speech production, including intonation, rhythm, and rate of speech.

A basic unit of prosody is the intonation group, consisting of the nucleus (stressed word) and a number of syllables before and after it. The nucleus is marked by one of several characteristic pitch patterns, or nuclear tones, including low rise, high rise, low fall, high fall, fall-rise, rise-fall, and in some accents a level tone, Cruttenden (1997). Other syllables may be given a secondary degree of stress, again marked largely by pitch prominence, although increased volume and length also play a part in stress.

There is no one to one mapping of intonation patterns and meaning, and the more subtle uses of intonation are still being acquired up to 11 years (Wells and Peppe, 2001). However, appropriate use of intonation, at a basic level, aids the intelligibility of speech, by marking clause boundaries, questions, and new information. It is also used to mark turns in conversation. This would appear to be the level at which 3-4 year olds are using intonation (Stackhouse and Wells, 1997). Discussion of disordered intonation can be found in Wells and Peppe (2001).

Intonation patterns are specific to particular languages and accents, and it is important to be aware of local variations when working on prosodics.

Children with DVD most commonly show the following types of prosodic difficulties:

- Irregular rate/rhythm
- Slow rate, with equalised stress
- Small (usually low) pitch range
- Little variety of intonation
- Inappropriate use of intonation
- Poor control of volume

Some basic strategies and activities, used with children attending the Nuffield Centre and Language Unit, are outlined below. Further ideas for developing prosodic skills may be found in texts focusing on voice disorders, hearing impairment, speech improvement and the teaching of English as a foreign language. A cross-disciplinary review of the literature on the treatment of prosody has been produced by Hargrove and McGarr (1994), and provides a comprehensive reference to the wide variety of techniques used.

a) Developing intonation

The following are presented as output activities, but concepts should be explained, as appropriate to the child's age, and input work is included by reversing roles. Modelling and visual cues (hand movements, or a toy aeroplane, or pictorial representations) are used extensively in work on prosody. See also suggestions on developing basic vocal control in Oro-motor section, above.

Establish a basic statement intonation pattern

- Start with a short phrase or sentence. The nucleus should be the last stressed word. Model the sentence with normal rhythm. Explain stressed and unstressed syllables if necessary. The nuclear tone will be a fall, if a standard English accent is used, but may be a rising or level tone in some accents.
- Support by using a visual cue to indicate the direction of pitch movement.
- If the child's voice sounds low or monotonous, model the sentence with a higher starting pitch. Some children start so low that they quickly hit the bottom of their range. See this chapter: oro-motor section for further suggestions on developing basic vocal control.

Experiment with different nuclear tones and a wider pitch range

- Encourage the child to make the sentence sound more exciting. See how many different ways you can say the same sentence (how many different "tunes" you can use), but differentiate between those which would be acceptable in the accent concerned and those which wouldn't. Model sentences with a wider pitch range. Given that a narrow pitch range is generally low, this means making more use of the top end of the pitch range: starting falls, and ending rises, higher. Demonstrate this with visual cues.
- Use IBM Speech Viewer III to experiment with pitch range.
- If the child has literacy skills, ask him to read a story, using his voice to make it sound more interesting.

Use intonation to indicate focus

- Using puppets, pretend that one has misheard a statement, repeating it back with an error. The other puppet must then correct the statement, emphasising the appropriate word, e.g.

Puppet A My dad's got a blue car.

Puppet B Your dad's got a blue train?

Puppet A No. My dad's got a blue **CAR**.

Puppet B Your dad's got a new car?

Puppet A No. My dad's got a **BLUE** car.

Puppet B Your gran's got a blue car?

Puppet A No. My **DAD'S** got a blue car.

Explain that the important word in the sentence needs the pitch movement. In a standard English accent, this will probably be a high fall.

Turn a statement into a question, using intonation

- Use the puppets again. This time one questions everything the other says, using a rising tone, e.g.
Puppet A I'm seven.
Puppet B You're ↗seven?
Puppet A My name's Henry.
Puppet B Your name's ↗Henry?
- Experiment with a high or wider rise, producing the effect of incredulity!

Indicate the end/continuation of an utterance, using intonation

- A contrast is made in standard English, between a falling tone at the end of a clause, indicating the end of the utterance, and a rising nuclear tone, which indicates that the speaker intends to continue. We are not sure whether this works in other accents!
- For younger children, this is practised by imitating a model, or producing complex sentences in a structured task (e.g. Danny went to the ↗shop, to buy some sweets.) Give visual support to show the direction of pitch movement.
- Older children can try a more complex activity. This time each puppet has something to say, but puppet B must wait until he hears the falling tone, to know that it's his turn. Puppet A must make sure he uses a rising tone if he still has more to say. Ask the child if he thinks the other puppet sounds as if he's finished talking.

Begin to use voice and prosody to convey emotions

- Act out familiar stories (e.g. the three bears, little red riding hood) thinking about how the characters feel, and experimenting with the voices.
- Repeat a message, as if you are frightened/happy/angry/sad etc. See if other children can identify the emotion.
- Encourage the children to vary pitch, voice quality and rate of speech, as well as volume!

b) Developing volume control

Shouting

Children who have difficulty with vocal control often benefit from being taught to shout or "call", using pitch rather than just increasing the airflow and laryngeal tension, risking vocal abuse. Explain that it's rather like singing, and model "coo-ee" or "tea-time" on level tones. Extend this to slightly longer phrases and sentences, but also show the child how to break up a longer message into short phrases to shout.

This method also allows the child to retain better control of articulation, and thus improves the likelihood of the message being understood.

Using graded volume

Help the child to develop control of volume, by asking him to practise short sentences with quiet voice (for the adult sitting next to him), middle voice (for a toy positioned across the room, and loud voice (for a family member downstairs). He can then be reminded to use an appropriate volume, when he yells at the friend standing next to him, or uses too quiet a voice in the classroom.

Some children also need to be taught how to whisper. Variations on "Chinese Whispers" are the obvious activity!

8 Maximising intelligibility, and developing functional speech

It is important to recognise that the child with DVD is working with an impaired speech processing system. While less severely impaired children can be expected to develop essentially normal speech, those with more severe difficulties, and those with a dysarthric element, will need to develop strategies to maximise intelligibility. This generally involves compromising on the rate of speech, and sometimes on normal rhythm, in order to maximise accuracy and consistency of articulation. In addition, it is often necessary to improve self-monitoring skills, and develop wider communication skills and awareness. The following specific strategies have been found to be helpful:

- Aim for a rate of speech at the "slow" end of the normal range, if possible.
- Encourage the child to divide utterances into short phrases, to maximise control of airflow and facilitate use of rhythm and intonation.
- Some children benefit from using a "filler", such as "yeah" between phrases, to give themselves more time to formulate and plan utterances.
- Ensure that the child understands, at an appropriate level, both the problem and the strategy used (e.g. that he needs to talk slowly to give his lips and tongue time to get to the right place for all his sounds).
- Work on motor planning and monitoring skills as above. Use of a tape recorder can help some children to learn to monitor their speech.
- Use materials such as the Social Use of Language Program, (Rinaldi, 1993) to develop awareness of too rapid or too loud/quiet speech.
- Encourage use of normal rhythm and intonation wherever possible, as this enhances intelligibility.
- Develop awareness and use of non-verbal communication skills, such as eye-contact, turn-taking, gesture, facial expression.
- Develop awareness and use of conversational skills, such as monitoring the listener's response, topic maintenance and clear signalling of topic change.
- Practise key words in order to enhance intelligibility.
- Look at specific situations where intelligibility is a problem, and identify appropriate ways of supporting the child, e.g.
 - smaller class group
 - 1:1 support
 - use of an amplifier
 - use of communication aid
 - signing
 - target the child's use of his skills in a particular situation
 - training of staff in using particular strategies.

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CHAPTER 6: Developing Literacy Skills in Children with developmental verbal dyspraxia

Learning to read and spell is a complex process, involving both visual and phonological skills, as well as wider language abilities. It is widely recognised that a deficit in phonological processing underlies many cases of specific literacy difficulty (dyslexia) (Goswami and Bryant 1990; Snowling 1996; Stackhouse 1996). Although not all children with early speech impairment go on to have difficulty in acquiring literacy skills, there is increasing evidence that those with severe and persisting speech disorders are at high risk of literacy difficulties (Bird, Bishop and Freeman 1995; Stackhouse, Goulondris and Snowling 2002).

Bird, Bishop and Freeman (1995) investigated the link between speech impairment, phonological awareness and literacy development in a group of 5 -7 year olds with speech impairments. They were able to demonstrate that these children had significantly poorer phonological awareness than controls, and when re-assessed roughly 1 year and 2 years later, also had significantly poorer reading/spelling skills. The pattern of literacy difficulties shown by the speech impaired group showed particular problems with tasks requiring alphabetic encoding/decoding.

Bird, Bishop and Freeman (1995) found that the severity of the speech impairment correlated significantly with literacy outcomes; the only other factor to do so was non-verbal ability. Other studies have found language skills to correlate with subsequent literacy development (Bishop and Adams 1990; Leitaio et al 1997). Leitaio et al compared groups of speech impaired, language impaired, and a group of children with both speech and language impairment with a normal control group. It was the mixed group that showed the poorest literacy skills. It appears, therefore, that the effect of the phonological processing deficit may be compounded by wider, verbal or non-verbal processing deficits.

Stackhouse and Wells (1997) reported longitudinal studies of several children with developmental verbal dyspraxia (DVD) and severe literacy difficulties. They proposed a model of speech development which could be linked to Frith's (1985) model of literacy development, to explain the link between speech impairment and literacy difficulties. These models will be outlined below, but for a detailed discussion readers are recommended to refer to the original text.

Stages of speech development

Stackhouse and Wells (1997) identified 5 phases in normal speech development, culminating in phonological awareness:

Prelexical phase (0-12 months), when the foundations of articulation are laid through refining of oro-motor skills in feeding, vocalisation, babble and sound play. On the input side, the infant's acute phonetic discrimination is shaped towards phonological recognition of the language he is exposed to.

Whole word phase (12 - 18 months), when phonological representations and motor programs for first words are unsegmented wholes. Words are recognised by the most acoustically salient features of the adult form. The child attempts to reproduce these features in his motor programs within the constraints of his immature output processing.

Systematic simplification phase (18 - 36 months), when “stable correspondences” emerge between the target phonological structure and the child’s production. Stackhouse and Wells suggest that this provides the basis for the child’s gradual segmentation of his phonological representations into a sequence of individual phonemes. This process is not complete until the metaphonological phase.

Assembly phase (36 - 48 months)

The child at this phase is attempting to “master the phonetic and phonological aspects of complex utterances” (Stackhouse and Wells 1997). These include junction (i.e. the ways in which articulation at word boundaries is modified to allow a smooth transition between words), intonation, complex consonant sequences, and establishing the remaining consonants.

Metaphonological phase (48 - 60 months)

At this phase, children are beginning to demonstrate awareness of, and an ability to manipulate, the internal phonological structure of words, by playing simple rhyme and “I Spy” games. It is these skills which underpin the acquisition of alphabetic (phonic) reading and spelling skills.

Stages of literacy development

Frith’s (1985) model of literacy development maps the integration of visual, then phonological and finally orthographic strategies into the skills of reading and spelling.

Logographic stage

The child’s first attempts at reading involve recognition of a limited set of familiar words, using visual cues only. The size of this “sight vocabulary” will depend on his visual processing and memory skills. No reference is made to sounds at this stage.

Attempted spelling at this stage involves the child trying to reproduce what he remembers of the visual features of the word (e.g. “ce” for come), and tends to be difficult to decipher.

Alphabetic stage

As letter sounds are learned, the child is able to draw upon his knowledge of the phonological structure of words to attempt to segment and spell words by sound (e.g. “cum” for come). He is also able to supplement his sight vocabulary by blending (“sounding out”) unfamiliar words from their letters.

Basic alphabetic skills are taught from Reception Year (4-5 years) in the UK, with many of the vowel digraphs covered by the end of Key Stage 1 (7 years) (National Literacy Strategy 1998). Consolidation of these skills takes place during Key Stage 2 (7-11years).

Orthographic stage

Once a basic alphabetic strategy is established, the child must learn to recognise (in reading) and reproduce (in spelling) the letter strings and digraphs associated with particular phonemes/phoneme sequences and grammatical morphemes (e.g. “sh”, “ay”, “tion”, “ly”).

However, the relationship between the sounds of a word, in English, and its spelling is complex. The correct spelling choices for particular words must be learned (e.g. "bought", not "baut" or "bort"), as well as common irregular spellings, such as "come", "friend" and "beautiful". These skills are taught at Key Stage 2 (7 -11 years).

Phonic and spelling skills, taught at Key Stage 1, are summarised in the National Literacy Strategy, List 3.

The acquisition of literacy skills in children with DVD

Stackhouse and Wells (1997) suggest that, in children with speech disorders, development may be arrested at a particular phase:

- children with motor or structural impairments, or prelingual deafness - at the prelexical phase
- children with dyspraxia - at the whole word phase
- those with phonological disorder/delay - at different points in the systematic phase.

Failure to move through the stages of speech acquisition means that these children will not have developed the stable and accurate phonological representations, and the ability to reflect on and manipulate these, which is required in order to learn alphabetic reading and spelling skills. They may therefore acquire a sight vocabulary with little difficulty, providing their visual skills are satisfactory, but are likely to have considerable difficulty moving beyond this to use letter/sound correspondences to attempt systematic spelling and decoding of unfamiliar words.

Using the NDP to support the development of literacy skills

Top Tips

- Represent sounds with symbol pictures
- Develop systematic speech at each level
- Work with the child's full range of sounds
- Develop awareness of sounds in words during speech work
- Develop segmentation and blending skills alongside speech work
- Link sounds and symbols to letters
- Work with teaching staff where possible

Developing systematic speech

According to the Stackhouse and Wells model, the child must successfully pass through the Systematic Simplification Phase of speech development, in order to move onto the later Metaphonological Phase, when he should develop the ability to reflect on and manipulate sounds in words. For children with severely restricted or disordered speech, therefore, the first priority must be to develop systematic speech. This is not as daunting as it sounds as, for the purposes of literacy development, stable, systematic and accurate articulation is only required at word level, rather than in connected speech.

From clinical experience, the simplest way to bring some order into the chaotic speech of the child with dyspraxia, is to tackle it at each level of complexity (single sounds; CV

words/syllables; CVCV words; CVC words; multisyllabics; clusters). Developing systematic speech implies building a system of contrasted sounds at each level, as opposed to only working on sounds/words which are troublesome. This is an important principle of the Nuffield approach (See Chapter 4: Nuffield therapy approach). Awareness of sounds in words and metaphonological skills can be tackled at each level of complexity, alongside articulation:

- include the child's full range of sounds in some activities at each level (sequencing worksheets are a useful way of doing this)
- don't forget the vowels
- incorporate new sounds at each level, as they are established
- include activities where sounds/words are contrasted or sequenced (e.g. a mixed set of CV pictures for posting, snap games etc)
- ensure that a wide range of words/syllables are included at each level

Developing awareness of sounds in words

By developing speech systematically, combining sounds to make syllables and words, and making this process explicit through the use of symbol pictures, children can develop their awareness of sounds in words while working on output speech skills. Younger children can develop more accurate phonological representations and some awareness of their structure, without making demands on metaphonological skills which they do not yet have. Children with very severe output difficulties, who need to spend most of their therapy time practising articulation tasks, can develop their awareness of sounds at the same time:

- make the phonological structure of words explicit by using symbol pictures, Cued Articulation (Passy 1990) etc
- use output practice to support the child's awareness of sounds in words e.g. "b + ee = bee" build up transition worksheets)
- clarify phonological representations, as necessary, during speech work, using symbol pictures/cues and the child's knowledge of sounds (e.g. Kate has a "t" at the end, not a "k")
- work on discrimination problems as they occur in speech work (e.g. s/sh)

Linking sounds/symbols to letters

Bradley and Bryant (1985) showed that awareness of sounds needed to be combined with teaching of letter sounds in order to promote the development of alphabetic reading/spelling skills. Clinical experience has also shown that children with severe speech difficulties do not readily make the link between metaphonological skills they have learned in therapy sessions and literacy. Links between sounds and letters therefore need to be made explicit. Although the teaching of phonics, in most circumstances, will be the responsibility of the school, basic links between symbol pictures and letters can be reinforced with therapy tasks and materials.

Letters and written words have been included on some of the materials in the current program, but omitted on others due to problems with the spelling of vowel digraphs and the use of some activities where the written word would "give the game away". We decided it would be easier to write on the pictures when required, rather than to remove unwanted words.

- write letters/words on pictures, as appropriate for each child.
- match letters to symbol pictures - use short vowel sounds only and make sure parents are using sounds not letter names
- discourage the addition of schwa to voiceless sounds
- make explicit the links between sounds and written words (e.g. "p - ar - k" those two together say "ar") but don't expect the child to remember vowel digraphs
- if the child is in school, liaise with teachers, sharing cues/symbols pictures and letter formations, where possible.

Developing specific metaphonological skills

Clinical experience suggests that many children with typical dyspraxic features have great difficulty with metaphonological skills. Rhyme judgement, in particular, although occurring in normal development at around 4 years, may need to be taught specifically, after phoneme segmentation and blending has been established.

A wide range of metaphonological tasks are described in the literature, generally for the purpose of identifying phonological processing difficulties in subjects of various ages (Gorrie and Parkinson 1995; Frederickson 1995; North and Parker 1993). For the purposes of therapy, it is important to target the simplest and most useful processing skills. Recent research has identified phoneme segmentation and blending as the skill most consistently correlated with literacy skills, with rhyming correlating to a lesser extent (Muter 2003).

At the Nuffield speech and Language Unit, we have a carefully structured programme to develop phoneme segmentation and blending skills, which provides a basis for developing an alphabetic reading and spelling strategy. The sequence and integration of work on articulation, metaphonological skills and wider literacy skills, at the Unit, is shown in figure 1, overleaf. Using this approach, many children with severe speech and language impairment develop adequate literacy skills (see figure 2). The Unit phoneme segmentation/blending programme is described below.

It is important to bear in mind the child's age when attempting to work on metaphonological skills. For age norms, please refer to Appendix 7: normative information. Children under 4 years of age would not generally be expected to attempt segmentation/blending tasks. A range of games and activities targeting these skills can be found in Appendix 6: games.

Fig. 1 Nuffield Speech & Language Unit: Integration of work on speech & literacy

Therapy		Classroom
Articulation	Phonological Skills	
single sounds C + V → CV range of CV words early CVCV CV + CV → CVCV range of CVCV range of CVC	sounds/symbol pictures begin identification of C in CV sorting CV words to C or V segmenting & blending CV words early syllable segmentation segmenting & blending CVC words (long V's & diph's with pictures) (short vowels with letters)	letter formations for own name sight vocab letters/symbol pictures & letter formations
multisyllabics clusters extend vocab of complex words	start CVC onset + rime CVCV suffixes ing, er, y, le syllable segmentation segment & blend words with clusters segment complex words → invented spelling	segment & write CVC words segment & write CVCV words teach digraphs & begin high frequency irregular spellings integrating visual & phonic strategies

Fig 2 Reading, spelling & phonological skills of children at the Unit

Child	Time	CA	Lang	Sp	Phonological/phonics skills	Reading	RA	Spelling	SA
S	3½ yrs	7;9	++	+++	Good range of digraphs. Still makes segmentation errors, especially on multisyllabics and clusters. Now manages struc'd rhyme prod'n	Good sight vocab, mainly visual plus partial phonic strategy. Fluent reader.	8;3	Uses a combination of learned spellings + phonics, but some segmentation errors. Spellings are decipherable.	7;8
K	4 yrs	7;5	++++	+++	Segments/blends CVC words, with effort and errors. Struggles with structured rhyme prod'n and non words.	Fair sight vocab (200+ words). Fluency & comprehension limited by language problems. No phonic strategy.	7;4	Just beginning to attempt some of own spelling. plus knows 12 high frequency words.	6;7
J	7 ms	6;10	++	++	(None initially). Now segments/blends CVC words, with effort and errors. Struggles with structured rhyme prod'n.	70+ word sight vocab		Just beginning to attempt some of own spelling	
P	2 yrs	6;9	++	++	Good segmentation/blending of CVC words + range of digraphs, clusters and first suffixes. + structured rhyme prod'n.	Uses sight vocab (200+ words) + phonics Becoming fluent	7;5	Good phonic speller! Knows 12 high frequency words.	7;1
M	2 yrs	6;3	++	+++	Segments/blends CVC words with effort & errors. Struggles with non words and structured rhyme production.	100 + word sight vocab	6;10	Just beginning to attempt own spelling. Knows 6 high frequency words.	6;1
Jn	2 yrs	6;1	+++	+++	Segments/blends CV words and has started work on CVC. Unable to attempt structured rhyme production.	25 word sight vocab		Copies from model	
LP	1 yr	6;0	–	++	Segments/blends CVC words, with effort. Started clusters, digraphs & suffixes. Manages struct'd rhyme prod'n.	100+ word sight vocab	6;10	Beginning to attempt own spelling. plus knows 10 high frequency words.	6;5
LS	1 yr	5;11	++	+	Segments/blends CV words. Started work on CVC. Unable to attempt structured rhyme production.	30 word sight vocab		Copies from model	
C	1 yr	5;9	++(+)	+++	Segments/blends CV words. Started work on CVC. Unable to attempt structured rhyme production.	40 word sight vocab		Copies from model	
H	1 yr	5;8	++	+++	Segments/blends CV words. Started work on CVC. Unable to attempt structured rhyme production.	30 word sight vocab		Copies from model	
W	1 yr	5;8	+	++	Segments/blends CVC words, with effort and errors. Struggles with non words. Shows spont. rhyme prod'n!	Good sight vocab (200+ words) Becoming fluent.		Beginning to attempt own spelling.	
A	1 yr	5;6	+	++	CV seg/blend & started work on CVC. unable to attempt str'd rhyme prod'n.	20 word sight vocab		Beginning to copy from model. Writing dyspraxia.	

Key to figure 2

Time: Length of time at the Unit

CA: Chronological Age

Lang: Severity of language impairment

- + Delay, or disorder of only one aspect of language (vocabulary, syntax or pragmatics).
- ++ Disordered expressive language. Initially, limited range of 2-4 word utterances. Residual difficulties only after 2-3 years. Comprehension within normal range.
- +++ Severely disordered expressive language. Initially only small number of single words. After 2-3 years, showing limited auditory memory and associated comprehension difficulties, word retrieval difficulties, grammatical errors.
- ++++ Very severe disorder. Probably no expressive language initially and severely impaired comprehension. Additional attention and processing difficulties. After 2-3 years, marked comprehension problems, telegraphic expressive language with severe retrieval difficulties.

Sp: Severity of speech impairment

- + Delay, or disorder already largely resolved on admission.
- ++ Dyspraxia/severe phonological disorder. Initially, unintelligible speech, limited range of sounds and syllable structures, vowel errors. After 2-3 years, clear speech, using strategies to pace himself and learn new words.
- +++ Severe dyspraxia. Few or no sounds initially, and usually oral dyspraxia. Still problems with unfamiliar and complex words, after 2-3 years.
- ++++ Very severe oral and verbal dyspraxia, often plus dysarthria. Speech may still be unintelligible after 2-3 years.

RA: Reading Age, calculated from Schonell Graded Word Reading Test.

SA: Spelling Age, calculated from Vernon Graded Word Spelling Test.

The Nuffield Speech and Language Unit Programme for Phoneme segmentation and blending

1 Sound/symbol links

Children first learn to select a range of consonant and vowel symbols to sound.

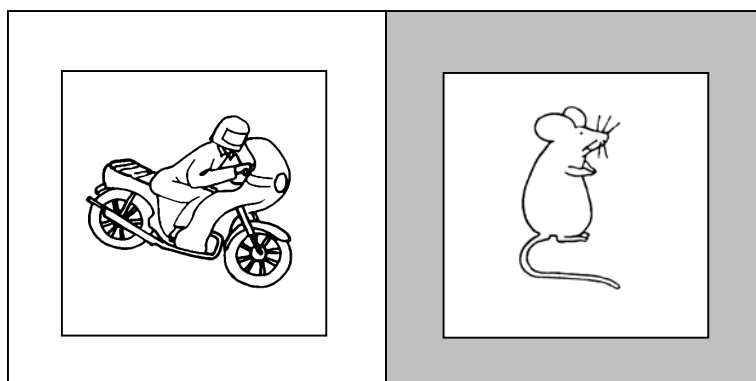
2 CV segmentation and blending

They then begin to identify the consonant in CV words, selecting the appropriate symbol picture. This is from an adult's production at first. The cue "begins with..." is associated with this task (e.g. "me begins with...(m)"). We usually start just with m/s, as these sounds are perceptually very different and are also easy to prolong. Other consonants are added, until the child is choosing from a full range of plosives and nasals, plus easier glides and fricatives /p b m t d n k g h w l f s/. These have their own place on a home grid, to make it easier for the child to locate the picture he wants.

	p	b	m	f	
	t	d	n	s	l
	k	g	w	h	

Table 1: Consonant Home Grid

Children then begin to select the vowel as well, starting just with a choice of / ee ah oo /, then adding the other long vowels and diphthongs. Vowel symbol pictures are kept separate from the consonants, and are on different coloured card (e.g. white for consonants and blue for vowels). The child places the consonant and vowel pictures on a coloured strip (white + blue), to reinforce the sequence of the task. Vowels are cued with "sounds like..." (e.g. "me sounds like...(ee)").



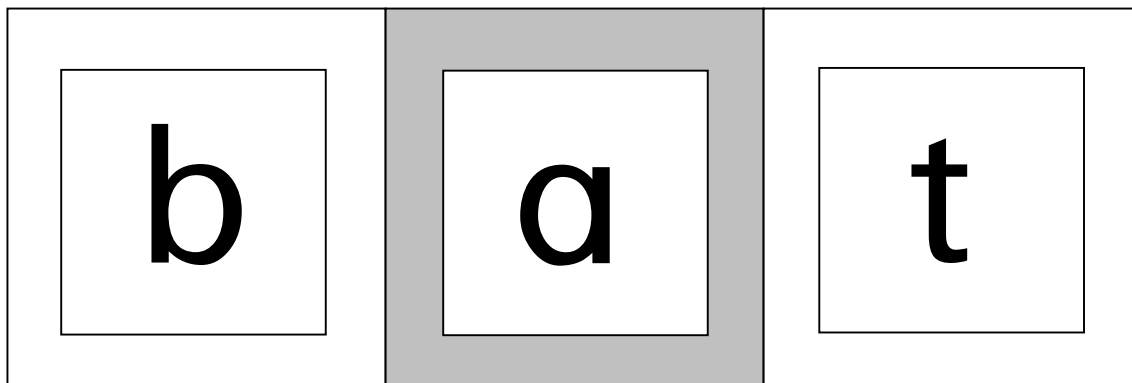
Example of CV segmentation - 'me'

CV pictures with the appropriate sound symbols on the back allow objective feedback. Blending is practised by reversing these cards, and asking the child to say the 2 sounds and work out the hidden word. Non words can be practised to consolidate skills.

The remaining consonants are added to the consonant grid, as appropriate for each child.

3 CVC segmentation and blending

Once children are confident with CV words, CVC words can be attempted. A three box strip is used (white-blue-white), and a third cue "What's the sound at the end?" is added.



Above: example of CVC segmentation - 'bat'.

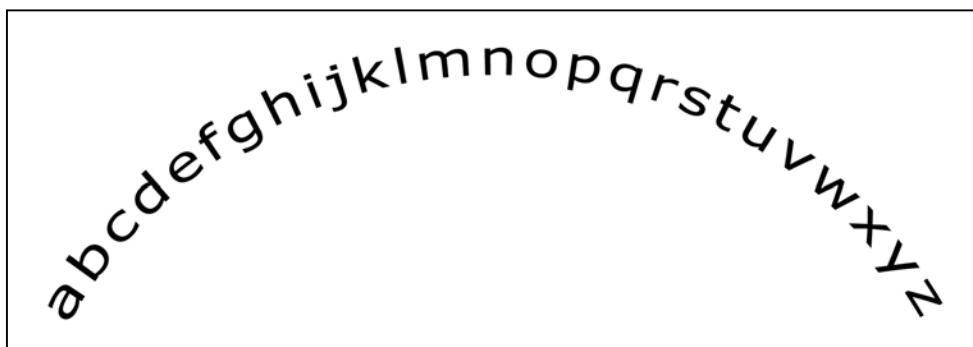
CVC words with short vowels are worked on using letters rather than symbol pictures, as children by this stage have learned a basic range of letters/sounds.

Non-words and simple manipulation tasks can be used to extend and consolidate skills. For example, a child can be asked to make a word (e.g. came) with his symbol pictures, and then change one sound to make it say "cake", then another to make "lake" and another to make "like". This is the equivalent of some of the harder metaphonological tasks, but with the visual support of the symbol pictures.

4 Consonant clusters

Once CVC can be segmented and blended, and the child has begun to work on clusters in his speech, CCVC and CVCC words can be attempted. Consonants are arranged as above, with additional consonants placed above the grid. Vowel symbols can be grouped in terms of long vowels, diphthongs and short vowels, or a limited selection can be used. A four box strip, (white/white/blue/white for CCVC), is used to prompt the child, and he is reminded that there are two sounds to listen for at the beginning/end.

CCVC and CVCC words with short vowels can also be worked on using letters. Reinforce the sequence of the alphabet, by setting the letters out in a rainbow shape, or use letters from an alphabet jigsaw.



5 CVCV words with -ing, -er, -y, -le suffixes

These suffixes are not covered until Y3 in the National Literacy Strategy, but we usually teach them, once short vowel CVC words can be read and spelt, and the child is making a reasonable attempt at saying CVCV words (don't worry about immature production of -ing /ɪn/ and -le /ʊ/). We generally work on these using letters, and introduce a basic doubling rule at the same time (i.e. tipping), explaining that "you usually need two letters in the middle of a word".

Start just with -ing and -er, by sorting pictures to the appropriate suffix. Then read and match words to pictures, and finally spell the words. Most children need to be reminded to segment and write the first syllable, before selecting and copying the appropriate suffix. We use a cue "what little word can you hear at the beginning?" to help them focus on the first syllable.

6 Digraphs

We teach digraphs in the following sequence, building up a set of key cards for each child, showing the digraph on one side and the picture of his key word for that digraph on the other, plus a small number of familiar words with that digraph. The key cards are regularly revised and used as a reference for spelling in the classroom.

- sh, ch, short oo (shop, chip, book)
- ee, ar, long oo (see, car, moon)
- or, ir (fork, bird)
- oa/ow, ai/ay, oi/oy, y (boat, snow, rain, play, coin, boy, sky)
- Magic e (snake, pipe, rope, cube)
- ou/ow, ur, ea (house, clown, nurse, sea)
- igh, th, qu (night, thunder, quick)
- au/aw, air/are/ear (autumn, claw, chair, stare, wear)
- ear, tch, dge, alk (hear, match, bridge, walk)

7 Spelling multisyllabic words

Once a child knows a range of digraphs and makes a good attempt at saying multisyllabic words, he can attempt to segment and spell them, using his phonic knowledge. This is a good way of consolidating metaphonological skills, extending spelling skills, and developing strategies for learning new complex words, (see Chapter 5: Therapy/Connected Speech).

Syllable segmentation and blending

Syllable segmentation/blending is worked on alongside articulation of bisyllabic, and later multisyllabic words:

- CV + CV = CVCV (pay + purr = paper) build up transition worksheets.
- Dividing pictures into the appropriate number of syllables.
- Counting, tapping, clapping, or jumping according to the number of syllables.
- Segmenting multisyllabics to attempt spelling, for children with a higher level of speech/literacy.

Onset/Rime segmentation and blending

At the simplest level, onset/rime skills can be practised by segmenting and blending CV words:

- C + V = CV build-up transition worksheets, and hiding pictures (e.g. p + ee = pea). Use without covering the CV word at first to demonstrate the task. Cover the CV picture for a true blending task.
- Sorting CV pictures to consonant (onset). Mixed CV pictures with 2 or 3 different onsets. Ask the child to listen to the adult's production, listening for which sound the word "begins with". He must then place the CV picture in the appropriate container, marked with the consonant symbol picture.
- Sorting CV pictures to vowel (rime). In the same way, CV pictures with 2 or 3 different rimes can be sorted to containers with the appropriate vowel symbol pictures.

VC pictures and rhyming sets of CVC words have been included in the NDP, to permit segmentation and blending tasks at CVC level.

- Sort 2 sets of CVC pictures to rime, (e.g. bat, cat, hat, fat, pig, big, wig, dig). Rimes can be represented by a picture, if a real word, or by symbol pictures or letters, according to the child's skills.
- Select a short vowel VC rime (e.g. "at"). Put the letters in the appropriate places on the white/blue/white strip if he still needs this. Lay the letters of the alphabet in a rainbow, as above. Ask the child to select initial consonants and blend to make words. Decide whether each word is real or funny/pretend. Encourage him to blend onset to rime (c-at → cat) rather than 3 separate sounds (c-a-t → cat). Keep reminding him that these are all "at" words. This is the structured rhyme production task referred to in figure 1.
- Onset tasks are generally easier than rhyme. A simple structured "I Spy" game can be played using words that the child can say and pictures to support the organisation of the task. In an unstructured "I Spy", beware of the child who cheerfully picks an initial sound and then agrees with whichever guess he likes the sound of!
- Tasks involving sorting pictures to onset can also be attempted. Problems are generally related to initial clusters, longer words, confusion of initial/final consonants and articulation errors. Tasks involving matching words with the same onset make greater demands on processing, as the onsets of both words must be segmented and the target onset also retained.

Once a concept of rhyme has been established, either through the structured work above or where a child has less difficulty, simple games such as rhyme lotto can be attempted. Beware of children who learn these games without really having the skills!

- Simple verbal rhyming games, such as the following (gleaned from a local nursery class) support the retrieval of words by rhyme by specifying the semantic field the child must access:

"Wibbly Wobbly Wenny, The elephant's looking at.....(Jenny)".

This can easily be modified to use different vocabulary. The elephant could buy food, or go to the farm:

"Wibbly Wobbly Wow, The elephant saw a....(cow)".

- Simple rhyming texts can be used, with the child supplying the rhyming word from the picture/memory or thinking of an alternative rhyming word/nonsense word. Some excellent rhyming picture books have been produced over the last few years, in response to the emphasis on rhyme for pre-schoolers and Key Stage 1.
- See Appendix 6: Games for further activities to develop onset/rime skills.

Implications for education and working with schools

The involvement of speech and language therapists in supporting literacy development is a matter of current debate, with policy decisions made by individual SLT services. However, there are clearly arguments when working closely with a child of school age with severe speech difficulties, for including the development of metaphonological skills as well as articulation/expressive phonology in therapy targets:

- work on developing phonological representations and awareness of sounds in words will be necessary in order to develop speech skills.
- work on segmentation/blending, and onset/rime when appropriate, can take place alongside speech work.
- work on metaphonological skills must take into account the child's incomplete speech system - the SLT is the only professional with the skills to devise an appropriate programme.

The assessment and teaching of literacy skills will, however, generally be the responsibility of the school. Children with DVD and associated phonological processing deficits are likely to require specialist teaching of literacy skills, as recommended for children with specific learning difficulties (dyslexia). Teaching will need to be systematic and thorough, especially in relation to phonics and spelling, with a good deal of practice and revision.

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Chapter 7: Case studies

The case studies below have been selected to illustrate treatment and outcomes, with children (covering an age range of 4 – 12 years) who show a range of different profiles, and who were treated using different models of service provision.

Chrissie was first seen as an unintelligible 4 year old, with some expressive language delay, as well as a moderately severe oral and verbal dyspraxia. She received weekly therapy over a period of two years, with daily practice at home.

Joseph presented with a severe speech disorder, with both dyspraxic and phonological components. He had blocks of weekly therapy, supported by practice at home and at school. For him, the acquisition of literacy skills was an ongoing problem, once his speech difficulties had been overcome.

Rickie showed an unusual speech pattern, including the use of clicks and ejective consonants. Developing appropriate motor programs for these sounds accounted for a good part of his therapy programme.

Paul and **Hannah** had particularly severe difficulties and attended the Nuffield Speech and Language Unit, in Paul's case for 18 months, and in Hannah's for 3 years. They both received intensive therapy. Their profiles, however, were quite different, necessitating a different focus in therapy, while working through the 4 Stages of the Treatment Plan (see Chapter 4: Treatment Planning). Paul had major word-finding difficulties, related to a speech processing problem which particularly affected the levels of phonological representations, motor programming and motor planning. Hannah had a severe oral and verbal dyspraxia, with respiration-laryngeal involvement.

Liam was first seen at the Nuffield Centre at 8;4 years, having had therapy since the age of 5 years, but still showing significant speech difficulties, including hypernasal resonance. He finally completed his treatment at 10 years, and when reviewed 2 years later, had normal speech.

Chrissie

Chrissie was first seen at the Nuffield Hearing and Speech Centre, aged 4 years 4 months. She had been referred to her local Speech and Language Therapy service at the age of 3 years, and had received two blocks of group therapy for five weeks each block. Her mother reported that she had made very little progress since the age of 3.

Her verbal comprehension was assessed and confirmed as age appropriate, she used a fair range of vocabulary and was able to convey information reasonably well. Her expressive language demonstrated some syntactical errors. She was unintelligible for most utterances, as a result of articulatory and phonological errors.

Speech Assessment

Oro-Motor Skills

She had no specific difficulty with respiration and breath control for speech production was developing normally. Facial expression was limited, although she had a lovely smile, which she used spontaneously. Lip shapes could be made accurately, but there was poor fluency of movement from one shape to another, especially for production of diphthongs. She demonstrated a range of movements and placements with her tongue, but was unable to control these and placement for sound production was not accurate.

Single Sounds

She was able to imitate all vowels and diphthongs, but was not able to produce them accurately without an immediate model.

Chrissie imitated the consonants /p – b – d – m – n – f – v – s – ʃ – tʃ – w – j/. She was able to sequence two consonants from her system, but was unable to sequence vowels.

CV Words

She attempted CV words with her consonant system, but these were variable in accuracy and her vowels were usually distorted as well. She was able to repeat a CV word, but was unable to produce a contrasting sequence. Most plosive consonants were voiced.

More Complex Words/Phrases and Sentences

Chrissie was unable to sustain any accuracy for vowels or consonants beyond the CV word level. Most sounds were replaced by [b] and [d].

"And she got go back to the shop." [æn di də dəʊ bʌ? du də dɒb]

"Then a lady dropping down again." [lɛn ə lædi dɒdi daʊn əde?]

Connected speech was usually unintelligible to everybody apart from her mother and brother.

Speech Processing Skills

Chrissie's auditory discrimination was developing fairly well. Work on this area had been done in her group therapy sessions, and she was able to discriminate between most minimal pairs. She was unable to imitate sounds or words more accurately than her spontaneous production, demonstrating some motor programming difficulties. Although she was very aware of which sounds were more difficult for her to produce, she was not

able to correct these without an immediate model. Her poor lip and tongue placements for sounds indicated that she was having difficulty at the motor planning level.

Diagnosis

Chrissie was found to have a moderately severe oral and verbal dyspraxia, with phonological delay. She showed evidence of difficulty with motor programming and motor planning.

Intervention

She was seen for one session per week, of one hour for speech and language therapy, over a two year period. Her parents worked with her, at home, for a minimum of five days per week, for 20 – 30 minutes per day. The therapist also visited her school, approximately once per month, to train the Learning Support Assistant, and provide new, additional work. The LSA worked with her for 30 minutes, three times per week.

Treatment Programme

Early therapy targets were aimed at:

- Improving lip shaping and fluency of movement, for more accurate diphthong production.
- Tongue exercises, to improve independent movements of her tongue and more accurate placement for both front and back tongue sounds.
- Widening her use of Facial Expressions.
- Improving voice quality, especially control of volume, and pitch.
- Long Vowel sequencing, and introduction of diphthongs in sequences.
- Accurate production and sequencing of consonants already in her system, with particular attention to voicing contrasts for /p – b/ and /t – d/.

Once these early targets were achieved with a good level of consistency, work was done to elicit the velar sounds /k/ and /g/. Chrissie was very aware of her difficulty, and having been an eager and confident worker during the sessions, she became tearful and anxious, as soon as the sounds were mentioned. Her mother suggested that if the therapist demonstrated how the sound might be elicited, she and Chrissie would try at home! The following week, Chrissie was able to produce a /k/ spontaneously, by holding her tongue down at the front, with her finger. Within two weeks, she was able to produce the sound in isolation, and in repetitive sequences (e.g. /k – k – k/), without the aid of her finger.

After this initial difficulty, Chrissie gained enough confidence to be able to work with the therapist on eliciting other sounds for her system.

After six months of therapy, she was able to use all plosives, nasals, fricatives and approximants accurately in CV, CVCV and CVC words and was beginning to self-correct in her spontaneous speech.

Progress was a little slower over the next six months, and her difficulties with motor planning became very obvious. A great deal of work was done to help her overcome these problems and continue to build up her sequencing skills. In addition, her facial expression and her voice production remained areas of difficulty for her.

After 18 months of therapy, Chrissie was beginning to use consonant clusters with fair accuracy, and medial and final consonants, including affricates, were being used with a fair level of competence in her spontaneous, connected speech. She was becoming a fluent reader, was writing quite nicely and developing excellent phonological awareness and early spelling skills.

The final six months of therapy were used to help her coordinate her facial expression, with her speech and voice production, and to help correct articulation of 'th' and 'r'.

Chrissie's reading, writing and spelling are 'above average' for her age. Her speech is entirely accurate, however, her voice production is still a little poor, with a high pitched voice and some limitation of intonation patterning. It has been decided to allow her time to mature, before any further voice work is considered.

Outcome

Chrissie worked through the 4 stages of the NDP treatment plan. After two years, she was intelligible at all times, used a complete, accurate system of vowels and consonants in all positions in words and in long complex sequences. Her literacy skills, including spelling were developing normally. She continued to have a high-pitched voice, with immature sounding intonation patterns used.

Joseph

At age 5 years 5 months, Joseph presented with a severe speech disorder with both dyspraxic and phonological components. He was unintelligible to all but familiar listeners and very aware of his speech production difficulties.

Speech Assessment

Oro-motor skills

- Adequate attempts at the four basic lip postures (open, closed, smile, round) but difficulties in producing half-open lip postures. Sequencing lip/jaw movements also proved difficult.
- Able to protrude, elevate and attempt lateral tongue movements and a range of tongue body postures for most vowels. Difficulties observed with attempts at alveolar placement (frequently realised interdentally) and velar postures.
- No specific concerns were noted with palatal function or respiration/larynx e.g. voice/vocal tone.

Single Sounds

- A restricted sound inventory consisting of the following consonants: /p, b, t, d, m, n, f, s, ʃ, h, w/.
- Imitated vowel production: long vowels: /ɑ i ɔ ɜ/, short vowels: /ɪ æ ʌ/ and diphthongs: /eə, ɪə, aɪ, aʊ, əʊ/.
- Sound sequencing difficulties e.g. C + C, V + V etc.

Single Words

- A restricted range of CV, CVC, CVCV and multisyllabic/complex words using the consonants /b, d, m, n/.
- Sound substitutions/processes e.g. context sensitive voicing, stopping, fronting, final consonant deletion.
- Typical dyspraxic features: medial/word final glottal stop insertion, vowel distortions and omission of weak syllables.

Speech processing skills

Joseph's auditory discrimination skills were not a major area of concern but he had little idea of phonological awareness even at the basic syllabic level.

Diagnosis

Joseph presented with a severe speech disorder with both dyspraxic and phonological components.

Intervention

Joseph attended for blocks of weekly therapy sessions at his local Health Centre, supported by practice at home and in school.

Treatment programme

A multi-level approach was adopted during therapy, targeting oro-motor, single sound and word level production with a further aim of combining words into short phrases with improved accuracy.

Early therapy sessions included work on oro-motor skills, single sound production, single word level production and phrase level work:

Oro-motor: to extend control of voice, lips, jaw and tongue.

Several minutes of each session were spent developing Joseph's control of his voice when producing long and diphthong vowels e.g. vowel length, volume, pitch. Joseph was encouraged to copy lip/jaw and tongue postures produced by the therapist and then produce corresponding vowel or consonant sounds e.g. wide open mouth for /a/; licking chocolate spread from behind the upper incisors to assist in facilitating alveolar-lateral placement for /l/.

Single sound production: to establish consistent and accurate production of all vowels and to extend Joseph's range of consonants to include the velar consonants /k and g/ and lateral approximant /l/.

Joseph was introduced to the Nuffield symbol pictures for vowels and consonants and cued articulation gestures. A mirror was used and tactile cues were given to provide him with feedback about placement, voice and manner. For example: for /k/, visual cues were used: Nuffield Dyspraxia Programme symbol, articulogram for voiceless (puff of air) and the cued articulation gesture together with tactile/oral support involving placing a small tongue depressor coated in chocolate spread at the back of the mouth to encourage differentiated back of tongue movement.

Single Word level: to establish consistent and accurate production of: CV, CVCV, CVC, CVCing verbs with established sounds.

In addition to the above aims, Joseph also worked on consolidating CV production for /b, d, m, n/ consonants with easier vowels e.g. to produce words: bee, do, me, no etc. Gradually a wider range of vowel and voiced consonants were introduced to increase his repertoire of CV words. For those voiceless consonants that were stimulable e.g. /p, t, f, s,ʃ/, Joseph was introduced to C+V = CV build up transition worksheets.

CV + CV contrasts were then introduced to develop accurate CVCV words. Particular attention was needed to produce the second syllable accurately using sound symbol cues and ending cards e.g. pea, tea, me etc to avoid glottal stop insertion.

As production attempts became more stable and accurate motor programs were established the following were introduced: more CVCVs, CVCs and CVCing verbs.

Phrase level: to combine the above single words into short 2-3 word phrases, maintaining articulation.

As therapy progressed, Joseph was encouraged to link his expanding repertoire of single words to form short 2-3 word phrases e.g. "funny cow", "funny car", "my daddy go" etc.

Phonological awareness skills - work on developing syllable awareness and basic rhyme skills were incorporated into Joseph's therapy programme.

Later therapy sessions focused on establishing the above skills and extending Joseph's speech production skills e.g. updating motor programs to include more complex words e.g. involving clusters and multisyllabics and incorporating these into phrases and sentences. Joseph also required support to develop the suprasegmental aspects of

speech e.g. reminders to slow his speech rate so that others could understand him fully and also to monitor intermittent hypernasality on vowels.

Outcome

Joseph received a number of therapy blocks over two years. At the end of this time, he was readily intelligible with a stable speech production system. His literacy acquisition was affected by his speech difficulties and Joseph required additional support in school to develop his reading and spelling skills.

Rickie

Rickie began a block of therapy at the Nuffield Hearing and Speech Centre, at 5 years. He had had two blocks of therapy at his local clinic previously. He had no difficulties with listening skills, comprehension or expressive language.

Rickie's speech was disordered, and was described by his father as "like a stammerer". His parents were not always able to understand what he said.

Speech Assessment

Oro-motor skills

Rickie achieved a range of lip and tongue movements, but had some difficulty with sequencing tasks. His oral/nasal resonance balance was appropriate, and he was able to sustain a vowel with good voice quality. Volume was appropriate, but pitch control was poor, with pitch suddenly shooting up in the middle of phrases.

Rickie's major oro-motor difficulty was in coordinating respiration and phonation, which affected his production of the voiceless plosives /p t k/.

Single sounds

Rickie was able to imitate all vowels and the following consonants:

/b d g m n w l j h ʃ tʃ dʒ/. Voiceless plosives were produced either as clicks or ejectives. The fricatives /f s/ were inaccurate, produced as [ɸ ç] respectively.

CV words

Rickie attempted CV words with his phonetic inventory: words beginning with voiceless plosives were produced with clicks/ejectives. The fricatives /f s/ were produced as [ɸ] and [ç] respectively.

Pea → [p'i]

Pie → [ɸaɪ]

Tea → [li]

Key → [k'i]

He managed to repeat the same word 3-5 times, but had difficulty with sequences of 2 different CV words.

More complex words

Rickie produced CVC, and CVCV words with the same range of sounds, except that /f s/ were deleted in initial and final positions, and replaced by glottal stop [ʔ] in medial position. Consonant harmony was also noted.

Kite → [k'aɪk'] sun → [ʌn]

Sock → [k'ɒk'] leaf → [li]

Garden → [dɒdɒn] bus → [bʌ]

Multisyllabic words were inaccurate, and also dysrhythmic, due to the effects of the abnormal airstream mechanisms used (i.e. clicks and ejectives). Rickie was able to use initial /bl/ and /gl/ clusters.

Speech processing skills

Rickie showed age appropriate discrimination and phonological awareness. He was able to identify syllable number, and onset and rime. He rejected his own speech production errors.

Diagnosis

Rickie was diagnosed as having a moderate developmental verbal dyspraxia, with an oro-motor component.

He showed a patchy sound system, where a group of early developing sounds had not been acquired. There was a dysrhythmic quality to his speech, due to his use of clicks, ejectives and glottal stops, plus some mild pitch breaks.

Intervention

Rickie attended a block of 20 weekly therapy sessions, at the Nuffield Hearing and Speech Centre, supported by daily practice at home.

Treatment programme

Rickie's patchy profile was well suited to the multi-level, multi-target approach of the Nuffield Dyspraxia Programme:

- Using the sounds Rickie could produce, a wide range of CV, CVCV and CVC words were established.
- These words were then practised in sequencing activities, and at phrase level, incorporating practice of intonation patterns (Stage 2-3 of treatment plans).
- In contrast, Rickie did not have accurate motor programs for /p t k f s/, so therapy for these sounds started at Stage 1, working first on underlying oro-motor skills. He worked on establishing gentle voice (initiating vocalisation with /h/), control of airflow, coordination of airflow and phonation. Intonation patterns were practised with different vowel sounds. Appropriate placement/postures for /p t/ were taught.
- At the same time, Rickie was taught correct motor programs for /f s/, using articulograms to support correct placement and manner.
- /f s/ were then practised in sequences, with little difficulty, and in CV and CVCV words.
- Once Rickie had achieved gentle plosion on /p t k/ he worked on sustaining his motor programs to produce repetitions of these sounds, before extending this to sequences of two sounds.
- A set of contrastive CV words was then established, with /p t k/.

Outcome

Rickie learned to produce /p t k f s/ with accurate motor programs. His use of clicks and ejectives reduced, and the rhythm of his speech improved. He generalised the target sounds to new words with ongoing practice at home. When reviewed 6 months later, he had consolidated his target sounds and his intelligibility was good.

Paul

At 6 years, Paul's speech was largely unintelligible, due to multiple and inconsistent phonological and articulatory errors. He had significant expressive language difficulties, including syntactic errors and omissions and severe word finding difficulties. His comprehension was age-appropriate, and he was acutely aware of his difficulties. He had no literacy skills, and was desperate to read.

Speech Assessment

Oro-motor skills

Paul had no specific problems with respiration/larynx or palate. He showed an adequate range of lip and tongue movements, but had difficulty organising sequences of movement, and sustaining postures. Placement for tongue tip sounds was imprecise and inconsistent. Tongue body postures for vowels were adequate. Velar consonants were achieved inconsistently.

Single sounds

Paul imitated /p b m t d n k g s f w j h/ with reasonable consistency and accuracy. He was able to imitate a full range of vowels but with inconsistent errors. He managed short sequences of 2 consonants, but was unable to sustain vowel sequences.

CV words

CV words were attempted with all the above sounds, but vowels were sometimes imprecise or misselected. He could generally produce repetitions of CV words, but could not sustain a sequence of 2 words (e.g. bow – go).

More Complex Words

Some familiar CVC and CVCV words were produced accurately, while attempts at others were severely disordered and inconsistent e.g.

/pɒp/ - pop	/dʌbu/ - table	/pæʔətrɪxə/ - caterpillar
/teɪk/ - cake	/sɪʔɪ/ - sooty	/tɛwɪfʌm/ - telephone
/dɒg/ - dog	/tʌrə/, /kʌrə/ - tiger	/sɛzɪ/ - television
/kæʔ/, /kæk/, /tæk/ - cat		

Phrases and Sentences

Paul had great difficulty imitating even short sentences. His responses frequently differed from the model in terms of syntax, and even vocabulary, as well as articulation, e.g.

/kʌʔɪz pɑpɪ/ -	Katy's party
/dædɪ jɪz bukin/ -	Daddy is talking (possibly "booking", as the picture showed Dad holding a book!)
/fɪsɪz ɪz ə gʊd fuʔbɔw ɪn ə ɛu/ - Christopher is the best footballer in our school.	

Speech Processing Skills

Paul was generally unable to imitate words more accurately than his spontaneous production, demonstrating motor programming difficulties. With some words, (generally shorter and more familiar), he commented on his errors, while still being unable to correct them, showing more accurate phonological representations than motor programs. However, for other words, his phonological representations were clearly inaccurate, for example, he was completely confused over *parrot*, *carrot* and *pirate*. His inconsistency of production of the same word (often with several different realisations in the space of a few minutes), suggested that he was also having difficulty with motor planning.

He made errors on an auditory discrimination task, when asked to judge whether 2 words/non-words were the same, demonstrating fragile auditory processing skills.

Paul also showed poor awareness of the phonological structure of words. He was unable to count the number of syllables in words, or to match words by onset. He managed to match a few words by rhyme, but was unable to extend this to unfamiliar materials.

In general, all aspects of processing showed greater difficulties at more complex levels.

Diagnosis

Paul was found to have a severe speech disorder, with both dyspraxic and phonological components. He showed particular difficulties with phonological representations, motor programming and motor planning.

Intervention

Paul was admitted to the language unit. He received 2 x 20 minute sessions per day, working on articulation, phonology and phonic skills, over the following 18 months. Language skills were worked on separately.

Treatment Programme

Paul worked on speech and phonological/phonic skills side by side. (See Chapter 6: Developing Literacy Skills for an outline of the Unit's approach to developing phonological skills and phonics.)

Paul covered the 4 stages of the treatment plan (see Chapter 4).

As he already had a good range of single sounds and no major oro-motor difficulties, Paul quickly progressed through Stage 1, consolidating his range of sounds by linking them to symbol pictures and practising sequencing activities. The remaining sounds were added to his system gradually, and incorporated in his sets of words (below).

A good deal of time was spent establishing consistent production of a wide range of CV, CVC and CVCV words (Stage 2-3), clarifying phonological representations, ensuring the accuracy of his motor programs and consistent production at the motor planning stage. Paul built up his own sets of words at each level, and practised these regularly. As his segmentation/blending skills improved, he was able to use symbol pictures on the reverse of his cards to check/cue his production. This approach was continued with multisyllabics and consonant clusters.

Motor programming remained a problem for Paul. By the time he left the Unit, his difficulties were with new complex words, such as topic vocabulary and unfamiliar names. By this point, he was able to split words in to syllables himself, and write a

“phonic” version to remind himself how to say the word. Lists of words were stuck on the fridge to be practised!

Paul’s motor planning difficulties meant that a good deal of work was required to incorporate the words established at single word level into sentences. This was supported through sequencing activities (e.g. cake-kate), phrase level practice (e.g. Peter digging), the use of carrier sentences (e.g. My _____ is red), and finally making up and repeating his own short sentences. Paul found repeating a short sentence 2 or 3 times, accurately, and without changing it, a real challenge!

His motor planning difficulties meant that connected speech was always vulnerable to breakdown. To overcome this, Paul worked on pacing his speech, monitoring and self-correcting.

Outcome

After 18 months, he was able to make himself understood easily, despite some errors, by monitoring and pacing his speech. He still needed to learn unfamiliar, complex words by splitting them into syllables and practising saying them. His literacy skills were within normal limits for his age. He transferred back to mainstream school, with some support.

Hannah

Hannah was admitted to the language unit at 4; 10 years, essentially non-verbal. She communicated only at a very basic level, using some gesture, vocalisation, 4 spoken words and a small number of signs. Her comprehension showed a moderate–severe delay. Hannah had a history of conductive hearing loss associated with glue ear, and a mild on-going unilateral loss. It was noted that she had made no progress in terms of speech over the previous 12 months, despite some speech and language therapy input.

Speech Assessment

Oro-motor skills

There was no history of feeding difficulties, although it was noted that Hannah struggled with chewy sweets, and there was some loss of liquid during drinking.

Muscle tone around her mouth and face appeared rather high, with limited mobility of lips and tongue. However, Hannah did manage a basic range of lip and tongue movements.

She was able to blow, but had great difficulty initiating voice and producing good vocal tone at a voluntary level.

Single Sounds

Her attempts at /p b m ɪ æ u/ were reasonably consistent. It was also possible to elicit /n f h/ and approximations to the long vowels /i ɑ u/, with support.

Single words

Hannah attempted 4 words spontaneously:

/dɛ/ - yes

/nʌ/ /nəʊ/ - no

/bʲi/ - me

/ʌləʊ/ - hello

Attempts at imitating other words, during assessment, were generally voiceless. Hannah managed some vowel differentiation, and some appropriate placement for consonants, but without airflow or voice.

Phonological Skills

Hannah was able to discriminate between minimally different words in her receptive vocabulary (e.g. tea/key, night/knife, nail/snail), showing that her input skills were at a significantly higher level than her output.

Diagnosis

Hannah was found to have a very severe oral and verbal dyspraxia, with a possible mild dysarthric element. She had particular difficulties with the motor programming/motor programs and motor execution levels of processing. Her respiration-laryngeal system was particularly severely affected.

The extent of Hannah's dyspraxia became more apparent during therapy, when she found it extremely difficult to establish voluntary control over oral movements and combine these with airstream and voice to establish motor programs for sounds, or to modify existing programs.

Her need to be taught to use her sounds and CV syllables to attempt the words she wanted to say, and the increase in her expressive language once some basic speech skills were established, confirmed the severity of her motor programming difficulties.

Intervention

Hannah received 2 x 20 minute sessions daily, working specifically on her articulation and a group session on phonological skills, during her first 2 years at the Unit. During her third year, this was reduced to a single speech session. Her language skills were worked on separately, initially using signing to support her very limited verbal skills.

Treatment Programme

Hannah worked through the 4 stages of the NDP treatment plan (see Chapter 4) during her 3 years at the Unit.

Hannah worked on extending oro-motor control, (including control of airstream and voice), and establishing/improving motor programs for single sounds, throughout her treatment programme.

First year

Initially, work focused on establishing the basic lip/jaw postures required for vowels, front tongue/dental and back tongue/velar closure, and developing control of her airstream and voice (see Chapter 5: Therapy: oro-motor skills and Appendix 4 on eliciting sounds).

Hannah had great difficulty combining postures with airstream and voice at a voluntary level, to create speech sounds. She was therefore encouraged to switch on either voice or airstream and experiment with sounds, by moving her tongue around in her mouth. The sounds she produced were reinforced by being remodelled/cued/associated with symbol pictures, until they could eventually be elicited by cues.

Once established, motor programs for single sounds were consolidated by means of repetition, contrasting and sequencing activities. All targets incorporated work on developing control of airstream and voice (see Chapter 5: Therapy/oro-motor skills).

Voiced consonants /b d g m n w j/ were first combined with easier vowels to form CV syllables. Hannah was taught to use these to represent words in her signed vocabulary, thus increasing her communication skills and maximising practice of her speech skills. Modified articulatory targets for words were devised on the basis of normal developmental processes: pre-vocalic voicing, final consonant deletion and cluster reduction (e.g. /gɑ/ for car, /bəʊ/ for boat, /neɪ/ for snake). These targets were modified as motor programs for voiceless CVs were established. The first very simple CVCV words were also introduced (Mummy, Daddy, baby, Nanny).

Hannah worked on her oro-motor skills, and establishing a basic range of single sounds and CV syllables (Stage 1) for the first year. At the same time she worked on segmenting and blending CV words (see Chapter 6: Developing Literacy Skills).

Second year

During the second year she consolidated her range of CV words/syllables, improving the accuracy of lip/jaw/tongue postures for vowels, long/short vowel contrasts, airflow and placement for consonants. She also practised a wide range of CVC and CVCV words (stage 2), monitoring the accuracy of articulation and use of voice and airstream.

Licking and blowing activities continued to be used to lead into work on articulation, and sequencing activities were used to extend articulatory skills.

Work on tongue control also focused on developing central airflow for the sibilants /s z ʃ tʃ dʒ/, which were being produced as laterals and detracting significantly from the intelligibility of Hannah's speech. She also worked on discrimination of /s, ʃ/ from [ʔ].

Towards the end of the year easier multisyllabics and short phrases/sentences were introduced (Stage 3).

In phonology/phonics sessions, she mastered segmentation and blending of CVC words, and simple onset/rime tasks. She also worked on discrimination of vowels she was struggling to produce accurately (e.g. /i - ɪ/, /u - ʊ/), and on clarifying phonological representations and meanings of new words.

Third year

Hannah worked on improving the accuracy of vowels and consonants, and sustaining accurate articulation at sentence level. Work on tongue control continued to support work on tongue placements for the sibilants, and blowing to support work on increasing airflow for consonants and volume. Sequencing of single sounds and words continued to be used to develop accuracy with greater speed, without the additional demands of formulating language.

A wider range of multisyllabics and consonant clusters were tackled, and acceptable articulation of /r θ ð/ established at word level.

Work on voice continued, at sentence level, to increase her pitch range and use of phrasing and intonation. Work was done on the co-ordination of airflow and articulation at higher volumes, so that she could shout.

Sentence level work focused on junction strategies (see Chapter 5: Therapy Approach/connected speech), pacing and monitoring (Stage 4).

She worked on phonic skills in the classroom, learning a wide range of consonant and vowel digraphs (for example: ee, oy, th). Phonological representations and word meanings continued to be clarified as new words were introduced in speech sessions and in the classroom.

Outcome

By the end of the first year she was talking in short sentences, although the intelligibility of her speech was very poor. After almost 3 years, her phonology was more or less accurate, but her speech could still be unclear, as a result of continuing motor execution difficulties, i.e. poor vocal control (low pitch range/creaky), inadequate airflow for consonants and a somewhat limited range of tongue movement. However, she was generally able to make herself understood, by pacing and monitoring her speech.

Hannah's literacy skills progressed well to be within normal limits for her age. Her language skills are around the 11th centile, and general ability in the low average range. Hannah transferred to mainstream school at the end of the year, with support.

Liam

Liam was first seen at the Nuffield Hearing and Speech Centre, aged 8 years 4 months. He had received ‘occasional blocks’ of 6 weekly sessions, of Speech and Language Therapy since the age of 5 years, and although his sound system had improved, he was still not easily intelligible. In addition, his voice had marked hypernasal resonance, adding to the poor intelligibility. He had been examined using videofluoroscopy but was unable to tolerate nasendoscopy.

His hearing, verbal comprehension and expressive language were developing normally.

Liam was in mainstream school, and having some difficulty with spelling. He was accepted by his peers and had friends. He had experienced some teasing and was very self-conscious about his speech.

Speech Assessment

Oro-Motor Skills

Facial Expression was developing but was a little limited at times. Lip shape and movement were well controlled and fluent. Tongue placement was good, apart from tongue tip to alveolar ridge, which was variable and inaccurate.

Palate control and coordination were markedly poor, with hypernasal resonance, some nasal turbulence and nasal escape demonstrated.

His voice was weak and husky.

Single Sounds

He imitated the full range of vowels and diphthongs, but with marked hypernasal resonance. Some consonants were affected by hypernasal resonance: /b – k – f/ were noted.

/θ ð / → [f v]

/ʃ ʒ / → palatal placement

/j/ → [i]

/r/ → [ʊ]

Liam was able to sequence established consonant sounds with accuracy, but at a reduced speed – he scored 66%. He was able to sequence vowels accurately but with hypernasal resonance and nasal escape evident.

CV Words

He was able to produce CV words, but with errors on both vowels and consonants.

e.g. go → [gɔ̃] shoe → [tʃ̃].

More Complex Words

CVCV words were produced, but with errors on fricatives and vowels.

Multi-syllabics were also attempted but with errors and more marked hypernasal resonance.

Consonant Clusters demonstrated problems with medial and final position and hypernasal resonance.

Phrases and Sentences

He was able to repeat phrases and sentences, but the accuracy of consonants and vowels was reduced and there was marked hypernasal resonance.

e.g. 'Caterpillars change into butterflies' [kætəpɪnəz tʃeɪn ɪntʊ blətəfəlaɪz]

Connected speech was often unintelligible outside family and friends, largely as a result of the hypernasal quality.

Speech Processing Skills

Auditory discrimination was excellent.

Syllable structure was maintained, he was able to identify sounds in both onset and rhyme tasks.

He was unable to imitate sounds or words more accurately, than those produced in his spontaneously, demonstrating his motor programming difficulty.

He was unable to correct his own production, and generally seemed unaware which productions were accurate, indicating difficulty with motor planning.

Diagnosis

Liam was found to have a moderate Oral and Verbal Dyspraxia, with marked palatal involvement. He showed evidence of motor programming and motor planning difficulties.

His parents had been advised that palatal surgery would be necessary, but were reluctant. It was suggested that a trial course of six months speech and language therapy should be given to address palate coordination, as well as his motor programming and planning difficulties, and then, the situation could be reviewed.

Intervention

Liam lived too far away to receive weekly speech and language therapy. He attended for three introductory sessions, over a four week period, and a Home Programme was then provided.

Liam and his family worked on this for two months. He was then reviewed, and a further programme set for three months. This regime was followed for eighteen months.

He worked, at home, for a minimum of six days per week, for 20 – 30 minutes per day throughout this period. Much of the work was done on the car journey to school and back.

Both Liam and the therapist were helped greatly by his mother, who was a speech and drama teacher, and actress, and understood the importance of speech exercises!

Treatment Programme

Initial Therapy targets were aimed at:

- Improving palatal awareness and coordination.
- Improving tongue awareness and accuracy of placement.
- Sound production and sequencing, using plosives and approximants, in contrast with nasal consonants, for improved palatal coordination and tongue accuracy – to establish motor programming patterns.

After four months, as his sound accuracy improved, Liam added fricatives and affricates, in contrast with nasal consonants at Single sound Sequencing and CV and CVCV Word Levels, aiming to increase palatal coordination at word level, to establish his motor planning skills. Basic palatal exercises were continued as well.

After six months, work was widened to include using the full range of consonants and vowels at Word, Phrase and Sentence Levels.

During the final six months of therapy, Liam generalised his skills, with both his sound system and his nasal/oral resonance, using rhymes, general language work, as well as reviewing his phonological awareness, spelling and self-monitoring skills. By this stage of therapy, although he had a slightly nasal quality, no nasal escape or marked hypernasality was apparent, and palatal surgery was certainly not indicated.

Therapy for a child, whom it is not possible to see regularly is, potentially extremely difficult, however, with Liam it was felt to be realistic because:

- a) He was mature and well-motivated.
- b) His mother had excellent understanding through her professional background, but worked to the specified programme.
- c) The family were able to keep themselves motivated and organised, so that the exercises were done for six days each week.

Outcome

Liam worked through all four stages of the NDP Treatment Plan, although the emphasis of the work was on basic oro-motor and sound sequencing skills, with especial attention to palate strengthening, and coordination exercises.

After eighteen months, Liam had a complete sound system of vowels and consonants. His score for sequencing complex and multisyllabic words was 100%. Consonant Clusters scored at 94% (from 77% on initial assessment). Phrases and Sentences also scored 100% for accuracy.

Hypernasal resonance was markedly reduced, and was not compromising intelligibility, although it was still apparent in his speech.

The family then moved abroad, so Liam's speech was not reviewed again at that stage.

Liam was seen again, aged 12 years 2 months, after a fifteen month break. His speech and language have become well established, there was no evidence of inaccurate sounds or hypernasal resonance.

Appendix 1: Speech error patterns in children with DVD

The aims of analysing speech data are:

- 1 To identify the range of sounds and sequences of sounds the child is using at each level of syllable structure.
- 2 To identify error patterns, which may indicate the nature of underlying processing difficulties, and indicate treatment strategies.

Children with DVD typically present with a limited sound system, limited syllable structure and inconsistent error patterns. Errors often affect vowels as well as consonants. Errors increase in longer structures, less familiar vocabulary and with increased rate of speech (see Chapter 1: Theory: Literature Review).

Common consonant error patterns

Error patterns include normal developmental processes, although the cause of these is likely to be motor programming difficulties rather than immature perception.

Developmental processes – paradigmatic (substitution)

- 1 Stopping – may indicate that the child lacks the fine motor coordination to maintain a narrow stricture, as opposed to complete closure.
- 2 Prevocalic voicing – difficulties in coordinating articulation and voice onset.
- 3 Fronting – lack of motor patterns for velars, labiodentals, palato-alveolars, or grooving for /s/.
- 4 Gliding – lack of motor programs for /r/ or /l/.

Developmental processes – syntagmatic (sequencing)

Processes which affect syllable structure and possible sequences of sounds are a common feature of the speech of dyspraxic children.

- 1 Assimilation – is seen in CVC, CVCV, multisyllabics, sentences and in connected speech. Velar/alveolar contrasts and voicing contrasts within the same word are most susceptible. Vowels may also be assimilated. For example: dog→/gɒg/; garden→/dɑdən/; tissue→/tʃʊtʃu/; caterpillar→/kækæpijæ/; Katy's party→/keɪkɪ pɑpɪ/. Typically, more assimilation is seen in more complex structures, and it is gradually overcome as skills are mastered at successive levels.
- 2 Final consonant deletion – children may be unaware of the final consonant in the target, but equally, the addition of a final consonant to the basic CV syllable makes additional motor demands.
- 3 Cluster reduction.
- 4 Weak syllable deletion – bisyllabic words may be reduced to one syllable, or multisyllabics may lose one or more unstressed syllable. In connected speech, unstressed syllables frequently correspond with grammatical morphemes: these may also be lost (e.g. baby→/beɪ/; banana→/nənə/; washing machine→/'wɒʃ də'ʃɪn/).

Atypical processes

A number of processes which are not common in normal development, or only in the earliest stages, are also seen frequently in children with DVD.

- 1 Backing – typically alveolar→velar, but other sounds may also be affected.
- 2 Glottal substitution – may indicate that the child is unable to produce the target sounds, or that he is unable to join them to the vowel to create an appropriate motor programme.
- 3 Favourite articulation – occurs when children have only a very limited range of sounds in their repertoire. One child with DVD attempted all but a few familiar words as [ʊʌ] or [ʊʌʊʌ]. Many children seem to use mainly /d/ in the early stages.
- 4 Metathesis – i.e. the switching of phonemes in a word, or sometimes across word boundaries (e.g. tap→/bæ/; ticket→/kɪʔkɪt/; ambulance→/æləməʌt/; Bobby's goggles→/bɒɡɪ bɒɡu/).
- 5 Chronological mismatch – i.e. early processes persist, while some later skills have been mastered. With experience, children with DVD may establish motor programs for a range of familiar words. However, their speech processing skills/phonological system remain impaired, and earlier skills may not have been mastered unless they have been specifically taught.
- 6 Small number of clearly articulated words – usually very familiar or well-practised words. Older children may have learned accurate motor programs for a range of words, but still have difficulty with motor programming for unfamiliar words.
- 7 Small number of very disordered words – usually reflecting earlier speech patterns, where the child has not updated motor programs.
- 8 Variable production of the same word – as a result of imprecise phonological representations, and/or motor programs, or motor planning difficulties.

Articulatory processes

Children with oro-motor difficulties, or difficulties at the level of producing individual sounds, may show articulatory errors. It is more important to describe as accurately as possible what the child is doing, than to be able to transcribe the errors. Visual as well as auditory information is necessary for this: it may therefore be useful to video assessment.

Although there are many possible errors, some patterns are common:

- 1 Imprecise placement:
Dental or interdental articulation of alveolar phonemes [t̪ d̪ n̪ s̪ ʃ̪]
Palatalised articulation of alveolars [tʲ dʲ nʲ sʲ ʃʲ]
Palatal articulation of alveolars or velars [c ɟ ɕ]
Uvular articulation of velars [q ɢ]
The diacritics “more forward” [ʲ] or [̪], and “more back” [-] or [̠] may be useful.
- 2 Weak articulation, changing plosives to fricatives, fricatives to approximants. Use a “more open than” diacritic [̞].
- 3 Inadequate airflow, resulting in reduced aspiration/friction, making consonants less audible. [t̚] indicates no plosion.

Analysing vowel errors

In order to analyse vowel errors, it is necessary to know the target vowel system, and the typical lip, jaw and tongue positions associated with each vowel phoneme. A full range of varieties of English is described in Wells (1982a, b, c). The analysis below is based on a standard Southern English accent.

Vowels can be thought of in groups, according to the following articulatory dimensions: high – low, front – back, round – spread, and long – short.

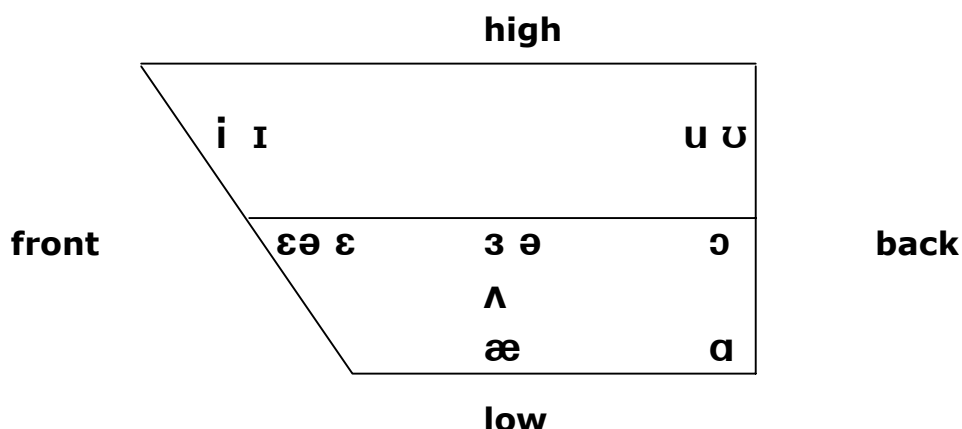
Vowels grouped by length

Long vowels: i ɑ u ɔ ɜ (ɛə)

Diphthongs: ɪə aɪ ɔɪ əʊ eɪ

Short vowels: ɪ æ ʊ ɒ ʌ ɛ

Long and short vowels, grouped by tongue posture



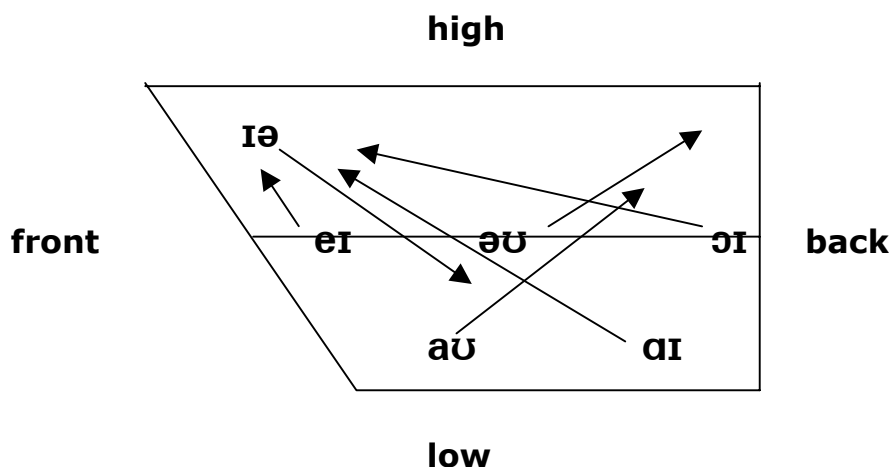
Vowel height is related to jaw and tongue position. Children with more limited tongue control tend to rely more on jaw movement to get the correct height.

In our somewhat simplified model of the S. English vowel system, we distinguish 3 heights: high (or close) – mid (half-open) – low (open). In order to make this simplification, we group the long/short vowel pairs as if their tongue positions were identical. This is not strictly the case: the short vowels tend to be a little less extreme than their long counterparts. However, we find this a useful starting point for working on vowels with children (see Appendix 4: Eliciting Sounds).

The front – back dimension is produced by tongue position. Again 3 positions are contrasted: front – central – back, also a somewhat simplified position.

In this accent, as in many accents of English, lip rounding is associated with back vowels and spreading with front vowels.

Diphthongs



Diphthongs are viewed as moving from one position towards another:

- towards a high front posture
- towards a high back posture
- towards a low/central posture

Other Vowel Systems

The vowel systems of other accents or varieties of English may differ only in terms of phonetics – i.e. the same number of vowel phonemes is used, with the same distribution, but with differences in pronunciation. For example, many accents do pronounce the phoneme / $\varepsilon\theta$ / as a diphthong [$\varepsilon\theta$], and this is a frequent variant in Southern British English (despite our model above, where it is treated as a long vowel [$\varepsilon:$]). Other accents, for example in the British West Country, rhotacise this vowel [$\varepsilon\lambda$].

Phonemic differences also occur, when one accent has more or less phonemes than another. For example, a Yorkshire accent uses only / u / for the Southern English phonemes / $\wedge$ / and / u /. However, it also has potentially two extra phonemes / $u\theta$ / and / $\partial\theta$ / for words which would be within the / ∂ / set in Southern English.

It should also be remembered that there are variations in pronunciation within local accents, between individual communities and between older people (who tend to have more traditional accents) and younger people.

May be related to perception and phonological representations, but often also related to difficulties with motor control and programming:

- For further information on vowel development and vowel disorders, see Ball and Gibbon (2002).

Problems with sound joining

- 1 Glottal coarticulation, insertion, substitution
Kipper – [kɪʔpə]; go away – [gəʊ ʔəweɪ]
- 2 Omission of a sound or part of a sound
down – [da:n]; bad cat – [bæ kæt]
- 3 Elision of weak syllables
Computer – [putə] ; go in the car – [gəʊ ɪ kɑ]
- 4 Pausing between words
- 5 Lack of normal assimilation and elision at word/syllable boundaries (See Chapter 3: Assessment: Phrases and Sentences).

Problems with suprasegmental features

Vocal tone	– breathy, hoarse, tense, hard attack, creaky
Volume	– too loud, too quiet, no variation
Pitch	– too low, too high, limited pitch range
Intonation	– only one contour, inappropriate use
Rate/rhythm	– too fast, uneven rhythm, elision – slow, equalised stress
Resonance	– hypernasal, hyponasal, mixed resonance

References

- Ball, M. and Gibbon, F. E.** (2002): Vowel Disorders. Butterworth-Heinemann.
- Wells, J.C.** (1982a): Accents of English (1) An Introduction. Cambridge University Press.
- Wells, J.C.** (1982a): Accents of English (2) The British Isles. Cambridge University Press.
- Wells, J.C.** (1982a): Accents of English (3) Beyond the British Isles. Cambridge University Press.

Appendix 2: Toys and equipment

Equipment for oro-motor and phonetic placement activities

Oro-motor activities are used to elicit features of sound production, and where appropriate to develop awareness and control of articulatory structures. See Chapter 5: Therapy, and Appendix 4: Eliciting sounds.

- Early sound making sheets (Therapy Resources 1, p6)
- Lip shapes (Therapy Resources 1, p23)
- Articulograms (Therapy Resources 1, p21)
- Mirror
- Torch
- Tongue depressor or spatula
- Straws of various sizes
- Plastic tubing
- Blowing toys with different mouthpieces
- Bubbles
- Candles in a secure holder
- Toothbrush
- **Sugary foods:** 100s and 1000s, chocolate flakes, sherbet, chocolate spread, honey, jam
- **Sugar-free foods:** marmite, yoghurt, grated apple, mashed banana
- Sterilising fluid to clean equipment

NB Oral Health Note: *It is recommended that sugary foods and drinks are used only at mealtimes. At other times, sugar-free foods should be used to avoid the risk of tooth decay. It is not recommended that teeth are cleaned immediately after eating sugary foods. Children's teeth should be brushed routinely for 2 minutes, morning and night, using a toothpaste containing no less than 1000ppm of fluoride. Children under 7 years should be supervised. It is important to arrange regular check ups with a dentist who will then be able to offer individual oral health advice. (Our thanks to: Mrs K Kerr, Senior Oral Health Promoter, Bucks Priority Dental Service.)*

Equipment for voice exercises

Voice work is included throughout the programme, as appropriate for an individual child.

For suggestions on developing control of vocal tone, pitch, length and volume, see Chapter 5: therapy, 98.

For suggestions on developing prosody at higher levels, see Chapter 5, p149.

- Early sound making sheets (Therapy Resources 1, p6)
- Length, pitch and volume voice worksheets (therapy Resources 1, p24)
- Playdough/blutack
- String tightrope, suspended between two posts
- A toy slide
- A cotton-reel or bead on a string
- Roads/paths drawn on paper
- Pictures of adult and baby animals
- Miniature farm and zoo animals
- A set of large and small objects
- Toy vehicles–cars, buses, fire engines, police cars, ambulances, trains, aeroplanes etc.
- Pictures of household items e.g. telephones, doorbells, washing machines, vacuum cleaner, that make noises
- Puppets
- Echo-microphone
- “Talking toys” (i.e. voice activated toys)
- Computerised equipment eg Laryngograph

Toys to facilitate practice

The following are some of our favourite toys for eliciting repeated productions of a target sound or word, or group of targets. The games are easily controlled by an adult, and motivating whilst not being too distracting for the child. There are plenty of similar games on the market. For a wide range of games to use in therapy, see Appendix 6: Tasks/Activities/Games.

- Pop-up Pirate (Tomy)
- Monkey Business (Early Learning Centre)
- Hooked (Chad Valley)
- Dolphin Basketball Chad Valley)
- Button Board (Early Learning Centre)
- Building blocks
- Miniature people
- Miniature animals
- Miniature dinosaurs
- Hopping frogs

Appendix 3: Articulograms

Articulograms are a system of simple diagrammatic forms, representing articulatory features (e.g. voice, place, manner and lip posture), which frequently require work for children with dyspraxic difficulties. These can be used alongside consistent verbal cues to support work on particular aspects of articulation, at oro-motor, single sound and word levels.

The articulograms and verbal cues are designed to provide the child with information about how to articulate a particular feature. For example, the child is introduced to the articulogram and told he needs to make a “puff of air” to produce the voiceless plosives.

Consonant articulograms

The consonant articulograms were originally published by Stephens and Elton (1986) and are reproduced in this volume with the permission of Hilary Stephens, Specialist Speech and Language Therapist, Nuffield Speech and Language Unit.

These can be combined to specify all consonants, apart from: /r/, /θ/ and /ð/.

The consonant articulograms are depicted below and described in terms of: feature specified, verbal cues and relevant phonemes.

Vowel articulograms

A set of seven lip shapes has been added to cue lip/jaw postures for vowels and for the glides /w/ and /j/. A sequence of two articulograms is used to specify diphthongs and glides.

These can be combined with verbal cues for vowel length and tongue posture to provide a framework for the articulation of a standard range of vowels. Both articulograms and verbal cues should be modified as required to take account of regional variations in vowel production.

Fine-tuning of vowel quality (i.e. lip, jaw and tongue postures) is achieved through modelling and practice.

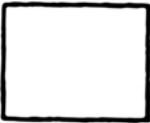
The vowel articulograms are depicted below and described in terms of feature specified, verbal cues and relevant phonemes.

The articulograms are referred to throughout Appendix 4: Eliciting single sounds in isolation.

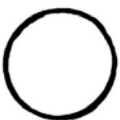
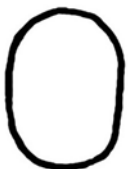
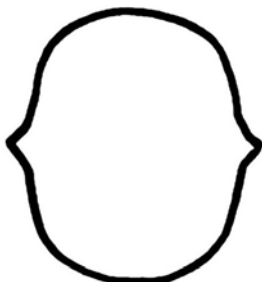
References

Stephens, H. and Elton, M. (1986) Description of Systematic Use of Articulograms. College of Speech and Language Therapists Bulletin: December.

CONSONANT ARTICULOGRAMS

Articulogram	Feature specified	Verbal cue	Relevant phonemes
	bilabial	lips together	/ p b m /
	alveolar	tongue on/behind your teeth	/ t d n s z /
	velar	mouth open, tongue up at the back	/ k g ŋ /
	labio-dental	bite your lip	/ f v /
	lateral	tongue up to touch your teeth	/ l /
	lip rounding	round mouth	/ ʃ w /
	open	mouth open	/ h w j /
	nasal	down your nose	/ m n ŋ /
	fricative	blow sound	/ f v s z ʃ ʒ /
	voice	voice on	/ b d g v z ʒ /
	aspiration	puff of air	/ p t k /

VOWEL ARTICULOGRAMS

Articulogram	Feature specified	Verbal cue	Relevant phonemes
	spread lips	smile	/ i ɪ ʊ ə aɪ eɪ oʊ /
	open, rounded	mouth open (long open)	/ ɑ ɒ /
	round lips	round mouth	/ u ʊ əʊ ɔʊ /
	neutral, half-open	just open a bit	/ ɜ ʌ əʊ ɪə /
	half-open, spread	open a bit and smile	/ ɛə ɛ eɪ /
	half-open, rounded	open a bit and round	/ ɔ ɔɪ /
	open, spread	wide open (smiley open)	/ æ ɔʊ aɪ /

Appendix 4: Eliciting single sounds in isolation

Children with speech difficulties use a restricted sound system in their everyday speech, but they can often imitate a fuller range of sounds in isolation. In teaching children sounds which are not currently in their repertoire, imitation of an adult model, supported by cues e.g. picture symbols, manual symbols etc, is in many cases a successful strategy. However, for some children, including those with dyspraxic difficulties, imitation can be very difficult and they need to be taught how to produce both consonant and vowel sounds.

This may involve a number of strategies, used singly or more commonly combined to provide maximum support: visual - the child watching the therapist producing the sound and/or himself in a mirror (perhaps with a torch to see inside the mouth); tactile and kinaesthetic - the therapist physically helping the child's articulators to move e.g. pushing the lips together and the child then experiencing the kinaesthetic sensation of that movement; verbal - the therapist describes required movements to the child and/or uses vocabulary cues established through work on specific articulatory features, eg: switch your voice on, tongue on your teeth; articulograms can be used to provide diagrammatic cues for articulatory features (see Appendix 3 above, and Therapy resources 1, p21).

Speech sounds are associated with NDP picture cues Therapy Resources 1: Single Sounds p43, and or with written letters or manual cues, as in Cued Articulation (Passy 1990).

Some strategies are more appropriate for some sounds than others and individual children respond to some strategies better than others. Therefore, when a number of different ideas are provided, therapists are advised to experiment and find what works best for their individual child.

Top Tips

- Use ideas in the early sound making sheets for children who have few sounds.
- Elicit the required articulatory feature(s) and link to cues.
- Use simple, accurate verbal explanations, supported by modelling.
- Use verbal cues and articulograms to support elicitation of individual articulatory features and to aid storage and retrieval.
- Reinforce correct motor patterns using the same cues (e.g. good try – you got your lips together)
- Utilise movements which the child can produce at an automatic level and contexts in which the target sound occurs at an automatic level.
- Emphasise movements/postures or articulatory feature required rather than the target sound, if the child has an inaccurate motor program for the sound.
- Make sure the child's position is stable (i.e. upright with feet flat on the floor). Be prepared to steady head/jaw as necessary.
- Be confident - don't be afraid to touch the child's face, mouth, nose, but build up the child's trust first and be sensitive to gagging etc.

- Acknowledge that trying to make sounds can be difficult. Sometimes ideas work and sometimes they don't! Sometimes you have to keep on trying!
- Aim to establish /p, b, m, t, d, n, k, g, f, s, h, w, j, l/ and all vowels at a single sound level, quite early in therapy. Accept approximations of targets at first and gradually shape these towards to accurate production.

Consonants

/p/ the child needs to (a) put his lips together (b) produce an oral blow airstream and (c) blow his lips apart.

Tactile cue – push lips together for the child; kinaesthetic cue – feel puff of air on back of hand; verbal description – “lips together and blow”; visual cue – look at adult/look in mirror and see lips tight together. Associate with articulograms (aspiration and lips together and with picture symbol “popper”.

/b/ the child needs to (a) put his lips together (b) produce an oral voiced airstream and (c) open mouth.

Tactile cue – push lips together and touch neck to stimulate voice production; verbal description – “lips together and switch your voice on”; visual cue – look at adult and in mirror. Associate with articulograms – voiced airstream and lips together and with picture symbol “ball”.

Other ideas

If the child can produce /m/, it is possible to elicit /b/ from this by gently blocking the nostrils.

If the child can produce /p/ but not /b/, explain that they need to produce a “shout” /p/+ vowel e.g. /p æ/.

It may be helpful to produce a babbled VCV sequence /æ b æ b æ b æ/ etc.

/m/ the child needs to (a) put his lips together (b) produce a nasal voiced airstream and (c) sustain the sound.

Tactile cue – push lips together and place finger on nose; verbal description – “lips together, voice on and let the air down your nose”; kinaesthetic cue – feel nasal vibration; visual cue – look at adult and in mirror. Associate with articulograms (nasal and lips together) and picture symbol “motorbike”.

Other ideas

Play making motorbike noises.

/t/ the child needs to (a) make complete tongue contact with the alveolar ridge and upper teeth (b) produce a blow oral airstream and (c) release the tongue contact.

Tactile and kinaesthetic cues – place a finger under the front of the tongue and push it upwards towards alveolar ridge; “tickle” the tip of the tongue and the alveolar ridge and see if the child can put the two tickled areas together; place 100s and 1000s or jam etc on alveolar ridge and ask child to try and raise tongue to get the food; visual cue – use mirror and torch; verbal cue – “tongue up behind your top teeth and blow” or “tongue on your teeth and blow”. Associate with articulograms – aspiration and alveolar and picture symbol “tap”.

Other ideas

For children unable to elevate their tongue to the alveolar ridge, /t/ can be elicited initially by making the sound interdental (which can later be modified).

Easiest vowel to accompany /t/ is /i/.

/d/ the child needs to (a) make complete tongue contact with the alveolar ridge (b) produce a voiced oral airstream and (c) release the tongue contact.

Use tactile, kinaesthetic, and visual cues similar to /t/. Verbal cue – “tongue up behind your teeth and switch your voice on”. Associate with articulograms – voiced airstream and alveolar and picture symbol /d/.

Other ideas

If the child can produce /t/, /d/ can be elicited by asking him to produce a loud or shouted /t/ + vowel /tæ/.

As for /t/, /d/ can also be elicited interdental.

/n/ the child needs to (a) make complete tongue contact with the alveolar ridge (b) produce a nasal voiced airstream and (c) sustain the produced sound.

Kinaesthetic cue – feel nose for nasal vibration; verbal cue – “tongue up behind your teeth, switch your voice on; air needs to come through your nose”. Visual cue – use mirror and torch. Associate with articulograms – nasal and alveolar; and with picture symbol “nose”.

Other ideas

Again can be produced interdental, if tongue elevation is difficult.

/k/ the child needs to (a) raise the back of his tongue to make contact with the soft palate (b) produce an oral blow airstream and (c) release the contact.

Tactile cues – in most cases, the child needs help to raise the back of the tongue. Strategies to encourage this include: push tongue tip down and back with finger/spatula etc; play with a toothbrush while babbling (holds tongue down); describe and demonstrate to the child to tuck tongue tip behind lower teeth; hold jaw open with one finger and with the other hand push under the child’s chin to encourage velar movement; verbal cues – “mouth open, tongue up at the back and blow.”; visual cue – use mirror and torch. Associate with articulograms – velar and blow airstream and with picture symbol “camera”.

Other ideas

Try to elicit from a back vowel e.g. /a/, /u/ or /ɔ/ ->/k/.

/g/ the child needs to (a) raise the back of his tongue to make contact with the soft palate (b) produce a voiced oral airstream and (c) release the sound.

Similar strategies can be used as for /k/, but this time the child needs to produce a voiced sound – “switch your voice on”.

If he can already say /k/, /g/ can be elicited by producing a shout /k/ + vowel e.g. /kæ/ etc.

Associate with articulograms – velar and voiced airstream and with picture symbol “geese.”

Other ideas

Back vowels e.g. /ɑ/, /u/ or /ɔ/ are most appropriate to support.

/ŋ/ the child needs to (a) raise the back of his tongue to make contact with the soft palate (b) produce a voiced nasal airstream and (c) sustain the produced sound.

Tactile cues – push tongue back with a small lolly stick from dental production of /n/; verbal cue – “like /n/ but your tongue needs to go up at the back, like /g/”. Associate with articulograms – nasal and velar and with picture symbol “sing”.

Other ideas

May be possible to elicit “ping”, “ting” or “king” verb endings and then isolate /ŋ/ from these.

/h/ the child needs to (a) open his mouth and (b) produce an oral blow airstream.

This is the simplest fricative sound to produce. Verbal cue – “open your mouth and blow”; Kinaesthetic cue – feel the resulting friction on back of hand.

Associate with articulograms – open mouth and aspiration and with picture symbol “hippo”.

Other ideas

“Sigh” or “make the sound of a panting dog”.

/f/ the child needs to (a) place his upper incisors on lower lip (b) smile (c) produce an oral blow airstream and (d) produce friction.

Kinaesthetic cue – gently bite lower lip; verbal cue – “smile (increase muscle tone), bite and blow”. Let him feel resulting friction on back of hand etc; visual cue – look at adult and in mirror. Associate with articulograms – fricative and labiodental and with picture symbol “firework”.

Other ideas

It is also possible to develop friction aspect by maintaining lip position and alternating ingressive and egressive airstreams (suck/blow).

/v/ the child needs to (a) place his upper incisors on lower lip (b) smile (c) produce a voiced oral airstream.

/v/ can be elicited from /f/ by asking the child to produce a shout /f/ + vowel e.g. /fæ/. Point out the resulting lip vibration produced and ask him to feel it. Use long cardboard tubes to help emphasise vibration.

Associate the sound with articulograms- fricative, voiced and labiodental and with picture symbol "van".

/s/ the child needs to (a) raise his tongue tip to almost touch the alveolar ridge (b) smile (c) produce an oral blow airstream and (d) channel the air through a narrow central groove between the tongue and alveolar ridge.

Verbal cue: "tongue on your teeth and blow".

Associate with articulograms –fricative and alveolar and with the picture symbol "snake" (stress hissing aspect of sound).

Other ideas

Lengthen a /t/ sound. Start by saying /t/ slowly and gradually lengthen the friction [ts, tss, tss]. Try to stop the child in the middle of his [tsss] production and start again. Model [tss...sss] and see if he can imitate. Repeat and practise.

Associate with verbal cues: a lazy /t/; a sleepy /t/ etc.

A smile lip shape tends to help as does association with close front vowels e.g. /i/ etc.

As you say /t/, try placing a short straw on the teeth to encourage the child to direct the airflow centrally.

Move a /j/ sound forward as you smile.

For children with lateral production [ɬ] encourage them to gently bite on the sides of the tongue (make a tongue sandwich) and push the cheeks in to encourage central production.

If an interdental substitution [θ] is produced, see if the child can put his teeth together first and then try to imitate /s/.

/z/ the child needs to (a) raise his tongue tip to almost touch the alveolar ridge (b) smile (c) produce a voiced oral airstream (d) channel the air through a narrow central groove between the tongue and the alveolar ridge.

Try to elicit from a shouted /s/ + vowel e.g. /sæ/.

Associate with articulograms –fricative, voiced and alveolar and picture symbol "zip"

Other ideas

Buzz like a bee!

/ʃ/ the child needs (a) to raise the tongue tip or blade to the palato-alveolar junction (b) to round and protrude his lips (c) produce an oral blow airstream (d) release air through a broader central grooved channel (compared to a narrow channel for /s/).

Tactile cue -may be achievable from /s/ by pushing tongue back a little, with a small lolly stick or straw and rounding lips; verbal cue – “tongue up behind your teeth, round mouth and blow”. Associate with articulograms – fricative and round mouth and with the picture symbol “hush” baby sleeping.

Other ideas

May be achievable from “sj” in medial position e.g. “miss you”.

If the child is able to produce /tʃ/, /ʃ/ can be achieved by prolonging and isolating the second element /tshhhh/ → /ʃ/.

/tʃ/ the child needs to (a) make tongue contact behind the alveolar ridge (b) produce an oral blow airstream (c) release the sound slowly through a central channel and (d) round lips.

Tactile cue – may be achievable from /t/ by pushing the tongue back with a small lollypop stick; verbal cue – “make a /t/ sound and round your lips. Let it go slowly”. Emphasise length and prolonged quality; kinaesthetic cue – feel friction on back of hand.

Associate with articulograms alveolar and aspiration + lip rounding and fricative and with picture symbol “chick”.

Other ideas

May also be achievable from repeated production of /tj/.

Blend /t/ and /ʃ/ together.

/dʒ/ the child needs to (a) make tongue contact behind the alveolar ridge (b) produce a voiced oral airstream (c) release the sound slowly through a central channel and (d) round lips.

May be achievable from /tj/ by producing a shouted /tj/ + vowel e.g. /tjæ/ or from repeated production of /dj/.

Associate with articulograms – alveolar and voiced + lip rounding and fricative and picture symbol “jack in the box”.

/w/ the child needs to (a) round his lips (b) produce a voiced oral airstream then (c) open his mouth.

Tactile cue – push lips into rounded shape; visual cue – look at adult model and in mirror; verbal cue – “round your mouth, switch your voice on, then open”. Vowel support - try to elicit from /u/

Associate with articulograms – round and open and the picture symbol “worm”.

/j/ the child needs to (a) raise the blade of the tongue towards the hard palate (b) smile (c) produce a voiced oral airstream and (d) open his mouth

Tactile cue – push lips into smile position; verbal cue - “smile, switch your voice on and open”; kinaesthetic cue – feel sides of tongue pressing against teeth. Vowel support - try to elicit from /i/.

Associate with articulogram – smile and open and picture symbol “yoyo”.

/l/ the child needs to (a) raise his tongue tip to make central contact with the alveolar ridge (b) produce a voiced oral airstream (c) smile a little (to allow better lateral airflow), and (d) then lower tongue. The articulation depends on a quick lowering of the tongue tip as well as on the tongue tip up phase.

Tactile cue – push the front of the tongue up to the alveolar ridge with a finger; verbal cues - “tongue up, little smile, voice on, tongue down”; visual cue - look at adult and in mirror.

Associate with articulograms – lateral and voiced and with picture symbol “lolly”.

Other ideas

For children unable to produce a tongue tip, /l/ may be achievable through a central interdental contact with bunched tongue shape and smile lip posture to encourage lateral airflow.

/r/ the child needs to (a) raise his tongue tip towards the alveolar ridge, curling back to produce a retroflexed tongue posture (b) produce a voiced oral airstream and (c) have jaw half-open and lips in a somewhat rounded/protruded posture.

This can be a very difficult sound to elicit. Try combining vowel /ɜ/ with a velar approximant [ɣ] with either a labio-dental approximant /ʋ/ or a tongue tip approximant (i.e. /l, z, or ʃ/ that doesn't quite touch).

Associate with picture symbol “roller skate”.

Other ideas

Growling noise!

It may be possible to achieve an acceptable articulation in clusters first and then try to isolate.

Vowels

The production of vowels involves a pulmonic egressive airstream passing through the vibrating vocal folds and through the oral cavity, unobstructed by movements of the articulators. The exact vowel produced is determined by the degree of jaw opening and associated tongue height, the position of the raised part of the tongue, the size and shape of lip opening and the duration of the vocal fold vibration.

Children can be taught to produce vowels by utilizing the parameters of: jaw opening (and associated tongue height)– open, half-open, close, tongue position (front vs. back), lip shape (open, round, smile) and length of vocal fold duration (long vs. short). In some cases it will be necessary for the therapist or parent to touch and manipulate the child's face and mouth to help him achieve correct production. Build up the child's trust before doing this.

Frequent consistent clear models should be given. Verbal cues and articulograms are used to elicit and support retrieval of movement patterns for specific articulatory features. Once a sound can be elicited reliably, it should be associated with the relevant NDP picture cue (Therapy Resources 1, p90). "Cued Vowels" (Passy 1990) can also be used to support articulatory features and retrieval.

For children with severe speech difficulties, virtually all vowels will need to be taught. In this case, the suggested order is: the long vowels /i/, /a/, /u/ and the corresponding short vowels /ɪ/, /æ/, /ʊ/. Then the three diphthongs /ɪə/, /aɪ/, /aʊ/ followed by the half-open vowels: /ɜ/, /ɔ/, /ɛə/. Finally the remaining short vowels and diphthongs.

See Early sound making sheets (Therapy Resources 1, p6).

Young children with very restricted vowel systems may also benefit from active "vocal play" and experimentation. It is sometimes much easier to elicit a vowel by accident in this way (and subsequently modify and shape it) than it is to try and elicit the target by building up component articulatory features. In addition, it can be helpful to use a proximate consonant e.g. /n/ etc as an articulatory launch pad for introducing vowels.

For children with less severe difficulties, only some vowels will need to be taught. In this case, utilize the child's strengths and teach new vowels from vowels already established.

The following suggestions are designed to elicit a set of 18 standard (Southern) English vowels. Regional variations will require adaptations. For further information on vowels, see Appendix 1: Speech error patterns.

Remember first "taught" attempts are likely to be approximations of vowels – work at achieving correct placement and gradually modify the acoustic accuracy through modelling and feedback. Utilize input work to develop discrimination skills and to help provide feedback.

Long vowels

/i/ Jaw: close; lips: smile; tongue: mid-tongue raised.

Verbal cue – “smile and switch your voice on”. Stress the length of the vowel, by sustaining voice. Associate with picture symbol “Eddie mouse” and with the articulogram “smile”.

/a/ Jaw: open; lips: long open; tongue: back of tongue retracted.

Verbal cue – “open your mouth and switch your voice on”. Again stress the length.

Tactile cue – get the child can push his tongue back with his finger.

Associate with picture symbol “Arthur crocodile” and with the articulogram “long open”.

Other ideas

Play at visiting the doctor, opening your mouth and saying “ah”.

/u/ Jaw: close; lips: round; tongue: back raised.

Verbal cue – “round mouth and switch your voice on”; Tactile cue – push the child’s lips into a rounded shape. Again stress length and try to sustain the vowel. Associate with the picture symbol – the wind blowing and with the articulogram “round”.

Other ideas

Playing ghosts can be great fun!

/ɜ/ Jaw: half-open; lips: neutral; Tongue: neutral.

Tactile cue - help the child achieve the correct degree of openness (about 1.5 cm) by placing a small toothbrush between his teeth and asking him to switch his voice on.

Remove the toothbrush as he says the vowel and try to maintain the jaw position. The toothbrush keeps the jaw open and encourages appropriate tongue height. Associate with the picture symbol “Ernie bear” and with the articulogram “open a bit”.

/ɔ/ Jaw: half-open; lips: round; tongue: back raised.

This vowel is another half-open vowel, more complicated to produce than /ɜ/ as it also requires rounded lips. Again use a small toothbrush placed between the child’s teeth to help him achieve the appropriate degree of openness. Verbal cue: “open a bit, round mouth and switch your voice on”. Remove the toothbrush as he says the sound.

Associate with the picture symbol “oar” and with the articulogram “open a bit and round”.

It can be helpful to start by producing /u/, to help achieve back tongue position and then opening jaw to half-open position.

Short Vowels

The corresponding short vowels can be taught by maintaining the jaw opening and lip shapes of the long vowels, but shortening the duration of the vocal fold vibration. Some slight modifications may also be necessary to lip shape e.g. for /æ/ lip shape is wide open rather than long open (see different articulogram). Tongue postures are slightly less extreme than for long vowels – positions can be fine-tuned by modelling once basic production is achieved.

Comparisons between long and short vowels can be helpful. Remember to involve discrimination work (child as the listener) as well as production work (child as the speaker).

/ɪ/ Jaw: close; lips: smile; tongue: middle of tongue raised.

Shorten /i/. Associate with picture symbol “incy” and with articulogram “smile”. Stress short (vs. long) aspect. Compare with /i/.

/æ/ Jaw: open; lips: wide open; tongue: forwards and low.

Shorten /a/ but lips need to be wide open rather than long. Associate with picture symbol “ant” and with articulogram “wide open”. Stress short duration of the sound.

/ʊ/ Jaw: close; lips: round; tongue: back raised.

Shorten /u/. Associate with picture symbol “Oofy monkey” (and play at making monkey sounds) and with the articulogram “round”. Compare with long vowel /u/.

/ʌ/ Jaw: half-open; lips: neutral; tongue: neutral.

Shorten /a/. Use toothbrush to help with jaw position. Stress short duration of the sound. Associate with picture symbol “umbrella” and with articulogram “open a bit”.

/ɒ/ Jaw: open; lips: long open; tongue: back of tongue retracted.

Shorten /a/ but lips do not need to be quite so open – experiment! Associate with picture symbol “octopus” and with articulogram “long open”.

/ɛ/ shorten first element of /ɛə/ diphthong. Use toothbrush to help with jaw position and smile. Associate with picture symbol “elephant” and with articulogram “open a bit and smile”.

Diphthongs

/ɪə/ Starts from /i/ and moves to open neutral /ə/.

Teach by modifying the long vowel /i/. As the child says /i/, the adult should place a forefinger on his chin to open his jaw. Associate with picture symbol "ear" and with the combined articuloagrams "smile and then open a bit". Stress moving aspect of the sound – use a toy slide or a picture of a slide to demonstrate dynamic movement and articuloagrams to show change of lip shape.

/aɪ/ Starts from /æ/ or /ɑ/ (depending on regional accent) and moves to close /i/.

Teach by modifying either the long vowel /ɑ/ or the short vowel /æ/. As the child says the long or short vowel, help him to move his lips to a smile shape by closing his jaw and pushing the corners of his mouth outwards. Emphasise the moving nature of the sound. Associate with the picture symbol "eye" and the articuloagrams "open and then smile".

/aʊ/ Starts from /æ/ and moves to rounded /u/.

Teach by modifying the short vowel /æ/. As the child says the short vowel help him to close his jaw and change his lip shape, by pushing his jaw closed and rounding his lips. Emphasise the moving nature of the sound. Associate with the picture symbol "bump on the head" and with the articuloagrams "wide open and then round".

/ɛə/ Starts from /ɛ/ and moves to a neutral /ə/.

This sound is often best introduced as a long vowel /ɛ:/, by omitting the second element (NB: in some regional accents, it is produced as a long vowel anyway). Jaw: half-open; lips: spread; tongue: mid tongue raised. Achieve jaw position by using a toothbrush (as for /ɜ/), but lips need to be spread. Gradually modify to a diphthong production, by moving to an open position. Associate with the picture symbol "aeroplane" and the articuloagrams "open a bit and smile, then open".

/eɪ/ Starts from /ɛ/ and moves to /i/.

Teach by modifying the half-open short vowel /ɛ/ or from the diphthong /ɛə/. As the child says the vowel /ɛ/, push his jaw closed and his lips into a smile position. Emphasise the moving nature of the sound. Associate with the picture symbol "angel" and the articuloagram "smile and open a bit, then smile".

/ɔɪ/ Starts from /ɔ/ and moves to /i/.

Teach by modifying the half-open long vowel /ɔ/. As the child says the vowel /ɔ/, push his jaw closed and his lips into a smile position. Emphasise the moving nature of the sound. Associate with the picture symbol "oily parrot" and with the articuloagrams "open a bit and round, then smile".

/əʊ/ Starts from /ɜ/ or /ʌ/ and moves to /u/.

Teach by modifying the long vowel /ɜ/ or from the short vowel /ʌ/. As the child says the vowel, push his jaw closer together and his lips into a rounded shape. Emphasise the moving nature of the sound. Associate with the picture symbol "Obie fish" and the articuloagrams "open a bit, then round".

References

Passy, J. (1990a) Cued Articulation. Ponteland, Northumberland. STASS Publications.

Appendix 5: Ideas for working with younger and older children

The Nuffield Dyspraxia Programme is most suitable for children aged 3-7 years of age, but it can be adapted for use with older or younger children. The following pages provide some ideas of how this might be achieved.

Ideas for working with younger children

Oral or verbal dyspraxia may be diagnosed or suspected in a child of under 3 years of age. Therefore, activities suitable to the developmental level of young children will be required. Try the following:

Sound making

Use suggestions in the Early Sound Making sheets (Therapy Resources 1, p6) to encourage general sound making: babble, animal sounds, environmental sounds, expressive sounds. etc. Ideas for eliciting speech sounds may be tried with young children, but should be left if unsuccessful.

Speech sound work

Most children approaching their third birthday cope with the picture symbols, providing that they are introduced through games and that activities are fun. However, speech sound work can be introduced to younger children by modifying the approach slightly to match their developmental level. Ideas include: adapted picture presentations; real object presentations; social routine presentations.

Suggested activities

- Enlarge the single sound pictures to A5 or even A4 size and colour them. Introduce 2-3 only, at a time. Stick them on the walls and play "listening" games (running to the correct picture when you hear the sound) and "talking" games (saying the sounds in response to the pictures).
- Make a speech sound bag of toys that match the picture symbols e.g. a ball /b/, a toy motorbike /m/, a toy plastic snake /s/, a toy drum /d/, a toy camera /k/, a cuddly worm /w/, a toy crocodile /a/, a toy mouse /i/ etc. Play games matching the toys to the large pictures.
- Use puppets and soft toys to support sound work.
- Use the character names and carrier phrases to introduce vowels e.g. Arthur crocodile says /ah/; I can see with my "eye" etc.
- Use nursery rhyme tunes to introduce speech sounds e.g. to the tune of "The wheels on the bus"- "The bouncing ball goes bbb, bbb, bbb, the bouncing ball goes bbb, all day long."
- Make simple open the flap books, hiding the picture symbols under flaps, which the child can open.

Word level work

Play games utilizing simple CV and VC structures. The adult will need to provide plenty of modelling and reinforcement. Ideas include:

- Building bricks and knocking down: "up"/"down"
- "Go" games - wait until you hear me say "go"
- "Boo" surprise games
- "Bye"/"gone" when posting
- "More" to ask for something you want
- "knock" on the door
- "two" when playing matching games
- "yes/no" choices

Introducing adapted first words

Whilst awaiting the child's developmental readiness to begin formal therapy, it is possible to introduce an adapted vocabulary of first words. The term 'adapted' refers to the way in which the sound pattern of words can be changed to fall within the child's limited articulatory abilities. It is a bid to 'get started' by making best use of 'what the child has got'. Here are some steps to follow:

- Identify any single sounds, VC and CV syllables that the child currently produces as vocalisations for speech or non-speech purposes.
- Identify a small set of "most-wanted" first words.
- Now simplify the phonological form of these conventional adult words so that they fall within the child's current repertoire of vocalisations, taking into consideration both syllable structure and segmental structure. You want to end up with an adapted target that is simple enough for the child to articulate when you present it to them as a model for repetition, e.g. water – [wawa]; smoke – [məʊ]; digger – [dɪdɪ]; apple – [ʌbə].
- Next, practise saying the adapted target the way the child would, i.e. imitate their overall phonetic style. It greatly encourages the child to repeat the model if they hear it spoken as they themselves would.
- Finally, introduce the adapted target alongside the conventional adult form during the course of natural spoken interaction. It is a matter of noticing a moment when the child has a contextual need for the word and 'plugging the gap'.

Speaker	Communication	Sequence of steps
Child	(<i>pointing to tractor in story book</i>) [dɛ]	[contextual need arises]
Therapist	(pointing to same picture) yes that's the farmer's tractor	[referent confirmed]
	(pause)	
	you can say	[invitation to talk]
	(pause)	
	[daʔda]	[adapted target presented]
	(pause)	
	tractor	[conventional form presented]
	(pause)	
	[daʔda]	[adapted target presented again]
Child	[.....]	[possible attempt]
Mother	that's right tractor	[adapted target confirmed]

- It may take you a couple of goes to find a target that is appropriately adapted, i.e. one that they are willing/able to repeat. Equally, the child may alter features of the adapted target that you present. This is all part of working it out together. The aim is to give the child a starting point for the expression of conventionalised meanings. For this reason, working with parents/carers is essential.

Ideas for working with older children

A number of children are not diagnosed with Oral or Verbal Dyspraxia until they are a little older – so it is necessary to have some ideas for working with older children.

These ideas have been used successfully, with children from the age of about 7 years, and can be adapted for a range of ages into the 'early 'teens'.

Basic Principles

Whatever the child's age, if therapy is needed for Oral and/or Verbal Dyspraxia, and this approach has not been used previously, it is important to follow developmental guidelines in planning therapy.

The aim of therapy remains the same – to establish motor programs and develop motor planning skills, in small stages. The challenge in working with older children is to make the basic work acceptable to them.

It is important to gain the child's understanding and cooperation – so an explanation of what is expected is the first step. Aim to explain the principles of therapy and your expectations. As with all therapy for Dyspraxia, unless regular practice is done, progress will not be made – use the analogy of a sportsman, or musician doing basic 'warm-up' exercises every day. Once the child understands, he is much more likely to cooperate.

Try to set a realistic goal for the amount of practice, a chart can be helpful: 7 – 8 year olds still like getting stars or stickers! We suggest that 15 minutes practice, a minimum of 5 times per week, will produce a satisfactory rate of progress.

Speech Sound Work

Working on sound production, can be varied according to the child's age and ability.

It may be more suitable to use letters rather than symbols, - on occasions, we have adapted the Nuffield Dyspraxia Programme using other symbol systems e.g. 'Letterland' and 'Jolly Phonics' as well as letters.

It is important, however, to remember that this is a Programme to develop speech production and sequencing skills, not to develop literacy. If the child's production and sequencing is slowed down by his effort to remember the letters, he will not be able to develop the appropriate motor programs for his speech or develop motor planning skills.

It is also difficult to develop the range of vowels and diphthongs used in English, using letters, so it may be helpful to use symbols alongside the letters, while he is still developing the accurate programs. Many parents and teachers have found that this is a positive advantage for the child's development of phonological awareness.

Consonants and Vowels need to be established in exactly the same way as for younger children, working through production at Single Sound Level, to the early contrast work in V + V, C + C, V + C and C + V.

Suggested activities

At first, the child will need to practise these slowly and carefully, using worksheets with the sounds set out in the desired sequence. Once a satisfactory level of competence is achieved, more challenges can be set. Popular activities include:

- Adding 'boxes' so that the child can monitor his own performance.
- Use a stop-watch, to time the sequence, and challenge for a quicker time by next week, whilst ensuring that accuracy comes before speed!
- Throw a dice, and repeat the sequence that number of times.
- Use a 'magic wand' to point at the sounds in random order.
- Turn the sheet over and use bricks or counters to represent the sequence, so that the child has to remember as well as sequence accurately.
- Ask the child to say the sequence, while you 'mark' him for performance – can be very helpful when you need better attention!
- Games, such as Dominoes and Pairs are fun and the complete sequence of sounds or words can be used at the end of the game, for extra practice.
- It is helpful with all children, but especially important for older children, to include some language/vocabulary aspect in the therapy, so that they see it will be of benefit for their communication. The later parts of the Nuffield Dyspraxia Programme are very useful for older children, especially the Phrase Level work. This can be used earlier in therapy than with a younger child.
- Give a clue or Guessing games, to identify words and phrases also adds some challenge to the work.

Parents or School staff should be involved in setting of work, practising exercises and monitoring progress.

Appendix 6: Tasks/activities/games

Tasks/activities/games for Speech production (output)

Lotto

Materials Lotto bases e.g. for single consonants, single vowels, CV or VC words, matching sets of small cards.

Procedure Players are given a Lotto base each. Matching small cards are mixed up and placed on the table face down between the players. As each person selects a card they must say the sound/word chosen and then give it to the person with the matching Lotto base. At the end of the game players must say all the cards on their board, i.e. four repetitions.

This activity can be made more complex by giving each player two bases.

Pairs

Materials Pairs of small cards (e.g. single sounds, CV, CVC,CVCV etc).

Procedure Place a selection of paired cards on the table, face down in random order. Players turn over two cards per turn and say each sound/word. Cards are kept if they match, turned back over if they do not. Continue until all cards are matched and removed.

Snap

Materials Pairs of small cards (e.g. single sounds, CV, CVC,CVCV etc).

Procedure Mix up the cards and deal them out to the players face down. Each player arranges cards into a pile face down. Take it in turns to lay a card down face up. When two cards the same appear, players shout SNAP. First person to shout picks up the cards and adds them to the back of his pile of cards.

Collect-a-set

Materials Sets (4) of small cards (e.g. single sounds, CV etc).

Procedure Give each player one card of a set (face up on the table) and put the others together in a shuffled pile, face down. Take it in turns to pick up a card and say the sound/word and give it to the person collecting that set, who places the card down next to the original card. The winner is the first person to collect all 4 pictures. At the end say the sequence of sounds/words made.

Marked out boards can be made to help the child understand that he is collecting 4 pictures.

Stepping stones

Materials Small cards (e.g. single sounds, CV, CVC, CVCV etc).

Procedure Place sound/word cards on the floor in an alternating or random sequence. Child steps on each card and produces the appropriate sound/word. If he makes a mistake, he goes back a step.

The same type of game can be played by placing the cards on outlines of footprints, walking to a finish point; lily pads on a pond; ladders etc.

Hunt the pictures

Materials Small cards (e.g. single sounds, CV, CVC, CVCV etc).

Procedure Adult hides pictures around the room. The child travels around the room, looking for a particular item or items. When he finds it/them, he returns to the adult and says the sounds/words depicted on his cards, either singly or in a sequence.

Roll the dice

Materials Single sound symbol cards, CV, CVC words etc and a dice.

Procedure Choose a card and throw the dice. The child has to say the chosen sound/word as many times as the number shown on the dice.

Skittles

Materials Large plastic skittles; small cards (e.g. single sounds, CV, CVC, CVCV etc); sellotape/blutack etc.

Procedure Stick the small cards onto the skittles. Child rolls the ball and tries to knock the skittles down, and then says the sounds/words on the knocked down skittles.

Build a tower

Materials Large building bricks; small cards (e.g. single sounds, CV, CVC, CVCV etc); sellotape/blutack etc.

Procedure Stick a different card on each face of the bricks. Let the child build a tower of bricks and say the sequence of sounds/words starting from the bottom. Knock them down and have another turn. The sequence produced could be "controlled" by putting out some matched cards face down on the table. The child picks up a card and has to find the same card on the bricks and add it to the tower.

Put your hand in the bag

Materials A cloth bag; small cards (e.g. single sounds, CV, CVC, CVCV etc) or real objects.

Procedure Shake the bag. Child puts his hand into the bag and takes out a card/object and says what he has found. After several turns, the child puts his cards/objects into a line and says the sounds/words in a sequence.

Kim's game

- Materials** Small cards (e.g. single sounds, CV, CVC, CVCV etc) or matched objects.
- Procedure** Put the cards out on the table in front of the child. Spend a few minutes looking at them and naming them. Child covers his eyes and adult takes one (or more) away. Child looks again at the cards and tries to say which one has gone. Adult – child take turns to take a card away.

Guess the picture

- Materials** Small cards (e.g. single sounds, CV, CVC, CVCV, multisyllabic etc).
- Procedure** Put some cards out in front of the child. Adult describes one of the pictures and the child tries to guess which one is being described and name the picture. Both adult and child can have turns, depending on the child's level of output.

Treasure hunt

- Materials:** Small counters; small cards (e.g. single sounds, CV, CVC, CVCV etc).
- Procedure:** Place cards with sound/word facing upwards on the table. Place small counters under a random selection of cards whilst players (child, parent etc) have their eyes closed. Players take it in turn to name and turn over the card they think the "treasure" is under. The Winner is the one with most treasure when all the cards have been chosen.

Post box

- Materials** Covered shoe box with slot; small cards (e.g. single sounds, CV, CVC, CVCV etc).
- Procedure** There are a number of variations to this game depending on the target. In its simplest form the players take it in turn to select a card from a single pile, say the appropriate sound/word and then post it.
- For sequencing work use two piles of cards. Players select a card from each, say the sequence of two sounds/words and then post them. For phrase and sentence work have two piles of nouns or one pile of nouns and one of verbs. Players must say the two words before posting. For children with appropriate language skills sentences can be posted on the basis of whether or not they "make sense" as well as correct articulation e.g. "puppy shopping" could not be posted. Depending on the child's articulation and expressive language skills the therapist may request only the two words on the cards or a complete sentence e.g. "the puppy is shopping".

Happy families

Materials Pairs of small cards (e.g. single sounds, CV, CVC etc) mixed in a pack.

Procedure Deal the cards. Players should have five cards each with ten spare to place in the middle of the table. Players take a look at their cards. Any pairs are removed from the hand and placed in front of the player. One player asks the person on the right for a card that they need i.e. "have you got a ...". If the player who has been asked has the card it must be given over to the player who requested it and he then makes a pair and puts it down in front of him. If the player asked does not have the card the player whose turn it is must pick up a card from the pile in the middle. The game finishes when one player gets rid of all the cards in his hand and the winner is the one with the most pairs. Players must say the sounds/words on all their cards at the end of the game.

Fishing

Materials Cardboard "pond" or floor space for a "pond"; stick, string and magnet made into a "fishing rod"; small cards (e.g. lip shapes, single sounds, CV, CVC, CVCV etc) with a large paper clip on each one.

Procedure Child "fishes" a card out of the pond and says each sound or the word "caught".

Colours

Materials Small coloured counters (six different colours); dice with six different coloured sides to match the counters; small cards (e.g. single sounds, CV, CVC etc).

Procedure Place the cards face down on the table. Place one counter on each card (make sure you have at least one of each colour). Players throw the dice and win the card that has the matching coloured counter on it. Players say the sound/word and keep the counter. For sequencing practice, player picks up all the cards of the colour shown on the dice and sequences the appropriate sound or word; or two cards are placed under the one counter.

Shopping

Materials Small cards (e.g. single sounds, CV, CVC, CVCV etc) two of each sound/word; a "shop".

Procedure Adult takes one set of cards and sets up a "shop". Child selects a card from his matching set and must come and "buy" the matching card from the adult. Include target word in a carrier phrase e.g. "I want to buy a".

Choose a brick

Materials Bricks of different colours; a bag; small cards (e.g. single sounds, CV, CVC, CVCV etc) with numbers on the back in different colours that match the coloured bricks.

Procedure Put the bricks into a bag and lay the cards out on the table face down (with the numbers showing). Let the child put his hand in the bag, take out a brick and match it to the coloured number. Turn the card over and say the sound/word the corresponding number of times.

Match the shapes

- Materials** Cards with 3-4 different shapes drawn on them e.g. square, triangle, circle etc; small cards (e.g. single sounds, CV, CVC, CVCV etc) with a shape drawn/stuck on the back with Bluetack.
- Procedure** Lay the cards out on the table face down, with the shapes showing. Give each child a card with a 3-4 shapes pattern on it. Choose from the small cards to make the pattern and say the sequence of sounds/words made.

Cards on backs

- Materials** Small cards (e.g. single sounds, CV words etc); sellotape.
- Procedure** Stick 2 – 3 picture cards on each child's back. Child stands up, with his back facing the other children. One of the children takes a turn to say the sounds he can see (from left to right) and blend them together to make a word e.g. at CV level /m/ + /i/=me; at CVC level /p/+/a/+/k/=park etc and to give a meaning for the word he has made. NB A high level of blending skill is needed for C+V+C. A simpler activity could be played using cards of a CV + C e.g. boo + /t/= boot.

Slide and blend

- Materials** Toy slide or a slide drawn on a large piece of paper; small single sound cards.
- Procedure** Put some CV pictures out face up on the table in front of the child. Put a single sound consonant card at the top of the slide and a vowel card at the bottom. Let the consonant card go down the slide and join the vowel card. Say the sounds as this happens, aiming for a smooth, blended CV sequence. Find the CV picture made from a choice of CV cards on the table.
- A similar activity could be used to support other blending e.g. 2 vowels to create a diphthong; CV+CV to produce a CVCV word etc.

Walk along the path

- Materials** Small cards (e.g. single sounds, CV, CVC, CVCV etc); toy farm or zoo animals; animal house.
- Procedure** Adult puts out 2/3 "paths" of 6+ pictures, a choice of toy animals on the left and an animal house at the end of the paths, on the right. Child chooses an animal and one of the paths and makes his animal walk along the path to his house. Each time the animal steps on a picture, the child says the sound/word.

Make a picture

Materials Small cards (e.g. single sounds, CV, CVC, CCV) or cut up pictures from lottos; worksheets etc.

Procedure Adult draws a large outline picture of an object or character e.g. a tap; a caterpillar etc. The child sticks on small pictures which start with the same onset sound e.g. tap picture: toe, tea, two, tar, toy etc, to create a large picture. He names each picture as he adds it and at the end names all the pictures.

Alternatively, the child can make a large themed picture made up of small pictures with common onsets e.g. a snow scene for initial /s/ clusters: snow, snowman, scarf, sledge, sky, star etc.

The same type of game can be played where the child sticks on pictures with the same rhyme e.g. a large flower outline, with a /i/ vowel picture in the centre and CV pictures in the petals: pea, bee, me, tea, key, sea, etc.

Make a cake

Materials Single sound cards; candle shapes drawn on pieces of paper, with the single sound cards stuck onto them; an outline of a cake drawn on a large sheet of paper.

Procedure Adult hides the candles around the room. Child finds the candles, one at a time and sticks them on the cake. The child says each sound as he adds it to the cake and the whole sequence once all the sounds have been added. Sing Happy Birthday and pretend to blow out the candles!

Puzzles

Materials Photocopied pictures of CV, CVC, CVCV etc words, selected by onset or rime.

Procedure Adult cuts up pictures into two pieces (e.g. a picture of a key), puts one of the pieces into a pile to give to the child and puts the other pieces out on the table, mixed up with other pieces. The child is given a puzzle piece and has to find the matching piece and say the word he has made.

Hungry animal

Materials Puppet/toy animal with large mouth and small picture cards (e.g. single sounds, CV, CVC etc).

Procedure Therapist has control of puppet/animal. Child has to say sound/word on a card and then feed it to the puppet/animal. When all the cards have been "eaten", therapist and child swap roles.

Jumping frog

- Materials** Small plastic frogs (or kangaroos, spiders etc); small cards of single sounds, CV, CVC pictures etc.
- Procedure** Place the cards in a line. As the child produces target sound/word, he can make the frog jump on a picture. Bluetack can be stuck to the frog so that the cards stick to him as he jumps along the line.

Make a book/poster

- Materials** Coloured sheets of paper; photocopied sets pictures-single sounds, CV, CVC etc.
- Procedure** Therapist and child make a small book and child then chooses a picture, names it and sticks it in the book. Repeat activity until all pictures are in the book. Alternatively, he may like to make a small poster, involving a number of pictures. When finished, the child can decorate the book or poster and then take it home for further practice.

Race board game

- Materials** A large sheet of paper with a race track drawn on it, divided into boxes; small cards (e.g. single sounds; CV, CVC, selected by onset, rime, child's system etc); dice and counters.
- Procedure** Play as a traditional board game: shake the dice and move your counter along the track, according to the number thrown. Name the sound/word when you land on a picture. At the end, go along the whole track and say each sound/word.

Television sequences

- Materials** A sheet of paper with a small TV screen drawn on it and two vertical slits one on each side of the TV; long strips of paper with small cards (e.g. single sounds, CV etc) stuck on in a sequence.
- Procedure** Thread the long strips of paper through the TV screen and say the sounds, words as they appear on the screen.

Hang out the washing

- Materials** String and toy clothes pegs/paper clips; small cards: single sounds, CV, CVCV, multisyllabic level etc.
- Procedure** Child and therapist hang up the washing line and take turns to say the sounds/words and hang the cards on the line. When the washing line is full, the child tries to say all the items in a sequence.

Aliens

- Materials** Small cards of single consonants and vowels; a spaceship drawn on paper or made from a bowl; plastic aliens.
- Procedure** Child is shown a single sound consonant picture and a single sound vowel picture, which when blended together do not make a real word e.g. /sh/+/ei/ etc. Child tries to say or repeats CV non-word after therapist. Name is then assigned to a particular alien. Child uses the name to help the alien to find his friends, call his mummy, go home in his spaceship etc.

Tasks/activities/games for input work

Many of the output activities described above, using the Nuffield Centre Dyspraxia Programme materials can be used as input tasks by reversing the child's role: the child becomes the listener, rather than the speaker or for silent tasks, the child becomes the "teacher". In many therapy situations, it is ideal for both input and output activities to be presented "back to back" – however, individual children will need therapy targeted at their particular profile of difficulty and may require more of either input or output activities.

In general, for children with dyspraxic type difficulties, it is not advised to spend time working on listening and discrimination before starting work on production. Start on production from the word "go", but support with input work. Remember that for young children input work can actually be more difficult, especially if there is a "metalinguistic" aspect, requiring the child to reflect and judge (NB A comparison could be drawn with rhyme acquisition in normal-development: production develops first, then judgement). Ensure tasks are explained clearly and are appropriate to the child's age and stage of development –see chapter 5 on therapy approach and chapter 6 on literacy development.

The suggestions below have been contributed by different therapists. Some activities use colours to identify consonants/vowels or onset/rime. Several different colour coding systems are suggested here. It is important to select an appropriate system and use it consistently with an individual child.

Auditory discrimination

Same-different judgements

Materials Pairs of small cards (e.g. single sounds, CV).

Procedure Establish that the child understands the concept of same –different by using pairs of same/different cuddly toys; coloured bricks; farm animals; vehicles; pictures etc. Explain to him that this is a listening game: the adult will say two sounds/words and he has to decide if they are the "same" or "different". The child's response can be: to say "same/different"; to nod his head for "same" and shake his head for "different"; to sign "same/different"; to point to pairs of same/different toys, bricks, animals etc depending on his level.

A similar game can be played using nonwords (to avoid the child accessing lexical representations) instead of real words, at an appropriate level of complexity e.g. CV, CVC etc.

As an introduction to the task, the therapist and child can "decide together" if the sounds/words were same or different.

A fun variation might be to call the pair of same cuddly toys/animals "twins" etc. The child decides if the word is a "twin word" or not.

Minimal pair judgements

- Materials** Small picture cards, involving minimal pairs (e.g. CV/CV; CV/CVC; CVC/CVC etc); coloured counters.
- Procedure** Put out one minimal pair set in front of the child and give him some counters. Explain that this is a listening game: you will say a word and he should put a counter on the picture that goes with the word he hears. Repeat lots of times and randomize the order. Increase the complexity by involving more sets of minimal pairs.
- Alternatively, the child points to a picture rather than putting a counter on it etc.

Barrier game

- Materials** 2 sets of minimal pair cards (e.g. CV/CV; CV/CVC etc) - total 4-8 cards in a set; a book/card etc to act as a barrier.
- Procedure** The child and therapist have a set of the cards each, which they put on the table face up in front of them. Place the barrier between the two players, so that the other player's cards are hidden. The therapist explains that she will choose 3 cards (or as appropriate) and put them next to the barrier and will say their names; the child has to listen and choose the same pictures from his cards and place them next to the barrier on his side. Take the barrier away to check if he has found the correct pictures.
- This game can also be used as an output task, by taking turns for the therapist and child to be listener and speaker.

Choose the right version of the word

- Materials** Small cards (e.g. CV, CVC, CCV etc)
- Procedure** Give the child one picture and explain that this is a listening game. You will say lots of different words and he has to say "yes" or "no", according to whether he thinks the word goes with the picture. This task has been described as "speech error detection" (Locke 1980). The therapist can include the child's error form within the choices presented: is it a fis? is it a tis?; is it a pish?; is it a fish? etc. The task can provide evidence of the accuracy of the child's phonological representations.

Variations on this theme include:

- Materials** Small cards (e.g. CV, CVC, CCV etc); a toy bunny/baby, etc.
- Procedure** The bunny/baby is learning to talk. Sometimes, she says words right and sometimes she gets them wrong. Help her learn to say words properly: she will try and say the name of the picture – you listen and tell her when she has got it right and when she has got it wrong (or thumb up for "right" and thumb down for "wrong") e.g. target: "star" -/da, sa, spa, sta/ etc.
- Materials** Small cards (e.g. CV, CVC, CCV etc); a happy and a sad face.
- Procedure** The child is going to be the "teacher" and will help you say some words properly. You say correct and incorrect versions of the words and the child

has to point to the happy or unhappy face according to whether he thinks you said it correctly or not. You can include both phonemic and phonetic errors according to the child's own error pattern.

Materials Small cards (e.g. CV, CVC, CCV etc); a whistle and a rubbish bin.

Procedure The child listens as you say different versions of a word. The child has to blow a whistle when he spots an error and puts the card in the bin.

Visual discrimination

Match the lip shape

Materials The 4 basic lip shapes – cut out into individual pictures; single sound consonant and vowel cards with appropriate lip shapes (lips together, lips open wide, lips rounded, lips smiling); a bag.

Procedure Put the lip shape pictures in front of the child and the single sound pictures in the bag. The child puts his hand in the bag and takes out a symbol picture. He then has to match it to one of the lip shapes e.g. lips together: /p, b, m/, lips open wide: /a/, lips rounded: /u/, /ʊ/, /w/, lips smiling: /i/.

Alternative presentations include: the adult takes out a picture symbol and says the sound. The child watches and tries to match the sound to the appropriate lip shape; the child says the sound himself and then tries to match it to the appropriate lip shape; once the child has matched the pictures, he says the sounds and checks his lip shape in the mirror.

Other variations on this activity include: using the articulograms and matching the sound symbols to the diagrammatic shapes and/or using pictures of articulatory patterns and matching the symbols to these patterns (once they have been taught to the child).

Segmentation - syllabic level

How many beats?

Materials Small cards of 1, 2 and 3 syllable words.

Procedure Look at the pictures and talk about words having different number of parts/beats to them. You say the words and together with the child, clap or tap out the number of syllables. NB Many children cannot count and clap at the same time!

Variations include: jumping in numbered hoops; stepping on numbered footprints or stepping stones or lily pads according to the number of syllables.

Guess what I'm clapping

- Materials** Set of CVCV and multisyllabic words involving 3 and 4 syllables.
- Procedure** Place the words into piles of 2, 3 and 4 syllables face down on the table. Turn over the top card from each pile, so that the child has a choice of 3 words/pictures. The therapist then claps out one of the words and the child's task is to select the correct picture. If the therapist has a second set of the target words, feedback can be provided to the child visually. To make the task easier, reduce the piles to two sets. To make it more difficult, put the piles out of numerical order.

Stick the picture on the caterpillar

- Materials** Draw a caterpillar outline on a large sheet of paper and number the body segments; small cards of words of 1-5 syllables; blutack.
- Procedure** Look at each card in turn with the child and talk about the number of beats in the word – count them out. Let the child stick the picture onto the appropriately numbered body segment e.g. 1=boy; 2=table; 3=telephone; 4=calculator; 5=hippopotamus etc.
- NB Encourage the child to listen to you saying the word, rather than trying to say it himself –his version may muddle him until his speech is well established at a multisyllabic level!

Split the pictures

- Materials** Pictures of words of 1-5 syllables, enlarged if possible.
- Procedure** Look at each picture with the child and say the word for him. Decide how many beats there are in each word. Draw lines or cut along the lines to divide the picture into the required number of syllables.
- This could be turned into an output blending task by putting the picture back together as you say the word.

Giving presents

- Materials** Small cards (e.g. CV, CVCV, multisyllabic level); sets of small toys e.g. coloured plastic teddies; sets of dinosaurs etc.
- Procedure** Put the toys into groups of: 1, 2, 3, 4, 5. Look at the pictures and listen to each word being spoken. Decide how many beats are in the word and give the card to the appropriate toy group.

Compound nouns

- Materials** Small cards of CVCV compound nouns e.g. cowboy; bowtie; car key etc and small cards of the matched CV words: cow, boy, bow, tie, car, key etc.
- Procedure** Put the pile of compound nouns face down on the table and a selection of CV cards face up in front of the child. Explain to the child that you are going to turn over one of the cards and will say a word e.g. cowboy, which has two parts. The child's task is to find the small words which combine together to make the word e.g. cow and boy. This task may be introduced first using toys to represent the parts e.g. a toy cow and a Playmobile boy etc.
- This segmentation task can be turned into a blending task, either spoken or silent, by reversing the activity. This time a selection of compound nouns is placed face up on the table in front of the child and the CV cards are put into two piles face down in front of the child. The child turns over two cards (either silently or naming each) and then tries to find the compound noun they make (again silently or by naming it).

Segmentation - onset-rime level

In the case or in the bin

- Materials** Small cards (e.g. CV, CVC with a selection of plosive onsets); a case; a bin.
- Procedure** Put the small cards into a pile and explain to the child that you are going to say some words for him. He has to listen and decide if the word starts with a /k/ or not. If it starts with a /k/, it goes in the case; if it doesn't, it gets thrown in the bin. Remember to check that the child understands the concept of "starts/begins with" first.
- Variations on a theme include: using a "sack" for target initial /s/; a "pot" for initial /p/ etc.

Posting boxes

- Materials** 2 posting boxes each with a single sound symbol picture stuck on the front e.g. /t/ and /k/; small cards (CV; CVC etc level) that have /t/ or /k/ onsets.
- Procedure** Put the postboxes on the table or in different parts of the room and the pictures in a pile face down on the table. Explain to the child that you will pick up a small card and will say its name. The child has to decide if it starts with /t/ or /k/ and post it in the appropriate postbox (make sure the child understands "starts with").
- Variations on this theme include: using 2 buckets /baskets /boats/houses etc with symbol pictures on the front and sorting the pictures as described above.
- The task can also be done silently – the child picks up a card, reflects on the onset sound and decides where to put it.
- Similar spoken and silent sorting tasks can also be used, where the target is the rime, rather than the onset e.g. at a CV level, the symbol pictures on the

boxes/buckets etc would be vowels. In this case, use the cue "what does this word sound like?" rather than "what does this word end with" to avoid confusions at CVC level.

Also Coda can be the target at CVC level – "what does the word end with?": listen for words which have /k/, /s/ on the end, and allocate them accordingly – (again check the child understands "ends with").

Secret cupboards

Materials C+V=CV Hiding pictures; a selection of single consonant and vowel cards.

Procedure Cut halfway into the middle line for each sound symbol and fold the paper over to meet in the centre, thereby hiding the single sound pictures. Draw doorknobs to make it look like a cupboard. Look at the CV picture together and talk about the word being made up of two sounds, which are hiding under the doors. Say the word for the child (or present it silently) and see if he can find the sound he thinks it starts with (consonant) and then the sound that it sounds like (vowel). Check his selection by opening the doors.

This activity could be introduced in stages i.e. cover the consonant picture only, but show the vowel picture. See if the child can select the appropriate onset sound from a choice. Alternatively hide the vowel picture only and show the consonant picture. See if the child can select the appropriate rime from a choice.

NB: These segmentation tasks are reversals of the original blending design, in which the CV word is covered and the child blends the C+V sounds together and tries to find the matched CV word.

Yes/no rime judgement - CVC

Materials Small CVC cards/pictures (words ending in: -at; -in; -ig; -og etc); large picture of a cat; blutack.

Procedure Muddle up the small cards and place them face down on the table in front of the child. He turns one over and the adult names the picture. The child then decides if it is an "at" word or if it rhymes with/sounds like: "cat". If it does, the child sticks the card onto the cat picture.

Alternatively this task can be presented silently. In this case, the child turns the card over, reflects about it and decides whether it rhymes with "cat". Use cues e.g. "you're not allowed to say anything" etc to try and keep the task silent.

Rime sorting - CVC

Materials Small CVC cards/pictures (words ending in: -at and -in); large pictures of a "cat" and a "bin"; blutack.

Procedure Muddle up the small cards and place them in a pile face down on the table. The child turns over a card and the adult names the picture. The child decides if it rhymes with "cat" or "bin" / if it is an "at" word or an "in" word. (NB Decide on your cue and be consistent.) He then sticks it on the appropriate large picture. This task can also be presented silently.

Nonword rime judgement - CVC

Materials Toys monsters/aliens with CVC nonword names; a bin.

Procedure The adult says the name of one of the monsters e.g. "Rin" and the child listens. He then decides if the monster's name rhymes with "bin" –if it does, he puts the monster in the bin.

Suitable large pictures to use for all the above tasks include:

Tap –"ap"; cat –"at"; bin- "in"; pig-"ig"; sock-"ock"; sack –"ack"; pop –"op"; dog –"og"; pot –"ot".

Depending on the child's level, they can be labelled by written word- perhaps with the rime in red e.g. "at" and the onset in green e.g. "c". Or by using the picture symbols – again the rime segment could be on red card e.g. picture symbols: "ant and tap", and the onset in green e.g. picture symbol "camera". Again, different levels of support can also be added to the small cards e.g. the word can be written under the picture or on the back of the picture (also in red/green combination); the picture symbols can be drawn under the picture or on the back of the picture (again in red/green); no support is given at all –only the picture is presented.

Rhyme postboxes - CVC

Materials 3 postboxes, labelled with picture symbols/word endings on red cards e.g. "in", "oat", "ack" etc. Small picture cards ending in "in, oat and ack", with picture symbols/words written on the back.

Procedure Put the pictures in a pile face up on the table. The adult names each picture and the child decides which postbox to post the picture in. Alternatively, this can be a silent task- the child looks at each picture, reflects on its name and posts the picture in the correct postbox. At the end of the game, the child can "check" his answers by seeing whether the symbols/word endings match the labels on the front of the postbox.

Matching pairs ABX - CV

Materials Small cards of CV words, with the component picture symbols drawn/stuck on the back: the rime segment is outlined in red and the onset in green.

Procedure Therapist puts out 3 cards in front of the child (two that rhyme and one alliterative distractor) e.g. tie, paw, pie. The child is asked to listen as she says the 3 words and then to choose the two pictures that rhyme. The child then turns the cards over and checks if the rime segment is the same: tie and pie. Use supportive cues e.g. red for rhyme.

The task can also be presented silently.

The rhyme postman - CV

Materials A sack; sets of rhyming picture cards.

Procedure Put one set of rhyme picture cards in the sack and keep the others aside to use as "rhyme detection" cards. Take it in turns to be the postman and deliver a letter to a house e.g. letter received is a picture of a "shoe". Explain that this is a quiet game – it is early in the morning and no one is talking! Put two detection picture cards out in front of the child who receives the letter – one that rhymes with shoe e.g. "two" and one that doesn't e.g. "tea". The child's task is to identify the rhyming pair. He then has a turn to be the postman.

The task can also be presented by adding verbal support i.e. the adult names the words e.g. "you've got shoe. Does that rhyme with two or tea?", but this changes the nature of the task. NB The distractors can be controlled according to the child's level: alliterative alternatives can be added e.g. in example above, choice might be "two (rhyme) and show (alliterative)".

Find the rhyme race track game - CV/CVC

Materials Draw a race track outline; 5/6 pairs of rhyming pictures (CV, CVC levels); dice and counters; tokens.

Procedure Place the pictures on the blank boxes of the track and the counters at the beginning. Explain to the child that this is a silent game –he doesn't have to talk. Take it in turns to throw the dice and move around the board. When the child lands on a picture e.g. "cat", he has to look for a picture that rhymes with it e.g. "hat" and moves to that square. Therapist reinforces with a spoken response e.g. yes "cat and hat" etc and gives the child a token. If the child cannot find the picture, the therapist prompts by saying the target word e.g. "cat" and directing the child's visual attention towards the appropriate picture e.g. "hat". The winner is the person who reaches the end with the most tokens.

This task can also be used for an output task. This time the board should only contain one picture of each of the rhyme pairs – the other squares should be left blank. Explain to the child that this is a talking game. When he lands on a picture, he has to try and say words that rhyme with it. He is given a token for each word he can think of. He is also able to draw a picture of one of the words he thinks of, in a blank box.

Rhyme puzzles - CVC

Materials CVC pictures cut into two parts; the word is written on the picture in onset-rime components (use coloured pens: green for onset; red for rime etc).

Procedure Put out the onset pictures face up on the left hand side of the table and the rime pictures face up on the right hand side of the table. Ask the child to choose one of the puzzle pieces on the left hand side and match it to one of the puzzle pieces on the right hand side. Point out to the child the letters which make up the word (in onset and rime parts).

This game could be extended by having two sets of puzzles: one with the pictures only and one with the letters only – see if the child can match both the picture and orthographic puzzles.

Segmentation - phoneme level

How many sounds are there?

Materials Small cards of CVC words; single sound consonant and vowel symbols.

Procedure Put out a CVC card e.g. "bat" in front of the child and explain that this word is made up of sounds. The aim of the game is to find out how many sounds are in the word and what they are. Help him to work it out, by breaking the task down, using the single sound symbol pictures (in each case, give a choice of 3-4 pictures): (a) which sound does the word start with? – find /b/; (b) which sound is at the end? – find /t/ (c) which sound is in the middle? – find /æ/. Put the single sound pictures underneath the CVC word and let the child say how many sounds are in "bat".

This can also become a spoken output blending task, by asking the child to say the sounds and join them together.

Split up the pictures

Materials Pictures of CVC words; single sound symbol pictures.

Procedure This task is similar to Split the pictures at a syllabic level, but in this case, the child is asked to draw lines/cut the pictures by sounds rather than syllables. As in the previous task, support him in identifying the component sounds by providing choices of 3-4 single sound pictures. At the end draw lines/cut the picture according to the number of sounds.

As in the previous task, the child can also be encouraged to say the sounds and blend them together e.g. "b-a-t" makes "bat" etc.

Find the missing sound

Materials CV+C=CVC hiding pictures, cut up into component parts: CV, C and CVC or small cards of CV and CVC words and single sounds and a question mark drawn on a card.

Procedure Explain to the child that this is a silent game – no talking is involved. Put out a CV card in front of the child e.g. four, followed by the question mark and then the CVC word e.g. fork. Under the question mark, put out 3 of the single sound symbol pictures e.g. for /p/, /t/ and /k/. Explain the activity to the child: to make this word (point to the CVC picture), we need to say this word (point to the CV picture) and add on a sound (point to the single sound pictures). Let's think about it in our heads: this is "four" – we want to make "fork" – the missing sound must be /k/ – so I'll put /k/ on the question mark.

Once the child has selected the sound, the task can be turned into an output task by the child blending the CV+C together to produce the CVC word.

Did I forget it again?

- Materials** 2 sets of CVC words (e.g. photocopied pictures cut up from CVC hiding pictures); leave one set whole and cut the “ends” off the other set; single sound cards.
- Procedure** Explain to the child that this is a listening game. Put out 2 columns of pictures on the table, with 3-4 pictures in a column. Put all the complete words in one column and the incomplete words in the other column. The adult says the words either complete or incomplete in a random order and the child listens and tries to select the appropriate picture. When the child identifies an incomplete word correctly, the therapist should reinforce –yes I said /hɔ/, I left off the /s/ sound (show single sound symbol card). I should have said /hɔs/ and point to the complete word.

Blending

Ribbon game

- Materials** Ribbons; small cards of C, V sounds.
- Procedure** Place some vowel cards spread out on the table. The child chooses a ribbon and a consonant card. Place one end of the ribbon near to the child’s lips and ask him to say the consonant. Move the card down the ribbon as he maintains a long consonant sound until it arrives at the vowel card which he then says e.g. /f/.../ɔ/ to make “four”. This works particularly well with continuant sounds-the ribbon helps to facilitate the blending.

Blending bunny

- Materials** C+V=CV Hiding pictures; small cards of single consonants and vowels; a teddy and a bunny; a barrier.
- Procedure** Put out a selection of the CV Hiding pictures with the component sound pictures folded back underneath the CV or in cupboard doors as above; put the single consonant and vowel pictures in front of the teddy face down. Explain to the child that this is a listening game: teddy will do the talking. The therapist turns over a consonant and vowel card secretly (behind a barrier) and makes teddy say 2 sounds e.g. k+i; the child has to make the bunny jump onto the word he thinks teddy has made: “key”. Remove the barrier and open the cupboard doors and see if the sounds match.
- Include minimal pairs in the CV selection to stretch the child e.g. tea-key etc.

Find the match

Materials C+V=CV hiding or build-up worksheets.

Procedure Cut out the pictures into individual C+V and CV presentations. Put the C+V pictures into a pile face down on the table and put the CV pictures face up on the table.

Explain to the child that you will do the talking: turn over a C+V card and say the sounds. See if the child can select the appropriate CV picture. Remember the task can also be done silently. In addition, the selection of CV words can be controlled to maximize the "distraction" e.g. for /s+ i/ the CV choice might be: "tea" (onset distraction), "saw" (rime distraction), and "sea" (correct item).

Find the picture - CVC

Materials Small cards of CVC words and sound symbol pictures/plastic letters.

Procedure Explain to the child that this is a listening game. Put out the CVC cards in front of the child. The adult says each as a blended onset+rime sequence e.g. "d-og", "c-at" etc. The child selects the appropriate whole CVC word.

To enable the child to check his answers: as the adult says a blended word, she should put out the sound symbols (onset on a green card and the rime on red cards) or plastic letter sequence behind a barrier. Once the child has made his selection, remove the barrier and reinforce the blend for him using the letters/symbols e.g. "d -og" makes "dog" etc.

This activity can also be used when the adult produces a phoneme sequence (rather than an onset-rime sequence) e.g. "d-o-g" and the child tries to find the appropriate CVC word.

Put the picture back together - CVC

Materials CVC pictures split up/cut up into 3 segments; single sound symbol pictures, as appropriate for each segment.

Procedure Keep the onset segments and mix up the other two segments of the pictures and spread them out on the table. Give the child an onset and ask him to find the other two segments to make the whole picture. Help him to identify the single sounds which make up the word and put these out underneath the whole picture. Reinforce for him: yes "b-a-t" makes "bat" etc.

Rhyming spin - CVC

- Materials** Single sound cards, VC word cards; CVC words which rhyme with the VC words e.g. at/cat; up/cup; arm/farm; a line drawing of an archery target with 3 circles; blank, red, green and yellow coloured cards.
- Procedure** Place the green card in the outer circle with the single sound cards face down on top, the yellow card in the next circle with the VC cards on top and the red card in the centre with the CVC cards on top. Check the child understands concept of starts/begins/ends with. Turn over the first single consonant sound and then the VC picture and see if the child can blend the C+VC together. He can try and guess the CVC word on the red card. Discuss with the child how the VC and CVC rhyme e.g. "at and cat" etc.
- This task can also be used for segmentation of either onset or rime. For onset: turn over the red CVC card e.g. 'farm' and the yellow VC card e.g. 'arm' – see if the child can work out what the green single sound card is: /f/. Or for rime: turn over the red CVC card e.g. 'cup' and the green onset single sound card e.g. /k/ – see if the child can work out what the yellow VC card is: 'up'.

Rhyme production

Spin the rhyme wheel - CV

- Materials** Single sound symbol pictures – consonants and vowels; a wooden spinner wheel or a wheel drawn on a large piece of paper, with a pointer attached by a paper fastener; blank red and green cards.
- Procedure** Arrange the single sound consonant pictures around the wheel on the green cards e.g. /b, t, d, ʃ, g, s/ etc; put one vowel picture in the centre on the red card e.g. /əʊ/. Let the child spin the wheel or the pointer and say the consonant selected. Add the vowel to make a word e.g. bow, toe, dough, show, go, sew etc. Spin again. Use cues to support the activity e.g. "remember red for rhyme – the red sound stays the same; the green sound changes".
- NB: depending on the consonants and vowels selected, both real and nonwords may be generated.

Rhyme lotto - CVC

- Materials** Single sound symbols in VC rimes e.g. /aɪ/, /aʊ/ - 2 of each; single sound onset consonants: red and green blank cards.
- Procedure** Put the VC rimes on red cards and the single consonant onsets on green cards. Put a green card in front of each of the red cards and ask the child to say the C+VC =CVC rhyming pairs he has made e.g.: /b +aɪ t/=bite, /f +aɪt/=fight; /m + aʊs/=mouse, /h+aʊs/=house. As the child says the words, remind him "red for rhyme". Check: are they all real words? If not change the onset sound to make a real word.

Blend and rhyme - CVC

Materials Small cards of single sound picture symbols; a bag.

Procedure Put out the picture symbols: ant + tap to represent "at" on a red card. Put the remaining single sound consonant pictures in the bag. Let the child put his hand in the bag, select a card and put it in front of the VC sequence "at". The child blends the C+VC together e.g. "m-at"; "l-at" etc. Both real and nonwords may be generated – the child can also be asked to judge whether he has made a real word or a silly/made up word.

Plastic letters could be used instead of the single sound symbol pictures or in addition.

Rhyme races - CVC

Materials Small cards/pictures of CVC rhyming words – hidden around room.

Procedure Adult shows the picture of a CVC word e.g. "hat". The child is asked to try and think of a word that rhymes with it e.g. "cat, mat, bat" etc. Once he has said a word, he sets off around the room to find the picture and brings it back and says the rhyming pair e.g. "hat –cat" etc.

This activity can be "controlled" by putting out some possible choices of rhyming words in front of the child. Ask him to choose one (but to keep it secret) and try to find the picture of it in the room.

Make a poem

Materials CVC pictures, with a common rime e.g. 'at', 'in' etc.

Procedure Adult and child together try to create poems using the pictures/words. The adult writes out the words on a sheet of paper and the child sticks on the relevant picture e.g.:

"Pat the cat

Went to see the rat"

"Pat the cat

Plays with the bat" etc

Either the adult reads and the child fills in the relevant words from the pictures or the child tries to say the whole poem after the adult.

Appendix 7: Normative issues

Normative information is crucial for practitioners in any field of atypical development. A sound knowledge of what occurs in normal development enables a practitioner to recognise atypical performance and to measure it in terms of how far removed from normal it is.

The following notes provide information from normative studies on aspects of speech development that have relevance for children with speech disorders and specifically those with developmental verbal dyspraxia. Further details can be accessed from the individual papers in the reference list provided at the end of this appendix.

Contents

- The relationship between early vocalisations and speech
- Speech sound emergence theories
- The development of praxis
- The development of oral skills
- The development of diadochokinetic skills (DDK)
- The development of auditory discrimination skills
- Naming and repetition skills
- The development of phonological awareness skills

The relationship between early vocalisations and speech

Children with dyspraxic difficulties often have a history of delayed speech development. Typically, they have not babbled or babbled very late; they have been late in producing their first words and have had difficulties in imitating speech models provided by adults. In addition, many children demonstrate difficulties with breathing for speech and with vocal control-skills, usually mastered in the early months of life.

Information on early speech skills should be obtained from a case history and an informal assessment of voice should be made during the speech assessment. Referral to an ENT Consultant may also be required.

Normative notes from the literature:

Stark (1980)

Speech production skills can be viewed as "occurring in sequences of stages, each of which is related to its predecessors in a coherent way." (CF Jakobson 1941, 1968 on dissociation between "babbling" and "speech").

Vocalisations → babble → jargon → 1st words → word joining.

Oller (1980)

Basic phonation begins at birth when the child cries. By 4 months, vocal play involving squealing, cooing, laughter etc occurs.

Crystal (1986)

From 6 months, the child can produce pitch patterns resembling their mother tongue – involving rising and falling contours. By 18 months, the child demonstrates control of tonicity i.e. they can indicate the nucleus in a phrase through stress.

Byers-Brown et al (1986)

Case history information on early vocalisations and babbling tells us about:- the state of the child's speech production mechanism, the development of motor control, the interaction between vocalising and listening (turn-taking) and the child's responsiveness to prosodic features.

Vihman (1986; 1996)

Phonological development occurs in stages from babbling to speech:

Pragmatic Stage	-	6 months
Cognitive Stage	-	12 months
Systematic Stage	-	18 months

Stackhouse and Wells (1997)

The child passes through a series of identifiable stages in developing normal speech:

The Prelexical Phase (1st year of life)

The Whole word phonology phase (12 months + i.e. 2nd year of life)

The Systematic simplification phase (during 3rd year of life)

The Assembly phase (around 4th year of life)

The Metalinguistic phase (5th year of life)

Children with speech difficulties can be viewed as having arrested, problematic or slow development through the phases. Children with developmental verbal dyspraxia are thought to have arrested at the whole word stage - utterances produced as "whole unit" with little evidence of systematic patterns occurring.

Speech sound emergence

Children with dyspraxic difficulties usually have difficulty in producing a full range of consonant and vowel sounds. It is not simply the case that they cannot use a full range of speech sounds in their speech, rather that they cannot even make some sounds at an isolated single sound level. This can affect vowel sounds as well as consonant sounds. Assessment of both speech sounds in words and in isolation should be included in a detailed speech assessment.

Normative notes on speech sound emergence:

Menn (1983)

Speech development involves the acquisition of phonetic control and phonemic contrasts. The order of acquisition of speech sounds is largely a trial and error affair but there are physiological constraints.

Hewlett (1990)

Model of phonological processing and speech production - accounts for the interface of speech motor control and phonological development.

Oller (1980)

In vocal play, velar consonants occur early - the "Goo" stage, where the infant lying on his/her back makes lingual/velar contact when vocalising. Other consonants occur in reduplicated babble from about 6 months e.g. d, b, m.

Grunwell (1987)

Stages of development for phonemes (consonants) and phonotactic (syllabic) structures.

From about 9 months the child can produce sounds from some or all of the nasal, plosive, fricative and approximant groups, initially in CV syllables and then CVCV syllables. In early words (from 1.0 year), the bilabial and alveolar plosives and nasals and the labial approximant occur (p, b, m, t, d, n, w). By 3 years, the child can produce a full range of phonotactic structures and by 4.6 years, a full range of phonemes.

Ball and Gibbon (eds) (2002) Far less written about vowels than consonants, but most vowels are established by 2 years of age and the system is complete by 3 years of age. Vowel emergence develops from feeding movements combined with vocalisation from 2 months of age.

Brereton (2003) Investigated the performance of 47 normally-developing children aged 2-4 years on the Nuffield Dyspraxia Programme Assessment: Single sounds section.

Results:

- (a) a significant developmental progression was found in the number of single sounds imitated correctly by 2, 3 and 4 year old children. 4 year olds imitated the most consonants and vowels correctly and 2 year olds imitated the least sounds correctly.
- (b) There was a significant difference between the vowel imitation skills of 3 and 4 year olds but not 2 and 3 year olds.
- (c) There was a different pattern of progression for the imitation of vowels and consonants. Vowels could be imitated correctly earlier: 2 year olds imitated 94% vowels correctly but only 69% consonants.
- (d) The most common vowel error was reduction of diphthongs e.g. /εə/->/ε/; /Iə/->/I/ made by 2 and 3 year olds. 2 year olds also made some monothong errors e.g. /ε/->/Λ/ and /I/->/i/.

Williams and Stackhouse (2000) The group of 3 year olds were more variable in their performance on single sound consonant repetition than the groups of 4 and 5 year olds. The most common errors were: /θ/->/f/; /ð/->/v/, /r/->/w/ and dentalization of /s/ and /z/.

Lund and Duchan (1993)

Considerable variation in the age at which children can produce individual speech sounds correctly in all positions in words. For example, /d/ can be produced correctly in all positions in words by 50% of 2 year olds and 90% of 4 year olds whereas /s/ can only be produced correctly in all positions in words by 50% of 3 year olds and 90% of 8 year olds.

Praxis

Children with dyspraxic difficulties have difficulties with praxis, by definition. The development of praxis affects all motor skills – see Ayres (1986) "Praxis is a uniquely human skill enabling the brain to conceptualise, organise and direct purposeful interaction with the physical world." Information on the child's motor development can be obtained through a case history. Referral may need to be made to an Occupational therapist.

Some authors have clarified the meaning of praxis when applied to speech development e.g.:

Kent (1976)

"The ability to integrate and organise lower functions (e.g. oro-motor movements) to achieve new ends (e.g. speech production)".

Kent (1984) on motor skills learning:

"How abstract linguistic units e.g. phonemes can make contact with the physical events of articulatory control."

Milloy (1990) on speech praxis:

"When motor skills reach the stage of automatic movement to enable them to produce reliable articulation on each attempt at production."

Assessment of diadochokinetic skill (DDK) - see below can provide information on the development of speech praxis.

Normative notes on the age by which praxis is achieved:

Martin (1981) A high level of verbal praxis by 3 years.

Kools and Tweedie (1975); Milloy (1990); Limb, oral and motor skills for articulation by 6 years.

Kent (1976); Miller (1986) Praxis is not fully achieved until 12 years.

Oral motor skills

Many children with verbal dyspraxia also have oro-motor dyspraxia. Sometimes they also have feeding difficulties. Such difficulties may have occurred at an earlier point in the child's development and have subsequently been overcome or they may still be current. Assessment of oro-motor skills should be a routine part of a detailed speech assessment.

Normative notes on oro-motor skills:

Sharkey and Folkins (1985) Study of variability of lower lip and jaw movements:

Major development in oro-motor abilities occurs up to age 4 years and then oro-motor system goes through a process of "fine tuning."

K. Sharples (1989) Pilot study of the Nuffield Centre Dyspraxia Programme assessment:

Oro-motor assessment easily achieved by 4 years olds (80% + level).

Ozanne (1992)

Study of sequenced oral movements and movements in context for children aged 3-5 years.

- Below 4 years - there is wide variability in oro-motor functioning.
- Above 4 years - most children can perform routine oro-motor tasks.
- The items assessed were not all at the same level of difficulty - those involving tongue elevation were particularly difficult for young children.
- Young children were willing to co-operate with the movements in context subtest (involving tasks from the NDP e.g. licking and blowing), but sometimes refused to co-operate with producing oral movements on command/demonstration.

Winstock (1994)

Tongue elevation develops in feeding between 6 and 12 months of age.

Williams and Stackhouse (2000)

Study of 30 normally-developing children aged 3-5 years, assessed on production of isolated and sequenced oral movements:

- Tongue elevation difficulties shown by:
3 year olds 4 year olds 5 year olds
7/10 4/10 7/10
- Difficulties on silent, lip and tongue DDK sequences (x 5):

	3 year olds	4 year olds	5 year olds
Lips	4/10	1/10	1/10
Tongue	7/10	3/10	5/10

Diadochokinetic skills (DDK)

Children with developmental verbal dyspraxia are frequently reported in the literature to have difficulties with diadochokinetic tasks. Difficulties present as slow rates of production and/or sequencing difficulties. Tasks involving DDK skills should be included in a routine speech assessment. They can be useful for assessing motor programming and planning skills.

Normative notes on DDK skills:

Canning and Rose (1974)

300 children between ages 4½-9½ years and 13½-14½ years. Assessed speeds of silent tongue movements, repetition of individual consonants and repetition of a sequence of 3 consonants.

- Speed of fine motor movements of the lips and tongue continues to grow with age.
- Wide range of speeds within each age group.

Fletcher (1972/8)

384 children between ages 6-13 years assessed on repetition of single consonants, 2 consonant sequences and 3 consonant sequences.

- Speed of performance increased with age on all 3 subtests.

Henry (1990)

60 normal children and 30 speech disordered children assessed on DDK rates (repetition of single consonants, 2 consonant sequences and 3 consonant sequences), a non-linguistic rhythm task and an auditory memory task.

- DDK skills are developmentally age related in normal children.
- Children performing with the greatest ease on DDK tasks organised their repetitions rhythmically.

Williams and Stackhouse (2000)

Study of 30 normally-developing children, aged 3-5 years, assessed on spoken DDK tasks - repetition of real words, nonwords and syllable sequences of 2 and 3 syllable length:

- score for accuracy, consistency as well as rate. Accuracy and consistency more meaningful measures than rate with young children. Individual 3 year olds vary in their performance but generally are highly consistent although they may be inaccurate; 4 year olds are generally both accurate and consistent;
- speed of performance increases gradually with age, but there is little difference in rates of production for 3, 4 and 5 year olds. All age groups could produce five repetitions of 2 syllable sequences in under 3 seconds and five repetitions of 3 syllable sequences in under 4.5 seconds.
- DDK assessments should ideally include different types of stimuli - real words, nonwords and syllable sequences. Performance similar across the stimuli for normally-developing children, but may not be the case for children with speech disorders.

Prathanee et al (2003) The findings on DDK rates differed for a group of English speaking children and a group of Thai children, with the Thai children taking longer to produce the DDK sequences. Possible reasons given by the authors include: (a) in the Thai study, the vowel /ʌ/ was used, which takes longer to articulate than the vowel /ə/ used in English-speaking studies; (b) Thai children are shorter and smaller on average than Western children and therefore have smaller lung volume. They concluded that normative data for DDK rates from specific language and racial groups (i.e. body builds) should be collected and made available for clinical comparisons.

Auditory discrimination

Children with speech production difficulties may also have auditory processing difficulties and assessment of auditory discrimination skills should be a routine part of a detailed speech assessment.

Notes on auditory discrimination norms:

Barton (1980) Speech sound discrimination skills have been studied far less than production skills. Most normally-developing children can correctly perceive the majority of contrasts in their ambient language by about 2 years and generally perceptual skills are ahead of production skills.

Templin (1957) Speech sound discrimination abilities continue to develop up to 8 years, but the majority are established much earlier.

Naming and repetition skills

Many speech assessments require a child to name pictures or to repeat a word or nonsense word after the tester. Normative information on children's abilities to carry out such tasks is essential to interpret an individual child's performance.

Normative information

Vance, Stackhouse and Wells (1995)

100 normally-developing children, aged 3-7 years (20 in each age group), assessed on 4 tasks: picture naming, real word repetition, non-word repetition and sentence repetition. The single words included items of 1, 2, 3 and 4 syllables.

Results:

- Performance on all tasks increased with age, suggesting they were developmentally sensitive.
- Real word repetition was the easiest task, for all the children.
- Accuracy was poorer on longer words of 3 and 4 syllables.
- The difference between real and non-word scores was less marked for the younger children (who are still in the process of learning new words). Children of 5 + years tended to have a different profile, with a greater discrepancy between real and non-word performance.

NB: These findings are of note for clinical assessment – look for patterns of performance, across a range of tasks.

Phonological awareness skill norms

Many children with speech disorders including developmental verbal dyspraxia have difficulty with phonological awareness skills and therefore assessment of this area should be included in a detailed speech assessment, using age appropriate assessment tools.

Phonological awareness refers to the ability to reflect on and manipulate the structure of an utterance as distinct from its meaning. It is tested through tasks e.g. segmentation tasks, blending tasks, manipulation tasks etc. The development of phonological awareness skills is crucial for literacy development. Studies have demonstrated that it is possible to improve PA through training, however, this will not lead to an automatic improvement in literacy development. In order to improve reading and spelling skills, children need to link their PA knowledge with other literacy skills such as letter knowledge and reading experience (Bradley and Bryant 1983; Hatcher 1994).

Segmentation

Words can be segmented in different ways i.e. into syllables, into onset + rimes and into phonemes (Goswami and Bryant 1990).

Segmentation skills develop gradually over time, in normally-developing children:

- Syllable segmentation pre-school, around 4 years.
- Onset-rime segmentation at around 5 years.
- Phoneme segmentation at around 6/7 years (once literacy skills have developed).
- More complex manipulation tasks e.g. phoneme deletion and spoonerisms 8 years+.

NB: Consider this normal development when using input therapy activities, especially with pre-school children e.g. sorting by first/last sound etc.

Children with speech disorders often find PA tasks very difficult, as a result of their underlying speech processing deficits.

Blending

Blending tasks can be silent (listen to spoken components e.g. c-a-t and match it to a picture or written word), or involve production (listen to the components and then say the word they make f-i-sh -> "fish").

Usually regarded as tasks used with school aged children i.e. at least 5 years+.

NB: Children with speech disorders tend to find production tasks very difficult, but may achieve on silent tasks.

Consider the child's age when using blending tasks in therapy e.g. spoken cue "m - ee" makes "me"; "p - ar - k" makes "park" etc.

Rhyme

Although other phonological awareness skills are thought to be more strongly predictive of later literacy development e.g. phoneme segmentation and manipulation, a sensitivity to rhyme has been identified as being a prerequisite to developing phoneme segmentation skills (Bryant, MacLean, Bradley and Crossland 1989) and to be a precursor to reading by analogy (Goswami 1986).

Normative notes on rhyme from the literature:

Vance, Stackhouse and Wells (1994)

100 normally-developing children, aged 3-7 years (20 in each age group) assessed on a range of input and output rhyme tasks.

Results:

- Performance on all tasks increased with age, indicating that they are developmentally sensitive.
- 3 year olds had tacit rhyme awareness e.g. showed nursery-rhyme knowledge. Findings more reliable from 4 years.
- Production skills were ahead of judgement skills (in line with other studies), only coming in line at 7 years.
- Verbal rehearsal is a common strategy on judgement tasks in 4, 5 and 6 year olds.
- Scores on real and non-word input tasks were non-significant, suggesting that normally-developing children dealt with familiar and novel material in a similar way, presumably without accessing their representations.

Clinical implications:

- Children with DVD have ongoing problems with rhyme production, after they have developed strong rhyme judgement skills.
- Verbal rehearsal unlikely to be useful to a child with severe speech difficulties.
- Children with speech difficulties may show a difference in performance on real and nonword judgements, finding novel material more difficult to process.

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Addendum

Please note that further information on the normative studies by: Vance, Stackhouse and Wells and Williams and Stackhouse, will be available in: Vance, Stackhouse and Wells (in press) "Compendium of Speech and Auditory Procedures for the Psycholinguistic Framework: Children's Speech and Literacy Difficulties Book 4". Wiley Publishers.

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